

## Project Data Page:





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## **DESIGN CALCULATION**

*In Accordance with ASME Section VIII Division 1*

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\C-14020peRev 0.pvdb

Date of Analysis : Jul 23,2026 9:26am

**PV Elite 28 U2, April 2026**



## Vessel Design Summary:

### ASME Code, Section VIII Division 1, 2025

|                                                   |               |
|---------------------------------------------------|---------------|
| Diameter Spec : 5906.000 mm ID                    |               |
| Vessel Design Length, Tangent to Tangent          | 25600.00 mm   |
| Distance of Bottom Tangent above Grade            | 41000.00 mm   |
| Distance of Base above Grade                      | 19000.00 mm   |
| Specified Datum Line Distance                     | 22000.00 mm   |
| Internal Design Temperature                       | 150 °C        |
| Internal Design Pressure                          | 0.343 MPa     |
| External Design Temperature                       | 150 °C        |
| External Design Pressure                          | 0.103 MPa     |
| Maximum Allowable Working Pressure                | 0 MPa         |
| External Max. Allowable Working Pressure          | 0 MPa         |
| Shop Test Pressure                                | 0.648 MPa     |
| Field Test Pressure                               | 0.648 MPa     |
| Required Minimum Design Metal Temperature         | -1.0 °C       |
| Warmest Computed Minimum Design Metal Temperature | -29.0 °C      |
| Wind Design Code                                  | ASCE/SEI 7-16 |
| Earthquake Design Code                            | ASCE/SEI 7-16 |

### Materials of Construction:

| Component Type | Material  | Class | Thickness | UNS #    | Material Treatment Options |
|----------------|-----------|-------|-----------|----------|----------------------------|
| Shell          | SA-516 70 | ...   | ...       | K02700   |                            |
| Head           | SA-516 70 | ...   | ...       | K02700   |                            |
| Flange         | SA-266 2  | ...   | ...       | K03506   |                            |
| Skirt          | E250BR    | ...   | t > 40    | SA/IS 20 |                            |
| Skirt          | SA-516 70 | ...   | ...       | K02700   |                            |
| Nozzle         | SA-516 70 | ...   | ...       | K02700   |                            |
| Nozzle         | SA-106 B  | ...   | ...       | K03006   |                            |
| Nozzle         | SA-105    | ...   | ...       | K03504   | +IT                        |
| Re-Pad         | SA-516 70 | ...   | ...       | K02700   |                            |
| Nozzle Flg     | SA-105    | ...   | ...       | K03504   |                            |
| Rings          | SA-516 70 | ...   | ...       | K02700   |                            |
| Basing         | E250BR    | ...   | t > 40    | SA/IS 20 |                            |
| Flg Bolting    | SA-193 B7 | ...   | <= 64     | G41400   |                            |
| Base Bolting   | SA-36     | ...   | ...       | K02600   | +IT                        |

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized  
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

### Element Pressures and MAWP (MPa & mm):

| Element Description or Type | Design Pressure + Stat. head | Ext. Press. | Element M.A.W.P | Total Corrosion Allowance | Str. Flg. Gov. | In Creep Range |
|-----------------------------|------------------------------|-------------|-----------------|---------------------------|----------------|----------------|
| Skirt-9                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-8                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-7                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-6                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-5                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-4                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-3                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-2                     | ...                          | ...         | ...             | ...                       | N/A            | N/S            |
| Skirt-1                     | ...                          | ...         | ...             | ...                       | N/A            | No             |
| Btm Dish End                | 0.392                        | 0.10        | 1.000           | 1.0000                    | No             | No             |



|              |       |      |       |        |     |    |
|--------------|-------|------|-------|--------|-----|----|
| Shell-1      | 0.382 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-2      | 0.353 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-3      | 0.343 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-4      | 0.343 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-5      | 0.343 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-6      | 0.343 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-7      | 0.343 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-8      | 0.343 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Shell-9      | 0.343 | 0.10 | 1.000 | 1.0000 | N/A | No |
| Top Dish End | 0.343 | 0.10 | 0.000 | 1.0000 | No  | No |
| N2A FLG      | 0.344 | 0.10 | 1.000 | 1.0000 | N/A | No |
| N2B FLG      | 0.344 | 0.10 | 1.000 | 1.0000 | N/A | No |

N/S in the Creep Range column specifies that the Creep Range Temperature was not specified in the ASME Code Section II Part D for this material.

Liquid Level: 5041.87 mm Dens.: 0.001 kg/cm<sup>3</sup> Sp. Gr.: 0.987

#### Stiffener Ring Specifications:

| Elevation mm | Selected Type | Ring Description |
|--------------|---------------|------------------|
| 1000.00      | W8X21         | SR 1 Fr 70       |
| 3000.00      | W8X21         | SR 2 Fr 70       |
| 5000.00      | W8X21         | SR 3 Fr 80       |
| 7000.00      | W8X21         | SR 4 Fr 90       |
| 9000.00      | W8X21         | SR 5 Fr 90       |
| 11000.00     | W8X21         | SR 6 Fr 100      |
| 13000.00     | W8X21         | SR 7 Fr 110      |
| 15000.00     | W8X21         | SR 8 Fr 110      |
| 17000.00     | W8X21         | SR 9 Fr 120      |
| 19000.00     | W8X21         | SR 10 Fr 130     |
| 21000.00     | W8X21         | SR 11 Fr 130     |
| 23000.00     | W8X21         | SR 12 Fr 140     |
| 24999.99     | W8X21         | SR 13 Fr 150     |

#### Element Types and Properties:

| Element Type | "To" Elev mm | Element Length mm | Nominal Thickness mm | Finished Thickness mm | Reqd Thk Internal mm | Reqd Thk External mm | Long Eff | Circ Eff |
|--------------|--------------|-------------------|----------------------|-----------------------|----------------------|----------------------|----------|----------|
| Skirt        | -20000.0     | 2000.0            | 36.0                 | 36.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | -17000.0     | 3000.0            | 22.0                 | 22.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | -14000.0     | 3000.0            | 22.0                 | 22.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | -11000.0     | 3000.0            | 20.0                 | 20.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | -8000.0      | 3000.0            | 20.0                 | 20.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | -5000.0      | 3000.0            | 18.0                 | 18.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | -2000.0      | 3000.0            | 18.0                 | 18.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | -1000.0      | 1000.0            | 16.0                 | 16.0                  | ...                  | ...                  | 0.70     | 0.70     |
| Skirt        | 0.0          | 1000.0            | 16.0                 | 16.0                  | ...                  | ...                  | 0.70     | 0.55     |
| Torisph      | 50.0         | 50.0              | 26.0                 | 22.0                  | 16.0                 | ...                  | 1.00     | 1.00     |
| Cylinder     | 3050.0       | 3000.0            | 16.0                 | 16.0                  | 9.2                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 6050.0       | 3000.0            | 16.0                 | 16.0                  | 8.6                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 9050.0       | 3000.0            | 14.0                 | 14.0                  | 8.4                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 12050.0      | 3000.0            | 14.0                 | 14.0                  | 8.4                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 15050.0      | 3000.0            | 14.0                 | 14.0                  | 8.4                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 18050.0      | 3000.0            | 14.0                 | 14.0                  | 8.4                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 21050.0      | 3000.0            | 14.0                 | 14.0                  | 8.4                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 24050.0      | 3000.0            | 14.0                 | 14.0                  | 8.4                  | ...                  | 1.00     | 1.00     |
| Cylinder     | 25550.0      | 1500.0            | 14.0                 | 14.0                  | 8.4                  | ...                  | 1.00     | 1.00     |
| Torisph      | 25600.0      | 50.0              | 24.0                 | 20.0                  | 14.1                 | ...                  | 1.00     | 1.00     |
| Body Flg     | 4000.0       | 148.0             | 104.0                | 98.0                  | 82.0                 | ...                  | 1.00     | 1.00     |
| Body Flg     | 4000.0       | 142.0             | 104.0                | 98.0                  | 82.0                 | ...                  | 1.00     | 1.00     |

#### External Pressure Calculations:

| From | To | External Actual T. mm | External Required T. mm | External Design Pressure MPa | External M.A.W.P. MPa |
|------|----|-----------------------|-------------------------|------------------------------|-----------------------|
|------|----|-----------------------|-------------------------|------------------------------|-----------------------|



|      |      |     |         |         |         |
|------|------|-----|---------|---------|---------|
| 10   | 20   | ... | No Calc | ...     | No Calc |
| 20   | 30   | ... | No Calc | ...     | No Calc |
| 30   | 40   | ... | No Calc | ...     | No Calc |
| 40   | 50   | ... | No Calc | ...     | No Calc |
| 50   | 60   | ... | No Calc | ...     | No Calc |
| 60   | 70   | ... | No Calc | ...     | No Calc |
| 70   | 80   | ... | No Calc | ...     | No Calc |
| 80   | 90   | ... | No Calc | ...     | No Calc |
| 90   | 100  | ... | No Calc | ...     | No Calc |
| 100  | 110  | 22  | 18.1871 | 0.10346 | 0.15448 |
| 110  | Ring | 16  | 11.1976 | 0.10346 | 0.26584 |
| Ring | Ring | 16  | 13.1185 | 0.10346 | 0.17755 |
| Ring | 120  | 16  | 13.1185 | 0.10346 | 0.17755 |
| 120  | Ring | 16  | 13.1185 | 0.10346 | 0.17755 |
| Ring | 130  | 16  | 13.1185 | 0.10346 | 0.17755 |
| 130  | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | 140  | 14  | 13.1136 | 0.10346 | 0.12469 |
| 140  | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | 150  | 14  | 13.1136 | 0.10346 | 0.12469 |
| 150  | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | 160  | 14  | 13.1136 | 0.10346 | 0.12469 |
| 160  | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | 170  | 14  | 13.1136 | 0.10346 | 0.12469 |
| 170  | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | 180  | 14  | 13.1136 | 0.10346 | 0.12469 |
| 180  | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | 190  | 14  | 13.1136 | 0.10346 | 0.12469 |
| 190  | Ring | 14  | 13.1136 | 0.10346 | 0.12469 |
| Ring | 220  | 14  | 9.68541 | 0.10346 | 0.24792 |
| 220  | 230  | 20  | 18.1699 | 0.10346 | 0.12671 |
| 230  | 240  | 98  | 81.9658 | 0.10346 | No Calc |
| 240  | 250  | 98  | 81.9658 | 0.10346 | No Calc |

**External Pressure Calculations:**

| From | To   | Actual Length<br>Bet. Stiffeners<br>mm | Allowable Length<br>Bet. Stiffeners<br>mm | Ring Inertia<br>Required<br>mm <sup>4</sup> | Ring Inertia<br>Available<br>mm <sup>4</sup> |
|------|------|----------------------------------------|-------------------------------------------|---------------------------------------------|----------------------------------------------|
| 10   | 20   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 20   | 30   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 30   | 40   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 40   | 50   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 50   | 60   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 60   | 70   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 70   | 80   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 80   | 90   | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 90   | 100  | 22000                                  | No Calc                                   | No Calc                                     | No Calc                                      |
| 100  | 110  | No Calc                                | No Calc                                   | No Calc                                     | No Calc                                      |
| 110  | Ring | 1332.02                                | 3317.59                                   | No Calc                                     | No Calc                                      |
| Ring | Ring | 2000                                   | 3317.34                                   | 24735462                                    | 59346300                                     |
| Ring | 120  | 2000                                   | 3317.34                                   | 29783172                                    | 59346300                                     |
| 120  | Ring | 2000                                   | 3317.34                                   | No Calc                                     | No Calc                                      |
| Ring | 130  | 2000                                   | 3317.34                                   | 29783164                                    | 59346300                                     |
| 130  | Ring | 2000                                   | 2373.49                                   | No Calc                                     | No Calc                                      |
| Ring | Ring | 2000                                   | 2373.49                                   | 29630336                                    | 56163728                                     |
| Ring | 140  | 2000                                   | 2373.49                                   | 29630336                                    | 56163728                                     |
| 140  | Ring | 2000                                   | 2373.49                                   | No Calc                                     | No Calc                                      |
| Ring | 150  | 2000                                   | 2373.49                                   | 29630336                                    | 56163728                                     |
| 150  | Ring | 2000                                   | 2373.49                                   | No Calc                                     | No Calc                                      |
| Ring | Ring | 2000                                   | 2373.49                                   | 29630336                                    | 56163728                                     |
| Ring | 160  | 2000                                   | 2373.49                                   | 29630336                                    | 56163728                                     |
| 160  | Ring | 2000                                   | 2373.49                                   | No Calc                                     | No Calc                                      |
| Ring | 170  | 2000                                   | 2373.49                                   | 29630336                                    | 56163728                                     |
| 170  | Ring | 2000                                   | 2373.49                                   | No Calc                                     | No Calc                                      |
| Ring | Ring | 2000                                   | 2373.49                                   | 29630336                                    | 56163728                                     |



|      |      |         |         |          |          |
|------|------|---------|---------|----------|----------|
| Ring | 180  | 2000    | 2373.49 | 29630336 | 56163728 |
| 180  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | 190  | 2000    | 2373.49 | 29630336 | 56163728 |
| 190  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | 220  | 932.144 | 2374.16 | 21727174 | 56163728 |
| 220  | 230  | No Calc | No Calc | No Calc  | No Calc  |
| 230  | 240  | No Calc | No Calc | No Calc  | No Calc  |
| 240  | 250  | No Calc | No Calc | No Calc  | No Calc  |

#### Local Stress Analysis Results:

| Description | Analysis Type | Max Stress Ratio | High Stress Location | Pass Fail |
|-------------|---------------|------------------|----------------------|-----------|
| N1 36"      | WRC-107/537   | 0.621            | n/a                  | Passed    |
| N11 2"      | WRC-107/537   | 0.562            | n/a                  | Passed    |
| N9          | WRC-297       | 0.819            | Shell                | Passed    |
| N8 2"       | WRC-297       | 0.819            | Shell                | Passed    |
| N2a         | WRC-297       | 0.923            | Shell                | Passed    |
| N2b         | WRC-297       | 0.923            | Shell                | Passed    |
| N3A         | WRC-297       | 0.896            | Shell                | Passed    |
| N4a         | WRC-297       | 0.971            | Shell                | Passed    |
| N4b         | WRC-297       | 0.971            | Shell                | Passed    |
| N3B         | WRC-297       | 0.855            | Shell                | Passed    |
| N5          | WRC-297       | 0.947            | Shell                | Passed    |
| N6          | WRC-297       | 0.819            | Shell                | Passed    |
| N7          | WRC-107/537   | 0.604            | n/a                  | Passed    |
| N10         | WRC-107/537   | 0.565            | n/a                  | Passed    |

#### Wind/Earthquake Shear, Bending:

| From | To  | Distance to Support mm | Cumulative Wind Shear N | Earthquake Shear N | Wind Bending N-mm | Earthquake Bending N-mm |
|------|-----|------------------------|-------------------------|--------------------|-------------------|-------------------------|
| 10   | 20  | 1000                   | 203180                  | 616549             | 5.501E+09         | 22.35E+09               |
| 20   | 30  | 3500                   | 196182                  | 616331             | 5.102E+09         | 21.11E+09               |
| 30   | 40  | 6500                   | 185709                  | 615746             | 4.529E+09         | 19.26E+09               |
| 40   | 50  | 9500                   | 175236                  | 614465             | 3.987E+09         | 17.42E+09               |
| 50   | 60  | 12500                  | 164767                  | 612585             | 3.477E+09         | 15.58E+09               |
| 60   | 70  | 15500                  | 154298                  | 609926             | 2.998E+09         | 13.74E+09               |
| 70   | 80  | 18500                  | 143832                  | 606785             | 2.551E+09         | 11.92E+09               |
| 80   | 90  | 20500                  | 133366                  | 602856             | 2134635904        | 10.10E+09               |
| 90   | 100 | 21500                  | 129879                  | 601531             | 2002959872        | 9.498E+09               |
| 100  | 110 | 22025                  | 126392                  | 600124             | 1874772864        | 8.897E+09               |
| 110  | 120 | 23550                  | 126211                  | 582741             | 1868455424        | 8.868E+09               |
| 120  | 130 | 26550                  | 115387                  | 519413             | 1505911424        | 7.214E+09               |
| 130  | 140 | 29550                  | 104563                  | 449871             | 1175851904        | 5.759E+09               |
| 140  | 150 | 32550                  | 91195.3                 | 415806             | 882094848         | 4.460E+09               |
| 150  | 160 | 35550                  | 76089.6                 | 349824             | 631065664         | 3.311E+09               |
| 160  | 170 | 38550                  | 65272.9                 | 321477             | 418936064         | 2.304E+09               |
| 170  | 180 | 41550                  | 50324.4                 | 261621             | 245469776         | 1428925824              |
| 180  | 190 | 44550                  | 35305                   | 231306             | 116973728         | 689236864               |
| 190  | 220 | 46800                  | 20221.4                 | 102629             | 33650432          | 188132544               |
| 220  | 230 | 47575                  | 14813.1                 | 89087.1            | 7363879           | 44287000                |

Abs Max of the all of the Stress Ratio's : 0.7879

#### Basering Data : Continuous Top Ring W/Gussets

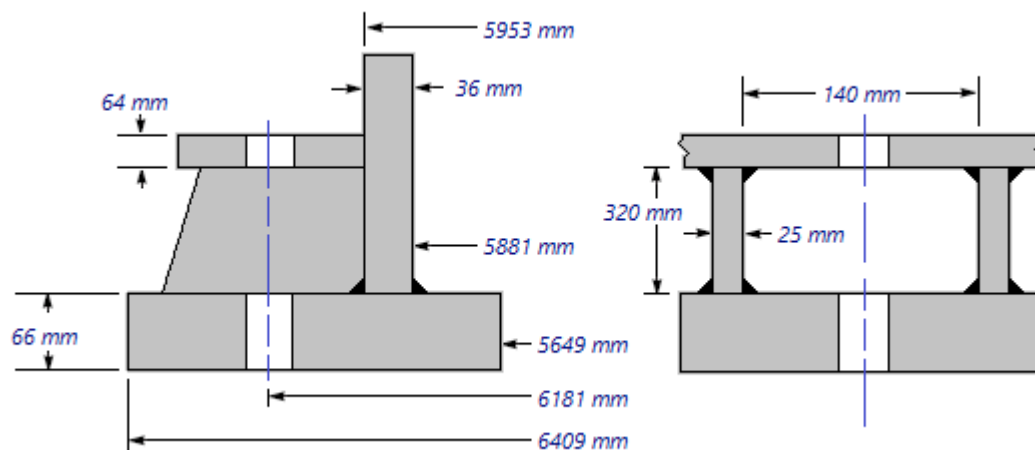
|                              |           |    |
|------------------------------|-----------|----|
| Thickness of Basering        | 66.0000   | mm |
| Inside Diameter of Basering  | 5649.0000 | mm |
| Outside Diameter of Basering | 6409.0000 | mm |
| Nominal Diameter of Bolts    | 80.0000   | mm |
| Diameter of Bolt Circle      | 6181.0000 | mm |
| Number of Bolts              | 36        |    |

|                            |         |    |
|----------------------------|---------|----|
| Thickness of Gusset Plates | 25.0000 | mm |
|----------------------------|---------|----|



|                                |          |    |
|--------------------------------|----------|----|
| Average Width of Gusset Plates | 218.0000 | mm |
| Height of Gussets              | 320.0000 | mm |
| Distance between Gussets       | 140.0000 | mm |
| Thickness of Top Plate or Ring | 64.0000  | mm |
| Radial Width of the Top Plate  | 228.0000 | mm |
| Number of Gussets              | 72       |    |

#### Basing Sketch for Cylindrical Skirts:



**Basing Cross Section View**

#### Loads for Foundation/Support Design:

|                                   |              |      |
|-----------------------------------|--------------|------|
| Total Wind Shear on Support       | 261517.      | N    |
| Total Earthquake Shear on Support | 674885.      | N    |
| Wind Moment on Support            | 10736592896. | N-mm |
| Earthquake Moment on Support      | 27581276160. | N-mm |
| Maximum Initial Bolt Load         | 330008.      | N    |

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

#### Weights:

|                                                |           |     |
|------------------------------------------------|-----------|-----|
| Fabricated - Bare w/o removable internals      | 204742.0  | kgm |
| Shop Test - Fabricated + water ( full )        | 958190.5  | kgm |
| Shipping - Fab. + rem. intls.+ shipping app.   | 224712.4  | kgm |
| Erected - Fab. + rem. intls.+ insul. (etc)     | 285281.2  | kgm |
| Empty - Fab. + intls. + details + wghts.       | 335949.8  | kgm |
| Operating - Empty + operating liquid (no ca)   | 621118.9  | kgm |
| Field Test - Empty weight + test medium (full) | 1018557.5 | kgm |



### Nozzle Calculation Summary (MPa & mm):

| Description | MAWP | Ext | MAPNC | UG-45 | [tr] | Weld Path | Areas or Stresses |
|-------------|------|-----|-------|-------|------|-----------|-------------------|
| N1 36"      | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| N11 2"      | 1    | ... | ...   | OK    | 3.42 | OK        | No Calc[*]        |
| K1a         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K2a         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K3a         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K4a         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| N9          | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| N8 2"       | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K1b         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K2b         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K3b         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K4b         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| N2a         | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| N2b         | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| N3A         | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| N4a         | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| N4b         | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| A1_30"      | 1    | OK  | ...   | ...   | ...  | OK        | Passed            |
| N3B         | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| K5a         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K6a         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K5b         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K6b         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| K7a         | 1    | ... | ...   | OK    | 4.80 | OK        | No Calc[*]        |
| A2_30"      | 1    | OK  | ...   | ...   | ...  | OK        | Passed            |
| A3_30"      | 1    | OK  | ...   | ...   | ...  | OK        | Passed            |
| K7b         | 1    | ... | ...   | OK    | 3.42 | OK        | No Calc[*]        |
| A4_30"      | 1    | OK  | ...   | ...   | ...  | OK        | Passed            |
| N5          | 1    | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| K8a         | 1    | ... | ...   | OK    | 3.42 | OK        | No Calc[*]        |
| A5_30"      | 1    | OK  | ...   | ...   | ...  | OK        | Passed            |
| K9a         | 1    | ... | ...   | OK    | 3.42 | OK        | No Calc[*]        |
| N6          | 1    | OK  | ...   | OK    | 5.26 | OK        | Passed            |
| K8b         | 1    | ... | ...   | OK    | 3.42 | OK        | No Calc[*]        |
| K9b         | 1    | ... | ...   | OK    | 3.42 | OK        | No Calc[*]        |
| A6_30"      | 1    | OK  | ...   | ...   | ...  | OK        | Passed            |
| A7_30"      | 1    | OK  | ...   | ...   | ...  | OK        | Passed            |
| N7          | ...  | OK  | ...   | OK    | 8.33 | OK        | Passed            |
| N10         | ...  | ... | ...   | OK    | 3.42 | OK        | No Calc[*]        |

### Nozzle MAWP Summary:

Minimum MAWP Nozzles : 0 Nozzle : N10  
 Minimum MAWP Shells/Flanges : 0 Element : Top Dish End  
 Minimum MAPnc Shells/Flanges : 1 Element : Top Dish End

Computed Vessel M.A.W.P. : 0 MPa

[\*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

### Check the spatial relationship between the nozzles:

| From Node | Nozzle Description | Y Coordinate<br>mm | Layout Angle<br>deg | Dia. Limit<br>mm |
|-----------|--------------------|--------------------|---------------------|------------------|
| 100       | N1 36"             | 0.000              | 0.000               | 1748.800         |
| 100       | N11 2"             | 0.000              | 180.000             | 102.325          |
| 110       | K1a                | 200.002            | 0.000               | 107.724          |
| 110       | K2a                | 200.002            | 90.000              | 107.724          |

|     |        |           |         |          |
|-----|--------|-----------|---------|----------|
| 110 | K3a    | 200.002   | 0.000   | 107.724  |
| 110 | K4a    | 200.002   | 0.000   | 107.724  |
| 110 | N9     | 200.002   | 315.000 | 107.724  |
| 110 | N8 2"  | 450.000   | 215.000 | 107.724  |
| 120 | K1b    | 6000.001  | 0.000   | 107.724  |
| 120 | K2b    | 4350.000  | 0.000   | 107.724  |
| 120 | K3b    | 4350.000  | 0.000   | 107.724  |
| 120 | K4b    | 4650.000  | 0.000   | 107.724  |
| 120 | N2a    | 4000.002  | -32.000 | 3085.600 |
| 120 | N2b    | 4000.002  | 32.000  | 3085.600 |
| 120 | N3A    | 5620.001  | 180.000 | 1489.600 |
| 120 | N4a    | 4000.002  | 242.000 | 1139.200 |
| 120 | N4b    | 4000.002  | 280.000 | 1139.200 |
| 120 | A1_30" | 5649.999  | 120.000 | 1460.000 |
| 120 | N3B    | 5620.001  | 240.000 | 1529.600 |
| 130 | K5a    | 8770.002  | 0.000   | 103.724  |
| 130 | K6a    | 8770.002  | 0.000   | 103.724  |
| 140 | K5b    | 9820.003  | 0.000   | 103.724  |
| 140 | K6b    | 9820.003  | 0.000   | 103.724  |
| 140 | K7a    | 10370.004 | 0.000   | 103.724  |
| 140 | A2_30" | 10100.004 | 30.000  | 1460.000 |
| 150 | A3_30" | 13700.006 | 30.000  | 1460.000 |
| 170 | K7b    | 18535.004 | 150.000 | 105.004  |
| 170 | A4_30" | 19570.006 | 120.000 | 1460.000 |
| 180 | N5     | 22049.998 | 180.000 | 2272.800 |
| 180 | K8a    | 22250.000 | 270.000 | 105.004  |
| 180 | A5_30" | 22349.998 | 45.000  | 1460.000 |
| 190 | K9a    | 25319.998 | 150.000 | 105.004  |
| 190 | N6     | 25309.998 | 300.000 | 204.521  |
| 190 | K8b    | 24710.004 | 270.000 | 105.004  |
| 190 | K9b    | 24050.002 | 230.000 | 105.004  |
| 190 | A6_30" | 24200.002 | 45.000  | 1460.000 |
| 190 | A7_30" | 24549.998 | 0.000   | 1460.000 |
| 220 | N7     | 0.000     | 0.000   | 1476.000 |
| 220 | N10    | 0.000     | 180.000 | 105.004  |

The nozzle spacing is computed by the following:

$$= \sqrt{(l^2 + l_c^2)} \text{ where}$$

l - Arc length along the inside vessel surface in the long. direction.  
 l<sub>c</sub> - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

*Diameter Limit clash between : N11 2" and N1 36".  
 The Centerline Dist. is 651.30 while the sum of the limit Radii is 925.56 mm.*

*Diameter Limit clash between : K3a and K1a.  
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 107.72 mm.*

*Diameter Limit clash between : K4a and K1a.  
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 107.72 mm.*

*Diameter Limit clash between : K4a and K3a.  
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 107.72 mm.*

*Diameter Limit clash between : K3b and K2b.  
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 107.72 mm.*

*Diameter Limit clash between : K6a and K5a.  
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 103.72 mm.*

*Diameter Limit clash between : K6b and K5b.  
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 103.72 mm.*

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### Nozzle Schedule:

| Description | Nominal or Actual Size | Schd or FVC Type | Flg Type | Nozzle O/Dia in | Wall Thk mm | Reinforcing Pad Diameter | Pad Thk mm | Cut Length mm | Flg Class |
|-------------|------------------------|------------------|----------|-----------------|-------------|--------------------------|------------|---------------|-----------|
| N11 2"      | 2.000 in               | 80               | WNF      | 2.375           | 5.537       | 160.32                   | 22.00      | 231.63        | 600       |
| K1a         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| K2a         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| K3a         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| K4a         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| N9          | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| N8 2"       | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 600       |
| K1b         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| K2b         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| K3b         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| K4b         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 225.81        | 150       |
| K5a         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 223.81        | 150       |
| K6a         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 223.81        | 150       |
| K5b         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 223.81        | 150       |
| K6b         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 223.81        | 150       |
| K7a         | 2.000 in               | Actual           | LWN      | 3.060           | 13.462      | ...                      | ...        | 223.81        | 150       |
| K7b         | 2.000 in               | STD              | WNF      | 2.375           | 3.912       | ...                      | ...        | 164.15        | 300       |
| K8a         | 2.000 in               | STD              | WNF      | 2.375           | 3.912       | ...                      | ...        | 164.15        | 300       |
| K9a         | 2.000 in               | STD              | WNF      | 2.375           | 3.912       | ...                      | ...        | 164.15        | 300       |
| K8b         | 2.000 in               | STD              | WNF      | 2.375           | 3.912       | ...                      | ...        | 164.15        | 300       |
| K9b         | 2.000 in               | STD              | WNF      | 2.375           | 3.912       | ...                      | ...        | 164.15        | 300       |
| N10         | 2.000 in               | STD              | WNF      | 2.375           | 3.912       | ...                      | ...        | 228.58        | 600       |
| N6          | 4.000 in               | STD              | WNF      | 4.500           | 6.020       | 314.30                   | 16.00      | 164.55        | 300       |
| N4a         | 24.000 in              | Actual           | WNF      | 24.000          | 20.000      | ...                      | ...        | 563.67        | 150       |
| N4b         | 24.000 in              | Actual           | WNF      | 24.000          | 20.000      | ...                      | ...        | 563.67        | 150       |
| A1_30"      | 30.000 in              | Actual           | WNF      | 30.000          | 16.000      | 1382.00                  | 16.00      | 440.68        | 150       |
| A2_30"      | 30.000 in              | Actual           | WNF      | 30.000          | 16.000      | 1382.00                  | 14.00      | 438.68        | 150       |
| A3_30"      | 30.000 in              | Actual           | WNF      | 30.000          | 16.000      | 1382.00                  | 14.00      | 438.68        | 150       |
| A4_30"      | 30.000 in              | Actual           | WNF      | 30.000          | 16.000      | 1382.00                  | 14.00      | 438.68        | 150       |
| A5_30"      | 30.000 in              | Actual           | WNF      | 30.000          | 16.000      | 1382.00                  | 14.00      | 438.68        | 150       |
| A6_30"      | 30.000 in              | Actual           | WNF      | 30.000          | 16.000      | 1500.00                  | 14.00      | 438.68        | 150       |
| A7_30"      | 30.000 in              | Actual           | WNF      | 30.000          | 16.000      | 1500.00                  | 14.00      | 438.68        | 150       |
| N7          | 30.000 in              | Actual           | WNF      | 30.000          | 12.000      | 1062.00                  | 18.00      | 432.18        | 150       |
| N3A         | 32.000 in              | Actual           | WNF      | 32.000          | 34.000      | ...                      | ...        | 456.10        | 150       |
| N3B         | 32.000 in              | Actual           | WNF      | 32.000          | 24.000      | ...                      | ...        | 456.10        | 150       |
| N1 36"      | 36.000 in              | Actual           | WNF      | 36.000          | 20.000      | 1050.00                  | 20.00      | 439.53        | 150       |
| N5          | 46.000 in              | Actual           | WNF      | 46.000          | 16.000      | ...                      | ...        | 488.36        | 150       |
| N2a         | 62.000 in              | Actual           | None     | 62.000          | 16.000      | ...                      | ...        | 1444.12       | ...       |
| N2b         | 62.000 in              | Actual           | None     | 62.000          | 16.000      | ...                      | ...        | 1444.12       | ...       |

### General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

### Nozzle Material and Weld Fillet Leg Size Details (mm):

| Description | Material | Shl Grve Weld | Noz Shl/Pad Weld | Pad OD Weld | Pad Grve Weld | Inside Weld |
|-------------|----------|---------------|------------------|-------------|---------------|-------------|
| N11 2"      | SA-106 B | 22.000        | 10.000           | 18.000      | 22.000        | ...         |
| K1a         | SA-105   | 16.000        | 10.000           | ...         | ...           | ...         |
| K2a         | SA-105   | 16.000        | 10.000           | ...         | ...           | ...         |
| K3a         | SA-105   | 16.000        | 10.000           | ...         | ...           | ...         |
| K4a         | SA-105   | 16.000        | 10.000           | ...         | ...           | ...         |
| N9          | SA-105   | 16.000        | 10.000           | ...         | ...           | ...         |
| N8 2"       | SA-106 B | 16.000        | 10.000           | ...         | ...           | ...         |



|        |           |        |        |        |        |     |
|--------|-----------|--------|--------|--------|--------|-----|
| K1b    | SA-105    | 16.000 | 10.000 | ...    | ...    | ... |
| K2b    | SA-105    | 16.000 | 10.000 | ...    | ...    | ... |
| K3b    | SA-105    | 16.000 | 10.000 | ...    | ...    | ... |
| K4b    | SA-105    | 16.000 | 10.000 | ...    | ...    | ... |
| K5a    | SA-105    | 14.000 | 10.000 | ...    | ...    | ... |
| K6a    | SA-105    | 14.000 | 10.000 | ...    | ...    | ... |
| K5b    | SA-105    | 14.000 | 10.000 | ...    | ...    | ... |
| K6b    | SA-105    | 14.000 | 10.000 | ...    | ...    | ... |
| K7a    | SA-105    | 14.000 | 10.000 | ...    | ...    | ... |
| K7b    | SA-106 B  | 14.000 | 10.000 | ...    | ...    | ... |
| K8a    | SA-106 B  | 14.000 | 10.000 | ...    | ...    | ... |
| K9a    | SA-106 B  | 14.000 | 10.000 | ...    | ...    | ... |
| K8b    | SA-106 B  | 14.000 | 10.000 | ...    | ...    | ... |
| K9b    | SA-106 B  | 14.000 | 10.000 | ...    | ...    | ... |
| N10    | SA-106 B  | 16.000 | 10.000 | ...    | ...    | ... |
| N6     | SA-106 B  | 14.000 | 10.000 | 12.000 | 16.000 | ... |
| N4a    | SA-516 70 | 16.000 | 10.000 | ...    | ...    | ... |
| N4b    | SA-516 70 | 16.000 | 10.000 | ...    | ...    | ... |
| A1_30" | SA-516 70 | 16.000 | 10.000 | 12.000 | 16.000 | ... |
| A2_30" | SA-516 70 | 14.000 | 10.000 | 12.000 | 14.000 | ... |
| A3_30" | SA-516 70 | 14.000 | 10.000 | 12.000 | 14.000 | ... |
| A4_30" | SA-516 70 | 14.000 | 14.000 | 12.000 | 14.000 | ... |
| A5_30" | SA-516 70 | 14.000 | 10.000 | 12.000 | 14.000 | ... |
| A6_30" | SA-516 70 | 14.000 | 14.000 | 12.000 | 14.000 | ... |
| A7_30" | SA-516 70 | 14.000 | 14.000 | 12.000 | 14.000 | ... |
| N7     | SA-516 70 | 16.000 | 10.000 | 14.000 | 18.000 | ... |
| N3A    | SA-516 70 | 16.000 | 16.000 | ...    | ...    | ... |
| N3B    | SA-516 70 | 16.000 | 16.000 | ...    | ...    | ... |
| N1 36" | SA-516 70 | 18.000 | 10.000 | 18.000 | 20.000 | ... |
| N5     | SA-516 70 | 16.000 | 10.000 | ...    | ...    | ... |
| N2a    | SA-516 70 | 22.000 | 10.000 | ...    | ...    | ... |
| N2b    | SA-516 70 | 16.000 | 10.000 | ...    | ...    | ... |

Note: The Outside projections below do not include the flange thickness.

**Nozzle Miscellaneous Data:**

| Description | Elev/Distance<br>From Datum<br>mm | Layout<br>Angle<br>deg | Proj<br>Outside<br>mm | Proj<br>Inside<br>mm | Installed in<br>Component |
|-------------|-----------------------------------|------------------------|-----------------------|----------------------|---------------------------|
| N11 2"      | ...                               | 180.0                  | 209.55                | 0.00                 | Btm Dish End              |
| K1a         | 200.002                           | 0.0                    | 209.55                | 0.00                 | Shell-1                   |
| K2a         | 200.002                           | 90.0                   | 209.55                | 0.00                 | Shell-1                   |
| K3a         | 200.002                           | 0.0                    | 209.55                | 0.00                 | Shell-1                   |
| K4a         | 200.002                           | 0.0                    | 209.55                | 0.00                 | Shell-1                   |
| N9          | 200.002                           | 315.0                  | 209.55                | 0.00                 | Shell-1                   |
| N8 2"       | 450.000                           | 215.0                  | 209.55                | 0.00                 | Shell-1                   |
| K1b         | 6000.001                          | 0.0                    | 209.55                | 0.00                 | Shell-2                   |
| K2b         | 4350.000                          | 0.0                    | 209.55                | 0.00                 | Shell-2                   |
| K3b         | 4350.000                          | 0.0                    | 209.55                | 0.00                 | Shell-2                   |
| K4b         | 4650.000                          | 0.0                    | 209.55                | 0.00                 | Shell-2                   |
| K5a         | 8770.002                          | 0.0                    | 209.55                | 0.00                 | Shell-3                   |
| K6a         | 8770.002                          | 0.0                    | 209.55                | 0.00                 | Shell-3                   |
| K5b         | 9820.003                          | 0.0                    | 209.55                | 0.00                 | Shell-4                   |
| K6b         | 9820.003                          | 0.0                    | 209.55                | 0.00                 | Shell-4                   |
| K7a         | 10370.004                         | 0.0                    | 209.55                | 0.00                 | Shell-4                   |
| K7b         | 18535.002                         | 150.0                  | 150.00                | 0.00                 | Shell-7                   |
| K8a         | 22250.000                         | 270.0                  | 150.00                | 0.00                 | Shell-8                   |
| K9a         | 25319.998                         | 150.0                  | 150.00                | 0.00                 | Shell-9                   |
| K8b         | 24710.002                         | 270.0                  | 150.00                | 0.00                 | Shell-9                   |
| K9b         | 24050.000                         | 230.0                  | 150.00                | 0.00                 | Shell-9                   |
| N10         | ...                               | 180.0                  | 200.00                | 0.00                 | Top Dish End              |
| N6          | 25309.998                         | 300.0                  | 150.00                | 0.00                 | Shell-9                   |
| N4a         | 4000.002                          | 242.0                  | 347.60                | 0.00                 | Shell-2                   |
| N4b         | 4000.002                          | 280.0                  | 347.60                | 0.00                 | Shell-2                   |
| A1_30"      | 5649.999                          | 120.0                  | 400.00                | 0.00                 | Shell-2                   |
| A2_30"      | 10100.004                         | 30.0                   | 400.00                | 0.00                 | Shell-4                   |
| A3_30"      | 13700.005                         | 30.0                   | 400.00                | 0.00                 | Shell-5                   |
| A4_30"      | 19570.004                         | 120.0                  | 400.00                | 0.00                 | Shell-7                   |



|        |           |       |        |      |              |
|--------|-----------|-------|--------|------|--------------|
| A5_30" | 22349.998 | 45.0  | 400.00 | 0.00 | Shell-8      |
| A6_30" | 24200.000 | 45.0  | 400.00 | 0.00 | Shell-9      |
| A7_30" | 24549.998 | 0.0   | 400.00 | 0.00 | Shell-9      |
| N7     | ...       | 0.0   | 400.00 | 0.00 | Top Dish End |
| N3A    | 5620.001  | 180.0 | 400.00 | 0.00 | Shell-2      |
| N3B    | 5620.001  | 240.0 | 400.00 | 0.00 | Shell-2      |
| N1_36" | ...       | 0.0   | 400.00 | 0.00 | Btm Dish End |
| N5     | 22049.998 | 180.0 | 400.00 | 0.00 | Shell-8      |
| N2a    | 4000.002  | -32.0 | 363.27 | 0.00 | Shell-2      |
| N2b    | 4000.002  | 32.0  | 363.27 | 0.00 | Shell-2      |



**Bill of Materials:**

| QTY | DESCRIPTION                                                  | MATERIAL  |
|-----|--------------------------------------------------------------|-----------|
| 1   | TORISPHERICAL HEAD: 26.0mm THK X 5906.0mm ID X 50.0mm        | SA-516 70 |
| 1   | TORISPHERICAL HEAD: 24.0mm THK X 5906.0mm ID X 50.0mm        | SA-516 70 |
| 2   | WELD NECK FLANGE: 104.0mm THK X 1714.0mm OD                  | SA-266 2  |
| 1   | INSULATION: 1044mm X 100mm THK                               | insu      |
| 2   | CLADDING: 1028.390mm X 3.000mm THK                           | ...       |
| 5   | INSULATION: 3000mm X 100mm THK                               | Hot       |
| 8   | CLADDING: 3000.000mm X 3.000mm THK                           | ...       |
| 13  | SECTION RING STIFFENER: W8X21                                | SA-516 70 |
| 1   | PACKING: 579.998mm                                           | ...       |
| 3   | INSULATION: 3000mm X 100mm THK                               | insu      |
| 5   | PLATFORM: 1000mm X 1500mm WIDE                               |           |
| 1   | PACKING: 2999.998mm                                          | ...       |
| 4   | LADDER: 4000.0mm                                             | ...       |
| 1   | PACKING: 920.003mm                                           | ...       |
| 1   | PACKING: 1514.996mm                                          | ...       |
| 1   | PACKING: 2985.000mm                                          | ...       |
| 1   | PACKING: 1000.000mm                                          | ...       |
| 1   | INSULATION: 1500mm X 100mm THK                               | insu      |
| 1   | CLADDING: 1500.000mm X 3.000mm THK                           | ...       |
| 1   | PLATFORM: 1000mm X 11942mm WIDE                              |           |
| 1   | LADDER: 3000.0mm                                             | ...       |
| 1   | INSULATION: 1044mm X 100mm THK                               | Hot       |
| 1   | INSULATION: 148mm X 100mm THK                                | insu      |
| 1   | INSULATION: 142mm X 100mm THK                                | insu      |
| 7   | CLASS 150 GR 1.1, 30.0" BLIND FLANGE(S)                      | SA-105    |
| 1   | BASERING: 6409.0mm OD X 5649.0 ID X 66.0mm                   | E250BR    |
| 36  | BASERING BOLTS: 80.0mm DIA                                   | SA-36     |
| 36  | BASERING BOLT NUTS: 80.0mm DIA                               | ...       |
| 1   | BASERING TOP RING: 6409.0mm OD X 5953.0 ID X 64.0mm          | ...       |
| 72  | BASERING GUSSET PLATES: 218.0mm X 218.0mm X 320.0mm X 25.0mm | ...       |
| 1   | GASKET: 1598.0mm OD X 1566.0mm ID                            | ...       |
| 48  | BODY FLANGE BOLTS: 25.4mm DIA                                | SA-193 B7 |
| 96  | NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA                       | ...       |



|    |                                                             |           |
|----|-------------------------------------------------------------|-----------|
| 6  | CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 2514.7mm X 2000.0mm  | E250BR    |
| 1  | CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 2566.0mm X 2000.0mm  | E250BR    |
| 1  | CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 17654.3mm X 2000.0mm | E250BR    |
| 12 | CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 2572.1mm X 3000.0mm  | E250BR    |
| 2  | CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 2520.7mm X 3000.0mm  | E250BR    |
| 2  | CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 17953.5mm X 3000.0mm | E250BR    |
| 12 | CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 2573.0mm X 3000.0mm  | E250BR    |
| 2  | CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 2521.5mm X 3000.0mm  | E250BR    |
| 2  | CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 17959.6mm X 3000.0mm | E250BR    |
| 14 | CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2550.3mm X 3000.0mm  | E250BR    |
| 2  | CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 17852.3mm X 3000.0mm | E250BR    |
| 7  | CYLINDRICAL SKIRT SEGMENT 16.0mm THK X 2551.2mm X 1000.0mm  | E250BR    |
| 1  | CYLINDRICAL SKIRT SEGMENT 16.0mm THK X 17858.4mm X 1000.0mm | E250BR    |
| 6  | CYLINDRICAL SKIRT SEGMENT 16.0mm THK X 2574.8mm X 1000.0mm  | SA-516 70 |
| 1  | CYLINDRICAL SKIRT SEGMENT 16.0mm THK X 2523.3mm X 1000.0mm  | SA-516 70 |
| 1  | CYLINDRICAL SKIRT SEGMENT 16.0mm THK X 17971.8mm X 1000.0mm | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 16.0mm THK X 11015.7mm X 3000.0mm       | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 16.0mm THK X 7229.0mm X 3000.0mm        | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 16.0mm THK X 309.2mm X 3000.0mm         | SA-516 70 |
| 6  | CYLINDRICAL SEGMENT 16.0mm THK X 2370.8mm X 3000.0mm        | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 16.0mm THK X 16183.2mm X 3000.0mm       | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 16.0mm THK X 1958.5mm X 3000.0mm        | SA-516 70 |
| 30 | CYLINDRICAL SEGMENT 14.0mm THK X 2368.6mm X 3000.0mm        | SA-516 70 |
| 5  | CYLINDRICAL SEGMENT 14.0mm THK X 16185.4mm X 3000.0mm       | SA-516 70 |
| 5  | CYLINDRICAL SEGMENT 14.0mm THK X 1973.9mm X 3000.0mm        | SA-516 70 |
| 6  | CYLINDRICAL SEGMENT 14.0mm THK X 2370.8mm X 3000.0mm        | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 14.0mm THK X 16183.2mm X 3000.0mm       | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 14.0mm THK X 1958.5mm X 3000.0mm        | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 14.0mm THK X 16185.4mm X 1500.0mm       | SA-516 70 |
| 6  | CYLINDRICAL SEGMENT 14.0mm THK X 2368.6mm X 1500.0mm        | SA-516 70 |
| 1  | CYLINDRICAL SEGMENT 14.0mm THK X 1973.9mm X 1500.0mm        | SA-516 70 |
| 1  | NAMEPLATE                                                   | ...       |



**Minimum Design Metal Temperature Results Summary (°C):**

| Description  | Notes | Curve | Basic MDMT | Reduced MDMT | UG-20(f) MDMT | Thickness ratio | Gov Thk mm | E*   | PWHT reqd |
|--------------|-------|-------|------------|--------------|---------------|-----------------|------------|------|-----------|
| N2A FLG      | [11]  | B     | -14        | -104         | -29           | 0.176           | 16.000     | 1.00 | No        |
| N2B FLG      | [11]  | B     | -14        | -104         | -29           | 0.176           | 16.000     | 1.00 | No        |
| Btm Dish End | [10]  | B     | -5         | -5           | -29           | 0.994           | 22.000     | 1.00 | No        |
| Btm Dish End | [7]   | B     | 0          | -37          |               | 0.470           | 26.000     | 1.00 | No        |
| Shell-1      | [8]   | B     | -14        | -27          | -29           | 0.768           | 16.000     | 1.00 | No        |
| Shell-2      | [8]   | B     | -14        | -29          | -29           | 0.727           | 16.000     | 1.00 | No        |
| Shell-3      | [8]   | B     | -18        | -28          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Shell-4      | [8]   | B     | -18        | -28          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Shell-5      | [8]   | B     | -18        | -28          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Shell-6      | [8]   | B     | -18        | -28          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Shell-7      | [8]   | B     | -18        | -28          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Shell-8      | [8]   | B     | -18        | -28          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Shell-9      | [8]   | B     | -18        | -28          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Top Dish End | [10]  | B     | -7         | -7           | -29           | 1.000           | 20.000     | 1.00 | No        |
| Top Dish End | [7]   | B     | -2         | -40          | -29           | 0.465           | 24.000     | 1.00 | No        |
| N1 36"       | [1]   | B     | -7         | -33          | -29           | 0.562           | 20.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| N11 2"       | [1]   | B     | -5         | -30          | -29           | 0.562           | 22.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K1a          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K2a          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K3a          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K4a          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| N9           | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| N8 2"        | [1]   | B     | -19        | -104         | -29           | 0.009           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K1b          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K2b          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K3b          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K4b          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| N2a          | [1]   | B     | 7          | -104         |               | 0.330           | 34.000     | 1.00 | No        |
| N2b          | [1]   | B     | 7          | -104         |               | 0.330           | 34.000     | 1.00 | No        |
| N3A          | [1]   | B     | 2          | -48          |               | 0.404           | 28.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| N4a          | [1]   | B     | -2         | -38          | -29           | 0.474           | 24.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| N4b          | [1]   | B     | -2         | -38          | -29           | 0.474           | 24.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| A130"        | [1]   | B     | -14        | -29          | -29           | 0.727           | 16.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| N3B          | [1]   | B     | 18         | -104         |               | 0.062           | 24.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K5a          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K6a          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K5b          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K6b          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| K7a          | [1]   | B     | -19        | -104         | -29           | 0.007           | 13.462     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |
| A230"        | [1]   | B     | -18        | -29          | -29           | 0.822           | 14.000     | 1.00 | No        |
| Nozzle Flg   | [5]   | A     | -18        | -104         |               |                 |            |      |           |





|               |      |   |     |      |     |       |        |      |    |
|---------------|------|---|-----|------|-----|-------|--------|------|----|
| A330"         | [1]  | B | -18 | -29  | -29 | 0.822 | 14.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| K7b           | [1]  | B | -8  | -104 | -29 | 0.037 | 3.423  | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| A430"         | [1]  | B | -18 | -29  | -29 | 0.822 | 14.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| N5            | [1]  | B | 4   | -48  |     | 0.369 | 30.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| K8a           | [1]  | B | -8  | -104 | -29 | 0.037 | 3.423  | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| A530"         | [1]  | B | -18 | -29  | -29 | 0.822 | 14.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| K9a           | [1]  | B | -8  | -104 | -29 | 0.037 | 3.423  | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| N6            | [1]  | B | -18 | -29  | -29 | 0.822 | 14.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| K8b           | [1]  | B | -8  | -104 | -29 | 0.037 | 3.423  | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| K9b           | [1]  | B | -8  | -104 | -29 | 0.037 | 3.423  | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| A630"         | [1]  | B | -18 | -29  | -29 | 0.822 | 14.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| A7 30"        | [1]  | B | -18 | -29  | -29 | 0.822 | 14.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| N7            | [1]  | B | -11 | -36  | -29 | 0.565 | 18.000 | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| N10           | [1]  | B | -8  | -104 | -29 | 0.037 | 3.423  | 1.00 | No |
| Nozzle Flg    | [5]  | A | -18 | -104 |     |       |        |      |    |
| Nozl. Bolting | [22] |   | -48 |      |     | 1.000 | 64.000 | 1.00 | No |
| Bolting       | [21] |   | -48 |      |     |       |        |      |    |

Warmest MDMT: 18 -5

Required Minimum Design Metal Temperature -1.0 °C  
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

#### Notes:

- [ ! ] - This was an impact tested material.
- [ 1 ] - Governing Nozzle Weld.
- [ 4 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [ 5 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [ 6 ] - MDMT Calculations at the Shell/Head Joint.
- [ 7 ] - MDMT Calculations for the Straight Flange.
- [ 8 ] - Cylinder/Cone/Flange Junction MDMT.
- [ 9 ] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.  
 UCS-66(g) was not considered.  
 UCS-66(i) was not considered.

#### Notes:

Impact test temps were not entered in and not considered in the analysis.  
 UCS-66(i) applies to impact tested materials not by specification and  
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).  
 The Basic MDMT includes the (30F) PWHT credit if applicable.





Class From To : Basic Element Checks.

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Note   90   110   There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

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There were no geometry errors or warnings.

**PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.**

### Information Regarding Nozzle Loads

*Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.*



*One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.*



*The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.*



*The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.*

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### Units used in this Analysis (SIASME):

| Name           | System Unit | Constant    | User Unit          |
|----------------|-------------|-------------|--------------------|
| Length         | Feet        | 304.8000    | mm                 |
| Force          | Pounds      | 4.4480      | N                  |
| Mass           | Pounds      | 0.4536      | kgm                |
| Area           | sq. inches  | 645.1600    | mm <sup>2</sup>    |
| Moment         | ft. lbs.    | 1356.3000   | N-mm               |
| Stress         | lbs./sq.in. | 0.0069      | MPa                |
| Temperature    | Degrees F   | 0.5556      | °C                 |
| Pressure       | psig        | 0.0069      | MPa                |
| Elast. Modulus | lbs./sq.in. | 0.0069      | MPa                |
| Pipe Density   | lbs./cu.in. | 0.0277      | kg/cm <sup>3</sup> |
| Ins. Density   | lbs./cu.ft. | 0.1600E-04  | kg/cm <sup>3</sup> |
| Fluid Density  | lbs./cu.ft. | 0.1600E-04  | kg/cm <sup>3</sup> |
| Wind Speed     | miles/hr    | 1.6093      | km/hr              |
| Tray Weight    | lbs./sq.ft. | 0.0005      | kg/cm <sup>2</sup> |
| Inertia        | in.4        | 416231.0000 | mm <sup>4</sup>    |
| G Load         | G's         | 1.0000      | g's                |
| Wind Load      | lbs./sq.ft. | 0.0479      | kPa                |
| Elevation      | Feet        | 304.8000    | mm                 |
| Volume         | in.3        | 0.0164      | ltr                |
| Diameter       | inches      | 25.4000     | mm                 |
| Thickness      | inches      | 25.4000     | mm                 |

### PV Elite Vessel Analysis Program: Input Data

|                                          |        |     |
|------------------------------------------|--------|-----|
| Design Internal Pressure (for Hydrotest) | 0.3432 | MPa |
| Design Internal Temperature              | 150.0  | °C  |
| Projection of Nozzle from Vessel Top     | 0      | mm  |
| Projection of Nozzle from Vessel Bottom  | 0      | mm  |
| Minimum Design Metal Temperature         | -1.0   | °C  |
| Type of Construction                     | Welded |     |
| Special Service                          | Amine  |     |
| Degree of Radiography                    | RT-1   |     |
| Use Higher Longitudinal Stresses (Flag)  | Y      |     |
| Select t for Internal Pressure (Flag)    | N      |     |
| Select t for External Pressure (Flag)    | N      |     |
| Select t for Axial Stress (Flag)         | N      |     |
| Select Location for Stiff. Rings (Flag)  | N      |     |
| Consider Vortex Shedding                 | Y      |     |

### Shop Pressure Test:

|                                      |            |
|--------------------------------------|------------|
| Type of Pressure Test                | UG-99(b)   |
| Pressure Test Position               | Horizontal |
| Test Performed in Corroded Condition | No         |

### Field Pressure Test:

|                                      |          |
|--------------------------------------|----------|
| Type of Pressure Test                | UG-99(b) |
| Pressure Test Position               | Vertical |
| Test Performed in Corroded Condition | Yes      |

|              |                   |
|--------------|-------------------|
| Load Case 1  | NP+EW+1.1WI+FW+BW |
| Load Case 2  | NP+EW+EE+FS+BS    |
| Load Case 3  | NP+OW+1.1WI+FW+BW |
| Load Case 4  | NP+OW+EQ+FS+BS    |
| Load Case 5  | NP+HW+HI          |
| Load Case 6  | NP+HW+HE          |
| Load Case 7  | IP+OW+1.1WI+FW+BW |
| Load Case 8  | IP+OW+EQ+FS+BS    |
| Load Case 9  | EP+OW+1.1WI+FW+BW |
| Load Case 10 | EP+OW+EQ+FS+BS    |
| Load Case 11 | HP+HW+HI          |
| Load Case 12 | HP+HW+HE          |
| Load Case 13 | IP+WE+EW          |
| Load Case 14 | IP+WF+CW          |
| Load Case 15 | IP+VO+OW          |



|                                          |                                |                   |       |
|------------------------------------------|--------------------------------|-------------------|-------|
| Load Case 16                             |                                | IP+VE+EW          |       |
| Load Case 17                             |                                | NP+VO+OW          |       |
| Load Case 18                             |                                | FS+BS+IP+OW       |       |
| Load Case 19                             |                                | FS+BS+EP+OW       |       |
| Load Case 20                             |                                | BL+IP+OW          |       |
|                                          |                                |                   |       |
| Wind Design Code                         |                                | ASCE/SEI 7-16     |       |
| Wind Load Reduction Scale Factor         |                                | 1.100             |       |
| Basic Wind Speed                         | [V]                            | 97.2              | km/hr |
| Surface Roughness Category               |                                | C: Open Terrain   |       |
| Type of Surface                          |                                | Moderately Smooth |       |
| Base Elevation                           |                                | 19000             | mm    |
| Percent Wind for Hydrotest               |                                | 50.0              |       |
| Using User defined Wind Press. Vs Elev.  |                                | N                 |       |
| Height of Hill or Escarpment             | H or Hh                        | 0                 | mm    |
| Distance Upwind of Crest                 | Lh                             | 0                 | mm    |
| Distance from Crest to the Vessel        | x                              | 0                 | mm    |
| Type of Terrain ( Hill, Escarpment )     |                                | Flat              |       |
| Damping Factor (Beta) for Wind (Ope)     |                                | 0.0100            |       |
| Damping Factor (Beta) for Wind (Empty)   |                                | 0.0000            |       |
| Damping Factor (Beta) for Wind (Filled)  |                                | 0.0000            |       |
| Site Elevation above Sea Level           |                                | 0                 | mm    |
|                                          |                                |                   |       |
| Seismic Design Code                      |                                | ASCE/SEI 7-16     |       |
| Seismic Load Reduction Scale Factor      |                                | 1.000             |       |
| Importance Factor                        |                                | 1.250             |       |
| Table Value Fa                           |                                | 1.600             |       |
| Table Value Fv                           |                                | 2.400             |       |
| Max. Mapped Res. Acceleration            | [Ss]                           | 0.147             |       |
| Max. Eff. Ground Acceleration            | [S]                            | 0.065             |       |
| Force Modification Factor R              |                                | 1.250             |       |
| Site Class                               |                                | D                 |       |
| Component Elevation Ratio                | z/h                            | 0.000             |       |
| Amplification Factor                     | Ap                             | 0.000             |       |
| Force Factor                             |                                | 0.000             |       |
| Consider Vertical Acceleration           |                                | Yes               |       |
| Minimum Acceleration Multiplier          |                                | 0.000             |       |
| User Value of Sds (used if > 0 )         |                                | 0.000             |       |
| User Value of Sd1 (used if > 0 )         |                                | 0.000             |       |
| Moment Reduction Factor Tau              |                                | 1.000             |       |
|                                          |                                |                   |       |
| M.A.W.P. + Static Head to Element Bottom |                                | Y                 |       |
| Consider MAP New and Cold in Noz. Design |                                | N                 |       |
| Consider External Loads for Nozzle Des.  |                                | Y                 |       |
| Use ASME VIII-1 Appendix 1-9             |                                | N                 |       |
|                                          |                                |                   |       |
| Perform Blast Load Analysis              |                                | No                |       |
| Material Database Year                   | Current w/Addenda or Code Year |                   |       |

#### Configuration Directives:

|                                                          |     |
|----------------------------------------------------------|-----|
| g0 is taken as min( g0, attached shell thk )             | No  |
| Do not use Nozzle MDMT Interpretation VIII-1 01-37       | No  |
| Use Table G instead of exact equation for "A"            | Yes |
| Shell Head Joints are Tapered                            | Yes |
| Compute "K" in corroded condition                        | Yes |
| Use Code Case 2286                                       | No  |
| Use the MAWP to compute the MDMT                         | Yes |
| For thickness ratios <= 0.35, MDMT will be -155F (-104C) | Yes |
| For PWHT & P1 Materials the MDMT can be < -55F (-48C)    | No  |
|                                                          |     |
| Using Metric Material Databases, ASME II D               | Yes |
| Calculate B31.3 type stress for Nozzles with Loads       | Yes |
| Reduce the MDMT due to lower membrane stress             | Yes |
| Consider Longitudinal Stress in MDMT Calculations        | Yes |

#### Complete Listing of Vessel Elements and Details:

|                   |      |
|-------------------|------|
| Element#          | 1/22 |
| Element From Node | 10   |



|                                         |                    |
|-----------------------------------------|--------------------|
| Element To Node                         | 20                 |
| Element Type                            | Skirt Support      |
| Description                             | Skirt-9            |
| Distance "FROM" to "TO"                 | 2000 mm            |
| Skirt Outside Diameter                  | 5953 mm            |
| Diameter of Skirt at Base               | 5953 mm            |
| Skirt Thickness                         | 36 mm              |
| Internal Corrosion Allowance            | 0 mm               |
| Nominal Thickness                       | 36 mm              |
| External Corrosion Allowance            | 0 mm               |
| Design Temperature Internal Pressure    | 150 °C             |
| Design Temperature External Pressure    | 40 °C              |
| Effective Diameter Multiplier           | 1.22               |
| Material Name                           | E250BR             |
| Allowable Stress, Ambient               | 117 MPa            |
| Allowable Stress, Operating             | 117 MPa            |
| Allowable Stress, Hydrotest             | 207 MPa            |
| Material Density                        | 0.00775 kg/cm³     |
| P Number Thickness                      | 32.004 mm          |
| Yield Stress, Operating                 | 203 MPa            |
| UCS-66 Chart Curve Designation          | A                  |
| External Pressure Chart Name            | CS-2               |
| UNS Number                              | SA/IS 20           |
| Class / Thickness / Grade               | :: t > 40          |
| Product Form                            | Plate; bar; shapes |
| Efficiency, Longitudinal Seam           | 0.7                |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7                |
| Weld is pre-Heated                      | Yes                |

|                                         |               |
|-----------------------------------------|---------------|
| Element#                                | 2/22          |
| Element From Node                       | 20            |
| Element To Node                         | 30            |
| Element Type                            | Skirt Support |
| Description                             | Skirt-8       |
| Distance "FROM" to "TO"                 | 3000 mm       |
| Skirt Outside Diameter                  | 5939 mm       |
| Diameter of Skirt at Base               | 5939 mm       |
| Skirt Thickness                         | 22 mm         |
| Internal Corrosion Allowance            | 0 mm          |
| Nominal Thickness                       | 22 mm         |
| External Corrosion Allowance            | 0 mm          |
| Design Temperature Internal Pressure    | 150 °C        |
| Design Temperature External Pressure    | 40 °C         |
| Effective Diameter Multiplier           | 1.22          |
| Material Name                           | E250BR        |
| Efficiency, Longitudinal Seam           | 0.7           |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7           |
| Weld is pre-Heated                      | No            |

|                                      |               |
|--------------------------------------|---------------|
| Element#                             | 3/22          |
| Element From Node                    | 30            |
| Element To Node                      | 40            |
| Element Type                         | Skirt Support |
| Description                          | Skirt-7       |
| Distance "FROM" to "TO"              | 3000 mm       |
| Skirt Outside Diameter               | 5939 mm       |
| Diameter of Skirt at Base            | 5939 mm       |
| Skirt Thickness                      | 22 mm         |
| Internal Corrosion Allowance         | 0 mm          |
| Nominal Thickness                    | 22 mm         |
| External Corrosion Allowance         | 0 mm          |
| Design Temperature Internal Pressure | 150 °C        |
| Design Temperature External Pressure | 40 °C         |
| Effective Diameter Multiplier        | 1.22          |
| Material Name                        | E250BR        |
| Efficiency, Longitudinal Seam        | 0.7           |



Efficiency, Head-to-Skirt or Circ. Seam 0.7  
Weld is pre-Heated No

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Element# 4/22  
Element From Node 40  
Element To Node 50  
Element Type Skirt Support  
Description Skirt-6  
Distance "FROM" to "TO" 3000 mm  
Skirt Outside Diameter 5937 mm  
Diameter of Skirt at Base 5937 mm  
Skirt Thickness 20 mm  
Internal Corrosion Allowance 0 mm  
Nominal Thickness 20 mm  
External Corrosion Allowance 0 mm  
Design Temperature Internal Pressure 150 °C  
Design Temperature External Pressure 40 °C  
Effective Diameter Multiplier 1.22  
Material Name E250BR  
Efficiency, Longitudinal Seam 0.7  
Efficiency, Head-to-Skirt or Circ. Seam 0.7  
Weld is pre-Heated No

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Element# 5/22  
Element From Node 50  
Element To Node 60  
Element Type Skirt Support  
Description Skirt-5  
Distance "FROM" to "TO" 3000 mm  
Skirt Outside Diameter 5937 mm  
Diameter of Skirt at Base 5937 mm  
Skirt Thickness 20 mm  
Internal Corrosion Allowance 0 mm  
Nominal Thickness 20 mm  
External Corrosion Allowance 0 mm  
Design Temperature Internal Pressure 150 °C  
Design Temperature External Pressure 40 °C  
Effective Diameter Multiplier 1.22  
Material Name E250BR  
Efficiency, Longitudinal Seam 0.7  
Efficiency, Head-to-Skirt or Circ. Seam 0.7  
Weld is pre-Heated No

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Element# 6/22  
Element From Node 60  
Element To Node 70  
Element Type Skirt Support  
Description Skirt-4  
Distance "FROM" to "TO" 3000 mm  
Skirt Outside Diameter 5935 mm  
Diameter of Skirt at Base 5935 mm  
Skirt Thickness 18 mm  
Internal Corrosion Allowance 0 mm  
Nominal Thickness 18 mm  
External Corrosion Allowance 0 mm  
Design Temperature Internal Pressure 150 °C  
Design Temperature External Pressure 40 °C  
Effective Diameter Multiplier 1.22  
Material Name E250BR  
Efficiency, Longitudinal Seam 0.7  
Efficiency, Head-to-Skirt or Circ. Seam 0.7  
Weld is pre-Heated No

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|                                         |               |
|-----------------------------------------|---------------|
| Element#                                | 7/22          |
| Element From Node                       | 70            |
| Element To Node                         | 80            |
| Element Type                            | Skirt Support |
| Description                             | Skirt-3       |
| Distance "FROM" to "TO"                 | 3000 mm       |
| Skirt Outside Diameter                  | 5935 mm       |
| Diameter of Skirt at Base               | 5935 mm       |
| Skirt Thickness                         | 18 mm         |
| Internal Corrosion Allowance            | 0 mm          |
| Nominal Thickness                       | 18 mm         |
| External Corrosion Allowance            | 0 mm          |
| Design Temperature Internal Pressure    | 150 °C        |
| Design Temperature External Pressure    | 40 °C         |
| Effective Diameter Multiplier           | 1.22          |
| Material Name                           | E250BR        |
| Efficiency, Longitudinal Seam           | 0.7           |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7           |
| Weld is pre-Heated                      | No            |

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|                                         |               |
|-----------------------------------------|---------------|
| Element#                                | 8/22          |
| Element From Node                       | 80            |
| Element To Node                         | 90            |
| Element Type                            | Skirt Support |
| Description                             | Skirt-2       |
| Distance "FROM" to "TO"                 | 1000 mm       |
| Skirt Outside Diameter                  | 5933 mm       |
| Diameter of Skirt at Base               | 5933 mm       |
| Skirt Thickness                         | 16 mm         |
| Internal Corrosion Allowance            | 0 mm          |
| Nominal Thickness                       | 16 mm         |
| External Corrosion Allowance            | 0 mm          |
| Design Temperature Internal Pressure    | 150 °C        |
| Design Temperature External Pressure    | 40 °C         |
| Effective Diameter Multiplier           | 1.22          |
| Material Name                           | E250BR        |
| Efficiency, Longitudinal Seam           | 0.7           |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.7           |
| Weld is pre-Heated                      | No            |

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|                                      |                |
|--------------------------------------|----------------|
| Element#                             | 9/22           |
| Element From Node                    | 90             |
| Element To Node                      | 100            |
| Element Type                         | Skirt Support  |
| Description                          | Skirt-1        |
| Distance "FROM" to "TO"              | 1000 mm        |
| Skirt Outside Diameter               | 5933 mm        |
| Diameter of Skirt at Base            | 5933 mm        |
| Skirt Thickness                      | 16 mm          |
| Internal Corrosion Allowance         | 0 mm           |
| Nominal Thickness                    | 16 mm          |
| External Corrosion Allowance         | 0 mm           |
| Design Temperature Internal Pressure | 150 °C         |
| Design Temperature External Pressure | 150 °C         |
| Effective Diameter Multiplier        | 1.22           |
| Material Name                        | SA-516 70      |
| Allowable Stress, Ambient            | 138 MPa        |
| Allowable Stress, Operating          | 138 MPa        |
| Allowable Stress, Hydrotest          | 234 MPa        |
| Material Density                     | 0.00775 kg/cm³ |
| P Number Thickness                   | 32.004 mm      |
| Yield Stress, Operating              | 232 MPa        |
| UCS-66 Chart Curve Designation       | B              |
| External Pressure Chart Name         | CS-2           |
| UNS Number                           | K02700         |





|                                         |       |
|-----------------------------------------|-------|
| Product Form                            | Plate |
| Efficiency, Longitudinal Seam           | 0.7   |
| Efficiency, Head-to-Skirt or Circ. Seam | 0.55  |
| Weld is pre-Heated                      | No    |

|                                      |               |
|--------------------------------------|---------------|
| Element#                             | 10/22         |
| Element From Node                    | 100           |
| Element To Node                      | 110           |
| Element Type                         | Torispherical |
| Description                          | Btm Dish End  |
| Distance "FROM" to "TO"              | 50 mm         |
| Inside Diameter                      | 5906 mm       |
| Element Thickness                    | 22 mm         |
| Internal Corrosion Allowance         | 1 mm          |
| Nominal Thickness                    | 26 mm         |
| External Corrosion Allowance         | 0 mm          |
| Design Internal Pressure             | 0.3432 MPa    |
| Design Temperature Internal Pressure | 150 °C        |
| Design External Pressure             | 0.1035 MPa    |
| Design Temperature External Pressure | 150 °C        |
| Effective Diameter Multiplier        | 1.22          |
| Material Name                        | SA-516 70     |
| Efficiency, Longitudinal Seam        | 1.0           |
| Efficiency, Circumferential Seam     | 1.0           |
| Tori Head Crown Radius               | 5949 mm       |
| Tori Head Knuckle Radius             | 356.94 mm     |
| Weld is pre-Heated                   | No            |

|                                      |                              |
|--------------------------------------|------------------------------|
| Element From Node                    | 100                          |
| Detail Type                          | Liquid                       |
| Detail ID                            | Liquid 10                    |
| Dist. from "FROM" Node / Offset dist | -979.87 mm                   |
| Height/Length of Liquid              | 1029.9 mm                    |
| Liquid Density                       | 0.0009857 kg/cm <sup>3</sup> |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Element From Node                    | 100                         |
| Detail Type                          | Insulation                  |
| Detail ID                            | insu                        |
| Dist. from "FROM" Node / Offset dist | -994.3 mm                   |
| Height/Length of Insulation          | 1044.3 mm                   |
| Thickness of Insulation              | 100 mm                      |
| Density                              | 0.000352 kg/cm <sup>3</sup> |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Element From Node                    | 100                         |
| Detail Type                          | Lining                      |
| Detail ID                            | Clad                        |
| Dist. from "FROM" Node / Offset dist | -978.39 mm                  |
| Height/Length of Lining              | 1028.4 mm                   |
| Thickness of Lining                  | 3 mm                        |
| Density                              | 0.008027 kg/cm <sup>3</sup> |

|                                      |           |
|--------------------------------------|-----------|
| Element From Node                    | 100       |
| Detail Type                          | Nozzle    |
| Detail ID                            | N1 36"    |
| Dist. from "FROM" Node / Offset dist | 0 mm      |
| Nozzle Diameter                      | 36 in.    |
| Nozzle Schedule                      | None      |
| Nozzle Class                         | 150       |
| Layout Angle                         | 0.0       |
| Blind Flange (Y/N)                   | N         |
| Weight of Nozzle ( Used if > 0 )     | 25921 N   |
| Grade of Attached Flange             | GR 1.1    |
| Nozzle Matl                          | SA-516 70 |

|                                      |        |
|--------------------------------------|--------|
| Element From Node                    | 100    |
| Detail Type                          | Nozzle |
| Detail ID                            | N11 2" |
| Dist. from "FROM" Node / Offset dist | 650 mm |



|                                  |          |     |
|----------------------------------|----------|-----|
| Nozzle Diameter                  | 2        | in. |
| Nozzle Schedule                  | 80       |     |
| Nozzle Class                     | 600      |     |
| Layout Angle                     | 180.0    |     |
| Blind Flange (Y/N)               | N        |     |
| Weight of Nozzle ( Used if > 0 ) | 0        | N   |
| Grade of Attached Flange         | GR 1.1   |     |
| Nozzle Matl                      | SA-106 B |     |

|                                      |            |
|--------------------------------------|------------|
| Element#                             | 11/22      |
| Element From Node                    | 110        |
| Element To Node                      | 120        |
| Element Type                         | Cylinder   |
| Description                          | Shell-1    |
| Distance "FROM" to "TO"              | 3000 mm    |
| Inside Diameter                      | 5906 mm    |
| Element Thickness                    | 16 mm      |
| Internal Corrosion Allowance         | 1 mm       |
| Nominal Thickness                    | 16 mm      |
| External Corrosion Allowance         | 0 mm       |
| Design Internal Pressure             | 0.3432 MPa |
| Design Temperature Internal Pressure | 150 °C     |
| Design External Pressure             | 0.1035 MPa |
| Design Temperature External Pressure | 150 °C     |
| Effective Diameter Multiplier        | 1.22       |
| Material Name                        | SA-516 70  |
| Efficiency, Longitudinal Seam        | 1.0        |
| Efficiency, Circumferential Seam     | 1.0        |
| Weld is pre-Heated                   | No         |

|                                      |                  |
|--------------------------------------|------------------|
| Element From Node                    | 110              |
| Detail Type                          | Liquid           |
| Detail ID                            | Liquid 20        |
| Dist. from "FROM" Node / Offset dist | 0 mm             |
| Height/Length of Liquid              | 3000 mm          |
| Liquid Density                       | 0.0009857 kg/cm³ |

|                                      |                 |
|--------------------------------------|-----------------|
| Element From Node                    | 110             |
| Detail Type                          | Insulation      |
| Detail ID                            | insu            |
| Dist. from "FROM" Node / Offset dist | 0 mm            |
| Height/Length of Insulation          | 3000 mm         |
| Thickness of Insulation              | 100 mm          |
| Density                              | 0.000352 kg/cm³ |

|                                      |                 |
|--------------------------------------|-----------------|
| Element From Node                    | 110             |
| Detail Type                          | Lining          |
| Detail ID                            | Clad            |
| Dist. from "FROM" Node / Offset dist | 0 mm            |
| Height/Length of Lining              | 3000 mm         |
| Thickness of Lining                  | 3 mm            |
| Density                              | 0.008027 kg/cm³ |

|                                      |              |
|--------------------------------------|--------------|
| Element From Node                    | 110          |
| Detail Type                          | Ring         |
| Detail ID                            | SR 1 Fr 70   |
| Dist. from "FROM" Node / Offset dist | 950 mm       |
| Stiffening Ring Moment of Inertia    | 31342198 mm⁴ |
| Distance from Shell Surface to Centr | 105.16 mm    |
| Stiffening Ring Cross Sectional Area | 3974.2 mm²   |
| Material Name                        | SA-516 70    |
| Stiffening Ring Section Name         | W8X21        |
| Height of Section Ring               | 210.31 mm    |
| Using Custom Stiffener Section       | No           |

|                   |            |
|-------------------|------------|
| Element From Node | 110        |
| Detail Type       | Ring       |
| Detail ID         | SR 2 Fr 70 |



|                                      |            |                 |
|--------------------------------------|------------|-----------------|
| Dist. from "FROM" Node / Offset dist | 2950       | mm              |
| Stiffening Ring Moment of Inertia    | 31342198   | mm <sup>4</sup> |
| Distance from Shell Surface to Centr | 105.16     | mm              |
| Stiffening Ring Cross Sectional Area | 3974.2     | mm <sup>2</sup> |
| Material Name                        | SA-516 70  |                 |
| Stiffening Ring Section Name         | W8X21      |                 |
| Height of Section Ring               | 210.31     | mm              |
| Using Custom Stiffener Section       | No         |                 |
|                                      |            |                 |
| Element From Node                    | 110        |                 |
| Detail Type                          | Nozzle     |                 |
| Detail ID                            | K1a        |                 |
| Dist. from "FROM" Node / Offset dist | 150        | mm              |
| Nozzle Diameter                      | 2          | in.             |
| Nozzle Schedule                      | None       |                 |
| Nozzle Class                         | 150        |                 |
| Layout Angle                         | 0.0        |                 |
| Blind Flange (Y/N)                   | N          |                 |
| Weight of Nozzle ( Used if > 0 )     | 0          | N               |
| Grade of Attached Flange             | GR 1.1     |                 |
| Nozzle Matl                          | SA-105 +IT |                 |
|                                      |            |                 |
| Element From Node                    | 110        |                 |
| Detail Type                          | Nozzle     |                 |
| Detail ID                            | K2a        |                 |
| Dist. from "FROM" Node / Offset dist | 150        | mm              |
| Nozzle Diameter                      | 2          | in.             |
| Nozzle Schedule                      | None       |                 |
| Nozzle Class                         | 150        |                 |
| Layout Angle                         | 90.0       |                 |
| Blind Flange (Y/N)                   | N          |                 |
| Weight of Nozzle ( Used if > 0 )     | 0          | N               |
| Grade of Attached Flange             | GR 1.1     |                 |
| Nozzle Matl                          | SA-105 +IT |                 |
|                                      |            |                 |
| Element From Node                    | 110        |                 |
| Detail Type                          | Nozzle     |                 |
| Detail ID                            | K3a        |                 |
| Dist. from "FROM" Node / Offset dist | 150        | mm              |
| Nozzle Diameter                      | 2          | in.             |
| Nozzle Schedule                      | None       |                 |
| Nozzle Class                         | 150        |                 |
| Layout Angle                         | 0.0        |                 |
| Blind Flange (Y/N)                   | N          |                 |
| Weight of Nozzle ( Used if > 0 )     | 0          | N               |
| Grade of Attached Flange             | GR 1.1     |                 |
| Nozzle Matl                          | SA-105 +IT |                 |
|                                      |            |                 |
| Element From Node                    | 110        |                 |
| Detail Type                          | Nozzle     |                 |
| Detail ID                            | K4a        |                 |
| Dist. from "FROM" Node / Offset dist | 150        | mm              |
| Nozzle Diameter                      | 2          | in.             |
| Nozzle Schedule                      | None       |                 |
| Nozzle Class                         | 150        |                 |
| Layout Angle                         | 0.0        |                 |
| Blind Flange (Y/N)                   | N          |                 |
| Weight of Nozzle ( Used if > 0 )     | 0          | N               |
| Grade of Attached Flange             | GR 1.1     |                 |
| Nozzle Matl                          | SA-105 +IT |                 |
|                                      |            |                 |
| Element From Node                    | 110        |                 |
| Detail Type                          | Nozzle     |                 |
| Detail ID                            | N9         |                 |
| Dist. from "FROM" Node / Offset dist | 150        | mm              |
| Nozzle Diameter                      | 2          | in.             |
| Nozzle Schedule                      | None       |                 |
| Nozzle Class                         | 150        |                 |
| Layout Angle                         | 315.0      |                 |
| Blind Flange (Y/N)                   | N          |                 |



|                                      |        |     |
|--------------------------------------|--------|-----|
| Weight of Nozzle ( Used if > 0 )     | 0      | N   |
| Grade of Attached Flange             | GR 1.1 |     |
| Nozzle Matl                          | SA-105 | +IT |
| Element From Node                    | 110    |     |
| Detail Type                          | Nozzle |     |
| Detail ID                            | N8     | 2"  |
| Dist. from "FROM" Node / Offset dist | 400    | mm  |
| Nozzle Diameter                      | 2      | in. |
| Nozzle Schedule                      | None   |     |
| Nozzle Class                         | 600    |     |
| Layout Angle                         | 215.0  |     |
| Blind Flange (Y/N)                   | N      |     |
| Weight of Nozzle ( Used if > 0 )     | 0      | N   |
| Grade of Attached Flange             | GR 1.1 |     |
| Nozzle Matl                          | SA-106 | B   |

|                                      |          |     |
|--------------------------------------|----------|-----|
| Element#                             | 12/22    |     |
| Element From Node                    | 120      |     |
| Element To Node                      | 130      |     |
| Element Type                         | Cylinder |     |
| Description                          | Shell-2  |     |
| Distance "FROM" to "TO"              | 3000     | mm  |
| Inside Diameter                      | 5906     | mm  |
| Element Thickness                    | 16       | mm  |
| Internal Corrosion Allowance         | 1        | mm  |
| Nominal Thickness                    | 16       | mm  |
| External Corrosion Allowance         | 0        | mm  |
| Design Internal Pressure             | 0.3432   | MPa |
| Design Temperature Internal Pressure | 150      | °C  |
| Design External Pressure             | 0.1035   | MPa |
| Design Temperature External Pressure | 150      | °C  |
| Effective Diameter Multiplier        | 1.22     |     |
| Material Name                        | SA-516   | 70  |
| Efficiency, Longitudinal Seam        | 1.0      |     |
| Efficiency, Circumferential Seam     | 1.0      |     |
| Weld is pre-Heated                   | Yes      |     |

|                                      |                |        |
|--------------------------------------|----------------|--------|
| Element From Node                    | 120            |        |
| Detail Type                          | Packing        |        |
| Detail ID                            | Pack Bed#1-(1) |        |
| Dist. from "FROM" Node / Offset dist | 2420           | mm     |
| Height of Packed Section             | 580            | mm     |
| Density                              | 0.000185       | kg/cm³ |
| Percent Volume Holdup                | 10.0           |        |
| Specific Gravity of Packing Liquid   | 0.98799998     |        |

|                                      |           |        |
|--------------------------------------|-----------|--------|
| Element From Node                    | 120       |        |
| Detail Type                          | Liquid    |        |
| Detail ID                            | Btm Liq-2 |        |
| Dist. from "FROM" Node / Offset dist | 0         | mm     |
| Height/Length of Liquid              | 1012      | mm     |
| Liquid Density                       | 0.0009857 | kg/cm³ |

|                                      |            |        |
|--------------------------------------|------------|--------|
| Element From Node                    | 120        |        |
| Detail Type                          | Insulation |        |
| Detail ID                            | insu       |        |
| Dist. from "FROM" Node / Offset dist | 0          | mm     |
| Height/Length of Insulation          | 3000       | mm     |
| Thickness of Insulation              | 100        | mm     |
| Density                              | 0.000352   | kg/cm³ |

|                                      |        |    |
|--------------------------------------|--------|----|
| Element From Node                    | 120    |    |
| Detail Type                          | Lining |    |
| Detail ID                            | Clad   |    |
| Dist. from "FROM" Node / Offset dist | 0      | mm |
| Height/Length of Lining              | 3000   | mm |
| Thickness of Lining                  | 3      | mm |



Density 0.008027 kg/cm<sup>3</sup>

Element From Node 120  
 Detail Type Ring  
 Detail ID SR 3 Fr 80  
 Dist. from "FROM" Node / Offset dist 1950 mm  
 Stiffening Ring Moment of Inertia 31342198 mm<sup>4</sup>  
 Distance from Shell Surface to Centr 105.16 mm  
 Stiffening Ring Cross Sectional Area 3974.2 mm<sup>2</sup>  
 Material Name SA-516 70  
 Stiffening Ring Section Name W8X21  
 Height of Section Ring 210.31 mm  
 Using Custom Stiffener Section No

Element From Node 120  
 Detail Type Nozzle  
 Detail ID K1b  
 Dist. from "FROM" Node / Offset dist 2950 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 150  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 0 N  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-105 +IT

Element From Node 120  
 Detail Type Nozzle  
 Detail ID K2b  
 Dist. from "FROM" Node / Offset dist 1300 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 150  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 0 N  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-105 +IT

Element From Node 120  
 Detail Type Nozzle  
 Detail ID K3b  
 Dist. from "FROM" Node / Offset dist 1300 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 150  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 0 N  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-105 +IT

Element From Node 120  
 Detail Type Nozzle  
 Detail ID K4b  
 Dist. from "FROM" Node / Offset dist 1600 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 150  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 0 N  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-105 +IT

Element From Node 120  
 Detail Type Nozzle  
 Detail ID N2a  
 Dist. from "FROM" Node / Offset dist 950 mm



|                                  |           |     |
|----------------------------------|-----------|-----|
| Nozzle Diameter                  | 62        | in. |
| Nozzle Schedule                  | None      |     |
| Nozzle Class                     | None      |     |
| Layout Angle                     | -32.0     |     |
| Blind Flange (Y/N)               | N         |     |
| Weight of Nozzle ( Used if > 0 ) | 5322.6    | N   |
| Grade of Attached Flange         | None      |     |
| Nozzle Matl                      | SA-516 70 |     |

|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element From Node                    | 120       |     |
| Detail Type                          | Nozzle    |     |
| Detail ID                            | N2b       |     |
| Dist. from "FROM" Node / Offset dist | 950       | mm  |
| Nozzle Diameter                      | 62        | in. |
| Nozzle Schedule                      | None      |     |
| Nozzle Class                         | None      |     |
| Layout Angle                         | 32.0      |     |
| Blind Flange (Y/N)                   | N         |     |
| Weight of Nozzle ( Used if > 0 )     | 5322.6    | N   |
| Grade of Attached Flange             | None      |     |
| Nozzle Matl                          | SA-516 70 |     |

|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element From Node                    | 120       |     |
| Detail Type                          | Nozzle    |     |
| Detail ID                            | N3A       |     |
| Dist. from "FROM" Node / Offset dist | 2570      | mm  |
| Nozzle Diameter                      | 32        | in. |
| Nozzle Schedule                      | None      |     |
| Nozzle Class                         | 150       |     |
| Layout Angle                         | 180.0     |     |
| Blind Flange (Y/N)                   | N         |     |
| Weight of Nozzle ( Used if > 0 )     | 3687.8    | N   |
| Grade of Attached Flange             | GR 1.1    |     |
| Nozzle Matl                          | SA-516 70 |     |

|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element From Node                    | 120       |     |
| Detail Type                          | Nozzle    |     |
| Detail ID                            | N4a       |     |
| Dist. from "FROM" Node / Offset dist | 950       | mm  |
| Nozzle Diameter                      | 24        | in. |
| Nozzle Schedule                      | None      |     |
| Nozzle Class                         | 150       |     |
| Layout Angle                         | 242.0     |     |
| Blind Flange (Y/N)                   | N         |     |
| Weight of Nozzle ( Used if > 0 )     | 2748      | N   |
| Grade of Attached Flange             | GR 1.1    |     |
| Nozzle Matl                          | SA-516 70 |     |

|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element From Node                    | 120       |     |
| Detail Type                          | Nozzle    |     |
| Detail ID                            | N4b       |     |
| Dist. from "FROM" Node / Offset dist | 950       | mm  |
| Nozzle Diameter                      | 24        | in. |
| Nozzle Schedule                      | None      |     |
| Nozzle Class                         | 150       |     |
| Layout Angle                         | 280.0     |     |
| Blind Flange (Y/N)                   | N         |     |
| Weight of Nozzle ( Used if > 0 )     | 2748      | N   |
| Grade of Attached Flange             | GR 1.1    |     |
| Nozzle Matl                          | SA-516 70 |     |

|                                      |        |     |
|--------------------------------------|--------|-----|
| Element From Node                    | 120    |     |
| Detail Type                          | Nozzle |     |
| Detail ID                            | A130"  |     |
| Dist. from "FROM" Node / Offset dist | 2600   | mm  |
| Nozzle Diameter                      | 30     | in. |
| Nozzle Schedule                      | None   |     |
| Nozzle Class                         | 150    |     |
| Layout Angle                         | 120.0  |     |
| Blind Flange (Y/N)                   | Y      |     |



|                                      |                 |     |
|--------------------------------------|-----------------|-----|
| Weight of Nozzle ( Used if > 0 )     | 6700            | N   |
| Grade of Attached Flange             | GR 1.1          |     |
| Nozzle Matl                          | SA-516 70       |     |
| Element From Node                    | 120             |     |
| Detail Type                          | Nozzle          |     |
| Detail ID                            | N3B             |     |
| Dist. from "FROM" Node / Offset dist | 2570            | mm  |
| Nozzle Diameter                      | 32              | in. |
| Nozzle Schedule                      | None            |     |
| Nozzle Class                         | 150             |     |
| Layout Angle                         | 240.0           |     |
| Blind Flange (Y/N)                   | N               |     |
| Weight of Nozzle ( Used if > 0 )     | 2862.7          | N   |
| Grade of Attached Flange             | GR 1.1          |     |
| Nozzle Matl                          | SA-516 70       |     |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | N2A SCHOPENTOET |     |
| Dist. from "FROM" Node / Offset dist | 1500            | mm  |
| Miscellaneous Weight                 | 14219           | N   |
| Offset from Element Centerline       | 1900            | mm  |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | N2B SCHOPENTOET |     |
| Dist. from "FROM" Node / Offset dist | 1500            | mm  |
| Miscellaneous Weight                 | 14219           | N   |
| Offset from Element Centerline       | 1900            | mm  |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | N2A INSERT PLAT |     |
| Dist. from "FROM" Node / Offset dist | 950             | mm  |
| Miscellaneous Weight                 | 16670           | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | N2B INSERT PLAT |     |
| Dist. from "FROM" Node / Offset dist | 950             | mm  |
| Miscellaneous Weight                 | 16670           | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | N3 INSERT PLAT  |     |
| Dist. from "FROM" Node / Offset dist | 2570            | mm  |
| Miscellaneous Weight                 | 3432.2          | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | CTDEAD WT       |     |
| Dist. from "FROM" Node / Offset dist | 1950            | mm  |
| Miscellaneous Weight                 | 43637           | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | CT LIQ WT       |     |
| Dist. from "FROM" Node / Offset dist | 1950            | mm  |
| Miscellaneous Weight                 | 102963          | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 120             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | btm Grid-1 DEAD |     |
| Dist. from "FROM" Node / Offset dist | 2420            | mm  |



|                                      |                 |    |
|--------------------------------------|-----------------|----|
| Miscellaneous Weight                 | 19612           | N  |
| Offset from Element Centerline       | 0               | mm |
| Element From Node                    | 120             |    |
| Detail Type                          | Weight          |    |
| Detail ID                            | N4AOVHPP        |    |
| Dist. from "FROM" Node / Offset dist | 950             | mm |
| Miscellaneous Weight                 | 82478           | N  |
| Offset from Element Centerline       | 4476            | mm |
| Element From Node                    | 120             |    |
| Detail Type                          | Weight          |    |
| Detail ID                            | N4BOVHPP        |    |
| Dist. from "FROM" Node / Offset dist | 950             | mm |
| Miscellaneous Weight                 | 82478           | N  |
| Offset from Element Centerline       | 4476            | mm |
| Element From Node                    | 120             |    |
| Detail Type                          | Weight          |    |
| Detail ID                            | N3B INSERT PLAT |    |
| Dist. from "FROM" Node / Offset dist | 2570            | mm |
| Miscellaneous Weight                 | 3432.1          | N  |
| Offset from Element Centerline       | 0               | mm |

|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element#                             | 13/22     |     |
| Element From Node                    | 130       |     |
| Element To Node                      | 140       |     |
| Element Type                         | Cylinder  |     |
| Description                          | Shell-3   |     |
| Distance "FROM" to "TO"              | 3000      | mm  |
| Inside Diameter                      | 5906      | mm  |
| Element Thickness                    | 14        | mm  |
| Internal Corrosion Allowance         | 1         | mm  |
| Nominal Thickness                    | 14        | mm  |
| External Corrosion Allowance         | 0         | mm  |
| Design Internal Pressure             | 0.3432    | MPa |
| Design Temperature Internal Pressure | 150       | °C  |
| Design External Pressure             | 0.1035    | MPa |
| Design Temperature External Pressure | 150       | °C  |
| Effective Diameter Multiplier        | 1.22      |     |
| Material Name                        | SA-516 70 |     |
| Efficiency, Longitudinal Seam        | 1.0       |     |
| Efficiency, Circumferential Seam     | 1.0       |     |
| Weld is pre-Heated                   | No        |     |

|                                      |          |                    |
|--------------------------------------|----------|--------------------|
| Element From Node                    | 130      |                    |
| Detail Type                          | Platform |                    |
| Detail ID                            | PLT6     |                    |
| Dist. from "FROM" Node / Offset dist | 1950     | mm                 |
| Platform Start Angle (degrees)       | 180.0    |                    |
| Platform End Angle (degrees)         | 0.0      |                    |
| Platform Wind Area                   | 2999999  | mm <sup>2</sup>    |
| Platform Weight                      | 30276    | N                  |
| Platform Railing Weight              | 0        | N/mm               |
| Platform Grating Weight              | 0.017    | kg/cm <sup>2</sup> |
| Platform Width                       | 1500     | mm                 |
| Platform Height                      | 1000     | mm                 |
| Platform Clearance or End Offset     | 135      | mm                 |
| Platform Force Coefficient           | 1.0      |                    |
| Ladder Layout Angle                  | 0.0      |                    |
| Ladder Start Elevation               | 0        | mm                 |
| Ladder End Elevation                 | 0        | mm                 |
| Unit Weight of Ladder                | 0        | N/mm               |
| Platform Length (top head platform)  | 0        | mm                 |

|                   |                |
|-------------------|----------------|
| Element From Node | 130            |
| Detail Type       | Packing        |
| Detail ID         | Pack Bed#1-(1) |





|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Dist. from "FROM" Node / Offset dist | 0.002326   | mm                 |
| Height of Packed Section             | 3000       | mm                 |
| Density                              | 0.000185   | kg/cm <sup>3</sup> |
| Percent Volume Holdup                | 10.0       |                    |
| Specific Gravity of Packing Liquid   | 0.98799998 |                    |
|                                      |            |                    |
| Element From Node                    | 130        |                    |
| Detail Type                          | Insulation |                    |
| Detail ID                            | insu       |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Height/Length of Insulation          | 3000       | mm                 |
| Thickness of Insulation              | 100        | mm                 |
| Density                              | 0.000352   | kg/cm <sup>3</sup> |
|                                      |            |                    |
| Element From Node                    | 130        |                    |
| Detail Type                          | Lining     |                    |
| Detail ID                            | Clad       |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Height/Length of Lining              | 3000       | mm                 |
| Thickness of Lining                  | 3          | mm                 |
| Density                              | 0.008027   | kg/cm <sup>3</sup> |
|                                      |            |                    |
| Element From Node                    | 130        |                    |
| Detail Type                          | Ring       |                    |
| Detail ID                            | SR 4 Fr 90 |                    |
| Dist. from "FROM" Node / Offset dist | 950        | mm                 |
| Stiffening Ring Moment of Inertia    | 31342198   | mm <sup>4</sup>    |
| Distance from Shell Surface to Centr | 105.16     | mm                 |
| Stiffening Ring Cross Sectional Area | 3974.2     | mm <sup>2</sup>    |
| Material Name                        | SA-516 70  |                    |
| Stiffening Ring Section Name         | W8X21      |                    |
| Height of Section Ring               | 210.31     | mm                 |
| Using Custom Stiffener Section       | No         |                    |
|                                      |            |                    |
| Element From Node                    | 130        |                    |
| Detail Type                          | Ring       |                    |
| Detail ID                            | SR 5 Fr 90 |                    |
| Dist. from "FROM" Node / Offset dist | 2950       | mm                 |
| Stiffening Ring Moment of Inertia    | 31342198   | mm <sup>4</sup>    |
| Distance from Shell Surface to Centr | 105.16     | mm                 |
| Stiffening Ring Cross Sectional Area | 3974.2     | mm <sup>2</sup>    |
| Material Name                        | SA-516 70  |                    |
| Stiffening Ring Section Name         | W8X21      |                    |
| Height of Section Ring               | 210.31     | mm                 |
| Using Custom Stiffener Section       | No         |                    |
|                                      |            |                    |
| Element From Node                    | 130        |                    |
| Detail Type                          | Nozzle     |                    |
| Detail ID                            | K5a        |                    |
| Dist. from "FROM" Node / Offset dist | 2720       | mm                 |
| Nozzle Diameter                      | 2          | in.                |
| Nozzle Schedule                      | None       |                    |
| Nozzle Class                         | 150        |                    |
| Layout Angle                         | 0.0        |                    |
| Blind Flange (Y/N)                   | N          |                    |
| Weight of Nozzle ( Used if > 0 )     | 0          | N                  |
| Grade of Attached Flange             | GR 1.1     |                    |
| Nozzle Matl                          | SA-105 +IT |                    |
|                                      |            |                    |
| Element From Node                    | 130        |                    |
| Detail Type                          | Nozzle     |                    |
| Detail ID                            | K6a        |                    |
| Dist. from "FROM" Node / Offset dist | 2720       | mm                 |
| Nozzle Diameter                      | 2          | in.                |
| Nozzle Schedule                      | None       |                    |
| Nozzle Class                         | 150        |                    |
| Layout Angle                         | 0.0        |                    |
| Blind Flange (Y/N)                   | N          |                    |
| Weight of Nozzle ( Used if > 0 )     | 0          | N                  |
| Grade of Attached Flange             | GR 1.1     |                    |



Nozzle Matl SA-105 +IT

Element From Node 130  
 Detail Type Weight  
 Detail ID N4A INSERT PLT  
 Dist. from "FROM" Node / Offset dist 1070 mm  
 Miscellaneous Weight 1470.9 N  
 Offset from Element Centerline 0 mm

Element From Node 130  
 Detail Type Weight  
 Detail ID N4B INSERT PLT  
 Dist. from "FROM" Node / Offset dist 1070 mm  
 Miscellaneous Weight 1470.9 N  
 Offset from Element Centerline 0 mm

Element# 14/22  
 Element From Node 140  
 Element To Node 150  
 Element Type Cylinder  
 Description Shell-4  
 Distance "FROM" to "TO" 3000 mm  
 Inside Diameter 5906 mm  
 Element Thickness 14 mm  
 Internal Corrosion Allowance 1 mm  
 Nominal Thickness 14 mm  
 External Corrosion Allowance 0 mm  
 Design Internal Pressure 0.3432 MPa  
 Design Temperature Internal Pressure 150 °C  
 Design External Pressure 0.1035 MPa  
 Design Temperature External Pressure 150 °C  
 Effective Diameter Multiplier 1.22  
 Material Name SA-516 70  
 Efficiency, Longitudinal Seam 1.0  
 Efficiency, Circumferential Seam 1.0  
 Weld is pre-Heated No

Element From Node 140  
 Detail Type Platform  
 Detail ID PLT7  
 Dist. from "FROM" Node / Offset dist 2950 mm  
 Platform Start Angle (degrees) 0.0  
 Platform End Angle (degrees) 180.0  
 Platform Wind Area 4950720 mm<sup>2</sup>  
 Platform Weight 31727 N  
 Platform Railing Weight 0 N/mm  
 Platform Grating Weight 0.017 kg/cm<sup>2</sup>  
 Platform Width 1500 mm  
 Platform Height 1000 mm  
 Platform Clearance or End Offset 135 mm  
 Platform Force Coefficient 1.0  
 Ladder Layout Angle 0.0  
 Ladder Start Elevation 27000 mm  
 Ladder End Elevation 31000 mm  
 Unit Weight of Ladder 0.3628 N/mm  
 Platform Length (top head platform) 0 mm

Element From Node 140  
 Detail Type Packing  
 Detail ID Pack Bed#1-(1)  
 Dist. from "FROM" Node / Offset dist 0 mm  
 Height of Packed Section 920 mm  
 Density 0.000185 kg/cm<sup>3</sup>  
 Percent Volume Holdup 10.0  
 Specific Gravity of Packing Liquid 0.98799998

Element From Node 140  
 Detail Type Insulation



|                                      |             |                    |
|--------------------------------------|-------------|--------------------|
| Detail ID                            | insu        |                    |
| Dist. from "FROM" Node / Offset dist | 0           | mm                 |
| Height/Length of Insulation          | 3000        | mm                 |
| Thickness of Insulation              | 100         | mm                 |
| Density                              | 0.000352    | kg/cm <sup>3</sup> |
|                                      |             |                    |
| Element From Node                    | 140         |                    |
| Detail Type                          | Lining      |                    |
| Detail ID                            | Clad        |                    |
| Dist. from "FROM" Node / Offset dist | 0           | mm                 |
| Height/Length of Lining              | 3000        | mm                 |
| Thickness of Lining                  | 3           | mm                 |
| Density                              | 0.008027    | kg/cm <sup>3</sup> |
|                                      |             |                    |
| Element From Node                    | 140         |                    |
| Detail Type                          | Ring        |                    |
| Detail ID                            | SR 6 Fr 100 |                    |
| Dist. from "FROM" Node / Offset dist | 1950        | mm                 |
| Stiffening Ring Moment of Inertia    | 31342198    | mm <sup>4</sup>    |
| Distance from Shell Surface to Centr | 105.16      | mm                 |
| Stiffening Ring Cross Sectional Area | 3974.2      | mm <sup>2</sup>    |
| Material Name                        | SA-516 70   |                    |
| Stiffening Ring Section Name         | W8X21       |                    |
| Height of Section Ring               | 210.31      | mm                 |
| Using Custom Stiffener Section       | No          |                    |
|                                      |             |                    |
| Element From Node                    | 140         |                    |
| Detail Type                          | Nozzle      |                    |
| Detail ID                            | K5b         |                    |
| Dist. from "FROM" Node / Offset dist | 770         | mm                 |
| Nozzle Diameter                      | 2           | in.                |
| Nozzle Schedule                      | None        |                    |
| Nozzle Class                         | 150         |                    |
| Layout Angle                         | 0.0         |                    |
| Blind Flange (Y/N)                   | N           |                    |
| Weight of Nozzle ( Used if > 0 )     | 0           | N                  |
| Grade of Attached Flange             | GR 1.1      |                    |
| Nozzle Matl                          | SA-105 +IT  |                    |
|                                      |             |                    |
| Element From Node                    | 140         |                    |
| Detail Type                          | Nozzle      |                    |
| Detail ID                            | K6b         |                    |
| Dist. from "FROM" Node / Offset dist | 770         | mm                 |
| Nozzle Diameter                      | 2           | in.                |
| Nozzle Schedule                      | None        |                    |
| Nozzle Class                         | 150         |                    |
| Layout Angle                         | 0.0         |                    |
| Blind Flange (Y/N)                   | N           |                    |
| Weight of Nozzle ( Used if > 0 )     | 0           | N                  |
| Grade of Attached Flange             | GR 1.1      |                    |
| Nozzle Matl                          | SA-105 +IT  |                    |
|                                      |             |                    |
| Element From Node                    | 140         |                    |
| Detail Type                          | Nozzle      |                    |
| Detail ID                            | K7a         |                    |
| Dist. from "FROM" Node / Offset dist | 1320        | mm                 |
| Nozzle Diameter                      | 2           | in.                |
| Nozzle Schedule                      | None        |                    |
| Nozzle Class                         | 150         |                    |
| Layout Angle                         | 0.0         |                    |
| Blind Flange (Y/N)                   | N           |                    |
| Weight of Nozzle ( Used if > 0 )     | 0           | N                  |
| Grade of Attached Flange             | GR 1.1      |                    |
| Nozzle Matl                          | SA-105 +IT  |                    |
|                                      |             |                    |
| Element From Node                    | 140         |                    |
| Detail Type                          | Nozzle      |                    |
| Detail ID                            | A230"       |                    |
| Dist. from "FROM" Node / Offset dist | 1050        | mm                 |
| Nozzle Diameter                      | 30          | in.                |



|                                      |                |    |
|--------------------------------------|----------------|----|
| Nozzle Schedule                      | None           |    |
| Nozzle Class                         | 150            |    |
| Layout Angle                         | 30.0           |    |
| Blind Flange (Y/N)                   | Y              |    |
| Weight of Nozzle ( Used if > 0 )     | 0              | N  |
| Grade of Attached Flange             | GR 1.1         |    |
| Nozzle Matl                          | SA-516 70      |    |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | 10T Margin     |    |
| Dist. from "FROM" Node / Offset dist | 1205           | mm |
| Miscellaneous Weight                 | 98062          | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | Vane1 DEAD WT  |    |
| Dist. from "FROM" Node / Offset dist | 2000           | mm |
| Miscellaneous Weight                 | 29418          | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | Vane1 LIQ WT   |    |
| Dist. from "FROM" Node / Offset dist | 2000           | mm |
| Miscellaneous Weight                 | 34321          | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | dist.1 DEAD W  |    |
| Dist. from "FROM" Node / Offset dist | 920            | mm |
| Miscellaneous Weight                 | 23093          | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | dist.1 LIQ WT  |    |
| Dist. from "FROM" Node / Offset dist | 920            | mm |
| Miscellaneous Weight                 | 79429          | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | Cleat wt       |    |
| Dist. from "FROM" Node / Offset dist | 1205           | mm |
| Miscellaneous Weight                 | 93157          | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | Lifting wt     |    |
| Dist. from "FROM" Node / Offset dist | 1205           | mm |
| Miscellaneous Weight                 | 58836          | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | Cladded Nozzle |    |
| Dist. from "FROM" Node / Offset dist | 0.002325       | mm |
| Miscellaneous Weight                 | 5393.3         | N  |
| Offset from Element Centerline       | 0              | mm |
| Element From Node                    | 140            |    |
| Detail Type                          | Weight         |    |
| Detail ID                            | Manway davit   |    |
| Dist. from "FROM" Node / Offset dist | 1150           | mm |
| Miscellaneous Weight                 | 3432.1         | N  |
| Offset from Element Centerline       | -1.0101        | mm |



|                                      |                |        |
|--------------------------------------|----------------|--------|
| Element#                             | 15/22          |        |
| Element From Node                    | 150            |        |
| Element To Node                      | 160            |        |
| Element Type                         | Cylinder       |        |
| Description                          | Shell-5        |        |
| Distance "FROM" to "TO"              | 3000           | mm     |
| Inside Diameter                      | 5906           | mm     |
| Element Thickness                    | 14             | mm     |
| Internal Corrosion Allowance         | 1              | mm     |
| Nominal Thickness                    | 14             | mm     |
| External Corrosion Allowance         | 0              | mm     |
| Design Internal Pressure             | 0.3432         | MPa    |
| Design Temperature Internal Pressure | 150            | °C     |
| Design External Pressure             | 0.1035         | MPa    |
| Design Temperature External Pressure | 150            | °C     |
| Effective Diameter Multiplier        | 1.22           |        |
| Material Name                        | SA-516 70      |        |
| Efficiency, Longitudinal Seam        | 1.0            |        |
| Efficiency, Circumferential Seam     | 1.0            |        |
| Weld is pre-Heated                   | No             |        |
|                                      |                |        |
| Element From Node                    | 150            |        |
| Detail Type                          | Packing        |        |
| Detail ID                            | Pack Bed#2-(2) |        |
| Dist. from "FROM" Node / Offset dist | 1485           | mm     |
| Height of Packed Section             | 1515           | mm     |
| Density                              | 0.000185       | kg/cm³ |
| Percent Volume Holdup                | 10.0           |        |
| Specific Gravity of Packing Liquid   | 0.98799998     |        |
|                                      |                |        |
| Element From Node                    | 150            |        |
| Detail Type                          | Insulation     |        |
| Detail ID                            | insu           |        |
| Dist. from "FROM" Node / Offset dist | 0              | mm     |
| Height/Length of Insulation          | 3000           | mm     |
| Thickness of Insulation              | 100            | mm     |
| Density                              | 0.000352       | kg/cm³ |
|                                      |                |        |
| Element From Node                    | 150            |        |
| Detail Type                          | Lining         |        |
| Detail ID                            | Clad           |        |
| Dist. from "FROM" Node / Offset dist | 0              | mm     |
| Height/Length of Lining              | 3000           | mm     |
| Thickness of Lining                  | 3              | mm     |
| Density                              | 0.008027       | kg/cm³ |
|                                      |                |        |
| Element From Node                    | 150            |        |
| Detail Type                          | Ring           |        |
| Detail ID                            | SR 7 Fr 110    |        |
| Dist. from "FROM" Node / Offset dist | 950            | mm     |
| Stiffening Ring Moment of Inertia    | 31342198       | mm⁴    |
| Distance from Shell Surface to Centr | 105.16         | mm     |
| Stiffening Ring Cross Sectional Area | 3974.2         | mm²    |
| Material Name                        | SA-516 70      |        |
| Stiffening Ring Section Name         | W8X21          |        |
| Height of Section Ring               | 210.31         | mm     |
| Using Custom Stiffener Section       | No             |        |
|                                      |                |        |
| Element From Node                    | 150            |        |
| Detail Type                          | Ring           |        |
| Detail ID                            | SR 8 Fr 110    |        |
| Dist. from "FROM" Node / Offset dist | 2950           | mm     |
| Stiffening Ring Moment of Inertia    | 31342198       | mm⁴    |
| Distance from Shell Surface to Centr | 105.16         | mm     |
| Stiffening Ring Cross Sectional Area | 3974.2         | mm²    |
| Material Name                        | SA-516 70      |        |
| Stiffening Ring Section Name         | W8X21          |        |



|                                      |           |     |
|--------------------------------------|-----------|-----|
| Height of Section Ring               | 210.31    | mm  |
| Using Custom Stiffener Section       | No        |     |
| Element From Node                    | 150       |     |
| Detail Type                          | Nozzle    |     |
| Detail ID                            | A330"     |     |
| Dist. from "FROM" Node / Offset dist | 1650      | mm  |
| Nozzle Diameter                      | 30        | in. |
| Nozzle Schedule                      | None      |     |
| Nozzle Class                         | 150       |     |
| Layout Angle                         | 30.0      |     |
| Blind Flange (Y/N)                   | Y         |     |
| Weight of Nozzle ( Used if > 0 )     | 0         | N   |
| Grade of Attached Flange             | GR 1.1    |     |
| Nozzle Matl                          | SA-516 70 |     |

|                                      |                 |    |
|--------------------------------------|-----------------|----|
| Element From Node                    | 150             |    |
| Detail Type                          | Weight          |    |
| Detail ID                            | btm Grid-2 DEAD |    |
| Dist. from "FROM" Node / Offset dist | 1485            | mm |
| Miscellaneous Weight                 | 19612           | N  |
| Offset from Element Centerline       | 0               | mm |

|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element#                             | 16/22     |     |
| Element From Node                    | 160       |     |
| Element To Node                      | 170       |     |
| Element Type                         | Cylinder  |     |
| Description                          | Shell-6   |     |
| Distance "FROM" to "TO"              | 3000      | mm  |
| Inside Diameter                      | 5906      | mm  |
| Element Thickness                    | 14        | mm  |
| Internal Corrosion Allowance         | 1         | mm  |
| Nominal Thickness                    | 14        | mm  |
| External Corrosion Allowance         | 0         | mm  |
| Design Internal Pressure             | 0.3432    | MPa |
| Design Temperature Internal Pressure | 150       | °C  |
| Design External Pressure             | 0.1035    | MPa |
| Design Temperature External Pressure | 150       | °C  |
| Effective Diameter Multiplier        | 1.22      |     |
| Material Name                        | SA-516 70 |     |
| Efficiency, Longitudinal Seam        | 1.0       |     |
| Efficiency, Circumferential Seam     | 1.0       |     |
| Weld is pre-Heated                   | No        |     |

|                                      |          |                    |
|--------------------------------------|----------|--------------------|
| Element From Node                    | 160      |                    |
| Detail Type                          | Platform |                    |
| Detail ID                            | PLT8     |                    |
| Dist. from "FROM" Node / Offset dist | 950      | mm                 |
| Platform Start Angle (degrees)       | 180.0    |                    |
| Platform End Angle (degrees)         | 0.0      |                    |
| Platform Wind Area                   | 4950720  | mm <sup>2</sup>    |
| Platform Weight                      | 31727    | N                  |
| Platform Railing Weight              | 0        | N/mm               |
| Platform Grating Weight              | 0.017    | kg/cm <sup>2</sup> |
| Platform Width                       | 1500     | mm                 |
| Platform Height                      | 1000     | mm                 |
| Platform Clearance or End Offset     | 135      | mm                 |
| Platform Force Coefficient           | 1.0      |                    |
| Ladder Layout Angle                  | 0.0      |                    |
| Ladder Start Elevation               | 31000    | mm                 |
| Ladder End Elevation                 | 35000    | mm                 |
| Unit Weight of Ladder                | 0.3628   | N/mm               |
| Platform Length (top head platform)  | 0        | mm                 |

|                                      |                |    |
|--------------------------------------|----------------|----|
| Element From Node                    | 160            |    |
| Detail Type                          | Packing        |    |
| Detail ID                            | Pack Bed#2-(2) |    |
| Dist. from "FROM" Node / Offset dist | 0.004651       | mm |



|                                    |            |                    |
|------------------------------------|------------|--------------------|
| Height of Packed Section           | 2985       | mm                 |
| Density                            | 0.000185   | kg/cm <sup>3</sup> |
| Percent Volume Holdup              | 10.0       |                    |
| Specific Gravity of Packing Liquid | 0.98799998 |                    |

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Element From Node                    | 160        |                    |
| Detail Type                          | Insulation |                    |
| Detail ID                            | insu       |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Height/Length of Insulation          | 3000       | mm                 |
| Thickness of Insulation              | 100        | mm                 |
| Density                              | 0.000352   | kg/cm <sup>3</sup> |

|                                      |          |                    |
|--------------------------------------|----------|--------------------|
| Element From Node                    | 160      |                    |
| Detail Type                          | Lining   |                    |
| Detail ID                            | Clad     |                    |
| Dist. from "FROM" Node / Offset dist | 0        | mm                 |
| Height/Length of Lining              | 3000     | mm                 |
| Thickness of Lining                  | 3        | mm                 |
| Density                              | 0.008027 | kg/cm <sup>3</sup> |

|                                      |             |                 |
|--------------------------------------|-------------|-----------------|
| Element From Node                    | 160         |                 |
| Detail Type                          | Ring        |                 |
| Detail ID                            | SR 9 Fr 120 |                 |
| Dist. from "FROM" Node / Offset dist | 1950        | mm              |
| Stiffening Ring Moment of Inertia    | 31342198    | mm <sup>4</sup> |
| Distance from Shell Surface to Centr | 105.16      | mm              |
| Stiffening Ring Cross Sectional Area | 3974.2      | mm <sup>2</sup> |
| Material Name                        | SA-516 70   |                 |
| Stiffening Ring Section Name         | W8X21       |                 |
| Height of Section Ring               | 210.31      | mm              |
| Using Custom Stiffener Section       | No          |                 |

|                                      |               |    |
|--------------------------------------|---------------|----|
| Element From Node                    | 160           |    |
| Detail Type                          | Weight        |    |
| Detail ID                            | dist.2 DEAD W |    |
| Dist. from "FROM" Node / Offset dist | 2985          | mm |
| Miscellaneous Weight                 | 23093         | N  |
| Offset from Element Centerline       | 0             | mm |

|                                      |               |    |
|--------------------------------------|---------------|----|
| Element From Node                    | 160           |    |
| Detail Type                          | Weight        |    |
| Detail ID                            | dist.2 LIQ WT |    |
| Dist. from "FROM" Node / Offset dist | 2985          | mm |
| Miscellaneous Weight                 | 79429         | N  |
| Offset from Element Centerline       | 0             | mm |

|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element#                             | 17/22     |     |
| Element From Node                    | 170       |     |
| Element To Node                      | 180       |     |
| Element Type                         | Cylinder  |     |
| Description                          | Shell-7   |     |
| Distance "FROM" to "TO"              | 3000      | mm  |
| Inside Diameter                      | 5906      | mm  |
| Element Thickness                    | 14        | mm  |
| Internal Corrosion Allowance         | 1         | mm  |
| Nominal Thickness                    | 14        | mm  |
| External Corrosion Allowance         | 0         | mm  |
| Design Internal Pressure             | 0.3432    | MPa |
| Design Temperature Internal Pressure | 150       | °C  |
| Design External Pressure             | 0.1035    | MPa |
| Design Temperature External Pressure | 150       | °C  |
| Effective Diameter Multiplier        | 1.22      |     |
| Material Name                        | SA-516 70 |     |
| Efficiency, Longitudinal Seam        | 1.0       |     |
| Efficiency, Circumferential Seam     | 1.0       |     |
| Weld is pre-Heated                   | No        |     |



|                                      |              |                    |
|--------------------------------------|--------------|--------------------|
| Element From Node                    | 170          |                    |
| Detail Type                          | Platform     |                    |
| Detail ID                            | PLT9         |                    |
| Dist. from "FROM" Node / Offset dist | 1950         | mm                 |
| Platform Start Angle (degrees)       | 0.0          |                    |
| Platform End Angle (degrees)         | 180.0        |                    |
| Platform Wind Area                   | 4950720      | mm <sup>2</sup>    |
| Platform Weight                      | 31727        | N                  |
| Platform Railing Weight              | 0            | N/mm               |
| Platform Grating Weight              | 0.017        | kg/cm <sup>2</sup> |
| Platform Width                       | 1500         | mm                 |
| Platform Height                      | 1000         | mm                 |
| Platform Clearance or End Offset     | 135          | mm                 |
| Platform Force Coefficient           | 1.0          |                    |
| Ladder Layout Angle                  | 0.0          |                    |
| Ladder Start Elevation               | 35000        | mm                 |
| Ladder End Elevation                 | 39000        | mm                 |
| Unit Weight of Ladder                | 0.3628       | N/mm               |
| Platform Length (top head platform)  | 0            | mm                 |
|                                      |              |                    |
| Element From Node                    | 170          |                    |
| Detail Type                          | Insulation   |                    |
| Detail ID                            | insu         |                    |
| Dist. from "FROM" Node / Offset dist | 0            | mm                 |
| Height/Length of Insulation          | 3000         | mm                 |
| Thickness of Insulation              | 100          | mm                 |
| Density                              | 0.000352     | kg/cm <sup>3</sup> |
|                                      |              |                    |
| Element From Node                    | 170          |                    |
| Detail Type                          | Lining       |                    |
| Detail ID                            | Clad         |                    |
| Dist. from "FROM" Node / Offset dist | 0            | mm                 |
| Height/Length of Lining              | 3000         | mm                 |
| Thickness of Lining                  | 3            | mm                 |
| Density                              | 0.008027     | kg/cm <sup>3</sup> |
|                                      |              |                    |
| Element From Node                    | 170          |                    |
| Detail Type                          | Ring         |                    |
| Detail ID                            | SR 10 Fr 130 |                    |
| Dist. from "FROM" Node / Offset dist | 950          | mm                 |
| Stiffening Ring Moment of Inertia    | 31342198     | mm <sup>4</sup>    |
| Distance from Shell Surface to Centr | 105.16       | mm                 |
| Stiffening Ring Cross Sectional Area | 3974.2       | mm <sup>2</sup>    |
| Material Name                        | SA-516 70    |                    |
| Stiffening Ring Section Name         | W8X21        |                    |
| Height of Section Ring               | 210.31       | mm                 |
| Using Custom Stiffener Section       | No           |                    |
|                                      |              |                    |
| Element From Node                    | 170          |                    |
| Detail Type                          | Ring         |                    |
| Detail ID                            | SR 11 Fr 130 |                    |
| Dist. from "FROM" Node / Offset dist | 2950         | mm                 |
| Stiffening Ring Moment of Inertia    | 31342198     | mm <sup>4</sup>    |
| Distance from Shell Surface to Centr | 105.16       | mm                 |
| Stiffening Ring Cross Sectional Area | 3974.2       | mm <sup>2</sup>    |
| Material Name                        | SA-516 70    |                    |
| Stiffening Ring Section Name         | W8X21        |                    |
| Height of Section Ring               | 210.31       | mm                 |
| Using Custom Stiffener Section       | No           |                    |
|                                      |              |                    |
| Element From Node                    | 170          |                    |
| Detail Type                          | Nozzle       |                    |
| Detail ID                            | K7b          |                    |
| Dist. from "FROM" Node / Offset dist | 485          | mm                 |
| Nozzle Diameter                      | 2            | in.                |
| Nozzle Schedule                      | STD          |                    |
| Nozzle Class                         | 300          |                    |
| Layout Angle                         | 150.0        |                    |
| Blind Flange (Y/N)                   | N            |                    |
| Weight of Nozzle ( Used if > 0 )     | 0            | N                  |





|                                      |                 |
|--------------------------------------|-----------------|
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-106 B        |
| Element From Node                    | 170             |
| Detail Type                          | Nozzle          |
| Detail ID                            | A430"           |
| Dist. from "FROM" Node / Offset dist | 1520 mm         |
| Nozzle Diameter                      | 30 in.          |
| Nozzle Schedule                      | None            |
| Nozzle Class                         | 150             |
| Layout Angle                         | 120.0           |
| Blind Flange (Y/N)                   | Y               |
| Weight of Nozzle ( Used if > 0 )     | 0 N             |
| Grade of Attached Flange             | GR 1.1          |
| Nozzle Matl                          | SA-516 70       |
| Element From Node                    | 170             |
| Detail Type                          | Weight          |
| Detail ID                            | N5 SCHOPENTOETE |
| Dist. from "FROM" Node / Offset dist | 1850 mm         |
| Miscellaneous Weight                 | 5883.6 N        |
| Offset from Element Centerline       | 0 mm            |
| Element From Node                    | 170             |
| Detail Type                          | Weight          |
| Detail ID                            | Vane2 DEAD WT   |
| Dist. from "FROM" Node / Offset dist | 2000 mm         |
| Miscellaneous Weight                 | 29418 N         |
| Offset from Element Centerline       | 0 mm            |
| Element From Node                    | 170             |
| Detail Type                          | Weight          |
| Detail ID                            | Vane2 LIQ WT    |
| Dist. from "FROM" Node / Offset dist | 2000 mm         |
| Miscellaneous Weight                 | 34321 N         |
| Offset from Element Centerline       | 0 mm            |

|                                      |            |
|--------------------------------------|------------|
| Element#                             | 18/22      |
| Element From Node                    | 180        |
| Element To Node                      | 190        |
| Element Type                         | Cylinder   |
| Description                          | Shell-8    |
| Distance "FROM" to "TO"              | 3000 mm    |
| Inside Diameter                      | 5906 mm    |
| Element Thickness                    | 14 mm      |
| Internal Corrosion Allowance         | 1 mm       |
| Nominal Thickness                    | 14 mm      |
| External Corrosion Allowance         | 0 mm       |
| Design Internal Pressure             | 0.3432 MPa |
| Design Temperature Internal Pressure | 150 °C     |
| Design External Pressure             | 0.1035 MPa |
| Design Temperature External Pressure | 150 °C     |
| Effective Diameter Multiplier        | 1.22       |
| Material Name                        | SA-516 70  |
| Efficiency, Longitudinal Seam        | 1.0        |
| Efficiency, Circumferential Seam     | 1.0        |
| Weld is pre-Heated                   | No         |

|                                      |                          |
|--------------------------------------|--------------------------|
| Element From Node                    | 180                      |
| Detail Type                          | Platform                 |
| Detail ID                            | PLT10                    |
| Dist. from "FROM" Node / Offset dist | 2950 mm                  |
| Platform Start Angle (degrees)       | 180.0                    |
| Platform End Angle (degrees)         | 0.0                      |
| Platform Wind Area                   | 4950720 mm <sup>2</sup>  |
| Platform Weight                      | 31727 N                  |
| Platform Railing Weight              | 0 N/mm                   |
| Platform Grating Weight              | 0.017 kg/cm <sup>2</sup> |



|                                      |              |                    |
|--------------------------------------|--------------|--------------------|
| Platform Width                       | 1500         | mm                 |
| Platform Height                      | 1000         | mm                 |
| Platform Clearance or End Offset     | 135          | mm                 |
| Platform Force Coefficient           | 1.0          |                    |
| Ladder Layout Angle                  | 0.0          |                    |
| Ladder Start Elevation               | 35000        | mm                 |
| Ladder End Elevation                 | 39000        | mm                 |
| Unit Weight of Ladder                | 0.3628       | N/mm               |
| Platform Length (top head platform)  | 0            | mm                 |
|                                      |              |                    |
| Element From Node                    | 180          |                    |
| Detail Type                          | Packing      |                    |
| Detail ID                            | Pack Bed#3   |                    |
| Dist. from "FROM" Node / Offset dist | 1860         | mm                 |
| Height of Packed Section             | 1000         | mm                 |
| Density                              | 0.000185     | kg/cm <sup>3</sup> |
| Percent Volume Holdup                | 15.0         |                    |
| Specific Gravity of Packing Liquid   | 0.98799998   |                    |
|                                      |              |                    |
| Element From Node                    | 180          |                    |
| Detail Type                          | Insulation   |                    |
| Detail ID                            | insu         |                    |
| Dist. from "FROM" Node / Offset dist | 0            | mm                 |
| Height/Length of Insulation          | 3000         | mm                 |
| Thickness of Insulation              | 100          | mm                 |
| Density                              | 0.000352     | kg/cm <sup>3</sup> |
|                                      |              |                    |
| Element From Node                    | 180          |                    |
| Detail Type                          | Lining       |                    |
| Detail ID                            | Clad         |                    |
| Dist. from "FROM" Node / Offset dist | 0            | mm                 |
| Height/Length of Lining              | 3000         | mm                 |
| Thickness of Lining                  | 3            | mm                 |
| Density                              | 0.008027     | kg/cm <sup>3</sup> |
|                                      |              |                    |
| Element From Node                    | 180          |                    |
| Detail Type                          | Ring         |                    |
| Detail ID                            | SR 12 Fr 140 |                    |
| Dist. from "FROM" Node / Offset dist | 1950         | mm                 |
| Stiffening Ring Moment of Inertia    | 31342198     | mm <sup>4</sup>    |
| Distance from Shell Surface to Centr | 105.16       | mm                 |
| Stiffening Ring Cross Sectional Area | 3974.2       | mm <sup>2</sup>    |
| Material Name                        | SA-516 70    |                    |
| Stiffening Ring Section Name         | W8X21        |                    |
| Height of Section Ring               | 210.31       | mm                 |
| Using Custom Stiffener Section       | No           |                    |
|                                      |              |                    |
| Element From Node                    | 180          |                    |
| Detail Type                          | Nozzle       |                    |
| Detail ID                            | N5           |                    |
| Dist. from "FROM" Node / Offset dist | 1000         | mm                 |
| Nozzle Diameter                      | 46           | in.                |
| Nozzle Schedule                      | None         |                    |
| Nozzle Class                         | 150          |                    |
| Layout Angle                         | 180.0        |                    |
| Blind Flange (Y/N)                   | N            |                    |
| Weight of Nozzle ( Used if > 0 )     | 4142.2       | N                  |
| Grade of Attached Flange             | GR 1.1       |                    |
| Nozzle Matl                          | SA-516 70    |                    |
|                                      |              |                    |
| Element From Node                    | 180          |                    |
| Detail Type                          | Nozzle       |                    |
| Detail ID                            | K8a          |                    |
| Dist. from "FROM" Node / Offset dist | 1200         | mm                 |
| Nozzle Diameter                      | 2            | in.                |
| Nozzle Schedule                      | STD          |                    |
| Nozzle Class                         | 300          |                    |
| Layout Angle                         | 270.0        |                    |
| Blind Flange (Y/N)                   | N            |                    |
| Weight of Nozzle ( Used if > 0 )     | 0            | N                  |



|                                      |                 |     |
|--------------------------------------|-----------------|-----|
| Grade of Attached Flange             | GR 1.1          |     |
| Nozzle Matl                          | SA-106 B        |     |
| Element From Node                    | 180             |     |
| Detail Type                          | Nozzle          |     |
| Detail ID                            | A530"           |     |
| Dist. from "FROM" Node / Offset dist | 1300            | mm  |
| Nozzle Diameter                      | 30              | in. |
| Nozzle Schedule                      | None            |     |
| Nozzle Class                         | 150             |     |
| Layout Angle                         | 45.0            |     |
| Blind Flange (Y/N)                   | Y               |     |
| Weight of Nozzle ( Used if > 0 )     | 0               | N   |
| Grade of Attached Flange             | GR 1.1          |     |
| Nozzle Matl                          | SA-516 70       |     |
| Element From Node                    | 180             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | btm Grid-3 DEAD |     |
| Dist. from "FROM" Node / Offset dist | 1860            | mm  |
| Miscellaneous Weight                 | 19612           | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 180             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | dist.3 DEAD W   |     |
| Dist. from "FROM" Node / Offset dist | 2860            | mm  |
| Miscellaneous Weight                 | 23093           | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 180             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | dist. 3 LIQ WT  |     |
| Dist. from "FROM" Node / Offset dist | 2860            | mm  |
| Miscellaneous Weight                 | 79429           | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 180             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | N5 INSERT PLT   |     |
| Dist. from "FROM" Node / Offset dist | 50.002          | mm  |
| Miscellaneous Weight                 | 8139            | N   |
| Offset from Element Centerline       | 0               | mm  |
| Element From Node                    | 180             |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | N5OVHPP         |     |
| Dist. from "FROM" Node / Offset dist | 1000            | mm  |
| Miscellaneous Weight                 | 460852          | N   |
| Offset from Element Centerline       | 4755            | mm  |

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|                                      |           |     |
|--------------------------------------|-----------|-----|
| Element#                             | 19/22     |     |
| Element From Node                    | 190       |     |
| Element To Node                      | 220       |     |
| Element Type                         | Cylinder  |     |
| Description                          | Shell-9   |     |
| Distance "FROM" to "TO"              | 1500      | mm  |
| Inside Diameter                      | 5906      | mm  |
| Element Thickness                    | 14        | mm  |
| Internal Corrosion Allowance         | 1         | mm  |
| Nominal Thickness                    | 14        | mm  |
| External Corrosion Allowance         | 0         | mm  |
| Design Internal Pressure             | 0.3432    | MPa |
| Design Temperature Internal Pressure | 150       | °C  |
| Design External Pressure             | 0.1035    | MPa |
| Design Temperature External Pressure | 150       | °C  |
| Effective Diameter Multiplier        | 1.22      |     |
| Material Name                        | SA-516 70 |     |



|                                      |              |                    |
|--------------------------------------|--------------|--------------------|
| Efficiency, Longitudinal Seam        | 1.0          |                    |
| Efficiency, Circumferential Seam     | 1.0          |                    |
| Weld is pre-Heated                   | No           |                    |
|                                      |              |                    |
| Element From Node                    | 190          |                    |
| Detail Type                          | Insulation   |                    |
| Detail ID                            | insu         |                    |
| Dist. from "FROM" Node / Offset dist | 0            | mm                 |
| Height/Length of Insulation          | 1500         | mm                 |
| Thickness of Insulation              | 100          | mm                 |
| Density                              | 0.000352     | kg/cm <sup>3</sup> |
|                                      |              |                    |
| Element From Node                    | 190          |                    |
| Detail Type                          | Lining       |                    |
| Detail ID                            | Clad         |                    |
| Dist. from "FROM" Node / Offset dist | 0            | mm                 |
| Height/Length of Lining              | 1500         | mm                 |
| Thickness of Lining                  | 3            | mm                 |
| Density                              | 0.008027     | kg/cm <sup>3</sup> |
|                                      |              |                    |
| Element From Node                    | 190          |                    |
| Detail Type                          | Ring         |                    |
| Detail ID                            | SR 13 Fr 150 |                    |
| Dist. from "FROM" Node / Offset dist | 950          | mm                 |
| Stiffening Ring Moment of Inertia    | 31342198     | mm <sup>4</sup>    |
| Distance from Shell Surface to Centr | 105.16       | mm                 |
| Stiffening Ring Cross Sectional Area | 3974.2       | mm <sup>2</sup>    |
| Material Name                        | SA-516 70    |                    |
| Stiffening Ring Section Name         | W8X21        |                    |
| Height of Section Ring               | 210.31       | mm                 |
| Using Custom Stiffener Section       | No           |                    |
|                                      |              |                    |
| Element From Node                    | 190          |                    |
| Detail Type                          | Nozzle       |                    |
| Detail ID                            | K9a          |                    |
| Dist. from "FROM" Node / Offset dist | 1270         | mm                 |
| Nozzle Diameter                      | 2            | in.                |
| Nozzle Schedule                      | STD          |                    |
| Nozzle Class                         | 300          |                    |
| Layout Angle                         | 150.0        |                    |
| Blind Flange (Y/N)                   | N            |                    |
| Weight of Nozzle ( Used if > 0 )     | 0            | N                  |
| Grade of Attached Flange             | GR 1.1       |                    |
| Nozzle Matl                          | SA-106 B     |                    |
|                                      |              |                    |
| Element From Node                    | 190          |                    |
| Detail Type                          | Nozzle       |                    |
| Detail ID                            | N6           |                    |
| Dist. from "FROM" Node / Offset dist | 1260         | mm                 |
| Nozzle Diameter                      | 4            | in.                |
| Nozzle Schedule                      | STD          |                    |
| Nozzle Class                         | 300          |                    |
| Layout Angle                         | 300.0        |                    |
| Blind Flange (Y/N)                   | N            |                    |
| Weight of Nozzle ( Used if > 0 )     | 0            | N                  |
| Grade of Attached Flange             | GR 1.1       |                    |
| Nozzle Matl                          | SA-106 B     |                    |
|                                      |              |                    |
| Element From Node                    | 190          |                    |
| Detail Type                          | Nozzle       |                    |
| Detail ID                            | K8b          |                    |
| Dist. from "FROM" Node / Offset dist | 660          | mm                 |
| Nozzle Diameter                      | 2            | in.                |
| Nozzle Schedule                      | STD          |                    |
| Nozzle Class                         | 300          |                    |
| Layout Angle                         | 270.0        |                    |
| Blind Flange (Y/N)                   | N            |                    |
| Weight of Nozzle ( Used if > 0 )     | 0            | N                  |
| Grade of Attached Flange             | GR 1.1       |                    |
| Nozzle Matl                          | SA-106 B     |                    |



Element From Node 190  
 Detail Type Nozzle  
 Detail ID K9b  
 Dist. from "FROM" Node / Offset dist 0.004651 mm  
 Nozzle Diameter 2 in.  
 Nozzle Schedule STD  
 Nozzle Class 300  
 Layout Angle 230.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 0 N  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-106 B

Element From Node 190  
 Detail Type Nozzle  
 Detail ID A630"  
 Dist. from "FROM" Node / Offset dist 150.01 mm  
 Nozzle Diameter 30 in.  
 Nozzle Schedule None  
 Nozzle Class 150  
 Layout Angle 45.0  
 Blind Flange (Y/N) Y  
 Weight of Nozzle ( Used if > 0 ) 0 N  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-516 70

Element From Node 190  
 Detail Type Nozzle  
 Detail ID A7 30"  
 Dist. from "FROM" Node / Offset dist 500 mm  
 Nozzle Diameter 30 in.  
 Nozzle Schedule None  
 Nozzle Class 150  
 Layout Angle 0.0  
 Blind Flange (Y/N) Y  
 Weight of Nozzle ( Used if > 0 ) 0 N  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-516 70

Element From Node 190  
 Detail Type Weight  
 Detail ID Davit  
 Dist. from "FROM" Node / Offset dist 0.004651 mm  
 Miscellaneous Weight 12748 N  
 Offset from Element Centerline 3000 mm

Element From Node 190  
 Detail Type Weight  
 Detail ID N6 INT PIPE WT  
 Dist. from "FROM" Node / Offset dist 1260 mm  
 Miscellaneous Weight 4903 N  
 Offset from Element Centerline 0 mm

---

Element# 20/22  
 Element From Node 220  
 Element To Node 230  
 Element Type Torispherical  
 Description Top Dish End  
 Distance "FROM" to "TO" 50 mm  
 Inside Diameter 5906 mm  
 Element Thickness 20 mm  
 Internal Corrosion Allowance 1 mm  
 Nominal Thickness 24 mm  
 External Corrosion Allowance 0 mm  
 Design Internal Pressure 0.3432 MPa  
 Design Temperature Internal Pressure 150 °C  
 Design External Pressure 0.1035 MPa



|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Design Temperature External Pressure | 150        | °C                 |
| Effective Diameter Multiplier        | 1.22       |                    |
| Material Name                        | SA-516 70  |                    |
| Efficiency, Longitudinal Seam        | 1.0        |                    |
| Efficiency, Circumferential Seam     | 1.0        |                    |
| Tori Head Crown Radius               | 5945       | mm                 |
| Tori Head Knuckle Radius             | 356.7      | mm                 |
| Weld is pre-Heated                   | No         |                    |
|                                      |            |                    |
| Element From Node                    | 220        |                    |
| Detail Type                          | Platform   |                    |
| Detail ID                            | TOP PLT    |                    |
| Dist. from "FROM" Node / Offset dist | 1500       | mm                 |
| Platform Start Angle (degrees)       | 0.0        |                    |
| Platform End Angle (degrees)         | 270.0      |                    |
| Platform Wind Area                   | 13405039   | mm <sup>2</sup>    |
| Platform Weight                      | 238947     | N                  |
| Platform Railing Weight              | 0          | N/mm               |
| Platform Grating Weight              | 0.017      | kg/cm <sup>2</sup> |
| Platform Width                       | 11942      | mm                 |
| Platform Height                      | 1000       | mm                 |
| Platform Clearance or End Offset     | 5925       | mm                 |
| Platform Force Coefficient           | 1.0        |                    |
| Ladder Layout Angle                  | 90.0       |                    |
| Ladder Start Elevation               | 43000      | mm                 |
| Ladder End Elevation                 | 46000      | mm                 |
| Unit Weight of Ladder                | 0.3628     | N/mm               |
| Platform Length (top head platform)  | 11942      | mm                 |
|                                      |            |                    |
| Element From Node                    | 220        |                    |
| Detail Type                          | Insulation |                    |
| Detail ID                            | insu       |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Height/Length of Insulation          | 1044.5     | mm                 |
| Thickness of Insulation              | 100        | mm                 |
| Density                              | 0.000352   | kg/cm <sup>3</sup> |
|                                      |            |                    |
| Element From Node                    | 220        |                    |
| Detail Type                          | Lining     |                    |
| Detail ID                            | Clad       |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Height/Length of Lining              | 1028.4     | mm                 |
| Thickness of Lining                  | 3          | mm                 |
| Density                              | 0.008027   | kg/cm <sup>3</sup> |
|                                      |            |                    |
| Element From Node                    | 220        |                    |
| Detail Type                          | Nozzle     |                    |
| Detail ID                            | N7         |                    |
| Dist. from "FROM" Node / Offset dist | 0          | mm                 |
| Nozzle Diameter                      | 30         | in.                |
| Nozzle Schedule                      | None       |                    |
| Nozzle Class                         | 150        |                    |
| Layout Angle                         | 0.0        |                    |
| Blind Flange (Y/N)                   | N          |                    |
| Weight of Nozzle ( Used if > 0 )     | 0          | N                  |
| Grade of Attached Flange             | GR 1.1     |                    |
| Nozzle Matl                          | SA-516 70  |                    |
|                                      |            |                    |
| Element From Node                    | 220        |                    |
| Detail Type                          | Nozzle     |                    |
| Detail ID                            | N10        |                    |
| Dist. from "FROM" Node / Offset dist | 1500       | mm                 |
| Nozzle Diameter                      | 2          | in.                |
| Nozzle Schedule                      | STD        |                    |
| Nozzle Class                         | 600        |                    |
| Layout Angle                         | 180.0      |                    |
| Blind Flange (Y/N)                   | N          |                    |
| Weight of Nozzle ( Used if > 0 )     | 0          | N                  |
| Grade of Attached Flange             | GR 1.1     |                    |
| Nozzle Matl                          | SA-106 B   |                    |



Element From Node 220  
 Detail Type Weight  
 Detail ID N7QVHPP  
 Dist. from "FROM" Node / Offset dist 0 mm  
 Miscellaneous Weight 89941 N  
 Offset from Element Centerline 4552 mm

Element From Node 220  
 Detail Type Weight  
 Detail ID 2 PSVS  
 Dist. from "FROM" Node / Offset dist 1020 mm  
 Miscellaneous Weight 107868 N  
 Offset from Element Centerline 5905 mm

Element From Node 220  
 Detail Type For./Mom.  
 Detail ID FM at N7  
 Dist. from "FROM" Node / Offset dist 1044 mm  
 Force in X Direction 41250 N  
 Force in Y Direction -61875 N  
 Force in Z Direction 41250 N  
 Moment about X Axis 86625000 N-mm  
 Moment about Y Axis 0 N-mm  
 Moment about Z Axis 86625000 N-mm  
 Force/Moment Combination Method SRSS

Element# 21/22  
 Element From Node 230  
 Element To Node 240  
 Element Type Flange  
 Description N2A FLG  
 Distance "FROM" to "TO" 148 mm  
 Flange Inside Diameter 1543 mm  
 Element Thickness 98 mm  
 Internal Corrosion Allowance 1 mm  
 Nominal Thickness 104 mm  
 External Corrosion Allowance 0 mm  
 Design Internal Pressure 0.3432 MPa  
 Design Temperature Internal Pressure 150 °C  
 Design External Pressure 0.1035 MPa  
 Design Temperature External Pressure 150 °C  
 Effective Diameter Multiplier 1.22  
 Material Name SA-266 2  
 Allowable Stress, Ambient 138 MPa  
 Allowable Stress, Operating 138 MPa  
 Allowable Stress, Hydrotest 225 MPa  
 Material Density 0.00775 kg/cm³  
 P Number Thickness 32.004 mm  
 Yield Stress, Operating 219 MPa  
 UCS-66 Chart Curve Designation B  
 External Pressure Chart Name CS-2  
 UNS Number K03506  
 Product Form Forgings  
 Perform Flange Stress Calculation (Y/N) Y  
 Weld is pre-Heated No

Element From Node 230  
 Detail Type Insulation  
 Detail ID insu  
 Dist. from "FROM" Node / Offset dist 0 mm  
 Height/Length of Insulation 148 mm  
 Thickness of Insulation 100 mm  
 Density 0.000352 kg/cm³



|                                         |            |
|-----------------------------------------|------------|
| Element From Node                       | 240        |
| Element To Node                         | 250        |
| Element Type                            | Flange     |
| Description                             | N2B FLG    |
| Distance "FROM" to "TO"                 | 142 mm     |
| Flange Inside Diameter                  | 1543 mm    |
| Element Thickness                       | 98 mm      |
| Internal Corrosion Allowance            | 1 mm       |
| Nominal Thickness                       | 104 mm     |
| External Corrosion Allowance            | 0 mm       |
| Design Internal Pressure                | 0.3432 MPa |
| Design Temperature Internal Pressure    | 150 °C     |
| Design External Pressure                | 0.1035 MPa |
| Design Temperature External Pressure    | 150 °C     |
| Effective Diameter Multiplier           | 1.22       |
| Material Name                           | SA-266 2   |
| Perform Flange Stress Calculation (Y/N) | Y          |
| Weld is pre-Heated                      | No         |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Element From Node                    | 240                         |
| Detail Type                          | Insulation                  |
| Detail ID                            | insu                        |
| Dist. from "FROM" Node / Offset dist | 0 mm                        |
| Height/Length of Insulation          | 142 mm                      |
| Thickness of Insulation              | 100 mm                      |
| Density                              | 0.000352 kg/cm <sup>3</sup> |





# XY Coordinate Calculations:

| From         | To      | X (Horiz.)<br>mm | Y (Vert.)<br>mm | DX (Horiz.)<br>mm | DY (Vert.)<br>mm |
|--------------|---------|------------------|-----------------|-------------------|------------------|
|              | Skirt-9 | ...              | 2000            | ...               | 2000             |
|              | Skirt-8 | ...              | 5000            | ...               | 3000             |
|              | Skirt-7 | ...              | 8000            | ...               | 3000             |
|              | Skirt-6 | ...              | 11000           | ...               | 3000             |
|              | Skirt-5 | ...              | 14000           | ...               | 3000             |
|              | Skirt-4 | ...              | 17000           | ...               | 3000             |
|              | Skirt-3 | ...              | 20000           | ...               | 3000             |
|              | Skirt-2 | ...              | 21000           | ...               | 1000             |
|              | Skirt-1 | ...              | 22000           | ...               | 1000             |
| Btm Dish End |         | ...              | 22050           | ...               | 50               |
|              | Shell-1 | ...              | 25050           | ...               | 3000             |
|              | Shell-2 | ...              | 28050           | ...               | 3000             |
|              | Shell-3 | ...              | 31050           | ...               | 3000             |
|              | Shell-4 | ...              | 34050           | ...               | 3000             |
|              | Shell-5 | ...              | 37050           | ...               | 3000             |
|              | Shell-6 | ...              | 40050           | ...               | 3000             |
|              | Shell-7 | ...              | 43050           | ...               | 3000             |
|              | Shell-8 | ...              | 46050           | ...               | 3000             |
|              | Shell-9 | ...              | 47550           | ...               | 1500             |
| Top Dish End |         | ...              | 47600           | ...               | 50               |
|              | N2A FLG | ...              | 26000           | ...               | 148              |
|              | N2B FLG | ...              | 26000           | ...               | 142              |

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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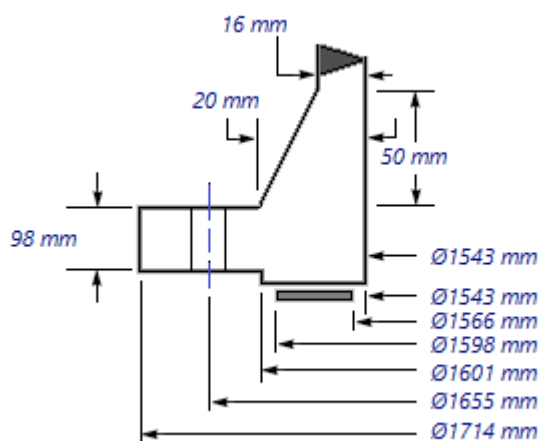


## Flange Input Data Values

Description: FLANGE :

### N2A FLG

|                                             |                    |              |      |
|---------------------------------------------|--------------------|--------------|------|
| Flange Type                                 | Integral Weld Neck |              |      |
| Design Pressure                             | P                  | 0.34         | MPa  |
| Design Temperature                          |                    | 150          | °C   |
| Internal Corrosion Allowance                | ci                 | 1.0000       | mm   |
| External Corrosion Allowance                | ce                 | 0.0000       | mm   |
| Use Corrosion Allowance in Thickness Calcs. |                    | Yes          |      |
| Flange Inside Diameter                      | B                  | 1543.000     | mm   |
| Flange Outside Diameter                     | A                  | 1714.000     | mm   |
| Flange Thickness                            | t                  | 98.0000      | mm   |
| Thickness of Hub at Small End               | go                 | 16.0000      | mm   |
| Thickness of Hub at Large End               | gl                 | 20.0000      | mm   |
| Length of Hub                               | h                  | 50.0000      | mm   |
| Flange Material                             |                    | SA-266 2     |      |
| Flange Material UNS number                  |                    | K03506       |      |
| Flange Allowable Stress At Temperature      | Sfo                | 138.00       | MPa  |
| Flange Allowable Stress At Ambient          | Sfa                | 138.00       | MPa  |
| Bolt Material                               |                    | SA-193 B7    |      |
| Bolt Allowable Stress At Temperature        | Sb                 | 172.00       | MPa  |
| Bolt Allowable Stress At Ambient            | Sa                 | 172.00       | MPa  |
| Diameter of Bolt Circle                     | C                  | 1655.000     | mm   |
| Nominal Bolt Diameter                       | a                  | 25.4000      | mm   |
| Type of Threads                             | TEMA Thread Series |              |      |
| Number of Bolts                             |                    | 48           |      |
| Flange Face Outside Diameter                | Fod                | 1601.000     | mm   |
| Flange Face Inside Diameter                 | Fid                | 1543.000     | mm   |
| Flange Facing Sketch                        |                    | Sketch 1a    |      |
| Gasket Outside Diameter                     | Go                 | 1598.000     | mm   |
| Gasket Inside Diameter                      | Gi                 | 1566.000     | mm   |
| Gasket Factor                               | m                  | 3.0000       |      |
| Gasket Design Seating Stress                | y                  | 68.95        | MPa  |
| Gasket Thickness                            | tg                 | 3.1750       | mm   |
| Raised face depth                           | Fd                 | 0.0000       | mm   |
| Axial Force on Flange                       |                    | 46088.19     | N    |
| Bending Moment on Flange                    |                    | 528747040.00 | N-mm |





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.34 \cdot (1543.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.34) + Ca$$

$$= 2.9277 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (772.5 + 0.6 \cdot 15.0)$$

$$= 2.649 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (771.5 + 0.6 \cdot 16.0)$$

$$= 2.827 \text{ MPa}$$

|                                          |             |
|------------------------------------------|-------------|
| Corroded Flange Thickness, tc = T-ci     | 97.000 mm   |
| Corroded Flange ID, Bcor = B+2*ci        | 1545.000 mm |
| Corroded Large Hub, g1Cor = g1-ci        | 19.000 mm   |
| Corroded Small Hub, g0Cor = go-ci        | 15.000 mm   |
| Code R Dimension, R = ((C-Bcor)/2)-g1cor | 36.000 mm   |
| Gasket Contact Width, N = (Go - Gi)/2    | 16.000 mm   |
| Basic Gasket Width, bo = N/2             | 8.000 mm    |
| Effective Gasket Width, b = Cb*sqrt(bo)  | 7.127 mm    |
| Gasket Reaction Diameter, G = Go - 2 * b | 1583.745 mm |

#### **Basic Flange and Bolt Loads:**

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1583.7451^2 \cdot 0.344$$

$$= 677307.375 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1274 \cdot \pi \cdot 1583.7451 \cdot 3.0 \cdot 0.34$$

$$= 73154.906 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1545.0^2 \cdot 0.3438$$

$$= 644573.062 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 677307 - 644573$$

$$= 32734.291 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max( Wo,1 + Wo,2 + H'p, 0 )$$

$$= \max( 677307 + 73155 + 0, 0 )$$

$$= 750462.312 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1274 \cdot \pi \cdot 1583.745 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 2444917.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max( ( Wo + Fa + 4Me/G ) / Sb, Wg/Sa )$$

$$= \max( ( 750462 + 46088 + 4 \cdot 52875E+09/1583.75 ) / 172, 2444918/172 )$$

$$= 14216.103 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 97.0/(3.0 + 0.5)$$

$$= 217.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin( \pi / n )$$



$$= 1655.0 \cdot \sin(\pi/48)$$

$$= 108.242 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{108.242/(2 \cdot 25.4 + 97.0)}, 1)$$

$$= 1.0000$$

#### Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

|                                         | Minimum   | Actual    | Maximum |
|-----------------------------------------|-----------|-----------|---------|
| Bolt Area:                              | 14216.103 | 17063.191 |         |
| Radial Distance between Hub and Bolts:  | 34.925    | 36.000    |         |
| Radial Distance between Bolts and Edge: | 26.987    | 29.500    |         |
| Circ. Spacing between the Bolts:        | 57.150    | 108.242   | 217.086 |

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 17063.191 \cdot 172.0 / (68.95 \cdot \pi(1598.0 + 1566.0))$$

$$= 4.282 \text{ mm}$$

#### Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm  
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 17063.1914$$

$$= 2934566.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 750462 - 677307$$

$$= 73154.94 \text{ N}$$

#### Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1655.0 - 1583.7451) / 2$$

$$= 35.6274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1655.0 - 1545.0) / 2 + 35.6274)$$

$$= 45.3137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1655.0 - 1545.0 - 19.0) / 2$$

$$= 45.5000 \text{ mm}$$

#### The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 84.5, Bb = 97.0, Cc = 50.0 \text{ and } Ddg = 17.0 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 84.5 \cdot 97.0^3 \cdot (1/3 - 0.210 \cdot (97.0/84.5) \cdot (1 - 1/12 \cdot (97.0/84.5)^4))$$

$$= 9806007.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 50.0 \cdot 17.0^3 \cdot (1/3 - 0.105 \cdot (17.0/50.0) \cdot (1 - 1/192 \cdot (17.0/50.0)^4))$$

$$= 73114.180 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:



$$= Kab + Kcd = 598.4 + 4.462$$

$$= 9879121.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left( \frac{1}{3} - 0.21 \left( \frac{t}{AR} \right) \left( 1 - \frac{1}{12} \left( \frac{t}{AR} \right)^4 \right) \right)$$

$$= 84.5 \cdot 97.0^3 \left( \frac{1}{3} - 0.21 \left( \frac{97.0}{84.5} \right) \left( 1 - \frac{1}{12} \left( \frac{97.0}{3.327} \right)^4 \right) \right)$$

$$= 9806007.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max( Ip,term1, Ip,term2 )$$

$$= \max( 9879121, 9806007 )$$

$$= 9879121.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.383 \cdot 15.0^2 \cdot 152.233 \cdot 1545.0 / 0.422$$

$$= 15168424.000 \text{ mm}^4$$

#### Summary of Moments for Internal Pressure: (N-mm)

| Loading              | Force    | Distance | Bolt Corr | Moment     |
|----------------------|----------|----------|-----------|------------|
| End Pressure, Md     | 644573.  | 45.5000  | 1.0000    | 29339956.  |
| Face Pressure, Mt    | 32734.   | 45.3137  | 1.0000    | 1483913.   |
| Gasket Load, Mg      | 73155.   | 35.6274  | 1.0000    | 2607376.   |
| Gasket Seating, Matm | 2934566. | 35.6274  | 1.0000    | 104593304. |
| Moment Load, Moe     | ...      | ...      | ...       | 51317520.  |

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[ \frac{I}{(0.384 \cdot IP + I)} \right] \cdot \left( \frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 5.2875E+09 \left[ \frac{15168424}{(0.384 \cdot 9879121 + 15168424)} \right] \cdot \left( \frac{45.5}{(1655.0 - 2 \cdot 45.5)} \right) +$$

$$46088 \cdot 45.5$$

$$= 51317520.000 \text{ N-mm}$$

Total Moment for Operation, Mo 84748760. N-mm

Total Moment for Gasket seating, Mg 104593304. N-mm

Effective Hub Length, ho = sqrt(Bcor\*goCor) 152.233 mm

Hub Ratio, h/h0 = HL / H0 0.328

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.267

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.889

Factor V per 4.16.5 0.422

Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1714.0 / 1545.0$$

$$= 1.109$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.109^2 \cdot \log_{10}(1.109))) / (1.109^2 - 1) / (1.109 - 1)$$

$$= 18.679$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.109^2(1 + 8.55246 \cdot \log_{10}(1.109)) - 1) / ((1.0472 + 1.9448 \cdot 1.109^2) \cdot (1.109 - 1))$$

$$= 1.874$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.109^2(1 + 8.55246 \cdot \log_{10}(1.109)) - 1) / (1.36136(1.109^2 - 1) \cdot (1.109 - 1))$$

$$= 20.526$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.109^2 + 1) / (1.109^2 - 1)$$

= 9.668

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 20.526 \cdot 15.0^2 \cdot 152.233 / 0.422 \\ &= 1666756.875 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.889 / 152.233 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (97.0 \cdot 0.006 + 1) / 1.874 + 97.0^3 / 1666757 \\ &= 1.383 \end{aligned}$$

### Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 84748760 / 1545.0) / (1.3834 \cdot 19.0^2) \\ &= 109.81 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 0.10459\text{E}+09 / 1545.0) / (1.3834 \cdot 19.0^2) \\ &= 135.52 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (1.7549 \cdot 84748760 / 1545.0) / (1.3834 \cdot 97.0^2) \\ &= 7.39 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (1.7549 \cdot 0.10459\text{E}+09 / 1545.0) / (1.3834 \cdot 97.0^2) \\ &= 9.12 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (18.6788 \cdot 84748760 / (97.0^2 \cdot 1545.0)) - 9.6679 \cdot 7 \\ &= 37.38 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (18.6788 \cdot 0.10459\text{E}+09 / (97.0^2 \cdot 1545.0)) - 9.6679 \cdot 9 \\ &= 46.14 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (110 + \max(7, 37)) / 2 \\ &= 73.59 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (136 + \max(9, 46)) / 2 \\ &= 90.83 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 2131444 / 17063.1914 \\ &= 124.93 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (2444918 / 17063.1914) \\ &= 143.30 \text{ MPa} \end{aligned}$$

**Flange Stress Analysis Results: MPa**



|                   | Actual | Operating Allowed | Gasket Seating Actual | Allowed |
|-------------------|--------|-------------------|-----------------------|---------|
| Longitudinal Hub  | 109.81 | 207.00            | 135.52                | 207.00  |
| Radial Flange     | 7.39   | 138.00            | 9.12                  | 138.00  |
| Tangential Flange | 37.38  | 138.00            | 46.14                 | 138.00  |
| Maximum Average   | 73.59  | 138.00            | 90.83                 | 138.00  |
| Bolting           | 124.93 | 172.00            | 143.30                | 172.00  |

Minimum Required Flange Thickness [Rigidity] 81.966 mm

Estimated M.A.W.P. ( Operating ) 1 MPa  
 Estimated Finished Weight of Flange at given Thk. 366.5 kgm  
 Estimated Unfinished Weight of Forging at given Thk 501.8 kgm

#### Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 0.10459E+09 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.104 \cdot 202713 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.999$  (should be  $\leq 1$ )

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 84748760 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.104 \cdot 195074 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.841$  (should be  $\leq 1$ )

#### Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 0.10459E+09 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.383 \cdot 202713 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.798$  (should be  $\leq 1$ )

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 84748760 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.383 \cdot 195074 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.672$  (should be  $\leq 1$ )

#### Minimum Design Metal Temperature Results:

Thickness Ratio = 0.176, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

#### Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

#### Flange Stress Analysis Results: MPa

|                   | Actual | Operating Allowed | Gasket Seating Actual | Allowed |
|-------------------|--------|-------------------|-----------------------|---------|
| Longitudinal Hub  | 105.29 | 207.00            | 127.73                | 207.00  |
| Radial Flange     | 7.64   | 138.00            | 9.26                  | 138.00  |
| Tangential Flange | 34.41  | 138.00            | 41.74                 | 138.00  |
| Maximum Average   | 69.85  | 138.00            | 84.74                 | 138.00  |
| Bolting           | 124.93 | 172.00            | 143.30                | 172.00  |

Minimum Req'd Thickness, ambient, no Corrosion: 72.746 mm

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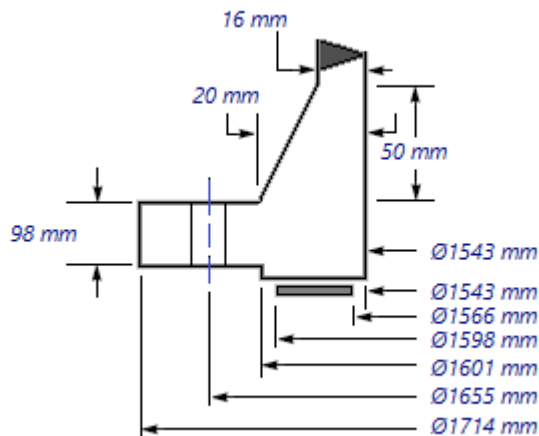


## Flange Input Data Values

Description: FLANGE :

### N2B FLG

|                                             |                    |              |      |
|---------------------------------------------|--------------------|--------------|------|
| Flange Type                                 | Integral Weld Neck |              |      |
| Design Pressure                             | P                  | 0.34         | MPa  |
| Design Temperature                          |                    | 150          | °C   |
| Internal Corrosion Allowance                | ci                 | 1.0000       | mm   |
| External Corrosion Allowance                | ce                 | 0.0000       | mm   |
| Use Corrosion Allowance in Thickness Calcs. |                    | Yes          |      |
| Flange Inside Diameter                      | B                  | 1543.000     | mm   |
| Flange Outside Diameter                     | A                  | 1714.000     | mm   |
| Flange Thickness                            | t                  | 98.0000      | mm   |
| Thickness of Hub at Small End               | go                 | 16.0000      | mm   |
| Thickness of Hub at Large End               | gl                 | 20.0000      | mm   |
| Length of Hub                               | h                  | 50.0000      | mm   |
| Flange Material                             |                    | SA-266 2     |      |
| Flange Material UNS number                  |                    | K03506       |      |
| Flange Allowable Stress At Temperature      | Sfo                | 138.00       | MPa  |
| Flange Allowable Stress At Ambient          | Sfa                | 138.00       | MPa  |
| Bolt Material                               |                    | SA-193 B7    |      |
| Bolt Allowable Stress At Temperature        | Sb                 | 172.00       | MPa  |
| Bolt Allowable Stress At Ambient            | Sa                 | 172.00       | MPa  |
| Diameter of Bolt Circle                     | C                  | 1655.000     | mm   |
| Nominal Bolt Diameter                       | a                  | 25.4000      | mm   |
| Type of Threads                             | TEMA Thread Series |              |      |
| Number of Bolts                             |                    | 48           |      |
| Flange Face Outside Diameter                | Fod                | 1601.000     | mm   |
| Flange Face Inside Diameter                 | Fid                | 1543.000     | mm   |
| Flange Facing Sketch                        |                    | Sketch 1a    |      |
| Gasket Outside Diameter                     | Go                 | 1598.000     | mm   |
| Gasket Inside Diameter                      | Gi                 | 1566.000     | mm   |
| Gasket Factor                               | m                  | 3.0000       |      |
| Gasket Design Seating Stress                | y                  | 68.95        | MPa  |
| Gasket Thickness                            | tg                 | 3.1750       | mm   |
| Raised face depth                           | Fd                 | 0.0000       | mm   |
| Axial Force on Flange                       |                    | 46088.19     | N    |
| Bending Moment on Flange                    |                    | 528747040.00 | N-mm |







Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.34 \cdot (1543.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.34) + Ca$$

$$= 2.9277 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (772.5 + 0.6 \cdot 15.0)$$

$$= 2.649 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (771.5 + 0.6 \cdot 16.0)$$

$$= 2.827 \text{ MPa}$$

|                                          |             |
|------------------------------------------|-------------|
| Corroded Flange Thickness, tc = T-ci     | 97.000 mm   |
| Corroded Flange ID, Bcor = B+2*ci        | 1545.000 mm |
| Corroded Large Hub, g1Cor = g1-ci        | 19.000 mm   |
| Corroded Small Hub, g0Cor = go-ci        | 15.000 mm   |
| Code R Dimension, R = ((C-Bcor)/2)-g1cor | 36.000 mm   |
| Gasket Contact Width, N = (Go - Gi)/2    | 16.000 mm   |
| Basic Gasket Width, bo = N/2             | 8.000 mm    |
| Effective Gasket Width, b = Cb*sqrt(bo)  | 7.127 mm    |
| Gasket Reaction Diameter, G = Go - 2 * b | 1583.745 mm |

#### **Basic Flange and Bolt Loads:**

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1583.7451^2 \cdot 0.344$$

$$= 677307.375 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.1274 \cdot \pi \cdot 1583.7451 \cdot 3.0 \cdot 0.34$$

$$= 73154.906 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1545.0^2 \cdot 0.3438$$

$$= 644573.062 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 677307 - 644573$$

$$= 32734.291 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max( Wo,1 + Wo,2 + H'p, 0 )$$

$$= \max( 677307 + 73155 + 0, 0 )$$

$$= 750462.312 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.1274 \cdot \pi \cdot 1583.745 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 2444918.000 \text{ N}$$

Required Bolt Area [Am]:

$$= \max( ( Wo + Fa + 4Me/G ) / Sb, Wg/Sa )$$

$$= \max( ( 750462 + 46088 + 4 \cdot 52875E+09/1583.75 ) / 172, 2444918/172 )$$

$$= 14216.104 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 97.0/(3.0 + 0.5)$$

$$= 217.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin( \pi / n )$$



$$= 1655.0 \cdot \sin(\pi/48)$$

$$= 108.242 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{108.242/(2 \cdot 25.4 + 97.0)}, 1)$$

$$= 1.0000$$

#### Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

|                                         | Minimum   | Actual    | Maximum |
|-----------------------------------------|-----------|-----------|---------|
| Bolt Area:                              | 14216.104 | 17063.191 |         |
| Radial Distance between Hub and Bolts:  | 34.925    | 36.000    |         |
| Radial Distance between Bolts and Edge: | 26.987    | 29.500    |         |
| Circ. Spacing between the Bolts:        | 57.150    | 108.242   | 217.086 |

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 17063.191 \cdot 172.0 / (68.95 \cdot \pi(1598.0 + 1566.0))$$

$$= 4.282 \text{ mm}$$

#### Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm  
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 17063.1914$$

$$= 2934566.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 750462 - 677307$$

$$= 73154.94 \text{ N}$$

#### Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1655.0 - 1583.7451) / 2$$

$$= 35.6274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1655.0 - 1545.0) / 2 + 35.6274)$$

$$= 45.3137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1655.0 - 1545.0 - 19.0) / 2$$

$$= 45.5000 \text{ mm}$$

#### The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 84.5, Bb = 97.0, Cc = 50.0 \text{ and } Ddg = 17.0 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 84.5 \cdot 97.0^3 \cdot (1/3 - 0.210 \cdot (97.0/84.5) \cdot (1 - 1/12 \cdot (97.0/84.5)^4))$$

$$= 9806007.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 50.0 \cdot 17.0^3 \cdot (1/3 - 0.105 \cdot (17.0/50.0) \cdot (1 - 1/192 \cdot (17.0/50.0)^4))$$

$$= 73114.180 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:



$$= Kab + Kcd = 598.4 + 4.462$$

$$= 9879121.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left( \frac{1}{3} - 0.21 \left( \frac{t}{AR} \right) \left( 1 - \frac{1}{12} \left( \frac{t}{AR} \right)^4 \right) \right)$$

$$= 84.5 \cdot 97.0^3 \left( \frac{1}{3} - 0.21 \left( \frac{97.0}{84.5} \right) \left( 1 - \frac{1}{12} \left( \frac{97.0}{3.327} \right)^4 \right) \right)$$

$$= 9806007.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max( Ip,term1, Ip,term2 )$$

$$= \max( 9879121, 9806007 )$$

$$= 9879121.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.383 \cdot 15.0^2 \cdot 152.233 \cdot 1545.0 / 0.422$$

$$= 15168424.000 \text{ mm}^4$$

#### Summary of Moments for Internal Pressure: (N-mm)

| Loading              | Force    | Distance | Bolt Corr | Moment     |
|----------------------|----------|----------|-----------|------------|
| End Pressure, Md     | 644573.  | 45.5000  | 1.0000    | 29339956.  |
| Face Pressure, Mt    | 32734.   | 45.3137  | 1.0000    | 1483913.   |
| Gasket Load, Mg      | 73155.   | 35.6274  | 1.0000    | 2607376.   |
| Gasket Seating, Matm | 2934566. | 35.6274  | 1.0000    | 104593304. |
| Moment Load, Moe     | ...      | ...      | ...       | 51317520.  |

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[ \frac{I}{(0.384 \cdot IP + I)} \right] \cdot \left( \frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 5.2875E+09 \left[ \frac{15168424}{(0.384 \cdot 9879121 + 15168424)} \right] \cdot \left( \frac{45.5}{(1655.0 - 2 \cdot 45.5)} \right) +$$

$$46088 \cdot 45.5$$

$$= 51317520.000 \text{ N-mm}$$

Total Moment for Operation, Mo 84748760. N-mm

Total Moment for Gasket seating, Mg 104593304. N-mm

Effective Hub Length, ho = sqrt(Bcor\*goCor) 152.233 mm

Hub Ratio, h/h0 = HL / H0 0.328

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.267

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.889

Factor V per 4.16.5 0.422

Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1714.0 / 1545.0$$

$$= 1.109$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.109^2 \cdot \log_{10}(1.109))) / (1.109^2 - 1) / (1.109 - 1)$$

$$= 18.679$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.109^2(1 + 8.55246 \cdot \log_{10}(1.109)) - 1) / ((1.0472 + 1.9448 \cdot 1.109^2) \cdot (1.109 - 1))$$

$$= 1.874$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.109^2(1 + 8.55246 \cdot \log_{10}(1.109)) - 1) / (1.36136(1.109^2 - 1) \cdot (1.109 - 1))$$

$$= 20.526$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.109^2 + 1) / (1.109^2 - 1)$$

= 9.668

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 20.526 \cdot 15.0^2 \cdot 152.233 / 0.422 \\ &= 1666756.875 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.889 / 152.233 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (97.0 \cdot 0.006 + 1) / 1.874 + 97.0^3 / 1666757 \\ &= 1.383 \end{aligned}$$

### Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 84748760 / 1545.0) / (1.3834 \cdot 19.0^2) \\ &= 109.81 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 0.10459\text{E}+09 / 1545.0) / (1.3834 \cdot 19.0^2) \\ &= 135.52 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (1.7549 \cdot 84748760 / 1545.0) / (1.3834 \cdot 97.0^2) \\ &= 7.39 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (1.7549 \cdot 0.10459\text{E}+09 / 1545.0) / (1.3834 \cdot 97.0^2) \\ &= 9.12 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (18.6788 \cdot 84748760 / (97.0^2 \cdot 1545.0)) - 9.6679 \cdot 7 \\ &= 37.38 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (18.6788 \cdot 0.10459\text{E}+09 / (97.0^2 \cdot 1545.0)) - 9.6679 \cdot 9 \\ &= 46.14 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (110 + \max(7, 37)) / 2 \\ &= 73.59 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (136 + \max(9, 46)) / 2 \\ &= 90.83 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 2131444 / 17063.1914 \\ &= 124.93 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (2444918 / 17063.1914) \\ &= 143.30 \text{ MPa} \end{aligned}$$

**Flange Stress Analysis Results: MPa**



|                   | Actual | Operating Allowed | Gasket Seating Actual | Allowed |
|-------------------|--------|-------------------|-----------------------|---------|
| Longitudinal Hub  | 109.81 | 207.00            | 135.52                | 207.00  |
| Radial Flange     | 7.39   | 138.00            | 9.12                  | 138.00  |
| Tangential Flange | 37.38  | 138.00            | 46.14                 | 138.00  |
| Maximum Average   | 73.59  | 138.00            | 90.83                 | 138.00  |
| Bolting           | 124.93 | 172.00            | 143.30                | 172.00  |

Minimum Required Flange Thickness [Rigidity] 81.966 mm

Estimated M.A.W.P. ( Operating ) 1 MPa  
 Estimated Finished Weight of Flange at given Thk. 366.5 kgm  
 Estimated Unfinished Weight of Forging at given Thk 501.8 kgm

#### Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 0.10459E+09 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.104 \cdot 202713 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.999$  (should be  $\leq 1$ )

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 84748760 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.104 \cdot 195074 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.841$  (should be  $\leq 1$ )

#### Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 0.10459E+09 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.383 \cdot 202713 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.798$  (should be  $\leq 1$ )

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / ( \lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki )$   
 $= 52.14 \cdot 84748760 / 1.0 \cdot 1.0 \cdot 0.422 / ( 1.383 \cdot 195074 \cdot 15.0^2 \cdot 152.233 \cdot 0.3 )$   
 $= 0.672$  (should be  $\leq 1$ )

#### Minimum Design Metal Temperature Results:

Thickness Ratio = 0.176, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

#### Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

#### Flange Stress Analysis Results: MPa

|                   | Actual | Operating Allowed | Gasket Seating Actual | Allowed |
|-------------------|--------|-------------------|-----------------------|---------|
| Longitudinal Hub  | 105.29 | 207.00            | 127.73                | 207.00  |
| Radial Flange     | 7.64   | 138.00            | 9.26                  | 138.00  |
| Tangential Flange | 34.41  | 138.00            | 41.74                 | 138.00  |
| Maximum Average   | 69.85  | 138.00            | 84.74                 | 138.00  |
| Bolting           | 124.93 | 172.00            | 143.30                | 172.00  |

Minimum Req'd Thickness, ambient, no Corrosion: 72.746 mm

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### Internal Pressure Results Summary:

#### Element Thickness, Pressure, Diameter and Allowable Stress :

| From | To       | Int. Press<br>+ Liq. Hd<br>MPa | Nominal<br>Thickness<br>mm | Total Corr<br>Allowance<br>mm | Element<br>Diameter<br>mm | Allowable<br>Stress(SE)<br>MPa | Element# |
|------|----------|--------------------------------|----------------------------|-------------------------------|---------------------------|--------------------------------|----------|
|      | Skirt-9  | ...                            | 36                         | ...                           | 5953                      | ...                            | 1        |
|      | Skirt-8  | ...                            | 22                         | ...                           | 5939                      | ...                            | 2        |
|      | Skirt-7  | ...                            | 22                         | ...                           | 5939                      | ...                            | 3        |
|      | Skirt-6  | ...                            | 20                         | ...                           | 5937                      | ...                            | 4        |
|      | Skirt-5  | ...                            | 20                         | ...                           | 5937                      | ...                            | 5        |
|      | Skirt-4  | ...                            | 18                         | ...                           | 5935                      | ...                            | 6        |
|      | Skirt-3  | ...                            | 18                         | ...                           | 5935                      | ...                            | 7        |
|      | Skirt-2  | ...                            | 16                         | ...                           | 5933                      | ...                            | 8        |
|      | Skirt-1  | ...                            | 16                         | ...                           | 5933                      | ...                            | 9        |
| Btm  | Dish End | 0.392                          | 26                         | 1                             | 5906                      | 138                            | 10       |
|      | Shell-1  | 0.3821                         | 16                         | 1                             | 5906                      | 138                            | 11       |
|      | Shell-2  | 0.353                          | 16                         | 1                             | 5906                      | 138                            | 12       |
|      | Shell-3  | 0.3432                         | 14                         | 1                             | 5906                      | 138                            | 13       |
|      | Shell-4  | 0.3432                         | 14                         | 1                             | 5906                      | 138                            | 14       |
|      | Shell-5  | 0.3432                         | 14                         | 1                             | 5906                      | 138                            | 15       |
|      | Shell-6  | 0.3432                         | 14                         | 1                             | 5906                      | 138                            | 16       |
|      | Shell-7  | 0.3432                         | 14                         | 1                             | 5906                      | 138                            | 17       |
|      | Shell-8  | 0.3432                         | 14                         | 1                             | 5906                      | 138                            | 18       |
|      | Shell-9  | 0.3432                         | 14                         | 1                             | 5906                      | 138                            | 19       |
| Top  | Dish End | 0.3432                         | 24                         | 1                             | 5906                      | 138                            | 20       |
|      | N2A FLG  | 0.3438                         | 104                        | 1                             | 1543                      | 138                            | 21       |
|      | N2B FLG  | 0.3438                         | 104                        | 1                             | 1543                      | 138                            | 22       |

#### Element Required Thickness and MAWP :

| From | To       | Design<br>Pressure<br>MPa | M.A.W.P.<br>Corroded<br>MPa | M.A.P.<br>New & Cold<br>MPa | Minimum<br>Thickness<br>mm | Required<br>Thickness<br>mm | Element# |
|------|----------|---------------------------|-----------------------------|-----------------------------|----------------------------|-----------------------------|----------|
|      | Skirt-9  | ...                       | No Calc                     | No Calc                     | 36                         | No Calc                     | 1        |
|      | Skirt-8  | ...                       | No Calc                     | No Calc                     | 22                         | No Calc                     | 2        |
|      | Skirt-7  | ...                       | No Calc                     | No Calc                     | 22                         | No Calc                     | 3        |
|      | Skirt-6  | ...                       | No Calc                     | No Calc                     | 20                         | No Calc                     | 4        |
|      | Skirt-5  | ...                       | No Calc                     | No Calc                     | 20                         | No Calc                     | 5        |
|      | Skirt-4  | ...                       | No Calc                     | No Calc                     | 18                         | No Calc                     | 6        |
|      | Skirt-3  | ...                       | No Calc                     | No Calc                     | 18                         | No Calc                     | 7        |
|      | Skirt-2  | ...                       | No Calc                     | No Calc                     | 16                         | No Calc                     | 8        |
|      | Skirt-1  | ...                       | No Calc                     | No Calc                     | 16                         | No Calc                     | 9        |
| Btm  | Dish End | 0.34324                   | 0.50155                     | 0.5762                      | 22                         | 15.9577                     | 10       |
|      | Shell-1  | 0.34324                   | 0.65978                     | 0.74528                     | 16                         | 9.19233                     | 11       |
|      | Shell-2  | 0.34324                   | 0.68881                     | 0.74528                     | 16                         | 8.56883                     | 12       |
|      | Shell-3  | 0.34324                   | 0.6057                      | 0.65238                     | 14                         | 8.35854                     | 13       |
|      | Shell-4  | 0.34324                   | 0.6057                      | 0.65238                     | 14                         | 8.35854                     | 14       |
|      | Shell-5  | 0.34324                   | 0.6057                      | 0.65238                     | 14                         | 8.35854                     | 15       |
|      | Shell-6  | 0.34324                   | 0.6057                      | 0.65238                     | 14                         | 8.35854                     | 16       |
|      | Shell-7  | 0.34324                   | 0.6057                      | 0.65238                     | 14                         | 8.35854                     | 17       |
|      | Shell-8  | 0.34324                   | 0.6057                      | 0.65238                     | 14                         | 8.35854                     | 18       |
|      | Shell-9  | 0.34324                   | 0.6057                      | 0.65238                     | 14                         | 8.35854                     | 19       |
| Top  | Dish End | 0.34324                   | 0.49828                     | 0.52419                     | 20                         | 14.0867                     | 20       |
|      | N2A FLG  | 0.34324                   | 0.71128                     | 0.71188                     | 98                         | 81.9658                     | 21       |
|      | N2B FLG  | 0.34324                   | 0.71128                     | 0.71188                     | 98                         | 81.9658                     | 22       |

Minimum 0 1

MAWP: 0 MPa, limited by: Top Dish End.

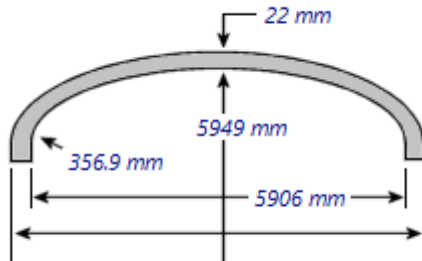
Elements Suitable for Design Internal Pressure.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Btm Dish End

| Torispherical Head Design Information: |                            |                           |            |
|----------------------------------------|----------------------------|---------------------------|------------|
| Design Pressure P                      | 0.392 MPa                  | Int. Design Temperature   | 150 °C     |
| Material                               | SA-516 70                  | Ext. Design Temperature   | 150 °C     |
| Length                                 | 50.000 mm                  | External Pressure Chart   | CS-2       |
|                                        |                            | UNS Number                | K02700     |
| Int. Corr. All. c                      | 1.0000 mm                  | Allowable Stress (ope) S  | 138.0 MPa  |
| Ext. Corr. All. ce                     | 0.0000 mm                  | Allowable Stress (amb) Sa | 138.0 MPa  |
| Inside Diameter Di                     | 5906.000 mm                | Actual Stress             | 98.3 MPa   |
| Spec. Min. Thk t <sub>e</sub> , or sa  | 22.0000 mm                 | Spec. Nominal Thk.        | 26.000 mm  |
| Surface Area                           | 34185060.0 mm <sup>2</sup> | SG of Contents            | 0.987      |
| Element Volume                         | 18028.6 ltr                | Weight of Contents        | 155727.0 N |
| Empty Weight                           | 66759.5 N                  | Operating Weight          | 222486.5 N |
|                                        |                            | Comp. All. Ope. Stress    | 97.5 MPa   |
| Ins. Crn. Rad. L or R                  | 5949.0000 mm               | Ins. Knuckle Radius r     | 356.940 mm |

| Radiography Information: |       |                         |       |
|--------------------------|-------|-------------------------|-------|
| User Defined             |       | User Defined            |       |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. E or v | 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 73.825 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 36.912 mm

#### Inside Corroded Head Depth [h]:

$$= L - \sqrt{\left(\left(L - \frac{D_i}{2}\right) \cdot \left(L + \frac{D_i}{2} - 2 \cdot r\right)\right)}$$

$$= 5950.0 - \sqrt{\left(\left(5950.0 - 5908.0/2\right) \cdot \left(5950.0 + 5908.0/2 - 2 \cdot 357.94\right)\right)}$$

$$= 997.061 \text{ mm}$$

#### M factor for Torispherical Heads ( Corroded ):

$$= (3 + \sqrt{\left(\left(L + C\right) / \left(r + C\right)\right)}) / 4 \text{ per Appendix 1-4 (b \& d)}$$

$$= (3 + \sqrt{\left(\left(5949.0 + 1.0\right) / \left(356.94 + 1.0\right)\right)}) / 4$$

$$= 1.7693$$

#### Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot L \cdot M) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4 (d)}$$

$$= (0.392 \cdot 5950.0 \cdot 1.7693) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.392)$$

$$= 14.9577 + 1.0000 = 15.9577 \text{ mm}$$

#### Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.049 MPa

$$= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t) \text{ per Appendix 1-4 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 21.0) / (1.7693 \cdot 5950.0 + 0.2 \cdot 21.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$



M factor for Torispherical Heads ( New & Cold ):  
 $= (3 + \sqrt{L/r})/4$  per Appendix 1-4 (b & d)  
 $= (3 + \sqrt{(5949.0/356.94)})/4$   
 $= 1.7706$

Maximum Allowable Pressure, New and Cold [MAPNC]:  
 $= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t)$  per Appendix 1-4 (d)  
 $= (2 \cdot 138.0 \cdot 1.0 \cdot 22.0) / (1.7706 \cdot 5949.0 + 0.2 \cdot 22.0)$   
 $= 1 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [Sact]:  
 $= (P \cdot (M \cdot L + 0.2 \cdot t)) / (2 \cdot E \cdot t)$   
 $= (0.392 \cdot (1.7693 \cdot 5950.0 + 0.2 \cdot 21.0)) / (2 \cdot 1.0 \cdot 21.0)$   
 $= 98.303 \text{ MPa}$

Straight Flange Required Thickness:  
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$  per UG-27 (c)(1)  
 $= (0.392 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.392) + 1.0$   
 $= 9.406 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:  
 Less Operating Hydrostatic Head Pressure of 0.049 MPa  
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$  per UG-27 (c)(1)  
 $= (138.0 \cdot 1.0 \cdot 25.0) / (2954.0 + 0.6 \cdot 25.0)$   
 $= 1 - 0 = 1 \text{ MPa}$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  5.271 %  
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

#### MDMT Calculations in the Knuckle Portion:

Govrn. thk,  $t_g = 22.0$ ,  $t_r = 20.875$ ,  $c = 1.0 \text{ mm}$ ,  $E^* = 1.0$   
 Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.994$ , Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -5 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### MDMT Calculations in the Head Straight Flange:

Govrn. thk,  $t_g = 26.0$ ,  $t_r = 11.739$ ,  $c = 1.0 \text{ mm}$ ,  $E^* = 1.0$   
 Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.47$ , Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.547 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 26.000 mm | Basic MDMT             | 0 °C      |
| Stress/Thk. Ratio        | 0.470     | Temperature Difference | 37 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -37 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency $E^*$ | 100 %     |

Note: Temperature values displayed above are for the location with the highest basic MDMT

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 73.825 mm  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 36.912 mm

#### EXTERNAL PRESSURE RESULTS for: Btm Dish End

#### Torispherical Head From 100 to 110 Ext. Chart: CS-2 at 150 °C





Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |        |           | MAEP     |
|---------------------------------------------|-----------|--------|-----------|----------|
| Tca                                         | Crown Rad | Ro/t   | Factor A  | Factor B |
| 21.000                                      | 5971.00   | 284.33 | 0.0004396 | 43.92    |

$$MAEP = B/(Ro/t) = 43.9227/284.3333 = 0.1545 \text{ MPa}$$

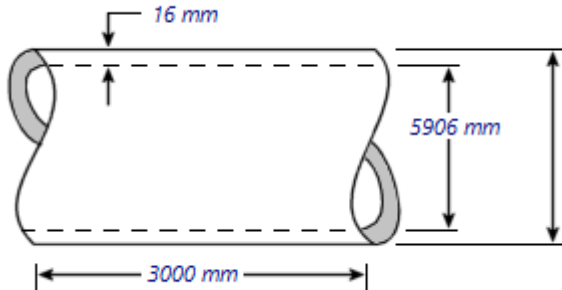
| Results for Required Thickness |           |        |           | Tca      |
|--------------------------------|-----------|--------|-----------|----------|
| Tca                            | Crown Rad | Ro/t   | Factor A  | Factor B |
| 17.187                         | 5971.00   | 347.41 | 0.0003598 | 35.95    |

$$MAEP = B/(Ro/t) = 35.9478/347.4118 = 0.1035 \text{ MPa}$$

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-1

| Cylindrical Shell Design Information:             |    |                            |                         |            |           |
|---------------------------------------------------|----|----------------------------|-------------------------|------------|-----------|
| Design Pressure                                   | P  | 0.382 MPa                  | Int. Design Temperature | 150 °C     |           |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature | 150 °C     |           |
| Length                                            |    | 3000.000 mm                | External Pressure Chart | CS-2       |           |
|                                                   |    |                            | UNS Number              | K02700     |           |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S          | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa         | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           | 75.5 MPa   |           |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 16.0000 mm                 | Spec. Nominal Thk.      | 16.000 mm  |           |
| Surface Area                                      |    | 55964332.0 mm <sup>2</sup> | SG of Contents          | 0.987      |           |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      | 793722.9 N |           |
| Empty Weight                                      |    | 67870.2 N                  | Operating Weight        | 861593.1 N |           |
|                                                   |    |                            | Comp. All. Ope. Stress  | 75.7 MPa   |           |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.382 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.382) \\
 &= 8.1923 + 1.0000 = 9.1923 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 15.0) / (2954.0 + 0.6 \cdot 15.0) \\
 &= 1 - 0 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 16.0) / (2953.0 + 0.6 \cdot 16.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.382 \cdot (2954.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0) \\
 &= 75.472 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.270 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, t<sub>g</sub> = 16.0, t<sub>r</sub> = 11.524, c = 1.0 mm, E\* = 1.0

Thickness Ratio =  $tr * E^*/(tg - c) = 0.768$ , Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.537 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 16.000 mm | Basic MDMT             | -14 °C    |
| Stress/Thk. Ratio        | 0.768     | Temperature Difference | 13 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -27 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-1

#### Cylindrical Shell From 110 to SR 1 Fr 70 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                                      | 5938.00   | 1332.02 | 395.87 | 0.2243 | 0.0008223 | 78.93    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*78.9273)/(3*395.8666) = 0.2658 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 10.198                         | 5938.00   | 1332.02 | 582.30 | 0.2243 | 0.0004523 | 45.19    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*45.1874)/(3*582.2954) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                               | 5938.00   | 3317.59 | 395.87 | 0.5587 | 0.0003076 | 30.74    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.7353)/(3*395.8666) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 1 Fr 70 to SR 2 Fr 70 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                                      | 5938.00   | 2000.00 | 395.87 | 0.3368 | 0.0005276 | 52.72    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*52.7157)/(3*395.8666) = 0.1776 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.118                         | 5938.00   | 2000.00 | 489.99 | 0.3368 | 0.0003806 | 38.02    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0233)/(3*489.9948) = 0.1035 \text{ MPa}$$



| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                               | 5938.00   | 3317.34 | 395.87 | 0.5587 | 0.0003077 | 30.74    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.7378)/(3*395.8666) = 0.1035 \text{ MPa}$$

### **Cylindrical Shell From SR 2 Fr 70 to 120 Ext. Chart: CS-2 at 150 °C**

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                                      | 5938.00   | 2000.00 | 395.87 | 0.3368 | 0.0005276 | 52.72    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*52.7157)/(3*395.8666) = 0.1776 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.118                         | 5938.00   | 2000.00 | 489.99 | 0.3368 | 0.0003806 | 38.02    |

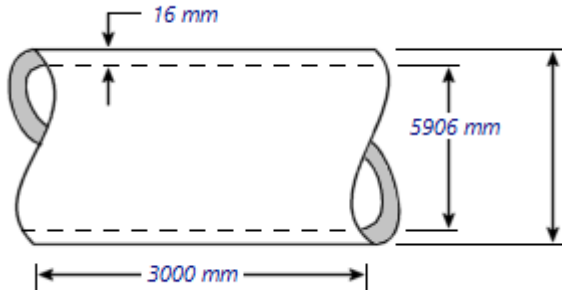
$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0233)/(3*489.9948) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                               | 5938.00   | 3317.34 | 395.87 | 0.5587 | 0.0003077 | 30.74    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.7378)/(3*395.8666) = 0.1035 \text{ MPa}$$

### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



### Internal Pressure Results for: Shell-2

| Cylindrical Shell Design Information:             |    |                            |                         |    |            |
|---------------------------------------------------|----|----------------------------|-------------------------|----|------------|
| Design Pressure                                   | P  | 0.353 MPa                  | Int. Design Temperature |    | 150 °C     |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C     |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2       |
|                                                   |    |                            | UNS Number              |    | K02700     |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa  |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa  |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 69.7 MPa   |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 16.0000 mm                 | Spec. Nominal Thk.      |    | 16.000 mm  |
| Surface Area                                      |    | 55964332.0 mm <sup>2</sup> | SG of Contents          |    | 0.987      |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      |    | 267743.6 N |
| Empty Weight                                      |    | 67870.2 N                  | Operating Weight        |    | 335613.9 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 75.7 MPa   |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.353 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.353)$$

$$= 7.5688 + 1.0000 = 8.5688 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.010 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (2954.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (2953.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.353 \cdot (2954.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 69.737 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.270 %

### Minimum Design Metal Temperature Results:

Govrn. thk, t<sub>g</sub> = 16.0, t<sub>r</sub> = 10.9, c = 1.0 mm, E\* = 1.0

Thickness Ratio =  $tr * E^* / (tg - c) = 0.727$ , Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.508 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 16.000 mm | Basic MDMT             | -14 °C    |
| Stress/Thk. Ratio        | 0.727     | Temperature Difference | 15 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -29 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-2

#### Cylindrical Shell From 120 to SR 3 Fr 80 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                                      | 5938.00   | 2000.00 | 395.87 | 0.3368 | 0.0005276 | 52.72    |

$$MAEP = (4*B) / (3*(Do/t)) = (4*52.7157) / (3*395.8666) = 0.1776 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.118                         | 5938.00   | 2000.00 | 489.99 | 0.3368 | 0.0003806 | 38.02    |

$$MAEP = (4*B) / (3*(Do/t)) = (4*38.0233) / (3*489.9948) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                               | 5938.00   | 3317.34 | 395.87 | 0.5587 | 0.0003077 | 30.74    |

$$MAEP = (4*B) / (3*(Do/t)) = (4*30.7378) / (3*395.8666) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 3 Fr 80 to 130 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 15.000                                      | 5938.00   | 2000.00 | 395.87 | 0.3368 | 0.0005276 | 52.72    |

$$MAEP = (4*B) / (3*(Do/t)) = (4*52.7157) / (3*395.8666) = 0.1776 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.118                         | 5938.00   | 2000.00 | 489.99 | 0.3368 | 0.0003806 | 38.02    |

$$MAEP = (4*B) / (3*(Do/t)) = (4*38.0233) / (3*489.9948) = 0.1035 \text{ MPa}$$

PV Elite 28 U2    Licensee: SL User  
FileName : C-1402\_Ope\_Rev 0  
Element: Shell-2

Step:    8    9:26am    Jul 23,2026

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HEXAGON

Results for Maximum Stiffened Length

Slen

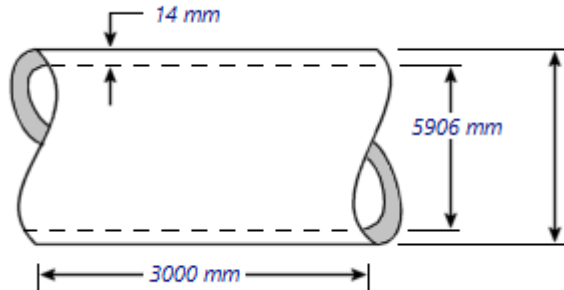
| Tca    | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
|--------|-----------|---------|--------|--------|-----------|----------|
| 15.000 | 5938.00   | 3317.34 | 395.87 | 0.5587 | 0.0003077 | 30.74    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.7378)/(3*395.8666) = 0.1035 \text{ MPa}$$

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-3

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 0.343 MPa                  | Int. Design Temperature |    | 150 °C    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C    |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 78.2 MPa  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 14.0000 mm                 | Spec. Nominal Thk.      |    | 14.000 mm |
| Surface Area                                      |    | 55926628.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 59366.4 N                  | Operating Weight        |    | 59366.4 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 65.7 MPa  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 7.3585 + 1.0000 = 8.3585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2954.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2953.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (2954.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 78.202 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.236 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\* / (tg - c) = 0.822, Temp. Reduction = 10 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 14.000 mm | Basic MDMT             | -18 °C    |
| Stress/Thk. Ratio        | 0.822     | Temperature Difference | 10 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -28 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-3

#### Cylindrical Shell From 130 to SR 4 Fr 90 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        |           | Slen     |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 4 Fr 90 to SR 5 Fr 90 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.688)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0131)/(3*489.8641) = 0.1035 \text{ MPa}$$



| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

### **Cylindrical Shell From SR 5 Fr 90 to 140 Ext. Chart: CS-2 at 150 °C**

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

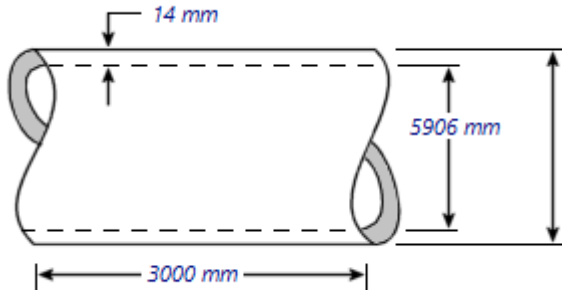
$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-4

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 0.343 MPa                  | Int. Design Temperature |    | 150 °C    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C    |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 78.2 MPa  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 14.0000 mm                 | Spec. Nominal Thk.      |    | 14.000 mm |
| Surface Area                                      |    | 55926628.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 59366.4 N                  | Operating Weight        |    | 59366.4 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 65.7 MPa  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 7.3585 + 1.0000 = 8.3585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2954.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2953.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (2954.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 78.202 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.236 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\* / (tg - c) = 0.822, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 14.000 mm | Basic MDMT             | -18 °C    |
| Stress/Thk. Ratio        | 0.822     | Temperature Difference | 10 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -28 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-4

#### Cylindrical Shell From 140 to SR 6 Fr 100 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        |           | Slen     |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 6 Fr 100 to 150 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

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FileName : C-1402\_Ope\_Rev 0  
Element: Shell-4

Step: 10    9:26am    Jul 23,2026

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Results for Maximum Stiffened Length

Slen

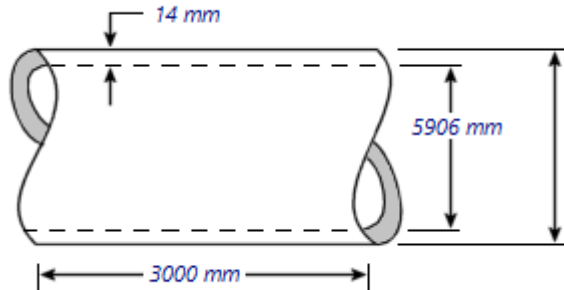
| Tca    | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
|--------|-----------|---------|--------|--------|-----------|----------|
| 13.000 | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-5

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 0.343 MPa                  | Int. Design Temperature |    | 150 °C    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C    |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 78.2 MPa  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 14.0000 mm                 | Spec. Nominal Thk.      |    | 14.000 mm |
| Surface Area                                      |    | 55926628.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 59366.4 N                  | Operating Weight        |    | 59366.4 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 65.7 MPa  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 7.3585 + 1.0000 = 8.3585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2954.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2953.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (2954.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 78.202 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.236 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, t<sub>g</sub> = 14.0, t<sub>r</sub> = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio = t<sub>r</sub> \* E\* / (t<sub>g</sub> - c) = 0.822, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 14.000 mm | Basic MDMT             | -18 °C    |
| Stress/Thk. Ratio        | 0.822     | Temperature Difference | 10 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -28 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-5

#### Cylindrical Shell From 150 to SR 7 Fr 110 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        |           | Slen     |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 7 Fr 110 to SR 8 Fr 110 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.688)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0131)/(3*489.8641) = 0.1035 \text{ MPa}$$



| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

**Cylindrical Shell From SR 8 Fr 110 to 160 Ext. Chart: CS-2 at 150 °C**

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

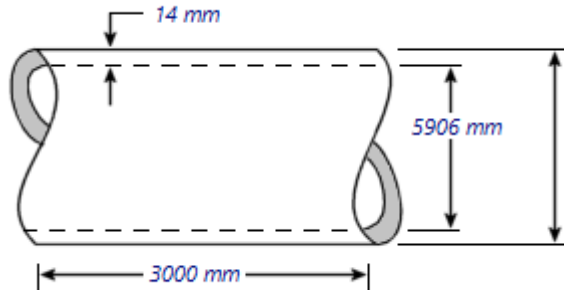
| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$



### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-6

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 0.343 MPa                  | Int. Design Temperature |    | 150 °C    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C    |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 78.2 MPa  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 14.0000 mm                 | Spec. Nominal Thk.      |    | 14.000 mm |
| Surface Area                                      |    | 55926628.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 59366.4 N                  | Operating Weight        |    | 59366.4 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 65.7 MPa  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 7.3585 + 1.0000 = 8.3585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2954.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2953.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (2954.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 78.202 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.236 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\* / (tg - c) = 0.822, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 14.000 mm | Basic MDMT             | -18 °C    |
| Stress/Thk. Ratio        | 0.822     | Temperature Difference | 10 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -28 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-6

#### Cylindrical Shell From 160 to SR 9 Fr 120 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        |           | Slen     |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 9 Fr 120 to 170 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$



Results for Maximum Stiffened Length

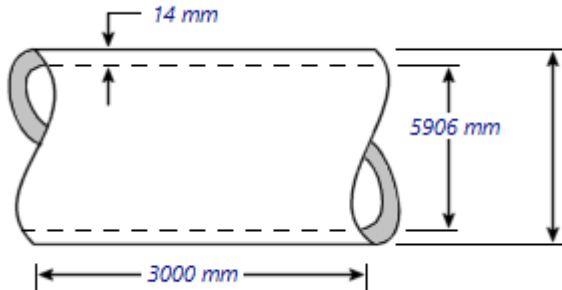
Slen

| Tca    | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
|--------|-----------|---------|--------|--------|-----------|----------|
| 13.000 | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-7

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 0.343 MPa                  | Int. Design Temperature |    | 150 °C    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C    |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 78.2 MPa  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 14.0000 mm                 | Spec. Nominal Thk.      |    | 14.000 mm |
| Surface Area                                      |    | 55926628.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 59366.4 N                  | Operating Weight        |    | 59366.4 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 65.7 MPa  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 7.3585 + 1.0000 = 8.3585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2954.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2953.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (2954.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 78.202 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.236 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\* / (tg - c) = 0.822, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 14.000 mm | Basic MDMT             | -18 °C    |
| Stress/Thk. Ratio        | 0.822     | Temperature Difference | 10 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -28 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-7

#### Cylindrical Shell From 170 to SR 10 Fr 130 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        |           | Slen     |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 10 Fr 130 to SR 11 Fr 130 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        |           | MAEP     |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.688)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        |           | Tca      |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0131)/(3*489.8641) = 0.1035 \text{ MPa}$$



| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

### **Cylindrical Shell From SR 11 Fr 130 to 180 Ext. Chart: CS-2 at 150 °C**

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

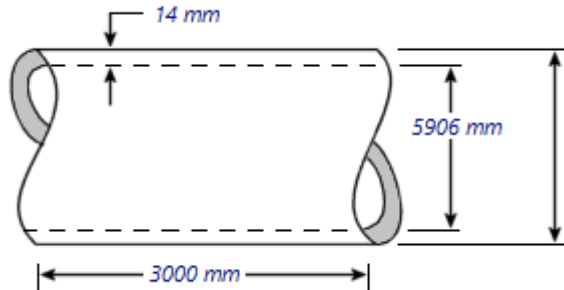
$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

### Internal Pressure Calculation Results:

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#### Internal Pressure Results for: Shell-8

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 0.343 MPa                  | Int. Design Temperature |    | 150 °C    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C    |
| Length                                            |    | 3000.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 78.2 MPa  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 14.0000 mm                 | Spec. Nominal Thk.      |    | 14.000 mm |
| Surface Area                                      |    | 55926628.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 82200.8 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 59366.4 N                  | Operating Weight        |    | 59366.4 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 65.7 MPa  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 7.3585 + 1.0000 = 8.3585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2954.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2953.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (2954.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 78.202 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.236 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\* / (tg - c) = 0.822, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 14.000 mm | Basic MDMT             | -18 °C    |
| Stress/Thk. Ratio        | 0.822     | Temperature Difference | 10 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -28 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-8

#### Cylindrical Shell From 180 to SR 12 Fr 140 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6881)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0132)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 12 Fr 140 to 190 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.688)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0131)/(3*489.8641) = 0.1035 \text{ MPa}$$





Results for Maximum Stiffened Length

Slen

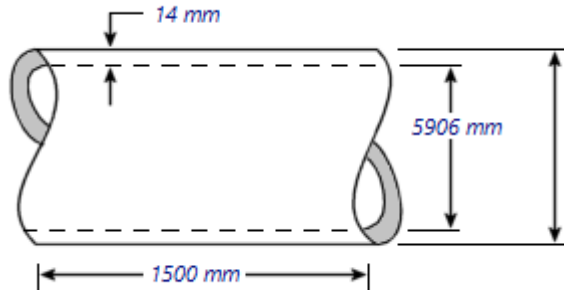
| Tca    | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
|--------|-----------|---------|--------|--------|-----------|----------|
| 13.000 | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Shell-9

| Cylindrical Shell Design Information:             |    |                            |                         |    |           |
|---------------------------------------------------|----|----------------------------|-------------------------|----|-----------|
| Design Pressure                                   | P  | 0.343 MPa                  | Int. Design Temperature |    | 150 °C    |
| Material                                          |    | SA-516 70                  | Ext. Design Temperature |    | 150 °C    |
| Length                                            |    | 1500.000 mm                | External Pressure Chart |    | CS-2      |
|                                                   |    |                            | UNS Number              |    | K02700    |
| Int. Corr. All.                                   | c  | 1.0000 mm                  | Allowable Stress (ope)  | S  | 138.0 MPa |
| Ext. Corr. All.                                   | ce | 0.0000 mm                  | Allowable Stress (amb)  | Sa | 138.0 MPa |
| Inside Diameter                                   | Di | 5906.000 mm                | Actual Stress           |    | 78.2 MPa  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 14.0000 mm                 | Spec. Nominal Thk.      |    | 14.000 mm |
| Surface Area                                      |    | 27963314.0 mm <sup>2</sup> | SG of Contents          |    | 0.000     |
| Element Volume                                    |    | 41100.4 ltr                | Weight of Contents      |    | 0.0 N     |
| Empty Weight                                      |    | 29683.2 N                  | Operating Weight        |    | 29683.2 N |
|                                                   |    |                            | Comp. All. Ope. Stress  |    | 65.7 MPa  |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 7.3585 + 1.0000 = 8.3585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2954.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2953.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (2954.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 78.202 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.236 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\* / (tg - c) = 0.822, Temp. Reduction = 10 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | Yes       | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 14.000 mm | Basic MDMT             | -18 °C    |
| Stress/Thk. Ratio        | 0.822     | Temperature Difference | 10 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -28 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

### EXTERNAL PRESSURE RESULTS for: Shell-9

#### Cylindrical Shell From 190 to SR 13 Fr 150 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |         |        |        | MAEP      |          |
|---------------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 2000.00 | 456.46 | 0.3370 | 0.0004273 | 42.69    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.688)/(3*456.4615) = 0.1247 \text{ MPa}$$

| Results for Required Thickness |           |         |        |        | Tca       |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 12.114                         | 5934.00   | 2000.00 | 489.86 | 0.3370 | 0.0003805 | 38.01    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.0131)/(3*489.8641) = 0.1035 \text{ MPa}$$

| Results for Maximum Stiffened Length |           |         |        |        | Slen      |          |
|--------------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                                  | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                               | 5934.00   | 2373.49 | 456.46 | 0.4000 | 0.0003548 | 35.45    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.448)/(3*456.4615) = 0.1035 \text{ MPa}$$

#### Cylindrical Shell From SR 13 Fr 150 to 220 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

| Results for Maximum Allowable Ext. Pressure |           |        |        |        | MAEP      |          |
|---------------------------------------------|-----------|--------|--------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen   | Do/t   | L/D    | Factor A  | Factor B |
| 13.000                                      | 5934.00   | 932.14 | 456.46 | 0.1571 | 0.0009915 | 84.87    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*84.8735)/(3*456.4615) = 0.2479 \text{ MPa}$$

| Results for Required Thickness |           |        |        |        | Tca       |          |
|--------------------------------|-----------|--------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen   | Do/t   | L/D    | Factor A  | Factor B |
| 8.685                          | 5934.00   | 932.14 | 683.22 | 0.1571 | 0.0005307 | 53.02    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*53.0191)/(3*683.215) = 0.1035 \text{ MPa}$$

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FileName : C-1402\_Ope\_Rev 0  
Element: Shell-9

Step: 15    9:26am    Jul 23,2026

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Results for Maximum Stiffened Length

Slen

| Tca    | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
|--------|-----------|---------|--------|--------|-----------|----------|
| 13.000 | 5934.00   | 2374.17 | 456.46 | 0.4001 | 0.0003547 | 35.44    |

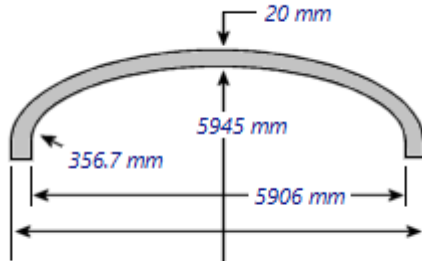
$$MAEP = (4*B)/(3*(Do/t)) = (4*35.4374)/(3*456.4615) = 0.1035 \text{ MPa}$$

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Top Dish End

| Torispherical Head Design Information: |    |                |                         |    |            |
|----------------------------------------|----|----------------|-------------------------|----|------------|
| Design Pressure                        | P  | 0.343 MPa      | Int. Design Temperature |    | 150 °C     |
| Material                               |    | SA-516 70      | Ext. Design Temperature |    | 150 °C     |
| Length                                 |    | 50.000 mm      | External Pressure Chart |    | CS-2       |
|                                        |    |                | UNS Number              |    | K02700     |
| Int. Corr. All.                        | c  | 1.0000 mm      | Allowable Stress (ope)  | S  | 138.0 MPa  |
| Ext. Corr. All.                        | ce | 0.0000 mm      | Allowable Stress (amb)  | Sa | 138.0 MPa  |
| Inside Diameter                        | Di | 5906.000 mm    | Actual Stress           |    | 95.1 MPa   |
| Spec. Min. Thk t,e, or sa              |    | 20.0000 mm     | Spec. Nominal Thk.      |    | 24.000 mm  |
| Surface Area                           |    | 34123380.0 mm² | SG of Contents          |    | 0.000      |
| Element Volume                         |    | 18031.4 ltr    | Weight of Contents      |    | 0.0 N      |
| Empty Weight                           |    | 61568.9 N      | Operating Weight        |    | 61568.9 N  |
|                                        |    |                | Comp. All. Ope. Stress  |    | 93.6 MPa   |
| Ins. Crn. Rad. L or R                  |    | 5945.0000 mm   | Ins. Knuckle Radius     | r  | 356.700 mm |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 73.825 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 36.912 mm

Inside Corroded Head Depth [h]:

$$= L - \sqrt{\left(\left(L - \frac{D_i}{2}\right) \cdot \left(L + \frac{D_i}{2} - 2 \cdot r\right)\right)}$$

$$= 5946.0 - \sqrt{\left(\left(5946.0 - 5908.0/2\right) \cdot \left(5946.0 + 5908.0/2 - 2 \cdot 357.7\right)\right)}$$

$$= 997.432 \text{ mm}$$

M factor for Torispherical Heads ( Corroded ):

$$= \left(3 + \sqrt{\left(\frac{L+C}{r+C}\right)}\right)/4 \text{ per Appendix 1-4 (b \& d)}$$

$$= \left(3 + \sqrt{\left(\frac{5945.0 + 1.0}{356.7 + 1.0}\right)}\right)/4$$

$$= 1.7693$$

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot L \cdot M) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4 (d)}$$

$$= (0.343 \cdot 5946.0 \cdot 1.7693) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.343)$$

$$= 13.0867 + 1.0000 = 14.0867 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t) \text{ per Appendix 1-4 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.0) / (1.7693 \cdot 5946.0 + 0.2 \cdot 19.0)$$

$$= 0 \text{ MPa}$$

M factor for Torispherical Heads ( New & Cold ):

$$= (3 + \sqrt{L/r})/4 \text{ per Appendix 1-4 (b \& d)}$$

$$= (3 + \sqrt{5945.0/356.7})/4$$

$$= 1.7706$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t) \text{ per Appendix 1-4 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 20.0) / (1.7706 \cdot 5945.0 + 0.2 \cdot 20.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (M \cdot L + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.343 \cdot (1.7693 \cdot 5946.0 + 0.2 \cdot 19.0)) / (2 \cdot 1.0 \cdot 19.0)$$

$$= 95.060 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 2954.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) + 1.0$$

$$= 8.359 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 23.0) / (2954.0 + 0.6 \cdot 23.0)$$

$$= 1 \text{ MPa}$$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  4.882 %

#### MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 20.0, tr = 19.0, c = 1.0 mm, E\* = 1.0

Thickness Ratio =  $tr \cdot E^* / (tg - c) = 1.0$ , Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 24.0, tr = 10.69, c = 1.0 mm, E\* = 1.0

Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.465$ , Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |           |                        |           |
|--------------------------|-----------|------------------------|-----------|
| MDMT Curve               | B         | Press. at MDMT         | 0.498 MPa |
| UCS-66(b) red.           | No        | Min. Des. Mtl. Temp.   | -1 °C     |
| Governing Thk.           | 24.000 mm | Basic MDMT             | -2 °C     |
| Stress/Thk. Ratio        | 0.465     | Temperature Difference | 37 °C     |
| UCS-66(c) red.           | No        | Computed Min. Temp.    | -40 °C    |
|                          |           | MDMT per UG-20(f)      | -29 °C    |
|                          |           | Joint Efficiency E*    | 100 %     |

Note: Temperature values displayed above are for the location with the highest basic MDMT

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 73.825 mm  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 36.912 mm

#### EXTERNAL PRESSURE RESULTS for: Top Dish End

#### Torispherical Head From 220 to 230 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa



| Results for Maximum Allowable Ext. Pressure |           |        |           | MAEP     |
|---------------------------------------------|-----------|--------|-----------|----------|
| Tca                                         | Crown Rad | Ro/t   | Factor A  | Factor B |
| 19.000                                      | 5965.00   | 313.95 | 0.0003982 | 39.78    |

$$\text{MAEP} = B / (\text{Ro/t}) = 39.7795 / 313.9474 = 0.1267 \text{ MPa}$$

| Results for Required Thickness |           |        |           | Tca      |
|--------------------------------|-----------|--------|-----------|----------|
| Tca                            | Crown Rad | Ro/t   | Factor A  | Factor B |
| 17.170                         | 5965.00   | 347.41 | 0.0003598 | 35.95    |

$$\text{MAEP} = B / (\text{Ro/t}) = 35.9478 / 347.411 = 0.1035 \text{ MPa}$$



### Hydrostatic Test Pressure Results:

|                 |                               |           |
|-----------------|-------------------------------|-----------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 0.648 MPa |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 0.446 MPa |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 0.624 MPa |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 0.548 MPa |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 0.491 MPa |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 0.491 MPa |
| App 27-4        | = M.A.W.P.                    | 0.498 MPa |

UG-99(b), Test Pressure Calculation:  
 = Test Factor • MAWP • Stress Ratio  
 = 1.3 • 0.498 • 1.0  
 = 0.648 MPa

### Horizontal Test performed per: UG-99b

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

### Stresses on Elements due to Test Pressure (MPa):

| From To      | Stress | Allowable | Ratio | Pressure |
|--------------|--------|-----------|-------|----------|
| Btm Dish End | 169.0  | 234.0     | 0.722 | 0.71     |
| Shell-1      | 130.7  | 234.0     | 0.558 | 0.71     |
| Shell-2      | 130.7  | 234.0     | 0.558 | 0.71     |
| Shell-3      | 149.3  | 234.0     | 0.638 | 0.71     |
| Shell-4      | 149.3  | 234.0     | 0.638 | 0.71     |
| Shell-5      | 149.3  | 234.0     | 0.638 | 0.71     |
| Shell-6      | 149.3  | 234.0     | 0.638 | 0.71     |
| Shell-7      | 149.3  | 234.0     | 0.638 | 0.71     |
| Shell-8      | 149.3  | 234.0     | 0.638 | 0.71     |
| Shell-9      | 149.3  | 234.0     | 0.638 | 0.71     |
| Top Dish End | 185.8  | 234.0     | 0.794 | 0.71     |

### Stress ratios for Nozzle and Pad Materials (MPa):

| Description | Pad/Nozzle | Ambient | Operating | Ratio |
|-------------|------------|---------|-----------|-------|
| N1 36"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| N1 36"      | Pad        | 138.00  | 138.00    | 1.000 |
| N11 2"      | Nozzle     | 118.00  | 118.00    | 1.000 |
| N11 2"      | Pad        | 138.00  | 138.00    | 1.000 |
| K1a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K2a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K3a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K4a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N9          | Nozzle     | 138.00  | 138.00    | 1.000 |
| N8 2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| K1b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K2b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K3b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K4b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N2a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N2b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N3A         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N4a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N4b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| A1_30"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| A1_30"      | Pad        | 138.00  | 138.00    | 1.000 |
| N3B         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K5a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K6a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K5b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K6b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K7a         | Nozzle     | 138.00  | 138.00    | 1.000 |





|        |        |        |        |       |
|--------|--------|--------|--------|-------|
| A2_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A2_30" | Pad    | 138.00 | 138.00 | 1.000 |
| A3_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A3_30" | Pad    | 138.00 | 138.00 | 1.000 |
| K7b    | Nozzle | 118.00 | 118.00 | 1.000 |
| A4_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A4_30" | Pad    | 138.00 | 138.00 | 1.000 |
| N5     | Nozzle | 138.00 | 138.00 | 1.000 |
| K8a    | Nozzle | 118.00 | 118.00 | 1.000 |
| A5_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A5_30" | Pad    | 138.00 | 138.00 | 1.000 |
| K9a    | Nozzle | 118.00 | 118.00 | 1.000 |
| N6     | Nozzle | 118.00 | 118.00 | 1.000 |
| N6     | Pad    | 138.00 | 138.00 | 1.000 |
| K8b    | Nozzle | 118.00 | 118.00 | 1.000 |
| K9b    | Nozzle | 118.00 | 118.00 | 1.000 |
| A6_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A6_30" | Pad    | 138.00 | 138.00 | 1.000 |
| A7_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A7_30" | Pad    | 138.00 | 138.00 | 1.000 |
| N7     | Nozzle | 138.00 | 138.00 | 1.000 |
| N7     | Pad    | 138.00 | 138.00 | 1.000 |
| N10    | Nozzle | 118.00 | 118.00 | 1.000 |

Minimum 1.000

#### Stress ratios for Stiffening Ring Materials (MPa):

| Description  | Ambient | Operating | Ratio |
|--------------|---------|-----------|-------|
| SR 1 Fr 70   | 137.90  | 137.90    | 1.000 |
| SR 2 Fr 70   | 137.90  | 137.90    | 1.000 |
| SR 3 Fr 80   | 137.90  | 137.90    | 1.000 |
| SR 4 Fr 90   | 137.90  | 137.90    | 1.000 |
| SR 5 Fr 90   | 137.90  | 137.90    | 1.000 |
| SR 6 Fr 100  | 137.90  | 137.90    | 1.000 |
| SR 7 Fr 110  | 137.90  | 137.90    | 1.000 |
| SR 8 Fr 110  | 137.90  | 137.90    | 1.000 |
| SR 9 Fr 120  | 137.90  | 137.90    | 1.000 |
| SR 10 Fr 130 | 137.90  | 137.90    | 1.000 |
| SR 11 Fr 130 | 137.90  | 137.90    | 1.000 |
| SR 12 Fr 140 | 137.90  | 137.90    | 1.000 |
| SR 13 Fr 150 | 137.90  | 137.90    | 1.000 |

Minimum 1.000

#### Stress ratios for Pressurized Vessel Elements (MPa):

| Description  | Ambient | Operating | Ratio |
|--------------|---------|-----------|-------|
| Btm Dish End | 138.00  | 138.00    | 1.000 |
| Shell-1      | 138.00  | 138.00    | 1.000 |
| Shell-2      | 138.00  | 138.00    | 1.000 |
| Shell-3      | 138.00  | 138.00    | 1.000 |
| Shell-4      | 138.00  | 138.00    | 1.000 |
| Shell-5      | 138.00  | 138.00    | 1.000 |
| Shell-6      | 138.00  | 138.00    | 1.000 |
| Shell-7      | 138.00  | 138.00    | 1.000 |
| Shell-8      | 138.00  | 138.00    | 1.000 |
| Shell-9      | 138.00  | 138.00    | 1.000 |
| Top Dish End | 138.00  | 138.00    | 1.000 |
| N2A FLG      | 138.00  | 138.00    | 1.000 |
| N2B FLG      | 138.00  | 138.00    | 1.000 |

Minimum 1.000

#### Hoop Stress in Nozzle Wall during Pressure Test (MPa):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
|-------------|---------|-----------|-------|



|        |       |        |       |
|--------|-------|--------|-------|
| N1 36" | 15.85 | 234.00 | 0.068 |
| N11 2" | 4.11  | 216.00 | 0.019 |
| K1a    | 1.75  | 225.00 | 0.008 |
| K2a    | 1.75  | 225.00 | 0.008 |
| K3a    | 1.75  | 225.00 | 0.008 |
| K4a    | 1.75  | 225.00 | 0.008 |
| N9     | 1.75  | 225.00 | 0.008 |
| N8 2"  | 1.75  | 216.00 | 0.008 |
| K1b    | 1.75  | 225.00 | 0.008 |
| K2b    | 1.75  | 225.00 | 0.008 |
| K3b    | 1.75  | 225.00 | 0.008 |
| K4b    | 1.75  | 225.00 | 0.008 |
| N2a    | 34.45 | 234.00 | 0.147 |
| N2b    | 34.45 | 234.00 | 0.147 |
| N3A    | 8.15  | 234.00 | 0.035 |
| N4a    | 10.47 | 234.00 | 0.045 |
| N4b    | 10.47 | 234.00 | 0.045 |
| A1_30" | 16.52 | 234.00 | 0.071 |
| N3B    | 11.67 | 234.00 | 0.050 |
| K5a    | 1.75  | 225.00 | 0.008 |
| K6a    | 1.75  | 225.00 | 0.008 |
| K5b    | 1.75  | 225.00 | 0.008 |
| K6b    | 1.75  | 225.00 | 0.008 |
| K7a    | 1.75  | 225.00 | 0.008 |
| A2_30" | 16.52 | 234.00 | 0.071 |
| A3_30" | 16.52 | 234.00 | 0.071 |
| K7b    | 5.16  | 216.00 | 0.024 |
| A4_30" | 16.52 | 234.00 | 0.071 |
| N5     | 25.48 | 234.00 | 0.109 |
| K8a    | 5.16  | 216.00 | 0.024 |
| A5_30" | 16.52 | 234.00 | 0.071 |
| K9a    | 5.16  | 216.00 | 0.024 |
| N6     | 6.42  | 216.00 | 0.030 |
| K8b    | 5.16  | 216.00 | 0.024 |
| K9b    | 5.16  | 216.00 | 0.024 |
| A6_30" | 16.52 | 234.00 | 0.071 |
| A7_30" | 16.52 | 234.00 | 0.071 |
| N7     | 22.12 | 234.00 | 0.095 |
| N10    | 5.16  | 216.00 | 0.024 |

Elements Suitable for Test Pressure.

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### Hydrostatic Test Pressure Results:

|                 |                               |           |
|-----------------|-------------------------------|-----------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 0.648 MPa |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 0.446 MPa |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 0.671 MPa |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 0.548 MPa |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 0.491 MPa |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 0.491 MPa |
| App 27-4        | = M.A.W.P.                    | 0.498 MPa |

UG-99(b), Test Pressure Calculation:  
 = Test Factor • MAWP • Stress Ratio  
 = 1.3 • 0.498 • 1.0  
 = 0.648 MPa

### Vertical Test performed per: UG-99b

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

### Stresses on Elements due to Test Pressure (MPa):

| From To      | Stress | Allowable | Ratio | Pressure |
|--------------|--------|-----------|-------|----------|
| Btm Dish End | 230.2  | 234.0     | 0.984 | 0.92     |
| Shell-1      | 179.4  | 234.0     | 0.767 | 0.91     |
| Shell-2      | 173.5  | 234.0     | 0.742 | 0.88     |
| Shell-3      | 193.5  | 234.0     | 0.827 | 0.85     |
| Shell-4      | 186.8  | 234.0     | 0.798 | 0.82     |
| Shell-5      | 180.1  | 234.0     | 0.770 | 0.79     |
| Shell-6      | 173.4  | 234.0     | 0.741 | 0.76     |
| Shell-7      | 166.7  | 234.0     | 0.712 | 0.73     |
| Shell-8      | 160.0  | 234.0     | 0.684 | 0.70     |
| Shell-9      | 153.3  | 234.0     | 0.655 | 0.67     |
| Top Dish End | 182.2  | 234.0     | 0.779 | 0.66     |

### Stress ratios for Nozzle and Pad Materials (MPa):

| Description | Pad/Nozzle | Ambient | Operating | Ratio |
|-------------|------------|---------|-----------|-------|
| N1 36"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| N1 36"      | Pad        | 138.00  | 138.00    | 1.000 |
| N11 2"      | Nozzle     | 118.00  | 118.00    | 1.000 |
| N11 2"      | Pad        | 138.00  | 138.00    | 1.000 |
| K1a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K2a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K3a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K4a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N9          | Nozzle     | 138.00  | 138.00    | 1.000 |
| N8 2"       | Nozzle     | 118.00  | 118.00    | 1.000 |
| K1b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K2b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K3b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K4b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N2a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N2b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N3A         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N4a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| N4b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| A1_30"      | Nozzle     | 138.00  | 138.00    | 1.000 |
| A1_30"      | Pad        | 138.00  | 138.00    | 1.000 |
| N3B         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K5a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K6a         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K5b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K6b         | Nozzle     | 138.00  | 138.00    | 1.000 |
| K7a         | Nozzle     | 138.00  | 138.00    | 1.000 |



|        |        |        |        |       |
|--------|--------|--------|--------|-------|
| A2_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A2_30" | Pad    | 138.00 | 138.00 | 1.000 |
| A3_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A3_30" | Pad    | 138.00 | 138.00 | 1.000 |
| K7b    | Nozzle | 118.00 | 118.00 | 1.000 |
| A4_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A4_30" | Pad    | 138.00 | 138.00 | 1.000 |
| N5     | Nozzle | 138.00 | 138.00 | 1.000 |
| K8a    | Nozzle | 118.00 | 118.00 | 1.000 |
| A5_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A5_30" | Pad    | 138.00 | 138.00 | 1.000 |
| K9a    | Nozzle | 118.00 | 118.00 | 1.000 |
| N6     | Nozzle | 118.00 | 118.00 | 1.000 |
| N6     | Pad    | 138.00 | 138.00 | 1.000 |
| K8b    | Nozzle | 118.00 | 118.00 | 1.000 |
| K9b    | Nozzle | 118.00 | 118.00 | 1.000 |
| A6_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A6_30" | Pad    | 138.00 | 138.00 | 1.000 |
| A7_30" | Nozzle | 138.00 | 138.00 | 1.000 |
| A7_30" | Pad    | 138.00 | 138.00 | 1.000 |
| N7     | Nozzle | 138.00 | 138.00 | 1.000 |
| N7     | Pad    | 138.00 | 138.00 | 1.000 |
| N10    | Nozzle | 118.00 | 118.00 | 1.000 |

Minimum

1.000

#### Stress ratios for Stiffening Ring Materials (MPa):

| Description  | Ambient | Operating | Ratio |
|--------------|---------|-----------|-------|
| SR 1 Fr 70   | 137.90  | 137.90    | 1.000 |
| SR 2 Fr 70   | 137.90  | 137.90    | 1.000 |
| SR 3 Fr 80   | 137.90  | 137.90    | 1.000 |
| SR 4 Fr 90   | 137.90  | 137.90    | 1.000 |
| SR 5 Fr 90   | 137.90  | 137.90    | 1.000 |
| SR 6 Fr 100  | 137.90  | 137.90    | 1.000 |
| SR 7 Fr 110  | 137.90  | 137.90    | 1.000 |
| SR 8 Fr 110  | 137.90  | 137.90    | 1.000 |
| SR 9 Fr 120  | 137.90  | 137.90    | 1.000 |
| SR 10 Fr 130 | 137.90  | 137.90    | 1.000 |
| SR 11 Fr 130 | 137.90  | 137.90    | 1.000 |
| SR 12 Fr 140 | 137.90  | 137.90    | 1.000 |
| SR 13 Fr 150 | 137.90  | 137.90    | 1.000 |

Minimum

1.000

#### Stress ratios for Pressurized Vessel Elements (MPa):

| Description  | Ambient | Operating | Ratio |
|--------------|---------|-----------|-------|
| Btm Dish End | 138.00  | 138.00    | 1.000 |
| Shell-1      | 138.00  | 138.00    | 1.000 |
| Shell-2      | 138.00  | 138.00    | 1.000 |
| Shell-3      | 138.00  | 138.00    | 1.000 |
| Shell-4      | 138.00  | 138.00    | 1.000 |
| Shell-5      | 138.00  | 138.00    | 1.000 |
| Shell-6      | 138.00  | 138.00    | 1.000 |
| Shell-7      | 138.00  | 138.00    | 1.000 |
| Shell-8      | 138.00  | 138.00    | 1.000 |
| Shell-9      | 138.00  | 138.00    | 1.000 |
| Top Dish End | 138.00  | 138.00    | 1.000 |
| N2A FLG      | 138.00  | 138.00    | 1.000 |
| N2B FLG      | 138.00  | 138.00    | 1.000 |

Minimum

1.000

#### Hoop Stress in Nozzle Wall during Pressure Test (MPa):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
|-------------|---------|-----------|-------|



|        |       |        |       |
|--------|-------|--------|-------|
| N1 36" | 20.62 | 234.00 | 0.088 |
| N11 2" | 5.35  | 216.00 | 0.025 |
| K1a    | 2.26  | 225.00 | 0.010 |
| K2a    | 2.26  | 225.00 | 0.010 |
| K3a    | 2.26  | 225.00 | 0.010 |
| K4a    | 2.26  | 225.00 | 0.010 |
| N9     | 2.26  | 225.00 | 0.010 |
| N8 2"  | 2.26  | 216.00 | 0.010 |
| K1b    | 2.18  | 225.00 | 0.010 |
| K2b    | 2.18  | 225.00 | 0.010 |
| K3b    | 2.18  | 225.00 | 0.010 |
| K4b    | 2.18  | 225.00 | 0.010 |
| N2a    | 42.89 | 234.00 | 0.183 |
| N2b    | 42.89 | 234.00 | 0.183 |
| N3A    | 10.15 | 234.00 | 0.043 |
| N4a    | 13.04 | 234.00 | 0.056 |
| N4b    | 13.04 | 234.00 | 0.056 |
| A1_30" | 20.57 | 234.00 | 0.088 |
| N3B    | 14.53 | 234.00 | 0.062 |
| K5a    | 2.11  | 225.00 | 0.009 |
| K6a    | 2.11  | 225.00 | 0.009 |
| K5b    | 2.04  | 225.00 | 0.009 |
| K6b    | 2.04  | 225.00 | 0.009 |
| K7a    | 2.04  | 225.00 | 0.009 |
| A2_30" | 19.19 | 234.00 | 0.082 |
| A3_30" | 18.50 | 234.00 | 0.079 |
| K7b    | 5.35  | 216.00 | 0.025 |
| A4_30" | 17.13 | 234.00 | 0.073 |
| N5     | 25.36 | 234.00 | 0.108 |
| K8a    | 5.13  | 216.00 | 0.024 |
| A5_30" | 16.44 | 234.00 | 0.070 |
| K9a    | 4.92  | 216.00 | 0.023 |
| N6     | 6.12  | 216.00 | 0.028 |
| K8b    | 4.92  | 216.00 | 0.023 |
| K9b    | 4.92  | 216.00 | 0.023 |
| A6_30" | 15.75 | 234.00 | 0.067 |
| A7_30" | 15.75 | 234.00 | 0.067 |
| N7     | 20.63 | 234.00 | 0.088 |
| N10    | 4.81  | 216.00 | 0.022 |

Elements Suitable for Test Pressure.

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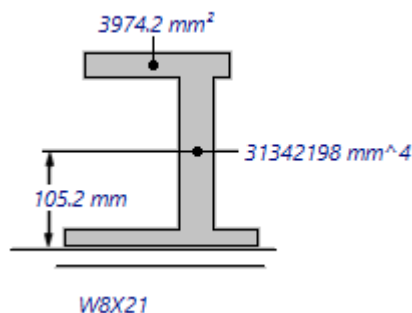


## External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

Stiffening Ring Calculations for: SR 1 Fr 70, W8X21, SA-516 70

Material treatment options:



Effective Length of Shell: 328 mm

|        | Area (mm²) | Distance (mm) | Area*Dist  |
|--------|------------|---------------|------------|
| Shell: | 4924.359   | 7.5000        | 36932.691  |
| Ring : | 3974.190   | 120.1560      | 477522.781 |
| Total: | 8898.548   |               | 514455.469 |

Centroid of Ring plus Shell: 58 mm

|        | Inertia      | Distance | A*Dist²      |
|--------|--------------|----------|--------------|
| Shell: | 92331.633    | 50.3134  | 12465718.000 |
| Ring : | 31342198.000 | -62.3426 | 15446079.000 |
| Total: | 31434530.000 |          | 27911796.000 |

Available Moment of Inertia, Ring + Shell: 0.593E+08 mm⁴

Required Stress in Ring plus Shell Breq 26.50 MPa  
 Required Strain in Ring plus Shell Areq 0.0002640

### Required Moment of Inertia, Ring plus Shell:

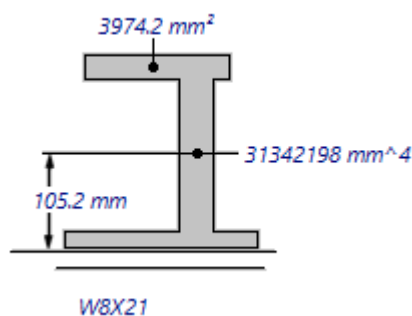
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5938.0^2 \cdot 1666.0103(15.0 + 3974.1899/1666.0103)0.000264) / 10.9$$

$$= 24735462 \text{ mm}^4$$

Stiffening Ring Calculations for: SR 2 Fr 70, W8X21, SA-516 70

Material treatment options:



Effective Length of Shell: 328 mm

|                                            | Area (mm <sup>2</sup> ) | Distance (mm) | Area*Dist                 |
|--------------------------------------------|-------------------------|---------------|---------------------------|
| Shell:                                     | 4924.359                | 7.5000        | 36932.691                 |
| Ring :                                     | 3974.190                | 120.1560      | 477522.781                |
| Total:                                     | 8898.548                |               | 514455.469                |
| Centroid of Ring plus Shell:               |                         |               | 58 mm                     |
|                                            | Inertia                 | Distance      | A*Dist <sup>2</sup>       |
| Shell:                                     | 92331.633               | 50.3134       | 12465718.000              |
| Ring :                                     | 31342198.000            | -62.3426      | 15446079.000              |
| Total:                                     | 31434530.000            |               | 27911796.000              |
| Available Moment of Inertia, Ring + Shell: |                         |               | 0.593E+08 mm <sup>4</sup> |
| Required Stress in Ring plus Shell         |                         |               | Breq 27.13 MPa            |
| Required Strain in Ring plus Shell         |                         |               | Areq 0.0002710            |

**Required Moment of Inertia, Ring plus Shell:**

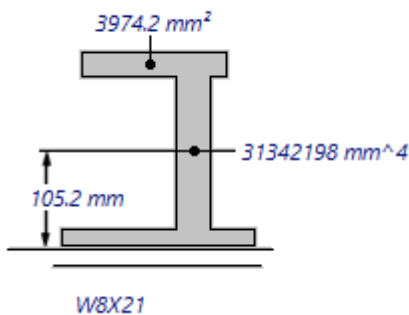
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5938.0^2 \cdot 2000.0(15.0 + 3974.1899/2000.0)0.000271) / 10.9$$

$$= 29783172 \text{ mm}^4$$

**Stiffening Ring Calculations for: SR 3 Fr 80, W8X21, SA-516 70**

**Material treatment options:**



| Effective Length of Shell:                 |                         |               | 328 mm                    |
|--------------------------------------------|-------------------------|---------------|---------------------------|
|                                            | Area (mm <sup>2</sup> ) | Distance (mm) | Area*Dist                 |
| Shell:                                     | 4924.359                | 7.5000        | 36932.691                 |
| Ring :                                     | 3974.190                | 120.1560      | 477522.781                |
| Total:                                     | 8898.548                |               | 514455.469                |
| Centroid of Ring plus Shell:               |                         |               | 58 mm                     |
|                                            | Inertia                 | Distance      | A*Dist <sup>2</sup>       |
| Shell:                                     | 92331.633               | 50.3134       | 12465718.000              |
| Ring :                                     | 31342198.000            | -62.3426      | 15446079.000              |
| Total:                                     | 31434530.000            |               | 27911796.000              |
| Available Moment of Inertia, Ring + Shell: |                         |               | 0.593E+08 mm <sup>4</sup> |
| Required Stress in Ring plus Shell         |                         |               | Breq 27.13 MPa            |
| Required Strain in Ring plus Shell         |                         |               | Areq 0.0002710            |

**Required Moment of Inertia, Ring plus Shell:**

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

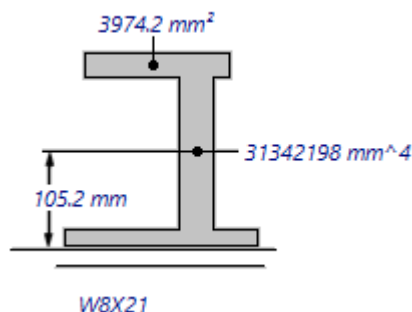
$$= (5938.0^2 \cdot 2000.0(15.0 + 3974.1899/2000.0)0.000271) / 10.9$$

$$= 29783164 \text{ mm}^4$$

**Stiffening Ring Calculations for: SR 4 Fr 90, W8X21, SA-516 70**



**Material treatment options:**



Effective Length of Shell: 306 mm

|        | Area (mm²) | Distance (mm) | Area*Dist  |
|--------|------------|---------------|------------|
| Shell: | 3971.747   | 6.5000        | 25816.359  |
| Ring : | 3974.190   | 118.1560      | 469574.406 |
| Total: | 7945.937   |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

|        | Inertia      | Distance | A*Dist²      |
|--------|--------------|----------|--------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000 |
| Ring : | 31342198.000 | -55.8108 | 12379001.000 |
| Total: | 31398134.000 |          | 24765620.000 |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm⁴

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

**Required Moment of Inertia, Ring plus Shell:**

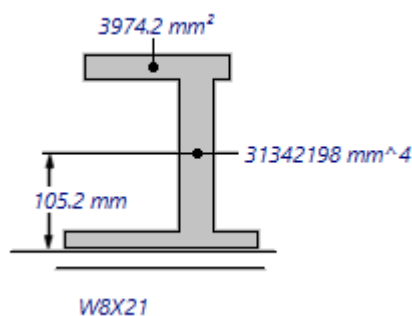
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

**Stiffening Ring Calculations for: SR 5 Fr 90, W8X21, SA-516 70**

**Material treatment options:**



Effective Length of Shell: 306 mm

|        | Area (mm²) | Distance (mm) | Area*Dist  |
|--------|------------|---------------|------------|
| Shell: | 3971.747   | 6.5000        | 25816.359  |
| Ring : | 3974.190   | 118.1560      | 469574.406 |
| Total: | 7945.937   |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm





|        | Inertia      | Distance | A*Dist <sup>2</sup> |
|--------|--------------|----------|---------------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000        |
| Ring : | 31342198.000 | -55.8108 | 12379001.000        |
| Total: | 31398134.000 |          | 24765620.000        |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm<sup>4</sup>

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

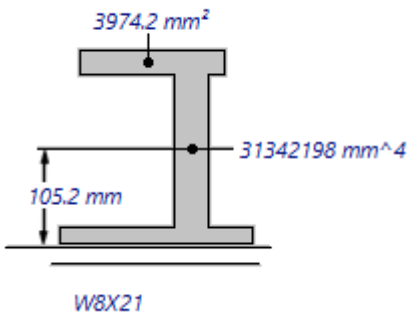
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 6 Fr 100, W8X21, SA-516 70

##### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm <sup>2</sup> ) | Distance (mm) | Area*Dist  |
|--------|-------------------------|---------------|------------|
| Shell: | 3971.747                | 6.5000        | 25816.359  |
| Ring : | 3974.190                | 118.1560      | 469574.406 |
| Total: | 7945.937                |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

|        | Inertia      | Distance | A*Dist <sup>2</sup> |
|--------|--------------|----------|---------------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000        |
| Ring : | 31342198.000 | -55.8108 | 12379001.000        |
| Total: | 31398134.000 |          | 24765620.000        |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm<sup>4</sup>

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

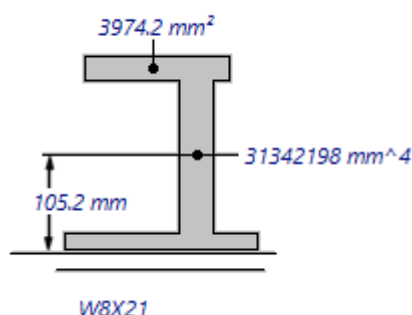
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 7 Fr 110, W8X21, SA-516 70

##### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm²) | Distance (mm) | Area*Dist  |
|--------|------------|---------------|------------|
| Shell: | 3971.747   | 6.5000        | 25816.359  |
| Ring : | 3974.190   | 118.1560      | 469574.406 |
| Total: | 7945.937   |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

|        | Inertia      | Distance | A*Dist²      |
|--------|--------------|----------|--------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000 |
| Ring : | 31342198.000 | -55.8108 | 12379001.000 |
| Total: | 31398134.000 |          | 24765620.000 |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm⁴

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

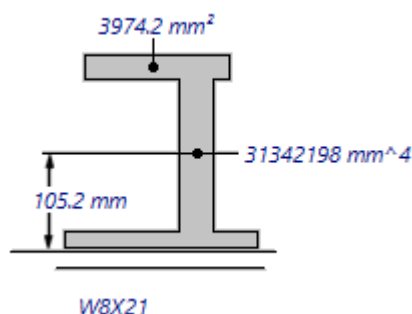
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 8 Fr 110, W8X21, SA-516 70

#### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm²) | Distance (mm) | Area*Dist  |
|--------|------------|---------------|------------|
| Shell: | 3971.747   | 6.5000        | 25816.359  |
| Ring : | 3974.190   | 118.1560      | 469574.406 |
| Total: | 7945.937   |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

| Inertia | Distance | A*Dist² |
|---------|----------|---------|
|---------|----------|---------|

|        |              |          |              |
|--------|--------------|----------|--------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000 |
| Ring : | 31342198.000 | -55.8108 | 12379001.000 |
| Total: | 31398134.000 |          | 24765620.000 |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm<sup>4</sup>

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

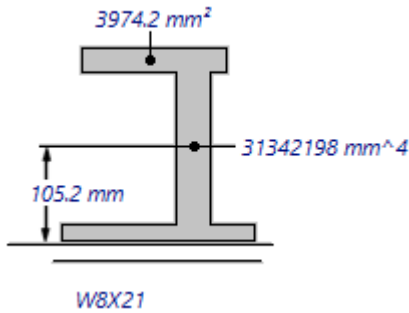
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 9 Fr 120, W8X21, SA-516 70

##### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm <sup>2</sup> ) | Distance (mm) | Area*Dist  |
|--------|-------------------------|---------------|------------|
| Shell: | 3971.747                | 6.5000        | 25816.359  |
| Ring : | 3974.190                | 118.1560      | 469574.406 |
| Total: | 7945.937                |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

|        | Inertia      | Distance | A*Dist <sup>2</sup> |
|--------|--------------|----------|---------------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000        |
| Ring : | 31342198.000 | -55.8108 | 12379001.000        |
| Total: | 31398134.000 |          | 24765620.000        |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm<sup>4</sup>

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

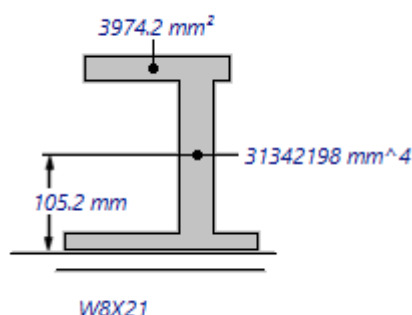
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 10 Fr 130, W8X21, SA-516 70

##### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm²) | Distance (mm) | Area*Dist  |
|--------|------------|---------------|------------|
| Shell: | 3971.747   | 6.5000        | 25816.359  |
| Ring : | 3974.190   | 118.1560      | 469574.406 |
| Total: | 7945.937   |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

|        | Inertia      | Distance | A*Dist²      |
|--------|--------------|----------|--------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000 |
| Ring : | 31342198.000 | -55.8108 | 12379001.000 |
| Total: | 31398134.000 |          | 24765620.000 |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm⁴

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

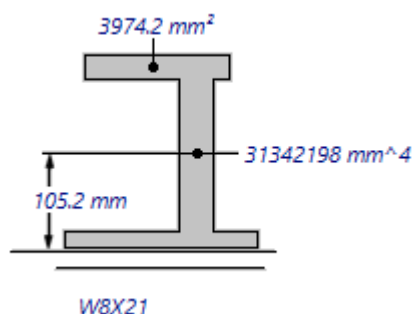
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 11 Fr 130, W8X21, SA-516 70

#### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm²) | Distance (mm) | Area*Dist  |
|--------|------------|---------------|------------|
| Shell: | 3971.747   | 6.5000        | 25816.359  |
| Ring : | 3974.190   | 118.1560      | 469574.406 |
| Total: | 7945.937   |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

| Inertia | Distance | A*Dist² |
|---------|----------|---------|
|---------|----------|---------|



|        |              |          |              |
|--------|--------------|----------|--------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000 |
| Ring : | 31342198.000 | -55.8108 | 12379001.000 |
| Total: | 31398134.000 |          | 24765620.000 |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm<sup>4</sup>

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

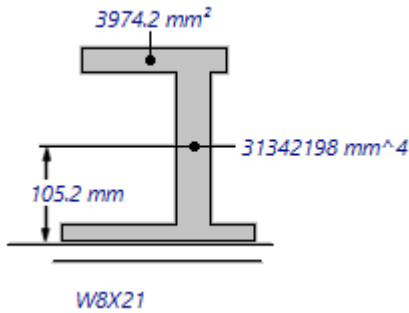
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 12 Fr 140, W8X21, SA-516 70

##### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm <sup>2</sup> ) | Distance (mm) | Area*Dist  |
|--------|-------------------------|---------------|------------|
| Shell: | 3971.747                | 6.5000        | 25816.359  |
| Ring : | 3974.190                | 118.1560      | 469574.406 |
| Total: | 7945.937                |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

|        | Inertia      | Distance | A*Dist <sup>2</sup> |
|--------|--------------|----------|---------------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000        |
| Ring : | 31342198.000 | -55.8108 | 12379001.000        |
| Total: | 31398134.000 |          | 24765620.000        |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm<sup>4</sup>

Required Stress in Ring plus Shell Breq 30.72 MPa  
 Required Strain in Ring plus Shell Areq 0.0003060

#### Required Moment of Inertia, Ring plus Shell:

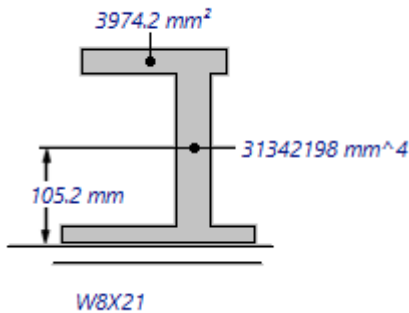
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 2000.0(13.0 + 3974.1899/2000.0)0.000306) / 10.9$$

$$= 29630336 \text{ mm}^4$$

#### Stiffening Ring Calculations for: SR 13 Fr 150, W8X21, SA-516 70

##### Material treatment options:



Effective Length of Shell: 306 mm

|        | Area (mm <sup>2</sup> ) | Distance (mm) | Area*Dist  |
|--------|-------------------------|---------------|------------|
| Shell: | 3971.747                | 6.5000        | 25816.359  |
| Ring : | 3974.190                | 118.1560      | 469574.406 |
| Total: | 7945.937                |               | 495390.750 |

Centroid of Ring plus Shell: 62 mm

|        | Inertia      | Distance | A*Dist <sup>2</sup> |
|--------|--------------|----------|---------------------|
| Shell: | 55935.391    | 55.8452  | 12386619.000        |
| Ring : | 31342198.000 | -55.8108 | 12379001.000        |
| Total: | 31398134.000 |          | 24765620.000        |

Available Moment of Inertia, Ring + Shell: 0.562E+08 mm<sup>4</sup>

Required Stress in Ring plus Shell Breq 29.31 MPa  
 Required Strain in Ring plus Shell Areq 0.0002920

#### Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5934.0^2 \cdot 1466.0721(13.0 + 3974.1899/1466.0721)0.000292) / 10.9$$

$$= 21727174 \text{ mm}^4$$

#### External Pressure Calculations:

| From | To   | Section Length<br>mm | Outside Diameter<br>mm | Corroded Thickness<br>mm | Factor A   | Factor B<br>MPa |
|------|------|----------------------|------------------------|--------------------------|------------|-----------------|
| 10   | 20   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 20   | 30   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 30   | 40   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 40   | 50   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 50   | 60   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 60   | 70   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 70   | 80   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 80   | 90   | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 90   | 100  | 22000                | ...                    | ...                      | No Calc    | No Calc         |
| 100  | 110  | No Calc              | 5971                   | 21                       | 0.00043962 | 43.9227         |
| 110  | Ring | 1332.02              | 5938                   | 15                       | 0.00082234 | 78.9273         |
| Ring | Ring | 2000                 | 5938                   | 15                       | 0.00052763 | 52.7157         |
| Ring | 120  | 2000                 | 5938                   | 15                       | 0.00052764 | 52.7157         |
| 120  | Ring | 2000                 | 5938                   | 15                       | 0.00052764 | 52.7157         |
| Ring | 130  | 2000                 | 5938                   | 15                       | 0.00052764 | 52.7157         |
| 130  | Ring | 2000                 | 5934                   | 13                       | 0.00042727 | 42.6881         |
| Ring | Ring | 2000                 | 5934                   | 13                       | 0.00042727 | 42.688          |
| Ring | 140  | 2000                 | 5934                   | 13                       | 0.00042727 | 42.6881         |
| 140  | Ring | 2000                 | 5934                   | 13                       | 0.00042727 | 42.6881         |
| Ring | 150  | 2000                 | 5934                   | 13                       | 0.00042727 | 42.6881         |
| 150  | Ring | 2000                 | 5934                   | 13                       | 0.00042727 | 42.6881         |
| Ring | Ring | 2000                 | 5934                   | 13                       | 0.00042727 | 42.688          |
| Ring | 160  | 2000                 | 5934                   | 13                       | 0.00042727 | 42.6881         |
| 160  | Ring | 2000                 | 5934                   | 13                       | 0.00042727 | 42.6881         |



|      |      |         |      |    |            |         |
|------|------|---------|------|----|------------|---------|
| Ring | 170  | 2000    | 5934 | 13 | 0.00042727 | 42.6881 |
| 170  | Ring | 2000    | 5934 | 13 | 0.00042727 | 42.6881 |
| Ring | Ring | 2000    | 5934 | 13 | 0.00042727 | 42.688  |
| Ring | 180  | 2000    | 5934 | 13 | 0.00042727 | 42.6881 |
| 180  | Ring | 2000    | 5934 | 13 | 0.00042727 | 42.6881 |
| Ring | 190  | 2000    | 5934 | 13 | 0.00042727 | 42.688  |
| 190  | Ring | 2000    | 5934 | 13 | 0.00042727 | 42.688  |
| Ring | 220  | 932.144 | 5934 | 13 | 0.00099148 | 84.8735 |
| 220  | 230  | No Calc | 5965 | 19 | 0.00039816 | 39.7795 |
| 230  | 240  | No Calc | ...  | 97 | No Calc    | No Calc |
| 240  | 250  | No Calc | ...  | 97 | No Calc    | No Calc |

**External Pressure Calculations:**

| From | To   | External Actual T.<br>mm | External Required T.<br>mm | External Design Pressure<br>MPa | External M.A.W.P.<br>MPa |
|------|------|--------------------------|----------------------------|---------------------------------|--------------------------|
| 10   | 20   | ...                      | No Calc                    | ...                             | No Calc                  |
| 20   | 30   | ...                      | No Calc                    | ...                             | No Calc                  |
| 30   | 40   | ...                      | No Calc                    | ...                             | No Calc                  |
| 40   | 50   | ...                      | No Calc                    | ...                             | No Calc                  |
| 50   | 60   | ...                      | No Calc                    | ...                             | No Calc                  |
| 60   | 70   | ...                      | No Calc                    | ...                             | No Calc                  |
| 70   | 80   | ...                      | No Calc                    | ...                             | No Calc                  |
| 80   | 90   | ...                      | No Calc                    | ...                             | No Calc                  |
| 90   | 100  | ...                      | No Calc                    | ...                             | No Calc                  |
| 100  | 110  | 22                       | 18.1871                    | 0.10346                         | 0.15448                  |
| 110  | Ring | 16                       | 11.1976                    | 0.10346                         | 0.26584                  |
| Ring | Ring | 16                       | 13.1185                    | 0.10346                         | 0.17755                  |
| Ring | 120  | 16                       | 13.1185                    | 0.10346                         | 0.17755                  |
| 120  | Ring | 16                       | 13.1185                    | 0.10346                         | 0.17755                  |
| Ring | 130  | 16                       | 13.1185                    | 0.10346                         | 0.17755                  |
| 130  | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | 140  | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| 140  | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | 150  | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| 150  | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | 160  | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| 160  | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | 170  | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| 170  | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | 180  | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| 180  | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | 190  | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| 190  | Ring | 14                       | 13.1136                    | 0.10346                         | 0.12469                  |
| Ring | 220  | 14                       | 9.68541                    | 0.10346                         | 0.24792                  |
| 220  | 230  | 20                       | 18.1699                    | 0.10346                         | 0.12671                  |
| 230  | 240  | 98                       | 81.9658                    | 0.10346                         | No Calc                  |
| 240  | 250  | 98                       | 81.9658                    | 0.10346                         | No Calc                  |

Minimum

0

**External Pressure Calculations:**

| From | To | Actual Length<br>Bet. Stiffeners<br>mm | Allowable Length<br>Bet. Stiffeners<br>mm | Ring Inertia<br>Required<br>mm^4 | Ring Inertia<br>Available<br>mm^4 |
|------|----|----------------------------------------|-------------------------------------------|----------------------------------|-----------------------------------|
| 10   | 20 | 22000                                  | No Calc                                   | No Calc                          | No Calc                           |
| 20   | 30 | 22000                                  | No Calc                                   | No Calc                          | No Calc                           |
| 30   | 40 | 22000                                  | No Calc                                   | No Calc                          | No Calc                           |
| 40   | 50 | 22000                                  | No Calc                                   | No Calc                          | No Calc                           |
| 50   | 60 | 22000                                  | No Calc                                   | No Calc                          | No Calc                           |
| 60   | 70 | 22000                                  | No Calc                                   | No Calc                          | No Calc                           |
| 70   | 80 | 22000                                  | No Calc                                   | No Calc                          | No Calc                           |



|      |      |         |         |          |          |
|------|------|---------|---------|----------|----------|
| 80   | 90   | 22000   | No Calc | No Calc  | No Calc  |
| 90   | 100  | 22000   | No Calc | No Calc  | No Calc  |
| 100  | 110  | No Calc | No Calc | No Calc  | No Calc  |
| 110  | Ring | 1332.02 | 3317.59 | No Calc  | No Calc  |
| Ring | Ring | 2000    | 3317.34 | 24735462 | 59346300 |
| Ring | 120  | 2000    | 3317.34 | 29783172 | 59346300 |
| 120  | Ring | 2000    | 3317.34 | No Calc  | No Calc  |
| Ring | 130  | 2000    | 3317.34 | 29783164 | 59346300 |
| 130  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | Ring | 2000    | 2373.49 | 29630336 | 56163728 |
| Ring | 140  | 2000    | 2373.49 | 29630336 | 56163728 |
| 140  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | 150  | 2000    | 2373.49 | 29630336 | 56163728 |
| 150  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | Ring | 2000    | 2373.49 | 29630336 | 56163728 |
| Ring | 160  | 2000    | 2373.49 | 29630336 | 56163728 |
| 160  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | 170  | 2000    | 2373.49 | 29630336 | 56163728 |
| 170  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | Ring | 2000    | 2373.49 | 29630336 | 56163728 |
| Ring | 180  | 2000    | 2373.49 | 29630336 | 56163728 |
| 180  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | 190  | 2000    | 2373.49 | 29630336 | 56163728 |
| 190  | Ring | 2000    | 2373.49 | No Calc  | No Calc  |
| Ring | 220  | 932.144 | 2374.16 | 21727174 | 56163728 |
| 220  | 230  | No Calc | No Calc | No Calc  | No Calc  |
| 230  | 240  | No Calc | No Calc | No Calc  | No Calc  |
| 240  | 250  | No Calc | No Calc | No Calc  | No Calc  |

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

| From  | To  | Element<br>Metal Wgt.<br>kgm | Element<br>ID Volume<br>ltr | Corroded<br>Metal Wgt.<br>kgm | Corroded<br>ID Volume<br>ltr | Extra due<br>Misc %<br>kgm |
|-------|-----|------------------------------|-----------------------------|-------------------------------|------------------------------|----------------------------|
| 10    | 20  | 17224.2                      | ...                         | 17224.2                       | ...                          | ...                        |
| 20    | 30  | 9508.83                      | ...                         | 9508.83                       | ...                          | ...                        |
| 30    | 40  | 9508.83                      | ...                         | 9508.83                       | ...                          | ...                        |
| 40    | 50  | 8644.29                      | ...                         | 8644.29                       | ...                          | ...                        |
| 50    | 60  | 8644.29                      | ...                         | 8644.29                       | ...                          | ...                        |
| 60    | 70  | 7779.94                      | ...                         | 7779.94                       | ...                          | ...                        |
| 70    | 80  | 7779.94                      | ...                         | 7779.94                       | ...                          | ...                        |
| 80    | 90  | 2305.13                      | ...                         | 2305.13                       | ...                          | ...                        |
| 90    | 100 | 2305.13                      | ...                         | 2305.13                       | ...                          | ...                        |
| 100   | 110 | 6808.03                      | 18028.6                     | 6548.98                       | 18062                        | ...                        |
| 110   | 120 | 6921.3                       | 82200.8                     | 6489.79                       | 82256.4                      | ...                        |
| 120   | 130 | 6921.3                       | 82200.8                     | 6489.79                       | 82256.4                      | ...                        |
| 130   | 140 | 6054.09                      | 82200.8                     | 5622.58                       | 82256.4                      | ...                        |
| 140   | 150 | 6054.09                      | 82200.8                     | 5622.58                       | 82256.4                      | ...                        |
| 150   | 160 | 6054.09                      | 82200.8                     | 5622.58                       | 82256.4                      | ...                        |
| 160   | 170 | 6054.09                      | 82200.8                     | 5622.58                       | 82256.4                      | ...                        |
| 170   | 180 | 6054.09                      | 82200.8                     | 5622.58                       | 82256.4                      | ...                        |
| 180   | 190 | 6054.09                      | 82200.8                     | 5622.58                       | 82256.4                      | ...                        |
| 190   | 220 | 3027.05                      | 41100.4                     | 2811.29                       | 41128.2                      | ...                        |
| 220   | 230 | 6278.7                       | 18031.4                     | 6019.84                       | 18064.8                      | ...                        |
| 230   | 240 | 386.795                      | 288.523                     | 381.006                       | 288.766                      | ...                        |
| 240   | 250 | 386.795                      | 288.523                     | 381.006                       | 288.766                      | ...                        |
| Total |     | 140755                       | 735343.44                   | 136557                        | 735884.12                    | 0                          |

**Weight of Details:**

| From | Type | Weight of<br>Detail<br>kgm | X Offset,<br>Dtl. Cent.<br>mm | Y Offset,<br>Dtl. Cent.<br>mm | Z Offset<br>Dtl. Cent.<br>mm | Description    |
|------|------|----------------------------|-------------------------------|-------------------------------|------------------------------|----------------|
| 100  | Liqd | 15880.8                    | ...                           | -489.937                      | ...                          | Liquid 10      |
| 100  | Insl | 1436.04                    | ...                           | -472.152                      | ...                          | insu           |
| 100  | Lini | 804.269                    | ...                           | -464.194                      | ...                          | Clad           |
| 100  | Nozl | 2643.36                    | ...                           | -1235.06                      | ...                          | N1 36"         |
| 100  | Nozl | 43.8743                    | -650                          | -3732.36                      | ...                          | N11 2"         |
| 110  | Liqd | 80942.6                    | ...                           | 1500                          | ...                          | Liquid 20      |
| 110  | Insl | 2005.46                    | ...                           | 1500                          | ...                          | insu           |
| 110  | Lini | 1341.41                    | ...                           | 1500                          | ...                          | Clad           |
| 110  | Ring | 594.955                    | ...                           | 950                           | ...                          | SR 1 Fr 70     |
| 110  | Ring | 594.955                    | ...                           | 2950                          | ...                          | SR 2 Fr 70     |
| 110  | Nozl | 7.92164                    | 3057.77                       | 150.002                       | ...                          | K1a            |
| 110  | Nozl | 7.92164                    | ...                           | 150.002                       | -3057.77                     | K2a            |
| 110  | Nozl | 7.92164                    | 3057.77                       | 150.002                       | ...                          | K3a            |
| 110  | Nozl | 7.92164                    | 3057.77                       | 150.002                       | ...                          | K4a            |
| 110  | Nozl | 7.92164                    | 2162.17                       | 150.002                       | 2162.17                      | N9             |
| 110  | Nozl | 9.75427                    | -2504.78                      | 400                           | 1753.87                      | N8 2"          |
| 120  | Pack | 2936.98                    | ...                           | 2710                          | ...                          | Pack Bed#1-(1) |
| 120  | Liqd | 27304.1                    | ...                           | 506                           | ...                          | Btm Liq-2      |
| 120  | Insl | 2005.46                    | ...                           | 1500                          | ...                          | insu           |
| 120  | Lini | 1341.41                    | ...                           | 1500                          | ...                          | Clad           |
| 120  | Ring | 594.955                    | ...                           | 1950                          | ...                          | SR 3 Fr 80     |
| 120  | Nozl | 7.92164                    | 3057.77                       | 2950                          | ...                          | K1b            |
| 120  | Nozl | 7.92164                    | 3057.77                       | 1300                          | ...                          | K2b            |
| 120  | Nozl | 7.92164                    | 3057.77                       | 1300                          | ...                          | K3b            |
| 120  | Nozl | 7.92164                    | 3057.77                       | 1600                          | ...                          | K4b            |
| 120  | Nozl | 929.588                    | 2658.32                       | 950.002                       | 1661.1                       | N2a            |
| 120  | Nozl | 929.588                    | 2658.32                       | 950.002                       | -1661.1                      | N2b            |
| 120  | Nozl | 376.075                    | -3153                         | 2570                          | ...                          | N3A            |
| 120  | Nozl | 280.233                    | -1467.94                      | 950.002                       | 2760.8                       | N4a            |
| 120  | Nozl | 280.233                    | 542.963                       | 950.002                       | 3079.3                       | N4b            |
| 120  | Nozl | 683.253                    | -1576.5                       | 2600                          | -2730.58                     | A1_30"         |
| 120  | Nozl | 291.934                    | -1576.5                       | 2570                          | 2730.58                      | N3B            |



|     |      |         |          |           |          |                 |
|-----|------|---------|----------|-----------|----------|-----------------|
| 120 | Wght | 1450    | ...      | 1500      | 1900     | N2A SCHOPENTOET |
| 120 | Wght | 1450    | ...      | 1500      | -1900    | N2B SCHOPENTOET |
| 120 | Wght | 1700.03 | ...      | 950.002   | ...      | N2A INSERT PLAT |
| 120 | Wght | 1700.03 | ...      | 950.002   | ...      | N2B INSERT PLAT |
| 120 | Wght | 350.006 | ...      | 2570      | ...      | N3 INSERT PLAT  |
| 120 | Wght | 4450    | ...      | 1950      | ...      | CT_DEAD WT      |
| 120 | Wght | 10500   | ...      | 1950      | ...      | CT LIQ WT       |
| 120 | Wght | 2000    | ...      | 2420      | ...      | btm Grid-1 DEAD |
| 120 | Wght | 8411    | -1383.16 | 950.002   | 4256.93  | N4A_OVH_PP      |
| 120 | Wght | 8411    | 390.109  | 950.002   | 4458.97  | N4B_OVH_PP      |
| 120 | Wght | 350     | ...      | 2570      | ...      | N3B INSERT PLAT |
| 130 | Plat | 3087.45 | ...      | 1950      | 2474.45  | PLT6            |
| 130 | Pack | 15222.2 | ...      | 1500      | ...      | Pack Bed#1-(1)  |
| 130 | Insl | 2004.14 | ...      | 1500      | ...      | insu            |
| 130 | Lini | 1341.41 | ...      | 1500      | ...      | Clad            |
| 130 | Ring | 594.568 | ...      | 950       | ...      | SR 4 Fr 90      |
| 130 | Ring | 594.568 | ...      | 2950      | ...      | SR 5 Fr 90      |
| 130 | Nozl | 7.87951 | 3057.77  | 2720      | ...      | K5a             |
| 130 | Nozl | 7.87951 | 3057.77  | 2720      | ...      | K6a             |
| 130 | Wght | 150.003 | ...      | 1070      | ...      | N4A INSERT PLT  |
| 130 | Wght | 150.003 | ...      | 1070      | ...      | N4B INSERT PLT  |
| 140 | Plat | 3235.44 | ...      | 2950      | -2474.45 | PLT7            |
| 140 | Pack | 4658.68 | ...      | 460.001   | ...      | Pack Bed#1-(1)  |
| 140 | Insl | 2004.14 | ...      | 1500      | ...      | insu            |
| 140 | Lini | 1341.41 | ...      | 1500      | ...      | Clad            |
| 140 | Ring | 594.568 | ...      | 1950      | ...      | SR 6 Fr 100     |
| 140 | Nozl | 7.87951 | 3057.77  | 770.002   | ...      | K5b             |
| 140 | Nozl | 7.87951 | 3057.77  | 770.002   | ...      | K6b             |
| 140 | Nozl | 7.87951 | 3057.77  | 1320      | ...      | K7a             |
| 140 | Nozl | 233.591 | 2730.58  | 1050      | -1576.5  | A2_30"          |
| 140 | Wght | 10000.2 | ...      | 1205      | ...      | 10T Margin      |
| 140 | Wght | 3000    | ...      | 2000      | ...      | Vane1 DEAD WT   |
| 140 | Wght | 3500    | ...      | 2000      | ...      | Vane1 LIQ WT    |
| 140 | Wght | 2355    | ...      | 920.003   | ...      | dist.1 DEAD W   |
| 140 | Wght | 8100    | ...      | 920.003   | ...      | dist.1 LIQ WT   |
| 140 | Wght | 9500    | ...      | 1205      | ...      | Cleat wt        |
| 140 | Wght | 6000    | ...      | 1205      | ...      | Lifting wt      |
| 140 | Wght | 550     | ...      | 0.0023254 | ...      | Cladded Nozzle  |
| 140 | Wght | 350     | -1.0101  | 1150      | ...      | Manway davit    |
| 150 | Pack | 7671.59 | ...      | 2242.5    | ...      | Pack Bed#2-(2)  |
| 150 | Insl | 2004.14 | ...      | 1500      | ...      | insu            |
| 150 | Lini | 1341.41 | ...      | 1500      | ...      | Clad            |
| 150 | Ring | 594.568 | ...      | 950       | ...      | SR 7 Fr 110     |
| 150 | Ring | 594.568 | ...      | 2950      | ...      | SR 8 Fr 110     |
| 150 | Nozl | 233.591 | 2730.58  | 1650      | -1576.5  | A3_30"          |
| 150 | Wght | 2000    | ...      | 1485      | ...      | btm Grid-2 DEAD |
| 160 | Plat | 3235.44 | ...      | 950.002   | 2474.45  | PLT8            |
| 160 | Pack | 15115.3 | ...      | 1492.5    | ...      | Pack Bed#2-(2)  |
| 160 | Insl | 2004.14 | ...      | 1500      | ...      | insu            |
| 160 | Lini | 1341.41 | ...      | 1500      | ...      | Clad            |
| 160 | Ring | 594.568 | ...      | 1950      | ...      | SR 9 Fr 120     |
| 160 | Wght | 2355    | ...      | 2985      | ...      | dist.2 DEAD W   |
| 160 | Wght | 8100    | ...      | 2985      | ...      | dist.2 LIQ WT   |
| 170 | Plat | 3235.44 | ...      | 1950      | -2474.45 | PLT9            |
| 170 | Insl | 2004.14 | ...      | 1500      | ...      | insu            |
| 170 | Lini | 1341.41 | ...      | 1500      | ...      | Clad            |
| 170 | Ring | 594.568 | ...      | 950       | ...      | SR 10 Fr 130    |
| 170 | Ring | 594.568 | ...      | 2950      | ...      | SR 11 Fr 130    |
| 170 | Nozl | 5.1683  | -2622.33 | 485.004   | -1514    | K7b             |
| 170 | Nozl | 666.489 | -1576.5  | 1520      | -2730.58 | A4_30"          |
| 170 | Wght | 600     | ...      | 1850      | ...      | N5 SCHOPENTOETE |
| 170 | Wght | 3000.05 | ...      | 2000      | ...      | Vane2 DEAD WT   |
| 170 | Wght | 3500    | ...      | 2000      | ...      | Vane2 LIQ WT    |
| 180 | Plat | 3235.44 | ...      | 2950      | 2474.45  | PLT10           |
| 180 | Pack | 5063.77 | ...      | 2360      | ...      | Pack Bed#3      |
| 180 | Insl | 2004.14 | ...      | 1500      | ...      | insu            |
| 180 | Lini | 1341.41 | ...      | 1500      | ...      | Clad            |
| 180 | Ring | 594.568 | ...      | 1950      | ...      | SR 12 Fr 140    |
| 180 | Nozl | 422.417 | -3153    | 1000      | ...      | N5              |
| 180 | Nozl | 5.1683  | ...      | 1200      | 3028.01  | K8a             |



|     |      |         |          |           |          |
|-----|------|---------|----------|-----------|----------|
| 180 | Nozl | 875.062 | 2229.51  | 1300      | -2229.51 |
| 180 | Wght | 2000    | ...      | 1860      | ...      |
| 180 | Wght | 2355    | ...      | 2860      | ...      |
| 180 | Wght | 8100    | ...      | 2860      | ...      |
| 180 | Wght | 830     | ...      | 50.0016   | ...      |
| 180 | Wght | 46997   | -4755    | 1000      | ...      |
| 190 | Insl | 1002.07 | ...      | 750       | ...      |
| 190 | Lini | 670.703 | ...      | 750       | ...      |
| 190 | Ring | 594.568 | ...      | 950       | ...      |
| 190 | Nozl | 5.1683  | -2622.33 | 1270      | -1514    |
| 190 | Nozl | 23.6511 | 1514     | 1260      | 2622.33  |
| 190 | Nozl | 5.1683  | ...      | 660.004   | 3028.01  |
| 190 | Nozl | 5.1683  | -1946.36 | 0.0046512 | 2319.59  |
| 190 | Nozl | 262.571 | 2229.51  | 150.005   | -2229.51 |
| 190 | Nozl | 904.042 | 3153     | 500       | ...      |
| 190 | Wght | 1300.02 | 3000     | 0.0046509 | ...      |
| 190 | Wght | 500     | ...      | 1260      | ...      |
| 220 | Plat | 24367.4 | 46.0003  | 1500      | ...      |
| 220 | Insl | 1456.35 | ...      | 522.245   | ...      |
| 220 | Lini | 804.269 | ...      | 514.195   | ...      |
| 220 | Nozl | 357.048 | ...      | 1236.43   | ...      |
| 220 | Nozl | 6.18885 | -1500    | 944.086   | ...      |
| 220 | Wght | 9172    | 4552     | ...       | ...      |
| 220 | Wght | 11000.2 | -4175.47 | 1020      | 4175.47  |
| 220 | Forc | ...     | ...      | 1044      | ...      |
| 230 | Insl | 20.1117 | ...      | 74        | ...      |
| 240 | Insl | 20.1117 | ...      | 71        | ...      |
| 120 | Pliq | 1569.18 | ...      | 2710.000  | ...      |
| 130 | Pliq | 8116.48 | ...      | 1500.001  | ...      |
| 140 | Pliq | 2489.06 | ...      | 460.001   | ...      |
| 150 | Pliq | 4098.81 | ...      | 2242.500  | ...      |
| 160 | Pliq | 8075.90 | ...      | 1492.505  | ...      |
| 180 | Pliq | 4058.24 | ...      | 2360.003  | ...      |

A5\_30"  
 btm Grid-3 DEAD  
 dist.3 DEAD W  
 dist. 3 LIQ WT  
 N5 INSERT PLT  
 N5\_OVH\_PP  
 insu  
 Clad  
 SR 13 Fr 150  
 K9a  
 N6  
 K8b  
 K9b  
 A6\_30"  
 A7\_30"  
 Davit  
 N6 INT PIPE WT  
 TOP PLT  
 insu  
 Clad  
 N7  
 N10  
 N7\_OVH\_PP  
 2 PSVS  
 FM at N7  
 insu  
 insu

#### Total Weight of Each Detail Type:

|                    |          |
|--------------------|----------|
| Platforms          | 40396.6  |
| Packing            | 50668.6  |
| Packing Liquid     | 28407.7  |
| Liquid             | 124127.5 |
| Insulation         | 19970.4  |
| Cladding & Overlay | 13010.5  |
| Stiffeners         | 7730.5   |
| Nozzles            | 10589.1  |
| Weights            | 186236.5 |

Sum of the Detail Weights 481137.4 kgm

#### Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

#### Weight Summation Results: (kgm)

|               | Fabricated | Shop Test | Shipping | Erected  | Empty    | Operating |
|---------------|------------|-----------|----------|----------|----------|-----------|
| Main Elements | 139981.5   | 139981.5  | 139981.5 | 139981.5 | 139981.5 | 139981.5  |
| Cladding      | 13010.5    | 13010.5   | 13010.5  | 13010.5  | 13010.5  | 13010.5   |
| Stif. Rings   | 7730.5     | 7730.5    | 7730.5   | 7730.5   | 7730.5   | 7730.5    |
| Nozzles       | 10589.1    | 10589.1   | 10589.1  | 10589.1  | 10589.1  | 10589.1   |
| Wld Weights   | 33430.3    | 33430.3   | 33430.3  | 33430.3  | 33430.3  | 33430.3   |
| Platforms     | ...        | ...       | ...      | 40396.6  | 40396.6  | 40396.6   |
| Packing       | ...        | ...       | ...      | ...      | 50668.6  | 50668.6   |
| Insulation    | ...        | ...       | 19970.4  | 19970.4  | 19970.4  | 19970.4   |
| Empty Weights | ...        | ...       | ...      | 20172.2  | 20172.2  | ...       |
| Test Weights  | ...        | 20172.2   | ...      | ...      | ...      | ...       |
| Ope Weights   | ...        | ...       | ...      | ...      | ...      | 152806.2  |
| Ope. Liquid   | ...        | ...       | ...      | ...      | ...      | 124127.5  |
| Packing Liq   | ...        | ...       | ...      | ...      | ...      | 28407.7   |

\*



|             |          |          |          |          |          |          |
|-------------|----------|----------|----------|----------|----------|----------|
| Test Liquid | ...      | 733276.4 | ...      | ...      | ...      | ...      |
| Totals      | 204742.0 | 958190.5 | 224712.4 | 285281.2 | 335949.8 | 621118.9 |

#### Field Installation Options:

Platforms installed before lifting.

\* Packing installed after lifting.

#### Weight Summary:

|                 |                                                     |               |
|-----------------|-----------------------------------------------------|---------------|
| Fabricated Wt.  | - Bare weight without removable internals           | 204742.0 kgm  |
| Shop Test Wt.   | - Fabricated weight + water ( full )                | 958190.5 kgm  |
| Shipping Wt.    | - Fab. weight + removable intls.+ shipping app.     | 224712.4 kgm  |
| Erected Wt.     | - Fab. wt + or - loose items (trays,platforms etc.) | 285281.2 kgm  |
| Ope. Wt. no Liq | - Fab. weight + internals. + details + weights      | 335949.8 kgm  |
| Operating Wt.   | - Empty weight + operating liq. uncorroded          | 621118.9 kgm  |
| Field Test Wt.  | - Empty weight + test medium (full)                 | 1018557.5 kgm |

#### Weights used in Vortex Shedding Calculations:

|                                                               |              |
|---------------------------------------------------------------|--------------|
| Mass of the Upper 1/3 of the Vertical Vessel (operating)      | 292258.2 kgm |
| Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) | 628446.9 kgm |
| Mass of the Upper 1/3 of the Vertical Vessel (empty)          | 181956.9 kgm |

#### Surface Areas of Elements:

| From | To  | Outside Surface Area<br>mm <sup>2</sup> | Inside Surface Area<br>mm <sup>2</sup> |
|------|-----|-----------------------------------------|----------------------------------------|
| 10   | 20  | 37403804                                | 36951412                               |
| 20   | 30  | 55973756                                | 55559064                               |
| 30   | 40  | 55973756                                | 55559064                               |
| 40   | 50  | 55954904                                | 55577916                               |
| 50   | 60  | 55954904                                | 55577916                               |
| 60   | 70  | 55936060                                | 55596760                               |
| 70   | 80  | 55936060                                | 55596760                               |
| 80   | 90  | 18639070                                | 18538540                               |
| 90   | 100 | 18639070                                | 18538540                               |
| 100  | 110 | 34185060                                | 33385778                               |
| 110  | 120 | 55964332                                | 55662740                               |
| 120  | 130 | 55964332                                | 55662740                               |
| 130  | 140 | 55926628                                | 55662740                               |
| 140  | 150 | 55926628                                | 55662740                               |
| 150  | 160 | 55926628                                | 55662740                               |
| 160  | 170 | 55926628                                | 55662740                               |
| 170  | 180 | 55926628                                | 55662740                               |
| 180  | 190 | 55926628                                | 55662740                               |
| 190  | 220 | 27963314                                | 27831370                               |
| 220  | 230 | 34123380                                | 33385890                               |
| 230  | 240 | 1246090                                 | 746512                                 |
| 240  | 250 | 1246090                                 | 746512                                 |

|       |             |                             |
|-------|-------------|-----------------------------|
| Total | 956663936.0 | 948893952.0 mm <sup>2</sup> |
| Total | 956.7       | 948.9 m <sup>2</sup>        |

#### Element and Detail Weights:

| From | To | Total Ele.<br>Empty Wgt.<br>kgm | Total. Ele.<br>Oper. Wgt.<br>kgm | Total. Ele.<br>Hydro. Wgt.<br>kgm | Total Dtl.<br>Offset Mom.<br>N-mm | Oper. Wgt.<br>No Liquid<br>kgm |
|------|----|---------------------------------|----------------------------------|-----------------------------------|-----------------------------------|--------------------------------|
| 10   | 20 | 17224.2                         | 17224.2                          | 17224.2                           | ...                               | 17224.2                        |
| 20   | 30 | 9508.83                         | 9508.83                          | 9508.83                           | ...                               | 9508.83                        |
| 30   | 40 | 9508.83                         | 9508.83                          | 9508.83                           | ...                               | 9508.83                        |
| 40   | 50 | 8644.29                         | 8644.29                          | 8644.29                           | ...                               | 8644.29                        |
| 50   | 60 | 8644.29                         | 8644.29                          | 8644.29                           | ...                               | 8644.29                        |
| 60   | 70 | 7779.94                         | 7779.94                          | 7779.94                           | ...                               | 7779.94                        |
| 70   | 80 | 7779.94                         | 7779.94                          | 7779.94                           | ...                               | 7779.94                        |
| 80   | 90 | 2305.13                         | 2305.13                          | 2305.13                           | ...                               | 2305.13                        |



|     |     |         |         |         |           |         |
|-----|-----|---------|---------|---------|-----------|---------|
| 90  | 100 | 2305.13 | 2305.13 | 2305.13 | ...       | 2305.13 |
| 100 | 110 | 11735.6 | 27616.4 | 29753.2 | 279764    | 11735.6 |
| 110 | 120 | 11507.4 | 92450   | 93658   | 641213    | 11507.4 |
| 120 | 130 | 18765.8 | 87248   | 100916  | 61760912  | 58374.8 |
| 130 | 140 | 10904.5 | 37330.7 | 96142.6 | 472718    | 29214.2 |
| 140 | 150 | 36651.6 | 63989.8 | 122038  | 6962784   | 61500.7 |
| 150 | 160 | 10822.4 | 24592.8 | 92972.9 | 6257175   | 20493.9 |
| 160 | 170 | 9994.2  | 46875.9 | 95380.2 | ...       | 38800   |
| 170 | 180 | 11260.4 | 21595.9 | 96646.5 | 10440486  | 21595.9 |
| 180 | 190 | 12126.8 | 83936.3 | 97512.9 | 2.186E+09 | 79878.1 |
| 190 | 220 | 8300.17 | 8300.17 | 49375.5 | 72084712  | 8300.17 |
| 220 | 230 | 29115   | 53482.4 | 71502.8 | 30100790  | 53482.4 |

Empty Support Force + the Sum of the Y forces 2184647.25 N  
 Operating Support Force + the Sum of the Y forces 6152565.50 N  
 Hydro Support Force + the Sum of the Y forces 10060060.00 N

#### Cumulative Vessel Weight

| From | To  | Cumulative Ope<br>Wgt. No Liquid<br>kgm | Cumulative<br>Oper. Wgt.<br>kgm | Cumulative<br>Hydro. Wgt.<br>kgm |
|------|-----|-----------------------------------------|---------------------------------|----------------------------------|
| 10   | 20  | 468584                                  | 621119                          | 1019599                          |
| 20   | 30  | 451360                                  | 603895                          | 1002375                          |
| 30   | 40  | 441851                                  | 594386                          | 992866                           |
| 40   | 50  | 432342                                  | 584877                          | 983357                           |
| 50   | 60  | 423698                                  | 576233                          | 974713                           |
| 60   | 70  | 415053                                  | 567588                          | 966069                           |
| 70   | 80  | 407273                                  | 559808                          | 958289                           |
| 80   | 90  | 399493                                  | 552029                          | 950509                           |
| 90   | 100 | 397188                                  | 549723                          | 948204                           |
| 100  | 110 | 394883                                  | 547418                          | 945899                           |
| 110  | 120 | 383148                                  | 519802                          | 916145                           |
| 120  | 130 | 371640                                  | 427352                          | 822487                           |
| 130  | 140 | 313265                                  | 340104                          | 721571                           |
| 140  | 150 | 284051                                  | 302773                          | 625428                           |
| 150  | 160 | 222550                                  | 238783                          | 503391                           |
| 160  | 170 | 202057                                  | 214191                          | 410418                           |
| 170  | 180 | 163257                                  | 167315                          | 315038                           |
| 180  | 190 | 141661                                  | 145719                          | 218391                           |
| 190  | 220 | 61782.6                                 | 61782.6                         | 120878                           |
| 220  | 230 | 53482.4                                 | 53482.4                         | 71502.8                          |

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

#### Cumulative Vessel Moment

| From | To  | Cumulative<br>Empty Mom.<br>N-mm | Cumulative<br>Oper. Mom.<br>N-mm | Cumulative<br>Hydro. Mom.<br>N-mm |
|------|-----|----------------------------------|----------------------------------|-----------------------------------|
| 10   | 20  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 20   | 30  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 30   | 40  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 40   | 50  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 50   | 60  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 60   | 70  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 70   | 80  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 80   | 90  | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 90   | 100 | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 100  | 110 | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 110  | 120 | 2.375E+09                        | 2.375E+09                        | 2.375E+09                         |
| 120  | 130 | 2.374E+09                        | 2.374E+09                        | 2.374E+09                         |
| 130  | 140 | 2.312E+09                        | 2.312E+09                        | 2.312E+09                         |
| 140  | 150 | 2.312E+09                        | 2.312E+09                        | 2.312E+09                         |
| 150  | 160 | 2.305E+09                        | 2.305E+09                        | 2.305E+09                         |



|     |     |           |           |           |
|-----|-----|-----------|-----------|-----------|
| 160 | 170 | 2.299E+09 | 2.299E+09 | 2.299E+09 |
| 170 | 180 | 2.299E+09 | 2.299E+09 | 2.299E+09 |
| 180 | 190 | 2.288E+09 | 2.288E+09 | 2.288E+09 |
| 190 | 220 | 102185496 | 102185496 | 102185496 |
| 220 | 230 | 30100790  | 30100790  | 30100790  |

### Nozzle Flange MAWP Results: (MPa & °C)

| Nozzle Description | Flange Ope. | Rating Ambient             | Des. Temp | Class | Grade/Group | Equiv. Press | - - - UG-44 | Max Pressure 50% | DNV   |
|--------------------|-------------|----------------------------|-----------|-------|-------------|--------------|-------------|------------------|-------|
| N1 36"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | 1.055        | 1.474       | 1.580            | 1.316 |
| N11 2"             | 9.01        | 10.20                      | 150       | 600   | GR 1.1      | 6.129        | 7.387       | 9.011            | 7.387 |
| K1a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K2a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K3a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K4a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N9                 | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N8 2"              | 9.01        | 10.20                      | 150       | 600   | GR 1.1      | 6.129        | 7.387       | 9.011            | 7.387 |
| K1b                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K2b                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K3b                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K4b                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N3A                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N4a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N4b                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| A1 30"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N3B                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | 1.184        | 1.344       | 1.580            | 1.186 |
| K5a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K6a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K5b                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K6b                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K7a                | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| A2 30"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| A3 30"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K7b                | 4.51        | 5.10                       | 150       | 300   | GR 1.1      | ...          | ...         | ...              | ...   |
| A4 30"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N5                 | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K8a                | 4.51        | 5.10                       | 150       | 300   | GR 1.1      | ...          | ...         | ...              | ...   |
| A5 30"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| K9a                | 4.51        | 5.10                       | 150       | 300   | GR 1.1      | ...          | ...         | ...              | ...   |
| N6                 | 4.51        | 5.10                       | 150       | 300   | GR 1.1      | 4.321        | 2.444       | 4.510            | 2.444 |
| K8b                | 4.51        | 5.10                       | 150       | 300   | GR 1.1      | ...          | ...         | ...              | ...   |
| K9b                | 4.51        | 5.10                       | 150       | 300   | GR 1.1      | ...          | ...         | ...              | ...   |
| A6 30"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| A7 30"             | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | ...          | ...         | ...              | ...   |
| N7                 | 1.58        | 1.96                       | 150       | 150   | GR 1.1      | 1.418        | 1.110       | 1.580            | 0.952 |
| N10                | 9.01        | 10.20                      | 150       | 600   | GR 1.1      | 8.575        | 4.941       | 0.435            | 4.940 |
| Min Rating         | 1.110       | 1.960, [for Core Elements] |           |       |             |              | ...         | ...              | ...   |

### Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

| Nozzle Description | Gasket Dimension G mm | Fm Value | Resolved Moment ME N-mm | Axial Force FE N | MAWP PD MPa | Maximum Pressure Pmax MPa | Pass Fail |
|--------------------|-----------------------|----------|-------------------------|------------------|-------------|---------------------------|-----------|
| N1 36"             | 947.801               | 0.60     | 160371776               | 67500            | 0.498       | 1.474                     | Pass      |
| N11 2"             | 77.851                | 0.50     | 494975                  | 3750             | 0.498       | 7.387                     | Pass      |
| N8 2"              | 77.851                | 0.50     | 494975                  | 3750             | 0.498       | 7.387                     | Pass      |
| N3B                | 843.026               | 0.60     | 126713584               | 60000            | 0.498       | 1.344                     | Pass      |
| N6                 | 138.176               | 0.50     | 1979899                 | 7500             | 0.498       | 2.444                     | Pass      |
| N7                 | 790.575               | 0.60     | 126501472               | 56250            | 0.498       | 1.110                     | Pass      |
| N10                | 77.851                | 0.50     | 707107                  | 4500             | 0.498       | 4.941                     | Pass      |



Evaluating Flange on Nozzle [N1 36"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(160371776) + 4 \cdot \text{abs}(67500)947.801 \leq \pi \cdot 947.801^3 [(1.580 - 0.498) + .600 \cdot 1.580] \\ &= 0.28220E+10 \leq 0.54315E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N1 36" [Pmax]:

$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 1.580, 0.600 \cdot 1.580 + 1.580 - 2821959168.0 / (\pi \cdot 947.801^3) ) \\ &= \min( 1.580, 1.474 ) \\ &= 1.474 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N11 2"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(494975) + 4 \cdot \text{abs}(3750)77.851 \leq \pi \cdot 77.851^3 [(9.011 - 0.498) + .500 \cdot 9.011] \\ &= 0.90878E+07 \leq 0.19302E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N11 2" [Pmax]:

$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 9.011, 0.500 \cdot 9.011 + 9.011 - 9087832.0 / (\pi \cdot 77.851^3) ) \\ &= \min( 9.011, 7.387 ) \\ &= 7.387 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N8 2"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(494975) + 4 \cdot \text{abs}(3750)77.851 \leq \pi \cdot 77.851^3 [(9.011 - 0.498) + .500 \cdot 9.011] \\ &= 0.90878E+07 \leq 0.19302E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N8 2" [Pmax]:

$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 9.011, 0.500 \cdot 9.011 + 9.011 - 9087832.0 / (\pi \cdot 77.851^3) ) \\ &= \min( 9.011, 7.387 ) \\ &= 7.387 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N3B]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(126713584) + 4 \cdot \text{abs}(60000)843.026 \leq \pi \cdot 843.026^3 [(1.580 - 0.498) + .600 \cdot 1.580] \\ &= 0.22298E+10 \leq 0.38220E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3B [Pmax]:

$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 1.580, 0.600 \cdot 1.580 + 1.580 - 2229825536.0 / (\pi \cdot 843.026^3) ) \\ &= \min( 1.580, 1.344 ) \\ &= 1.344 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N6]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1979899) + 4 \cdot \text{abs}(7500)138.176 \leq \pi \cdot 138.176^3 [(4.510 - 0.498) + .500 \cdot 4.510] \\ &= 0.35825E+08 \leq 0.51959E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N6 [Pmax]:

$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 4.510, 0.500 \cdot 4.510 + 4.510 - 35825308.0 / (\pi \cdot 138.176^3) ) \\ &= \min( 4.510, 2.444 ) \\ &= 2.444 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N7]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(126501472) + 4 \cdot \text{abs}(56250)790.575 \leq \pi \cdot 790.575^3 [(1.580 - 0.498) + .600 \cdot 1.580] \\ &= 0.22020E+10 \leq 0.31521E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N7 [Pmax]:

$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 1.580, 0.600 \cdot 1.580 + 1.580 - 2201974784.0 / (\pi \cdot 790.575^3) ) \\ &= \min( 1.580, 1.110 ) \\ &= 1.110 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N10]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$





$$= 16(707107) + 4 \cdot \text{abs}(4500) 77.851 \leq \pi \cdot 77.851^3 [(9.011 - 0.498) + .500 \cdot 9.011]$$
$$= 0.12716\text{E}+08 \leq 0.19302\text{E}+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N10 [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) )$$
$$= \min( 9.011, 0.500 \cdot 9.011 + 9.011 - 12715539.0 / (\pi \cdot 77.851^3) )$$
$$= \min( 9.011, 4.941 )$$
$$= 4.941 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.02966 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.97335 Hz.

The Natural Frequency for the Vessel (Filled) is 0.72166 Hz.

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### Forces/Moments Applied to Vessel (Combined w/Wind Loads)

| From | To  | X and Z Dir<br>Force Res.<br>N | X,Z Moment<br>and For Res<br>N-mm |
|------|-----|--------------------------------|-----------------------------------|
| 10   | 20  | ...                            | ...                               |
| 20   | 30  | ...                            | ...                               |
| 30   | 40  | ...                            | ...                               |
| 40   | 50  | ...                            | ...                               |
| 50   | 60  | ...                            | ...                               |
| 60   | 70  | ...                            | ...                               |
| 70   | 80  | ...                            | ...                               |
| 80   | 90  | ...                            | ...                               |
| 90   | 100 | ...                            | ...                               |
| 100  | 110 | ...                            | ...                               |
| 110  | 120 | ...                            | ...                               |
| 120  | 130 | ...                            | ...                               |
| 130  | 140 | ...                            | ...                               |
| 140  | 150 | ...                            | ...                               |
| 150  | 160 | ...                            | ...                               |
| 160  | 170 | ...                            | ...                               |
| 170  | 180 | ...                            | ...                               |
| 180  | 190 | ...                            | ...                               |
| 190  | 220 | ...                            | ...                               |
| 220  | 230 | 58336.3                        | 183434048                         |
| 230  | 240 | ...                            | ...                               |
| 240  | 250 | ...                            | ...                               |

### Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

| From | To  | X and Z Dir<br>Force Res.<br>N | X,Z Moment<br>and For Res<br>N-mm |
|------|-----|--------------------------------|-----------------------------------|
| 10   | 20  | ...                            | ...                               |
| 20   | 30  | ...                            | ...                               |
| 30   | 40  | ...                            | ...                               |
| 40   | 50  | ...                            | ...                               |
| 50   | 60  | ...                            | ...                               |
| 60   | 70  | ...                            | ...                               |
| 70   | 80  | ...                            | ...                               |
| 80   | 90  | ...                            | ...                               |
| 90   | 100 | ...                            | ...                               |
| 100  | 110 | ...                            | ...                               |
| 110  | 120 | ...                            | ...                               |
| 120  | 130 | ...                            | ...                               |
| 130  | 140 | ...                            | ...                               |
| 140  | 150 | ...                            | ...                               |
| 150  | 160 | ...                            | ...                               |
| 160  | 170 | ...                            | ...                               |
| 170  | 180 | ...                            | ...                               |
| 180  | 190 | ...                            | ...                               |
| 190  | 220 | ...                            | ...                               |
| 220  | 230 | 58336.3                        | 183434048                         |
| 230  | 240 | ...                            | ...                               |
| 240  | 250 | ...                            | ...                               |

### User Input Forces and Moments:

| From<br>Node | Distance<br>From | - - -<br>Fx | Forces - - -<br>Fy | - - -<br>Fz | - - -<br>Mx | Moments - - -<br>My | - - -<br>Mz |
|--------------|------------------|-------------|--------------------|-------------|-------------|---------------------|-------------|
| 220          | 1044.00          | 41250.      | -61875.            | 41250.      | 0.866E+08   |                     | 0.866E+08   |



## Wind Load Calculation Results:

### Input Values:

|                                         |                   |
|-----------------------------------------|-------------------|
| Wind Design Code                        | ASCE/SEI 7-16     |
| Wind Load Reduction Scale Factor        | 1.100             |
| Basic Wind Speed [V]                    | 97.2 km/hr        |
| Surface Roughness Category              | C: Open Terrain   |
| Type of Surface                         | Moderately Smooth |
| Base Elevation                          | 19000 mm          |
| Percent Wind for Hydrotest              | 50.0              |
| Using User defined Wind Press. Vs Elev. | N                 |
| Height of Hill or Escarpment H or Hh    | 0 mm              |
| Distance Upwind of Crest Lh             | 0 mm              |
| Distance from Crest to the Vessel x     | 0 mm              |
| Type of Terrain ( Hill, Escarpment )    | Flat              |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100            |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000            |
| Damping Factor (Beta) for Wind (Filled) | 0.0000            |
| Site Elevation above Sea Level          | 0 mm              |

### Wind Analysis Results

*The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.*

Dynamic Gust-Effect Factor, Operating Case [Gf]:  

$$= 0.925[(1+1.7 \cdot I_{zbar} \cdot \sqrt{(gQ^2 \cdot Q^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_{zbar})]$$

$$= 0.925[(1+1.7 \cdot 0.168 \cdot \sqrt{(3.4^2 \cdot 0.887^2 + 4.183^2 \cdot 0.442^2)}) / (1+1.7 \cdot 3.4 \cdot 0.168)]$$

$$= 0.943$$

|                                         |          |
|-----------------------------------------|----------|
| Natural Frequency of Vessel (Operating) | 0.973 Hz |
| Natural Frequency of Vessel (Empty)     | 1.030 Hz |
| Natural Frequency of Vessel (Test)      | 0.722 Hz |

|                                    |              |
|------------------------------------|--------------|
| User Entered Importance Factor is  | 1.000        |
| Force Coefficient [Cf]             | 0.606        |
| Structure Height to Diameter ratio | 8.106        |
| Height to top of Structure         | 48716.434 mm |

*This is classified as a flexible structure. Dynamic analysis performed.*

### Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of  $[\alpha]$  and  $[Zg]$ :  
 Exposure Category: C from Table 26.11-1  
 $\alpha = 9.5$ :  $Zg = 274320.0$  mm

Effective Height  $[z]$ :  
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$   
 $= 1000.0 + 19000.0 = 20000.0$  mm  
 $= 65.617$  ft. Imperial Units

Velocity Pressure coefficient evaluated at height  $z$   $[Kz]$ :  
 Because  $z$  (65.617 ft.)  $> 15$  ft.  
 $= 2.01(z / Zg)^{2/\alpha}$   
 $= 2.01(65.617/274320.0)^{2/9.5}$   
 $= 1.158$

Type of Hill: No Hill

Wind Directionality Factor  $[Kd]$ :  
 $= 1$ , per Table 26.6-1

As there is No Hill Present:  $[Kzt]$ :



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.158 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.943 \cdot 0.606 \cdot 156.349$$

$$= 1430.3 \text{ lbs. } [6362.1] \text{ N}$$

| Element      | Hgt (z)<br>mm | K1    | K2    | K3    | Kz    | Kzt   | qz<br>kPa |
|--------------|---------------|-------|-------|-------|-------|-------|-----------|
| Skirt-9      | 20000.0       | 0.000 | 0.000 | 0.000 | 1.158 | 1.000 | 0.766     |
| Skirt-8      | 22500.0       | 0.000 | 0.000 | 0.000 | 1.187 | 1.000 | 0.766     |
| Skirt-7      | 25500.0       | 0.000 | 0.000 | 0.000 | 1.219 | 1.000 | 0.766     |
| Skirt-6      | 28500.0       | 0.000 | 0.000 | 0.000 | 1.248 | 1.000 | 0.766     |
| Skirt-5      | 31500.0       | 0.000 | 0.000 | 0.000 | 1.274 | 1.000 | 0.766     |
| Skirt-4      | 34500.0       | 0.000 | 0.000 | 0.000 | 1.299 | 1.000 | 0.766     |
| Skirt-3      | 37500.0       | 0.000 | 0.000 | 0.000 | 1.322 | 1.000 | 0.766     |
| Skirt-2      | 39500.0       | 0.000 | 0.000 | 0.000 | 1.337 | 1.000 | 0.766     |
| Skirt-1      | 40500.0       | 0.000 | 0.000 | 0.000 | 1.344 | 1.000 | 0.766     |
| Btm Dish End | 41025.0       | 0.000 | 0.000 | 0.000 | 1.347 | 1.000 | 0.766     |
| Shell-1      | 42550.0       | 0.000 | 0.000 | 0.000 | 1.358 | 1.000 | 0.766     |
| Shell-2      | 45550.0       | 0.000 | 0.000 | 0.000 | 1.377 | 1.000 | 0.766     |
| Shell-3      | 48550.0       | 0.000 | 0.000 | 0.000 | 1.396 | 1.000 | 0.766     |
| Shell-4      | 51550.0       | 0.000 | 0.000 | 0.000 | 1.414 | 1.000 | 0.766     |
| Shell-5      | 54550.0       | 0.000 | 0.000 | 0.000 | 1.431 | 1.000 | 0.766     |
| Shell-6      | 57550.0       | 0.000 | 0.000 | 0.000 | 1.447 | 1.000 | 0.766     |
| Shell-7      | 60550.0       | 0.000 | 0.000 | 0.000 | 1.462 | 1.000 | 0.766     |
| Shell-8      | 63550.0       | 0.000 | 0.000 | 0.000 | 1.477 | 1.000 | 0.766     |
| Shell-9      | 65800.0       | 0.000 | 0.000 | 0.000 | 1.488 | 1.000 | 0.766     |
| Top Dish End | 67046.9       | 0.000 | 0.000 | 0.000 | 1.494 | 1.000 | 0.766     |

## Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

### Nomenclature

Cf - Correction factor for natural frequency  
 D - Average outer insulated diameter of vessel, mm  
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable  
 Dr - Average outer insulated of top half of vessel, mm  
 f - Natural frequency of vibration (Hertz)  
 f1 - Natural frequency of bare vessel based on a unit value of  $(D/L^2)(104)$   
 L - Total height of structure mm  
 Lc - Total length of conical section(s) of vessel, mm  
 tb - Uncorroded plate thickness at bottom of vessel, mm  
 V30 - Design Wind Speed provided by user, km/hr  
 Vc - Critical wind velocity, km/hr  
 Vw - Maximum wind speed at top of structure, km/hr  
 W - Total corroded weight of structure, N  
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N  
 Z - Maximum amplitude of vibration at top of vessel, mm  
 D1 - Logarithmic decrement ( taken as 0.03 for Welded Structures )  
 Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)  
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5  
 :: 0.0/47600.0 = 0.0



:: [ D / L<sup>2</sup> ] • 10<sup>4</sup> < 8 (English Units, ft.)  
 :: [ 19.83/156.17<sup>2</sup> ] • 10<sup>4</sup> = 8.131 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:  
 = W / ( L • Dr<sup>2</sup> )  
 = 6057117 / ( 47600.0 • 6017.563<sup>2</sup> )  
 = 0.35141E-05

Compute the damping factor Df which is a measure of instability [Df]:  
 = W • DI / ( L • Dr<sup>2</sup> )  
 = 6057117 • 0.03 / ( 47600.0 • 6017.563<sup>2</sup> )  
 = 0.10542E-06

Compute the critical wind velocity [Vc]:  
 = 3.4 • f • Dr  
 = 3.4 • 0.973 • 19.743  
 = 65.336 mile/hr [105.145 km/hr]

Compute the velocity at the top of the tower [Vw]:  
 = V30 • ( L / ( 30 + BaseHeight ) )<sup>0.143</sup>  
 = 60.4 • ( 156.17 / ( 30 + 62.3 ) )<sup>0.143</sup>  
 = 65.113 mile/hr [104.786 km/hr]

Compute the maximum gust velocity using the gust response factor Gh [Vg]:  
 = Vw • Gh  
 = 104.786 • 0.943  
 = 98.843 km/hr

Since Vc is greater than Vg the dynamic deflection Z, does not need to be computed.

#### **Platform Load Calculations**

| ID      | Wind Area<br>mm <sup>2</sup> | Elevation<br>mm | Pressure<br>kPa | Force<br>N | Cf   |
|---------|------------------------------|-----------------|-----------------|------------|------|
| PLT6    | 2999999.75                   | 49000.00        | 0.77            | 2298.14    | 1.00 |
| PLT7    | 4950720.00                   | 53000.00        | 0.77            | 3792.48    | 1.00 |
| PLT8    | 4950720.00                   | 57000.00        | 0.77            | 3792.48    | 1.00 |
| PLT9    | 4950720.00                   | 61000.00        | 0.77            | 3792.48    | 1.00 |
| PLT10   | 4950720.00                   | 65000.00        | 0.77            | 3792.48    | 1.00 |
| TOP PLT | 13405039.00                  | 68050.00        | 0.77            | 10268.88   | 1.00 |

#### **Wind Loads on Masses/Equipment/Piping**

| ID            | Wind Area<br>mm <sup>2</sup> | Elevation<br>mm | Pressure<br>kPa | Force<br>N |
|---------------|------------------------------|-----------------|-----------------|------------|
| N2A SCHOPENTO | 0.00                         | 45550.00        | 0.77            | 0.00       |
| N2B SCHOPENTO | 0.00                         | 45550.00        | 0.77            | 0.00       |
| N2A INSERT PL | 0.00                         | 45000.00        | 0.77            | 0.00       |
| N2B INSERT PL | 0.00                         | 45000.00        | 0.77            | 0.00       |
| N3 INSERT PLA | 0.00                         | 46620.00        | 0.77            | 0.00       |
| CTDEAD WT     | 0.00                         | 46000.00        | 0.77            | 0.00       |
| CT LIQ WT     | 0.00                         | 46000.00        | 0.77            | 0.00       |
| btm Grid-1 DE | 0.00                         | 46470.00        | 0.77            | 0.00       |
| N4AOVHPP      | 0.00                         | 45000.00        | 0.77            | 0.00       |
| N4BOVHPP      | 0.00                         | 45000.00        | 0.77            | 0.00       |
| N3B INSERT PL | 0.00                         | 46620.00        | 0.77            | 0.00       |
| N4A INSERT PL | 0.00                         | 48120.00        | 0.77            | 0.00       |
| N4B INSERT PL | 0.00                         | 48120.00        | 0.77            | 0.00       |
| 10T Margin    | 0.00                         | 51255.00        | 0.77            | 0.00       |
| Vane1 DEAD WT | 0.00                         | 52050.00        | 0.77            | 0.00       |
| Vane1 LIQ WT  | 0.00                         | 52050.00        | 0.77            | 0.00       |
| dist.1 DEAD W | 0.00                         | 50970.00        | 0.77            | 0.00       |
| dist.1 LIQ WT | 0.00                         | 50970.00        | 0.77            | 0.00       |
| Cleat wt      | 0.00                         | 51255.00        | 0.77            | 0.00       |



|               |      |          |      |      |
|---------------|------|----------|------|------|
| Lifting wt    | 0.00 | 51255.00 | 0.77 | 0.00 |
| Cladded Nozzl | 0.00 | 50050.00 | 0.77 | 0.00 |
| Manway davit  | 0.00 | 51200.00 | 0.77 | 0.00 |
| btm Grid-2 DE | 0.00 | 54535.00 | 0.77 | 0.00 |
| dist.2 DEAD W | 0.00 | 59035.00 | 0.77 | 0.00 |
| dist.2 LIQ WT | 0.00 | 59035.00 | 0.77 | 0.00 |
| N5 SCHOPENTOE | 0.00 | 60900.01 | 0.77 | 0.00 |
| Vane2 DEAD WT | 0.00 | 61050.01 | 0.77 | 0.00 |
| Vane2 LIQ WT  | 0.00 | 61050.01 | 0.77 | 0.00 |
| btm Grid-3 DE | 0.00 | 63910.00 | 0.77 | 0.00 |
| dist.3 DEAD W | 0.00 | 64910.00 | 0.77 | 0.00 |
| dist. 3 LIQ W | 0.00 | 64910.00 | 0.77 | 0.00 |
| N5 INSERT PLT | 0.00 | 62100.00 | 0.77 | 0.00 |
| N5OVHPP       | 0.00 | 63050.00 | 0.77 | 0.00 |
| Davit         | 0.00 | 65050.00 | 0.77 | 0.00 |
| N6 INT PIPE W | 0.00 | 66310.00 | 0.77 | 0.00 |
| N7OVHPP       | 0.00 | 66550.00 | 0.77 | 0.00 |
| 2 PSVS        | 0.00 | 67570.00 | 0.77 | 0.00 |

The Natural Frequency for the Vessel (Ope...) is 0.97335 Hz.

#### Wind Load Calculation:

| From | To  | Wind Height<br>mm | Wind Diameter<br>mm | Wind Area<br>mm <sup>2</sup> | Wind Pressure<br>kPa | Element Wind Load<br>N |
|------|-----|-------------------|---------------------|------------------------------|----------------------|------------------------|
| 10   | 20  | 20000             | 7262.66             | 14525321                     | 0.76606              | 6998.33                |
| 20   | 30  | 22500             | 7245.58             | 21736740                     | 0.76606              | 10472.8                |
| 30   | 40  | 25500             | 7245.58             | 21736740                     | 0.76606              | 10472.8                |
| 40   | 50  | 28500             | 7243.14             | 21729418                     | 0.76606              | 10469.3                |
| 50   | 60  | 31500             | 7243.14             | 21729418                     | 0.76606              | 10469.3                |
| 60   | 70  | 34500             | 7240.7              | 21722100                     | 0.76606              | 10465.8                |
| 70   | 80  | 37500             | 7240.7              | 21722100                     | 0.76606              | 10465.8                |
| 80   | 90  | 39500             | 7238.26             | 7238260                      | 0.76606              | 3487.41                |
| 90   | 100 | 40500             | 7238.26             | 7238260                      | 0.76606              | 3487.41                |
| 100  | 110 | 41025             | 7503                | 375150                       | 0.76606              | 180.748                |
| 110  | 120 | 42550             | 7488.36             | 22465078                     | 0.76606              | 10823.7                |
| 120  | 130 | 45550             | 7488.36             | 22465078                     | 0.76606              | 10823.7                |
| 130  | 140 | 48550             | 7483.48             | 22450440                     | 0.76606              | 13368                  |
| 140  | 150 | 51550             | 7483.48             | 22450440                     | 0.76606              | 15105.7                |
| 150  | 160 | 54550             | 7483.48             | 22450440                     | 0.76606              | 10816.7                |
| 160  | 170 | 57550             | 7483.48             | 22450440                     | 0.76606              | 14948.5                |
| 170  | 180 | 60550             | 7483.48             | 22450440                     | 0.76606              | 15019.4                |
| 180  | 190 | 63550             | 7483.48             | 22450440                     | 0.76606              | 15083.6                |
| 190  | 220 | 65800             | 7483.48             | 11225220                     | 0.76606              | 5408.33                |
| 220  | 230 | 67046.9           | 7498.12             | 6949585                      | 0.76606              | 14813.1                |

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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## Earthquake Load Calculation Results:

### Earthquake Load Calculation:

#### Input Values:

|                                     |               |
|-------------------------------------|---------------|
| Seismic Design Code                 | ASCE/SEI 7-16 |
| Seismic Load Reduction Scale Factor | 1.000         |
| Importance Factor                   | 1.250         |
| Table Value Fa                      | 1.600         |
| Table Value Fv                      | 2.400         |
| Max. Mapped Res. Acceleration [Ss]  | 0.147         |
| Max. Eff. Ground Acceleration [S]   | 0.065         |
| Force Modification Factor R         | 1.250         |
| Site Class                          | D             |
| Component Elevation Ratio z/h       | 0.000         |
| Amplification Factor Ap             | 0.000         |
| Force Factor                        | 0.000         |
| Consider Vertical Acceleration      | Yes           |
| Minimum Acceleration Multiplier     | 0.000         |
| User Value of Sds (used if > 0 )    | 0.000         |
| User Value of Sd1 (used if > 0 )    | 0.000         |
| Moment Reduction Factor Tau         | 1.000         |

#### Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:  
=  $C_t \cdot h_{nx}$  where  $C_t = 0.020$ ,  $x = 0.75$  and  $h_n$  = Structural Height (ft.)  
=  $0.020(159.437^{10.75})$   
= 0.897 seconds

The Coefficient  $C_u$  from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:  
= ( 1/Natural Frequency ) = ( 1/0.973 )  
= 1.027

Check the Value of T which is the smaller of  $C_u \cdot T_a$  and T:  
= Minimum Value of (  $1.7 \cdot 0.897$ , 1.027 ) per 12.8.2  
= 1.027

Seismic Response Coefficient per equation 12.8-2 [Cs]:  
=  $Sds / ( R / I )$   
=  $0.157 / ( 1.25 / 1.25 )$   
= 0.157

Maximum value of  $C_s$  per equation 12.8-3 [Cs]:  
=  $Sd1 / ( T( R / I_e ) )$   
=  $0.104 / ( 1.027( 1.25 / 1.25 ) )$   
= 0.101

Minimum value of  $C_s$  per equation 15.4-1 [Cs]:  
=  $\max( ( 0.044 \cdot SDS \cdot I_e ), 0.030 )$   
=  $\max( ( 0.044 \cdot 0.1568 \cdot 1.25 ), 0.030 )$   
= 0.030

Total Base Shear [V]:  
=  $C_s \cdot W$  (Equation 12.8-1):  
=  $0.1012 \cdot 6090691$   
= 616549.062 N

Vertical load per 12.4-4, [Ev]:  
=  $0.2 \cdot Sds \cdot W$   
=  $0.2 \cdot 0.157 \cdot 6090691$





= 191004.0 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$  (eqn. 12.8-11 ) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$  and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.2637

The Natural Frequency for the Vessel (Ope...) is 0.97335 Hz.

#### Earthquake Load Calculation:

| From | To  | Earthquake Height<br>mm | Earthquake Weight<br>N | Element Ope Load<br>N | Element Emp Load<br>N | Element Vertical Load<br>N |
|------|-----|-------------------------|------------------------|-----------------------|-----------------------|----------------------------|
| 10   | 20  | 1000                    | 168901                 | 217.799               | 286.281               | 129129                     |
| 20   | 30  | 3500                    | 93243.6                | 585.567               | 743.069               | 123832                     |
| 30   | 40  | 6500                    | 93243.6                | 1280.31               | 1596.67               | 120908                     |
| 40   | 50  | 9500                    | 84765.9                | 1880.12               | 2319.83               | 117984                     |
| 50   | 60  | 12500                   | 84765.9                | 2659.5                | 3256.28               | 115326                     |
| 60   | 70  | 15500                   | 76290.1                | 3141.26               | 3822.97               | 112668                     |
| 70   | 80  | 18500                   | 76290.1                | 3928.31               | 4757.12               | 110275                     |
| 80   | 90  | 20500                   | 22604.1                | 1325.14               | 1600.11               | 107883                     |
| 90   | 100 | 21500                   | 22604.1                | 1407.35               | 1697.1                | 107174                     |
| 100  | 110 | 22025                   | 270806                 | 17382.5               | 8901.47               | 106465                     |
| 110  | 120 | 23550                   | 906565                 | 63328                 | 9481.15               | 97972.4                    |
| 120  | 130 | 26550                   | 855554                 | 69542.2               | 17930.4               | 69542.6                    |
| 130  | 140 | 29550                   | 366064                 | 34065.3               | 11892.6               | 42712.4                    |
| 140  | 150 | 32550                   | 627484                 | 65981.7               | 45045.5               | 31232.6                    |
| 150  | 160 | 35550                   | 241156                 | 28346.9               | 14831.7               | 11554.7                    |
| 160  | 170 | 38550                   | 459665                 | 59856.3               | 15138.7               | 3992.06                    |
| 170  | 180 | 41550                   | 211769                 | 30315.2               | 18711.6               | -10423                     |
| 180  | 190 | 44550                   | 823079                 | 128677                | 21964.1               | -17064.1                   |
| 190  | 220 | 46800                   | 81391.5                | 13541.8               | 15976.9               | -42875.9                   |
| 220  | 230 | 47575                   | 524448                 | 89087.1               | 57192                 | -45428.3                   |



**Bending Moments due to user defined forces and moments.**

**User Force/Moment Shear and Bending**

| From | To  | Distance to Support<br>mm | Cumulative<br>Shr Wind Cas<br>N | Cumulative<br>Shr Eqk Cas<br>N | Wind<br>Bending<br>N-mm | Earthquake<br>Bending<br>N-mm |
|------|-----|---------------------------|---------------------------------|--------------------------------|-------------------------|-------------------------------|
| 10   | 20  | 1000                      | 58336.3                         | 58336.3                        | 2.958E+09               | 2.958E+09                     |
| 20   | 30  | 3500                      | 58336.3                         | 58336.3                        | 2.842E+09               | 2.842E+09                     |
| 30   | 40  | 6500                      | 58336.3                         | 58336.3                        | 2.667E+09               | 2.667E+09                     |
| 40   | 50  | 9500                      | 58336.3                         | 58336.3                        | 2.492E+09               | 2.492E+09                     |
| 50   | 60  | 12500                     | 58336.3                         | 58336.3                        | 2.316E+09               | 2.316E+09                     |
| 60   | 70  | 15500                     | 58336.3                         | 58336.3                        | 2141410560              | 2141410560                    |
| 70   | 80  | 18500                     | 58336.3                         | 58336.3                        | 1966330752              | 1966330752                    |
| 80   | 90  | 20500                     | 58336.3                         | 58336.3                        | 1791250944              | 1791250944                    |
| 90   | 100 | 21500                     | 58336.3                         | 58336.3                        | 1732891008              | 1732891008                    |
| 100  | 110 | 22025                     | 58336.3                         | 58336.3                        | 1674531072              | 1674531072                    |
| 110  | 120 | 23550                     | 58336.3                         | 58336.3                        | 1671613056              | 1671613056                    |
| 120  | 130 | 26550                     | 58336.3                         | 58336.3                        | 1496532992              | 1496532992                    |
| 130  | 140 | 29550                     | 58336.3                         | 58336.3                        | 1321453184              | 1321453184                    |
| 140  | 150 | 32550                     | 58336.3                         | 58336.3                        | 1146373248              | 1146373248                    |
| 150  | 160 | 35550                     | 58336.3                         | 58336.3                        | 971293440               | 971293440                     |
| 160  | 170 | 38550                     | 58336.3                         | 58336.3                        | 796213568               | 796213568                     |
| 170  | 180 | 41550                     | 58336.3                         | 58336.3                        | 621133632               | 621133632                     |
| 180  | 190 | 44550                     | 58336.3                         | 58336.3                        | 446053856               | 446053856                     |
| 190  | 220 | 46800                     | 58336.3                         | 58336.3                        | 270974080               | 270974080                     |
| 220  | 230 | 47575                     | 58336.3                         | 58336.3                        | 183434048               | 183434048                     |

**Note:**

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;  
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

**Wind/Earthquake Shear, Bending:**

| From | To  | Distance to Support<br>mm | Cumulative<br>Wind Shear<br>N | Earthquake<br>Shear<br>N | Wind<br>Bending<br>N-mm | Earthquake<br>Bending<br>N-mm |
|------|-----|---------------------------|-------------------------------|--------------------------|-------------------------|-------------------------------|
| 10   | 20  | 1000                      | 203180                        | 616549                   | 5.501E+09               | 22.35E+09                     |
| 20   | 30  | 3500                      | 196182                        | 616331                   | 5.102E+09               | 21.11E+09                     |
| 30   | 40  | 6500                      | 185709                        | 615746                   | 4.529E+09               | 19.26E+09                     |
| 40   | 50  | 9500                      | 175236                        | 614465                   | 3.987E+09               | 17.42E+09                     |
| 50   | 60  | 12500                     | 164767                        | 612585                   | 3.477E+09               | 15.58E+09                     |
| 60   | 70  | 15500                     | 154298                        | 609926                   | 2.998E+09               | 13.74E+09                     |
| 70   | 80  | 18500                     | 143832                        | 606785                   | 2.551E+09               | 11.92E+09                     |
| 80   | 90  | 20500                     | 133366                        | 602856                   | 2134635904              | 10.10E+09                     |
| 90   | 100 | 21500                     | 129879                        | 601531                   | 2002959872              | 9.498E+09                     |
| 100  | 110 | 22025                     | 126392                        | 600124                   | 1874772864              | 8.897E+09                     |
| 110  | 120 | 23550                     | 126211                        | 582741                   | 1868455424              | 8.868E+09                     |
| 120  | 130 | 26550                     | 115387                        | 519413                   | 1505911424              | 7.214E+09                     |
| 130  | 140 | 29550                     | 104563                        | 449871                   | 1175851904              | 5.759E+09                     |
| 140  | 150 | 32550                     | 91195.3                       | 415806                   | 882094848               | 4.460E+09                     |
| 150  | 160 | 35550                     | 76089.6                       | 349824                   | 631065664               | 3.311E+09                     |
| 160  | 170 | 38550                     | 65272.9                       | 321477                   | 418936064               | 2.304E+09                     |
| 170  | 180 | 41550                     | 50324.4                       | 261621                   | 245469776               | 1428925824                    |
| 180  | 190 | 44550                     | 35305                         | 231306                   | 116973728               | 689236864                     |
| 190  | 220 | 46800                     | 20221.4                       | 102629                   | 33650432                | 188132544                     |
| 220  | 230 | 47575                     | 14813.1                       | 89087.1                  | 7363879                 | 44287000                      |

**Note:**

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;  
 Wind: 1.100; Earthquake: 1.000.

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### Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

#### Wind Load Deflection:

| From | To  | Cumulative Shear<br>N | Centroid Deflection<br>mm | Elem. End Deflection<br>mm | Elem. Ang. Rotation |
|------|-----|-----------------------|---------------------------|----------------------------|---------------------|
| 10   | 20  | 58336.3               | 0.0024861                 | 0.0098785                  | 0.98127E-05         |
| 20   | 30  | 58336.3               | 0.033357                  | 0.073992                   | 0.32687E-04         |
| 30   | 40  | 58336.3               | 0.13124                   | 0.20455                    | 0.54108E-04         |
| 40   | 50  | 58336.3               | 0.29415                   | 0.40021                    | 0.76071E-04         |
| 50   | 60  | 58336.3               | 0.52216                   | 0.65937                    | 0.96434E-04         |
| 60   | 70  | 58336.3               | 0.81206                   | 0.98039                    | 0.00011728          |
| 70   | 80  | 58336.3               | 1.16369                   | 1.36129                    | 0.00013635          |
| 80   | 90  | 58336.3               | 1.43032                   | 1.50102                    | 0.00014306          |
| 90   | 100 | 58336.3               | 1.57337                   | 1.64734                    | 0.00014955          |
| 100  | 110 | 58336.3               | 1.65108                   | 1.65483                    | 0.00014979          |
| 110  | 120 | 58336.3               | 1.88699                   | 2.13358                    | 0.00016903          |
| 120  | 130 | 58336.3               | 2.3938                    | 2.66686                    | 0.00018614          |
| 130  | 140 | 58336.3               | 2.95286                   | 3.25184                    | 0.00020345          |
| 140  | 150 | 58336.3               | 3.56289                   | 3.88507                    | 0.0002183           |
| 150  | 160 | 58336.3               | 4.21747                   | 4.55917                    | 0.00023069          |
| 160  | 170 | 58336.3               | 4.90925                   | 5.26677                    | 0.00024063          |
| 170  | 180 | 58336.3               | 5.63084                   | 6.00052                    | 0.00024812          |
| 180  | 190 | 58336.3               | 6.37489                   | 6.75303                    | 0.00025315          |
| 190  | 220 | 58336.3               | 6.94323                   | 7.13402                    | 0.00025474          |
| 220  | 230 | 58336.3               | 7.14039                   | 7.14676                    | 0.00025474          |

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)  
 Allowable deflection : 238.000 Actual deflection : 7.147 mm

The following table is for Wind Deflection; Operating Case.

#### Wind Load Deflection:

| From | To  | Cumulative Shear<br>N | Centroid Deflection<br>mm | Elem. End Deflection<br>mm | Elem. Ang. Rotation |
|------|-----|-----------------------|---------------------------|----------------------------|---------------------|
| 10   | 20  | 203180                | 0.0047539                 | 0.018782                   | 0.18548E-04         |
| 20   | 30  | 196182                | 0.062724                  | 0.13767                    | 0.59888E-04         |
| 30   | 40  | 185709                | 0.24179                   | 0.37332                    | 0.96440E-04         |
| 40   | 50  | 175236                | 0.5318                    | 0.7167                     | 0.00013168          |
| 50   | 60  | 164767                | 0.92625                   | 1.15871                    | 0.00016224          |
| 60   | 70  | 154298                | 1.41358                   | 1.69026                    | 0.00019134          |
| 70   | 80  | 143832                | 1.98704                   | 2.30223                    | 0.00021591          |
| 80   | 90  | 133366                | 2.41122                   | 2.52225                    | 0.00022405          |
| 90   | 100 | 129879                | 2.63525                   | 2.75016                    | 0.00023169          |
| 100  | 110 | 126392                | 2.75596                   | 2.76176                    | 0.00023196          |
| 110  | 120 | 126211                | 3.11821                   | 3.49052                    | 0.00025312          |
| 120  | 130 | 115387                | 3.87703                   | 4.27614                    | 0.00026993          |
| 130  | 140 | 104563                | 4.68715                   | 5.10931                    | 0.00028481          |
| 140  | 150 | 91195.3               | 5.54108                   | 5.98103                    | 0.00029573          |
| 150  | 160 | 76089.6               | 6.42785                   | 6.88035                    | 0.00030331          |
| 160  | 170 | 65272.9               | 7.33742                   | 7.79805                    | 0.00030807          |
| 170  | 180 | 50324.4               | 8.26137                   | 8.7266                     | 0.00031065          |
| 180  | 190 | 35305                 | 9.19312                   | 9.6604                     | 0.00031168          |
| 190  | 220 | 20221.4               | 9.8942                    | 10.1281                    | 0.00031183          |
| 220  | 230 | 14813.1               | 10.1359                   | 10.1437                    | 0.00031183          |

#### Critical Wind Velocity for Tower Vibration:

| From | To | 1st Crit. Wind Speed | 2nd Crit. Wind Speed |
|------|----|----------------------|----------------------|
|------|----|----------------------|----------------------|



|     |     | km/hr   | km/hr   |
|-----|-----|---------|---------|
| 10  | 20  | 126.901 | 793.13  |
| 20  | 30  | 126.602 | 791.265 |
| 30  | 40  | 126.602 | 791.265 |
| 40  | 50  | 126.56  | 790.999 |
| 50  | 60  | 126.56  | 790.999 |
| 60  | 70  | 126.517 | 790.732 |
| 70  | 80  | 126.517 | 790.732 |
| 80  | 90  | 126.475 | 790.466 |
| 90  | 100 | 126.475 | 790.466 |
| 100 | 110 | 131.1   | 819.377 |
| 110 | 120 | 130.845 | 817.778 |
| 120 | 130 | 130.845 | 817.778 |
| 130 | 140 | 130.759 | 817.245 |
| 140 | 150 | 130.759 | 817.245 |
| 150 | 160 | 130.759 | 817.245 |
| 160 | 170 | 130.759 | 817.245 |
| 170 | 180 | 130.759 | 817.245 |
| 180 | 190 | 130.759 | 817.245 |
| 190 | 220 | 130.759 | 817.245 |
| 220 | 230 | 131.015 | 818.844 |

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)  
 Allowable deflection : 238.000 Actual deflection : 10.144 mm

Total Deflection in the Operating Condition + Applied Forces:  
 Allowable deflection : 238.000 Actual deflection : 17.290 mm

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**Longitudinal Stress Constants:**

| From | To  | Metal Area<br>New<br>mm <sup>2</sup> | Metal Area<br>Corroded<br>mm <sup>2</sup> | Section Modulus<br>New<br>mm <sup>3</sup> | Section Modulus<br>Corroded<br>mm <sup>3</sup> |
|------|-----|--------------------------------------|-------------------------------------------|-------------------------------------------|------------------------------------------------|
| 10   | 20  | 669194                               | 669194                                    | 983957696                                 | 983957696                                      |
| 20   | 30  | 408954                               | 408954                                    | 602711744                                 | 602711744                                      |
| 30   | 40  | 408954                               | 408954                                    | 602711744                                 | 602711744                                      |
| 40   | 50  | 371775                               | 371775                                    | 548102272                                 | 548102272                                      |
| 50   | 60  | 371775                               | 371775                                    | 548102272                                 | 548102272                                      |
| 60   | 70  | 334599                               | 334599                                    | 493459488                                 | 493459488                                      |
| 70   | 80  | 334599                               | 334599                                    | 493459488                                 | 493459488                                      |
| 80   | 90  | 297419                               | 297419                                    | 438776352                                 | 438776352                                      |
| 90   | 100 | 297419                               | 297419                                    | 438776352                                 | 438776352                                      |
| 100  | 110 | 409715                               | 391158                                    | 604959104                                 | 577755136                                      |
| 110  | 120 | 297673                               | 279117                                    | 439518560                                 | 412259616                                      |
| 120  | 130 | 297673                               | 279117                                    | 439518560                                 | 412259616                                      |
| 130  | 140 | 260374                               | 241818                                    | 384445728                                 | 357168416                                      |
| 140  | 150 | 260374                               | 241818                                    | 384445728                                 | 357168416                                      |
| 150  | 160 | 260374                               | 241818                                    | 384445728                                 | 357168416                                      |
| 160  | 170 | 260374                               | 241818                                    | 384445728                                 | 357168416                                      |
| 170  | 180 | 260374                               | 241818                                    | 384445728                                 | 357168416                                      |
| 180  | 190 | 260374                               | 241818                                    | 384445728                                 | 357168416                                      |
| 190  | 220 | 260374                               | 241818                                    | 384445728                                 | 357168416                                      |
| 220  | 230 | 372341                               | 353785                                    | 549772672                                 | 522550432                                      |



**Longitudinal Allowable Stresses:**

| From | To  | Tensile<br>MPa | Hydrotest<br>Tensile<br>MPa | Compressive<br>MPa | Hydrotest<br>Compressive<br>MPa |
|------|-----|----------------|-----------------------------|--------------------|---------------------------------|
| 10   | 20  | 98.2782        | 173.877                     | -116.129           | -116.391                        |
| 20   | 30  | 98.2782        | 173.877                     | -99.3025           | -99.4847                        |
| 30   | 40  | 98.2782        | 173.877                     | -99.3025           | -99.4847                        |
| 40   | 50  | 98.2782        | 173.877                     | -95.6495           | -95.8124                        |
| 50   | 60  | 98.2782        | 173.877                     | -95.6495           | -95.8124                        |
| 60   | 70  | 98.2782        | 173.877                     | -90.9032           | -90.9652                        |
| 70   | 80  | 98.2782        | 173.877                     | -90.9032           | -90.9652                        |
| 80   | 90  | 98.2782        | 173.877                     | -80.8301           | -80.8852                        |
| 90   | 100 | 91.0784        | 154.437                     | -80.8301           | -80.8852                        |
| 100  | 110 | 165.597        | 280.795                     | -97.4579           | -97.6305                        |
| 110  | 120 | 165.597        | 280.795                     | -75.7144           | -75.766                         |
| 120  | 130 | 165.597        | 280.795                     | -75.7144           | -75.766                         |
| 130  | 140 | 165.597        | 280.795                     | -65.6634           | -65.7082                        |
| 140  | 150 | 165.597        | 280.795                     | -65.6634           | -65.7082                        |
| 150  | 160 | 165.597        | 280.795                     | -65.6634           | -65.7082                        |
| 160  | 170 | 165.597        | 280.795                     | -65.6634           | -65.7082                        |
| 170  | 180 | 165.597        | 280.795                     | -65.6634           | -65.7082                        |
| 180  | 190 | 165.597        | 280.795                     | -65.6634           | -65.7082                        |
| 190  | 220 | 165.597        | 280.795                     | -65.6634           | -65.7082                        |
| 220  | 230 | 165.597        | 280.795                     | -93.5632           | -93.7149                        |



## Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

### Longitudinal Pressure Stresses due to:

| From | To  | Longitudinal Stress<br>Internal Pressure<br>MPa | Longitudinal Stress<br>External Pressure<br>MPa | Longitudinal Stress<br>Hydrotest Pressure<br>MPa |
|------|-----|-------------------------------------------------|-------------------------------------------------|--------------------------------------------------|
| 10   | 20  | ...                                             | ...                                             | ...                                              |
| 20   | 30  | ...                                             | ...                                             | ...                                              |
| 30   | 40  | ...                                             | ...                                             | ...                                              |
| 40   | 50  | ...                                             | ...                                             | ...                                              |
| 50   | 60  | ...                                             | ...                                             | ...                                              |
| 60   | 70  | ...                                             | ...                                             | ...                                              |
| 70   | 80  | ...                                             | ...                                             | ...                                              |
| 80   | 90  | ...                                             | ...                                             | ...                                              |
| 90   | 100 | ...                                             | ...                                             | ...                                              |
| 100  | 110 | 24.0729                                         | -7.3546                                         | 45.4302                                          |
| 110  | 120 | 33.7295                                         | -10.2653                                        | 63.6541                                          |
| 120  | 130 | 33.7295                                         | -10.2653                                        | 63.6541                                          |
| 130  | 140 | 38.9292                                         | -11.8327                                        | 73.467                                           |
| 140  | 150 | 38.9292                                         | -11.8327                                        | 73.467                                           |
| 150  | 160 | 38.9292                                         | -11.8327                                        | 73.467                                           |
| 160  | 170 | 38.9292                                         | -11.8327                                        | 73.467                                           |
| 170  | 180 | 38.9292                                         | -11.8327                                        | 73.467                                           |
| 180  | 190 | 38.9292                                         | -11.8327                                        | 73.467                                           |
| 190  | 220 | 38.9292                                         | -11.8327                                        | 73.467                                           |
| 220  | 230 | 26.6141                                         | -8.12061                                        | 50.226                                           |

### Longitudinal Stresses due to Weight Loads for these Conditions:

| From | To  | Wght. Str.<br>Empty<br>MPa | Wght. Str.<br>Operating<br>MPa | Wght. Str.<br>Hydrotest<br>MPa | Wght. Str.<br>Emp. Mom.<br>MPa | Wght. Str.<br>Opr. Mom.<br>MPa |
|------|-----|----------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| 10   | 20  | -6.86695                   | -9.1023                        | -14.9419                       | 2.41312                        | 2.41312                        |
| 20   | 30  | -10.8237                   | -14.4815                       | -24.0372                       | 3.93954                        | 3.93954                        |
| 30   | 40  | -10.5957                   | -14.2535                       | -23.8092                       | 3.93954                        | 3.93954                        |
| 40   | 50  | -11.4045                   | -15.4281                       | -25.9394                       | 4.33205                        | 4.33205                        |
| 50   | 60  | -11.1765                   | -15.2001                       | -25.7114                       | 4.33205                        | 4.33205                        |
| 60   | 70  | -12.1649                   | -16.6356                       | -28.3147                       | 4.81176                        | 4.81176                        |
| 70   | 80  | -11.9369                   | -16.4075                       | -28.0867                       | 4.81176                        | 4.81176                        |
| 80   | 90  | -13.1725                   | -18.2021                       | -31.3412                       | 5.41143                        | 5.41143                        |
| 90   | 100 | -13.0965                   | -18.1261                       | -31.2652                       | 5.41143                        | 5.41143                        |
| 100  | 110 | -9.90022                   | -10.6124                       | -9.90022                       | 4.10971                        | 4.10971                        |
| 110  | 120 | -13.462                    | -14.4601                       | -13.462                        | 5.75882                        | 5.75882                        |
| 120  | 130 | -13.0577                   | -14.0558                       | -13.0577                       | 5.75726                        | 5.75726                        |
| 130  | 140 | -12.7043                   | -13.7928                       | -12.7043                       | 6.47243                        | 6.47243                        |
| 140  | 150 | -11.5196                   | -12.2788                       | -11.5196                       | 6.4711                         | 6.4711                         |
| 150  | 160 | -9.02543                   | -9.68375                       | -9.02543                       | 6.45162                        | 6.45162                        |
| 160  | 170 | -8.19431                   | -8.68641                       | -8.19431                       | 6.4341                         | 6.4341                         |
| 170  | 180 | -6.6208                    | -6.78538                       | -6.6208                        | 6.4341                         | 6.4341                         |
| 180  | 190 | -5.74498                   | -5.90956                       | -5.74498                       | 6.40488                        | 6.40488                        |
| 190  | 220 | -2.50556                   | -2.50556                       | -2.50556                       | 0.28601                        | 0.28601                        |
| 220  | 230 | -1.48252                   | -1.48252                       | -1.48252                       | 0.057585                       | 0.057585                       |

### Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

| From | To | Wght. Str.<br>Hyd. Mom.<br>MPa | Bend. Str.<br>Oper. Wind<br>MPa | Bend. Str.<br>Oper. Equ.<br>MPa | Bend. Str.<br>Hyd. Wind<br>MPa | Bend. Str.<br>Hyd. Equ.<br>MPa |
|------|----|--------------------------------|---------------------------------|---------------------------------|--------------------------------|--------------------------------|
|------|----|--------------------------------|---------------------------------|---------------------------------|--------------------------------|--------------------------------|





|     |     |          |          |          |           |     |
|-----|-----|----------|----------|----------|-----------|-----|
| 10  | 20  | 2.41312  | 5.58911  | 22.7029  | 2.79456   | ... |
| 20  | 30  | 3.93954  | 8.46185  | 35.018   | 4.23092   | ... |
| 30  | 40  | 3.93954  | 7.51133  | 31.9514  | 3.75567   | ... |
| 40  | 50  | 4.33205  | 7.27183  | 31.7678  | 3.63591   | ... |
| 50  | 60  | 4.33205  | 6.34126  | 28.4094  | 3.17063   | ... |
| 60  | 70  | 4.81176  | 6.07348  | 27.8389  | 3.03674   | ... |
| 70  | 80  | 4.81176  | 5.16716  | 24.14    | 2.58358   | ... |
| 80  | 90  | 5.41143  | 4.86342  | 23.0129  | 2.43171   | ... |
| 90  | 100 | 5.41143  | 4.56341  | 21.6403  | 2.28171   | ... |
| 100 | 110 | 4.10971  | 3.24389  | 15.3947  | 1.62194   | ... |
| 110 | 120 | 5.75882  | 4.53078  | 21.503   | 2.26539   | ... |
| 120 | 130 | 5.75726  | 3.65165  | 17.4925  | 1.82583   | ... |
| 130 | 140 | 6.47243  | 3.2911   | 16.1196  | 1.64555   | ... |
| 140 | 150 | 6.4711   | 2.4689   | 12.4837  | 1.23445   | ... |
| 150 | 160 | 6.45162  | 1.76629  | 9.26797  | 0.88315   | ... |
| 160 | 170 | 6.4341   | 1.17256  | 6.44847  | 0.58628   | ... |
| 170 | 180 | 6.4341   | 0.68705  | 3.99942  | 0.34352   | ... |
| 180 | 190 | 6.40488  | 0.3274   | 1.92911  | 0.1637    | ... |
| 190 | 220 | 0.28601  | 0.094184 | 0.52656  | 0.047092  | ... |
| 220 | 230 | 0.057585 | 0.014088 | 0.084724 | 0.0070438 | ... |

#### Longitudinal Stresses due to these Conditions:

| From | To  | Vortex Shedding<br>Operating Case<br>MPa | Vortex Shedding<br>Empty Case<br>MPa | Vortex Shedding<br>Test Case<br>MPa | Earthquake<br>Empty Case<br>MPa |
|------|-----|------------------------------------------|--------------------------------------|-------------------------------------|---------------------------------|
| 10   | 20  | ...                                      | ...                                  | ...                                 | 9.52161                         |
| 20   | 30  | ...                                      | ...                                  | ...                                 | 14.6916                         |
| 30   | 40  | ...                                      | ...                                  | ...                                 | 13.4148                         |
| 40   | 50  | ...                                      | ...                                  | ...                                 | 13.3538                         |
| 50   | 60  | ...                                      | ...                                  | ...                                 | 11.967                          |
| 60   | 70  | ...                                      | ...                                  | ...                                 | 11.7686                         |
| 70   | 80  | ...                                      | ...                                  | ...                                 | 10.2667                         |
| 80   | 90  | ...                                      | ...                                  | ...                                 | 9.88634                         |
| 90   | 100 | ...                                      | ...                                  | ...                                 | 9.34031                         |
| 100  | 110 | ...                                      | ...                                  | ...                                 | 6.68167                         |
| 110  | 120 | ...                                      | ...                                  | ...                                 | 9.33572                         |
| 120  | 130 | ...                                      | ...                                  | ...                                 | 7.70973                         |
| 130  | 140 | ...                                      | ...                                  | ...                                 | 7.13725                         |
| 140  | 150 | ...                                      | ...                                  | ...                                 | 5.50085                         |
| 150  | 160 | ...                                      | ...                                  | ...                                 | 4.10359                         |
| 160  | 170 | ...                                      | ...                                  | ...                                 | 2.95783                         |
| 170  | 180 | ...                                      | ...                                  | ...                                 | 1.93793                         |
| 180  | 190 | ...                                      | ...                                  | ...                                 | 1.06022                         |
| 190  | 220 | ...                                      | ...                                  | ...                                 | 0.35334                         |
| 220  | 230 | ...                                      | ...                                  | ...                                 | 0.054391                        |

#### Longitudinal Stresses due to Applied Axial Forces:

| From | To  | Longitudinal Stress<br>Y Forces Wind<br>MPa | Longitudinal Stress<br>Y Forces Seismic<br>MPa |
|------|-----|---------------------------------------------|------------------------------------------------|
| 10   | 20  | -0.09247                                    | 0.19298                                        |
| 20   | 30  | -0.15131                                    | 0.30283                                        |
| 30   | 40  | -0.15131                                    | 0.29568                                        |
| 40   | 50  | -0.16645                                    | 0.31738                                        |
| 50   | 60  | -0.16645                                    | 0.31023                                        |
| 60   | 70  | -0.18494                                    | 0.33675                                        |
| 70   | 80  | -0.18494                                    | 0.3296                                         |
| 80   | 90  | -0.20806                                    | 0.36276                                        |
| 90   | 100 | -0.20806                                    | 0.36038                                        |
| 100  | 110 | -0.1582                                     | 0.2722                                         |
| 110  | 120 | -0.2217                                     | 0.35104                                        |
| 120  | 130 | -0.2217                                     | 0.24917                                        |
| 130  | 140 | -0.2559                                     | 0.17665                                        |



|     |     |          |           |
|-----|-----|----------|-----------|
| 140 | 150 | -0.2559  | 0.12917   |
| 150 | 160 | -0.2559  | 0.047787  |
| 160 | 170 | -0.2559  | 0.01651   |
| 170 | 180 | -0.2559  | -0.043106 |
| 180 | 190 | -0.2559  | -0.070572 |
| 190 | 220 | -0.2559  | -0.17732  |
| 220 | 230 | -0.17491 | -0.12842  |

#### Longitudinal Stresses due to User Forces and Moments:

| From | To  | Wind For/Mom<br>Corroded<br>MPa | Earthquake For/Mom<br>Corroded<br>MPa | Wind For/Mom<br>No Corrosion<br>MPa | Earthquake For/Mom<br>No Corrosion<br>MPa |
|------|-----|---------------------------------|---------------------------------------|-------------------------------------|-------------------------------------------|
| 10   | 20  | 3.00572                         | 3.00572                               | 3.00572                             | 3.00572                                   |
| 20   | 30  | 4.7134                          | 4.7134                                | 4.7134                              | 4.7134                                    |
| 30   | 40  | 4.423                           | 4.423                                 | 4.423                               | 4.423                                     |
| 40   | 50  | 4.54436                         | 4.54436                               | 4.54436                             | 4.54436                                   |
| 50   | 60  | 4.22503                         | 4.22503                               | 4.22503                             | 4.22503                                   |
| 60   | 70  | 4.3382                          | 4.3382                                | 4.3382                              | 4.3382                                    |
| 70   | 80  | 3.98351                         | 3.98351                               | 3.98351                             | 3.98351                                   |
| 80   | 90  | 4.08107                         | 4.08107                               | 4.08107                             | 4.08107                                   |
| 90   | 100 | 3.94811                         | 3.94811                               | 3.94811                             | 3.94811                                   |
| 100  | 110 | 2.89741                         | 2.89741                               | 2.76712                             | 2.76712                                   |
| 110  | 120 | 4.05346                         | 4.05346                               | 3.80206                             | 3.80206                                   |
| 120  | 130 | 3.62891                         | 3.62891                               | 3.40385                             | 3.40385                                   |
| 130  | 140 | 3.69862                         | 3.69862                               | 3.43619                             | 3.43619                                   |
| 140  | 150 | 3.20859                         | 3.20859                               | 2.98093                             | 2.98093                                   |
| 150  | 160 | 2.71856                         | 2.71856                               | 2.52567                             | 2.52567                                   |
| 160  | 170 | 2.22852                         | 2.22852                               | 2.07041                             | 2.07041                                   |
| 170  | 180 | 1.73849                         | 1.73849                               | 1.61514                             | 1.61514                                   |
| 180  | 190 | 1.24846                         | 1.24846                               | 1.15988                             | 1.15988                                   |
| 190  | 220 | 0.75843                         | 0.75843                               | 0.70462                             | 0.70462                                   |
| 220  | 230 | 0.35092                         | 0.35092                               | 0.33355                             | 0.33355                                   |

#### Bending Stresses due to Blast Load:

| From | To  | Bending<br>Stress<br>MPa |
|------|-----|--------------------------|
| 10   | 20  | ...                      |
| 20   | 30  | ...                      |
| 30   | 40  | ...                      |
| 40   | 50  | ...                      |
| 50   | 60  | ...                      |
| 60   | 70  | ...                      |
| 70   | 80  | ...                      |
| 80   | 90  | ...                      |
| 90   | 100 | ...                      |
| 100  | 110 | ...                      |
| 110  | 120 | ...                      |
| 120  | 130 | ...                      |
| 130  | 140 | ...                      |
| 140  | 150 | ...                      |
| 150  | 160 | ...                      |
| 160  | 170 | ...                      |
| 170  | 180 | ...                      |
| 180  | 190 | ...                      |
| 190  | 220 | ...                      |
| 220  | 230 | ...                      |

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure  
 EP = Longitudinal Stress due to External Pressure  
 HP = Longitudinal Stress due to Hydrotest Pressure  
 NP = No Pressure  
 EW = Longitudinal Stress due to Weight (No Liquid)  
 OW = Longitudinal Stress due to Weight (Operating)  
 HW = Longitudinal Stress due to Weight (Hydrotest)  
 WI = Bending Stress due to Wind Moment (Operating)  
 EQ = Bending Stress due to Earthquake Moment (Operating)  
 EE = Bending Stress due to Earthquake Moment (Empty)  
 HI = Bending Stress due to Wind Moment (Hydrotest)  
 HE = Bending Stress due to Earthquake Moment (Hydrotest)  
 WE = Bending Stress due to Wind Moment (Empty) (no CA)  
 WF = Bending Stress due to Wind Moment (Filled) (no CA)  
 CW = Longitudinal Stress due to Weight (Empty) (no CA)  
 VO = Bending Stress due to Vortex Shedding Loads ( Ope )  
 VE = Bending Stress due to Vortex Shedding Loads ( Emp )  
 VF = Bending Stress due to Vortex Shedding Loads ( Test No CA. )  
 FW = Axial Stress due to Vertical Forces for the Wind Case  
 FS = Axial Stress due to Vertical Forces for the Seismic Case  
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded  
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded  
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded  
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded  
 BL = Bending Stress due to Blast Loads, Corroded

**General Notes:**

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (\*) in the final column denotes overstress.

**Analysis of Load Case 1 : NP+EW+WI+FW+BW**

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 4.05           | 98.28             | -17.97       | 116.13            | 0.0412      | 0.1547      |
| 20        | 6.14           | 98.28             | -28.09       | 99.30             | 0.0625      | 0.2829      |
| 30        | 5.13           | 98.28             | -26.62       | 99.30             | 0.0522      | 0.2681      |
| 40        | 4.58           | 98.28             | -27.72       | 95.65             | 0.0466      | 0.2898      |
| 50        | 3.56           | 98.28             | -26.24       | 95.65             | 0.0362      | 0.2743      |
| 60        | 2.87           | 98.28             | -27.57       | 90.90             | 0.0292      | 0.3033      |
| 70        | 1.84           | 98.28             | -26.08       | 90.90             | 0.0187      | 0.2869      |
| 80        | 0.98           | 98.28             | -27.74       | 80.83             | 0.0099      | 0.3431      |
| 90        | 0.62           | 91.08             | -27.23       | 80.83             | 0.0068      | 0.3368      |
| 100       | 0.19           | 165.60            | -20.31       | 97.46             | 0.0012      | 0.2084      |
| 110       | 0.66           | 165.60            | -28.03       | 75.71             | 0.0040      | 0.3702      |
| 120       |                | 165.60            | -26.32       | 75.71             |             | 0.3476      |
| 130       | 0.50           | 165.60            | -26.42       | 65.66             | 0.0030      | 0.4024      |
| 140       | 0.37           | 165.60            | -23.92       | 65.66             | 0.0023      | 0.3643      |
| 150       | 1.66           | 165.60            | -20.22       | 65.66             | 0.0100      | 0.3079      |
| 160       | 1.38           | 165.60            | -18.29       | 65.66             | 0.0084      | 0.2785      |
| 170       | 1.98           | 165.60            | -15.74       | 65.66             | 0.0120      | 0.2397      |
| 180       | 1.98           | 165.60            | -13.98       | 65.66             | 0.0120      | 0.2129      |
| 190       |                | 165.60            | -3.90        | 65.66             |             | 0.0594      |
| 220       |                | 165.60            | -2.08        | 93.56             |             | 0.0222      |

**Analysis of Load Case 2 : NP+EW+EE+FS+BS**

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|



|     |       |        |        |        |        |        |
|-----|-------|--------|--------|--------|--------|--------|
| 10  | 8.27  | 98.28  | -21.61 | 116.13 | 0.0841 | 0.1861 |
| 20  | 12.82 | 98.28  | -33.87 | 99.30  | 0.1305 | 0.3410 |
| 30  | 11.48 | 98.28  | -32.08 | 99.30  | 0.1168 | 0.3230 |
| 40  | 11.14 | 98.28  | -33.32 | 95.65  | 0.1134 | 0.3483 |
| 50  | 9.66  | 98.28  | -31.39 | 95.65  | 0.0983 | 0.3282 |
| 60  | 9.09  | 98.28  | -32.75 | 90.90  | 0.0925 | 0.3602 |
| 70  | 7.45  | 98.28  | -30.67 | 90.90  | 0.0759 | 0.3374 |
| 80  | 6.57  | 98.28  | -32.19 | 80.83  | 0.0668 | 0.3982 |
| 90  | 5.96  | 91.08  | -31.44 | 80.83  | 0.0655 | 0.3889 |
| 100 | 4.06  | 165.60 | -23.32 | 97.46  | 0.0245 | 0.2393 |
| 110 | 6.04  | 165.60 | -32.26 | 75.71  | 0.0365 | 0.4261 |
| 120 | 4.29  | 165.60 | -29.90 | 75.71  | 0.0259 | 0.3950 |
| 130 | 4.78  | 165.60 | -29.84 | 65.66  | 0.0289 | 0.4544 |
| 140 | 3.79  | 165.60 | -26.57 | 65.66  | 0.0229 | 0.4047 |
| 150 | 4.30  | 165.60 | -22.25 | 65.66  | 0.0259 | 0.3389 |
| 160 | 3.44  | 165.60 | -19.80 | 65.66  | 0.0208 | 0.3015 |
| 170 | 3.45  | 165.60 | -16.77 | 65.66  | 0.0208 | 0.2555 |
| 180 | 2.90  | 165.60 | -14.53 | 65.66  | 0.0175 | 0.2213 |
| 190 |       | 165.60 | -4.08  | 65.66  |        | 0.0621 |
| 220 |       | 165.60 | -2.07  | 93.56  |        | 0.0222 |

## Analysis of Load Case 3 : NP+OW+WI+FW+BW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 1.81           | 98.28             | -20.20       | 116.13            | 0.0184      | 0.1740      |
| 20        | 2.48           | 98.28             | -31.75       | 99.30             | 0.0253      | 0.3197      |
| 30        | 1.47           | 98.28             | -30.28       | 99.30             | 0.0149      | 0.3049      |
| 40        | 0.55           | 98.28             | -31.74       | 95.65             | 0.0056      | 0.3319      |
| 50        |                | 98.28             | -30.26       | 95.65             |             | 0.3164      |
| 60        |                | 98.28             | -32.04       | 90.90             |             | 0.3525      |
| 70        |                | 98.28             | -30.55       | 90.90             |             | 0.3361      |
| 80        |                | 98.28             | -32.77       | 80.83             |             | 0.4054      |
| 90        |                | 91.08             | -32.26       | 80.83             |             | 0.3991      |
| 100       |                | 165.60            | -21.02       | 97.46             |             | 0.2157      |
| 110       |                | 165.60            | -29.02       | 75.71             |             | 0.3833      |
| 120       |                | 165.60            | -27.32       | 75.71             |             | 0.3608      |
| 130       |                | 165.60            | -27.51       | 65.66             |             | 0.4190      |
| 140       |                | 165.60            | -24.68       | 65.66             |             | 0.3759      |
| 150       | 1.00           | 165.60            | -20.88       | 65.66             | 0.0060      | 0.3179      |
| 160       | 0.89           | 165.60            | -18.78       | 65.66             | 0.0054      | 0.2860      |
| 170       | 1.82           | 165.60            | -15.90       | 65.66             | 0.0110      | 0.2422      |
| 180       | 1.82           | 165.60            | -14.15       | 65.66             | 0.0110      | 0.2154      |
| 190       |                | 165.60            | -3.90        | 65.66             |             | 0.0594      |
| 220       |                | 165.60            | -2.08        | 93.56             |             | 0.0222      |

## Analysis of Load Case 4 : NP+OW+EQ+FS+BS

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 19.21          | 98.28             | -37.03       | 116.13            | 0.1955      | 0.3189      |
| 20        | 29.49          | 98.28             | -57.85       | 99.30             | 0.3001      | 0.5826      |
| 30        | 26.36          | 98.28             | -54.27       | 99.30             | 0.2682      | 0.5465      |
| 40        | 25.53          | 98.28             | -55.75       | 95.65             | 0.2598      | 0.5829      |
| 50        | 22.08          | 98.28             | -51.86       | 95.65             | 0.2246      | 0.5422      |
| 60        | 20.69          | 98.28             | -53.29       | 90.90             | 0.2105      | 0.5862      |
| 70        | 16.86          | 98.28             | -49.01       | 90.90             | 0.1715      | 0.5392      |
| 80        | 14.67          | 98.28             | -50.34       | 80.83             | 0.1492      | 0.6228      |
| 90        | 13.23          | 91.08             | -48.77       | 80.83             | 0.1453      | 0.6033      |
| 100       | 12.06          | 165.60            | -32.74       | 97.46             | 0.0728      | 0.3360      |
| 110       | 17.21          | 165.60            | -45.42       | 75.71             | 0.1039      | 0.5999      |
| 120       | 13.07          | 165.60            | -40.69       | 75.71             | 0.0789      | 0.5374      |
| 130       | 12.67          | 165.60            | -39.91       | 65.66             | 0.0765      | 0.6077      |
| 140       | 10.01          | 165.60            | -34.31       | 65.66             | 0.0605      | 0.5226      |
| 150       | 8.80           | 165.60            | -28.07       | 65.66             | 0.0532      | 0.4275      |
| 160       | 6.44           | 165.60            | -23.78       | 65.66             | 0.0389      | 0.3622      |
| 170       | 5.34           | 165.60            | -19.00       | 65.66             | 0.0323      | 0.2894      |
| 180       | 3.60           | 165.60            | -15.56       | 65.66             | 0.0218      | 0.2370      |
| 190       |                | 165.60            | -4.25        | 65.66             |             | 0.0648      |



220 165.60 -2.10 93.56 0.0225

Analysis of Load Case 5 : NP+HW+HI

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 173.88            | -20.15       | 116.39            |             | 0.1731      |
| 20        |                | 173.88            | -32.21       | 99.48             |             | 0.3237      |
| 30        |                | 173.88            | -31.50       | 99.48             |             | 0.3167      |
| 40        |                | 173.88            | -33.91       | 95.81             |             | 0.3539      |
| 50        |                | 173.88            | -33.21       | 95.81             |             | 0.3467      |
| 60        |                | 173.88            | -36.16       | 90.97             |             | 0.3975      |
| 70        |                | 173.88            | -35.48       | 90.97             |             | 0.3901      |
| 80        |                | 173.88            | -39.18       | 80.89             |             | 0.4844      |
| 90        |                | 154.44            | -38.96       | 80.89             |             | 0.4816      |
| 100       |                | 280.79            | -15.63       | 97.63             |             | 0.1601      |
| 110       |                | 280.79            | -21.49       | 75.77             |             | 0.2836      |
| 120       |                | 280.79            | -20.64       | 75.77             |             | 0.2724      |
| 130       |                | 280.79            | -20.82       | 65.71             |             | 0.3169      |
| 140       |                | 280.79            | -19.23       | 65.71             |             | 0.2926      |
| 150       |                | 280.79            | -16.36       | 65.71             |             | 0.2490      |
| 160       |                | 280.79            | -15.21       | 65.71             |             | 0.2315      |
| 170       | 0.16           | 280.79            | -13.40       | 65.71             | 0.0006      | 0.2039      |
| 180       | 0.82           | 280.79            | -12.31       | 65.71             | 0.0029      | 0.1874      |
| 190       |                | 280.79            | -2.84        | 65.71             |             | 0.0432      |
| 220       |                | 280.79            | -1.55        | 93.71             |             | 0.0165      |

Analysis of Load Case 6 : NP+HW+HE

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 173.88            | -17.36       | 116.39            |             | 0.1491      |
| 20        |                | 173.88            | -27.98       | 99.48             |             | 0.2812      |
| 30        |                | 173.88            | -27.75       | 99.48             |             | 0.2789      |
| 40        |                | 173.88            | -30.27       | 95.81             |             | 0.3159      |
| 50        |                | 173.88            | -30.04       | 95.81             |             | 0.3136      |
| 60        |                | 173.88            | -33.13       | 90.97             |             | 0.3642      |
| 70        |                | 173.88            | -32.90       | 90.97             |             | 0.3617      |
| 80        |                | 173.88            | -36.75       | 80.89             |             | 0.4544      |
| 90        |                | 154.44            | -36.68       | 80.89             |             | 0.4534      |
| 100       |                | 280.79            | -14.01       | 97.63             |             | 0.1435      |
| 110       |                | 280.79            | -19.22       | 75.77             |             | 0.2537      |
| 120       |                | 280.79            | -18.81       | 75.77             |             | 0.2483      |
| 130       |                | 280.79            | -19.18       | 65.71             |             | 0.2918      |
| 140       |                | 280.79            | -17.99       | 65.71             |             | 0.2738      |
| 150       |                | 280.79            | -15.48       | 65.71             |             | 0.2355      |
| 160       |                | 280.79            | -14.63       | 65.71             |             | 0.2226      |
| 170       |                | 280.79            | -13.05       | 65.71             |             | 0.1987      |
| 180       | 0.66           | 280.79            | -12.15       | 65.71             | 0.0024      | 0.1849      |
| 190       |                | 280.79            | -2.79        | 65.71             |             | 0.0425      |
| 220       |                | 280.79            | -1.54        | 93.71             |             | 0.0164      |

Analysis of Load Case 7 : IP+OW+WI+FW+BW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 1.81           | 98.28             | -20.20       | 116.13            | 0.0184      | 0.1740      |
| 20        | 2.48           | 98.28             | -31.75       | 99.30             | 0.0253      | 0.3197      |
| 30        | 1.47           | 98.28             | -30.28       | 99.30             | 0.0149      | 0.3049      |
| 40        | 0.55           | 98.28             | -31.74       | 95.65             | 0.0056      | 0.3319      |
| 50        |                | 98.28             | -30.26       | 95.65             |             | 0.3164      |
| 60        |                | 98.28             | -32.04       | 90.90             |             | 0.3525      |
| 70        |                | 98.28             | -30.55       | 90.90             |             | 0.3361      |
| 80        |                | 98.28             | -32.77       | 80.83             |             | 0.4054      |
| 90        |                | 91.08             | -32.26       | 80.83             |             | 0.3991      |
| 100       | 23.55          | 165.60            |              | 97.46             | 0.1422      |             |
| 110       | 33.39          | 165.60            |              | 75.71             | 0.2016      |             |
| 120       | 32.49          | 165.60            |              | 75.71             | 0.1962      |             |
| 130       | 38.34          | 165.60            |              | 65.66             | 0.2315      |             |
| 140       | 38.54          | 165.60            |              | 65.66             | 0.2328      |             |



|     |       |        |       |        |
|-----|-------|--------|-------|--------|
| 150 | 39.93 | 165.60 | 65.66 | 0.2411 |
| 160 | 39.82 | 165.60 | 65.66 | 0.2405 |
| 170 | 40.75 | 165.60 | 65.66 | 0.2461 |
| 180 | 40.74 | 165.60 | 65.66 | 0.2460 |
| 190 | 37.31 | 165.60 | 65.66 | 0.2253 |
| 220 | 25.38 | 165.60 | 93.56 | 0.1533 |

## Analysis of Load Case 8 : IP+OW+EQ+FS+BS

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 19.21          | 98.28             | -37.03       | 116.13            | 0.1955      | 0.3189      |
| 20        | 29.49          | 98.28             | -57.85       | 99.30             | 0.3001      | 0.5826      |
| 30        | 26.36          | 98.28             | -54.27       | 99.30             | 0.2682      | 0.5465      |
| 40        | 25.53          | 98.28             | -55.75       | 95.65             | 0.2598      | 0.5829      |
| 50        | 22.08          | 98.28             | -51.86       | 95.65             | 0.2246      | 0.5422      |
| 60        | 20.69          | 98.28             | -53.29       | 90.90             | 0.2105      | 0.5862      |
| 70        | 16.86          | 98.28             | -49.01       | 90.90             | 0.1715      | 0.5392      |
| 80        | 14.67          | 98.28             | -50.34       | 80.83             | 0.1492      | 0.6228      |
| 90        | 13.23          | 91.08             | -48.77       | 80.83             | 0.1453      | 0.6033      |
| 100       | 36.13          | 165.60            | -8.67        | 97.46             | 0.2182      | 0.0890      |
| 110       | 50.94          | 165.60            | -11.69       | 75.71             | 0.3076      | 0.1545      |
| 120       | 46.80          | 165.60            | -6.96        | 75.71             | 0.2826      | 0.0919      |
| 130       | 51.60          | 165.60            | -0.98        | 65.66             | 0.3116      | 0.0149      |
| 140       | 48.94          | 165.60            |              | 65.66             | 0.2956      |             |
| 150       | 47.73          | 165.60            |              | 65.66             | 0.2882      |             |
| 160       | 45.37          | 165.60            |              | 65.66             | 0.2740      |             |
| 170       | 44.27          | 165.60            |              | 65.66             | 0.2674      |             |
| 180       | 42.53          | 165.60            |              | 65.66             | 0.2568      |             |
| 190       | 37.82          | 165.60            |              | 65.66             | 0.2284      |             |
| 220       | 25.50          | 165.60            |              | 93.56             | 0.1540      |             |

## Analysis of Load Case 9 : EP+OW+WI+FW+BW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 1.81           | 98.28             | -20.20       | 116.39            | 0.0184      | 0.1736      |
| 20        | 2.48           | 98.28             | -31.75       | 99.48             | 0.0253      | 0.3191      |
| 30        | 1.47           | 98.28             | -30.28       | 99.48             | 0.0149      | 0.3044      |
| 40        | 0.55           | 98.28             | -31.74       | 95.81             | 0.0056      | 0.3313      |
| 50        |                | 98.28             | -30.26       | 95.81             |             | 0.3159      |
| 60        |                | 98.28             | -32.04       | 90.97             |             | 0.3523      |
| 70        |                | 98.28             | -30.55       | 90.97             |             | 0.3359      |
| 80        |                | 98.28             | -32.77       | 80.89             |             | 0.4051      |
| 90        |                | 91.08             | -32.26       | 80.83             |             | 0.3991      |
| 100       |                | 165.60            | -28.38       | 97.46             |             | 0.2912      |
| 110       |                | 165.60            | -39.29       | 75.71             |             | 0.5189      |
| 120       |                | 165.60            | -37.58       | 75.71             |             | 0.4963      |
| 130       |                | 165.60            | -39.34       | 65.66             |             | 0.5992      |
| 140       |                | 165.60            | -36.52       | 65.66             |             | 0.5561      |
| 150       |                | 165.60            | -32.71       | 65.66             |             | 0.4981      |
| 160       |                | 165.60            | -30.61       | 65.66             |             | 0.4662      |
| 170       |                | 165.60            | -27.73       | 65.66             |             | 0.4224      |
| 180       |                | 165.60            | -25.98       | 65.66             |             | 0.3956      |
| 190       |                | 165.60            | -15.73       | 65.66             |             | 0.2396      |
| 220       |                | 165.60            | -10.20       | 93.56             |             | 0.1090      |

## Analysis of Load Case 10 : EP+OW+EQ+FS+BS

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        | 19.21          | 98.28             | -37.03       | 116.39            | 0.1955      | 0.3182      |
| 20        | 29.49          | 98.28             | -57.85       | 99.48             | 0.3001      | 0.5815      |
| 30        | 26.36          | 98.28             | -54.27       | 99.48             | 0.2682      | 0.5455      |
| 40        | 25.53          | 98.28             | -55.75       | 95.81             | 0.2598      | 0.5819      |
| 50        | 22.08          | 98.28             | -51.86       | 95.81             | 0.2246      | 0.5412      |
| 60        | 20.69          | 98.28             | -53.29       | 90.97             | 0.2105      | 0.5858      |
| 70        | 16.86          | 98.28             | -49.01       | 90.97             | 0.1715      | 0.5388      |
| 80        | 14.67          | 98.28             | -50.34       | 80.89             | 0.1492      | 0.6224      |
| 90        | 13.23          | 91.08             | -48.77       | 80.83             | 0.1453      | 0.6033      |



|     |      |        |        |       |        |        |
|-----|------|--------|--------|-------|--------|--------|
| 100 | 4.71 | 165.60 | -40.10 | 97.46 | 0.0284 | 0.4114 |
| 110 | 6.94 | 165.60 | -55.69 | 75.71 | 0.0419 | 0.7355 |
| 120 | 2.81 | 165.60 | -50.95 | 75.71 | 0.0169 | 0.6729 |
| 130 | 0.84 | 165.60 | -51.74 | 65.66 | 0.0051 | 0.7879 |
| 140 |      | 165.60 | -46.15 | 65.66 |        | 0.7028 |
| 150 |      | 165.60 | -39.91 | 65.66 |        | 0.6077 |
| 160 |      | 165.60 | -35.61 | 65.66 |        | 0.5424 |
| 170 |      | 165.60 | -30.83 | 65.66 |        | 0.4696 |
| 180 |      | 165.60 | -27.40 | 65.66 |        | 0.4172 |
| 190 |      | 165.60 | -16.09 | 65.66 |        | 0.2450 |
| 220 |      | 165.60 | -10.22 | 93.56 |        | 0.1093 |

Analysis of Load Case 11 : HP+HW+HI

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 173.88            | -20.15       | 116.39            |             | 0.1731      |
| 20        |                | 173.88            | -32.21       | 99.48             |             | 0.3237      |
| 30        |                | 173.88            | -31.50       | 99.48             |             | 0.3167      |
| 40        |                | 173.88            | -33.91       | 95.81             |             | 0.3539      |
| 50        |                | 173.88            | -33.21       | 95.81             |             | 0.3467      |
| 60        |                | 173.88            | -36.16       | 90.97             |             | 0.3975      |
| 70        |                | 173.88            | -35.48       | 90.97             |             | 0.3901      |
| 80        |                | 173.88            | -39.18       | 80.89             |             | 0.4844      |
| 90        |                | 154.44            | -38.96       | 80.89             |             | 0.4816      |
| 100       | 41.26          | 280.79            |              | 97.63             | 0.1469      |             |
| 110       | 58.22          | 280.79            |              | 75.77             | 0.2073      |             |
| 120       | 58.18          | 280.79            |              | 75.77             | 0.2072      |             |
| 130       | 68.88          | 280.79            |              | 65.71             | 0.2453      |             |
| 140       | 69.65          | 280.79            |              | 65.71             | 0.2481      |             |
| 150       | 71.78          | 280.79            |              | 65.71             | 0.2556      |             |
| 160       | 72.29          | 280.79            |              | 65.71             | 0.2575      |             |
| 170       | 73.62          | 280.79            |              | 65.71             | 0.2622      |             |
| 180       | 74.29          | 280.79            |              | 65.71             | 0.2646      |             |
| 190       | 71.29          | 280.79            |              | 65.71             | 0.2539      |             |
| 220       | 48.81          | 280.79            |              | 93.71             | 0.1738      |             |

Analysis of Load Case 12 : HP+HW+HE

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 173.88            | -17.36       | 116.39            |             | 0.1491      |
| 20        |                | 173.88            | -27.98       | 99.48             |             | 0.2812      |
| 30        |                | 173.88            | -27.75       | 99.48             |             | 0.2789      |
| 40        |                | 173.88            | -30.27       | 95.81             |             | 0.3159      |
| 50        |                | 173.88            | -30.04       | 95.81             |             | 0.3136      |
| 60        |                | 173.88            | -33.13       | 90.97             |             | 0.3642      |
| 70        |                | 173.88            | -32.90       | 90.97             |             | 0.3617      |
| 80        |                | 173.88            | -36.75       | 80.89             |             | 0.4544      |
| 90        |                | 154.44            | -36.68       | 80.89             |             | 0.4534      |
| 100       | 39.64          | 280.79            |              | 97.63             | 0.1412      |             |
| 110       | 55.95          | 280.79            |              | 75.77             | 0.1993      |             |
| 120       | 56.35          | 280.79            |              | 75.77             | 0.2007      |             |
| 130       | 67.24          | 280.79            |              | 65.71             | 0.2394      |             |
| 140       | 68.42          | 280.79            |              | 65.71             | 0.2437      |             |
| 150       | 70.89          | 280.79            |              | 65.71             | 0.2525      |             |
| 160       | 71.71          | 280.79            |              | 65.71             | 0.2554      |             |
| 170       | 73.28          | 280.79            |              | 65.71             | 0.2610      |             |
| 180       | 74.13          | 280.79            |              | 65.71             | 0.2640      |             |
| 190       | 71.25          | 280.79            |              | 65.71             | 0.2537      |             |
| 220       | 48.80          | 280.79            |              | 93.71             | 0.1738      |             |

Analysis of Load Case 13 : IP+WE+EW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 98.28             | -9.28        | 116.13            |             | 0.0799      |
| 20        |                | 98.28             | -14.76       | 99.30             |             | 0.1487      |
| 30        |                | 98.28             | -14.54       | 99.30             |             | 0.1464      |
| 40        |                | 98.28             | -15.74       | 95.65             |             | 0.1645      |



|     |       |        |        |       |        |
|-----|-------|--------|--------|-------|--------|
| 50  |       | 98.28  | -15.51 | 95.65 | 0.1621 |
| 60  |       | 98.28  | -16.98 | 90.90 | 0.1868 |
| 70  |       | 98.28  | -16.75 | 90.90 | 0.1842 |
| 80  |       | 98.28  | -18.58 | 80.83 | 0.2299 |
| 90  |       | 91.08  | -18.51 | 80.83 | 0.2290 |
| 100 | 18.28 | 165.60 |        | 97.46 | 0.1104 |
| 110 | 26.03 | 165.60 |        | 75.71 | 0.1572 |
| 120 | 26.43 | 165.60 |        | 75.71 | 0.1596 |
| 130 | 32.70 | 165.60 |        | 65.66 | 0.1975 |
| 140 | 33.88 | 165.60 |        | 65.66 | 0.2046 |
| 150 | 36.36 | 165.60 |        | 65.66 | 0.2195 |
| 160 | 37.17 | 165.60 |        | 65.66 | 0.2245 |
| 170 | 38.74 | 165.60 |        | 65.66 | 0.2340 |
| 180 | 39.59 | 165.60 |        | 65.66 | 0.2391 |
| 190 | 36.71 | 165.60 |        | 65.66 | 0.2217 |
| 220 | 25.19 | 165.60 |        | 93.56 | 0.1521 |

Analysis of Load Case 14 : IP+WF+CW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 98.28             | -9.10        | 116.13            |             | 0.0784      |
| 20        |                | 98.28             | -14.48       | 99.30             |             | 0.1458      |
| 30        |                | 98.28             | -14.25       | 99.30             |             | 0.1435      |
| 40        |                | 98.28             | -15.43       | 95.65             |             | 0.1613      |
| 50        |                | 98.28             | -15.20       | 95.65             |             | 0.1589      |
| 60        |                | 98.28             | -16.64       | 90.90             |             | 0.1830      |
| 70        |                | 98.28             | -16.41       | 90.90             |             | 0.1805      |
| 80        |                | 98.28             | -18.20       | 80.83             |             | 0.2252      |
| 90        |                | 91.08             | -18.13       | 80.83             |             | 0.2242      |
| 100       | 13.94          | 165.60            |              | 97.46             | 0.0842      |             |
| 110       | 20.17          | 165.60            |              | 75.71             | 0.1218      |             |
| 120       | 20.55          | 165.60            |              | 75.71             | 0.1241      |             |
| 130       | 26.12          | 165.60            |              | 65.66             | 0.1577      |             |
| 140       | 27.53          | 165.60            |              | 65.66             | 0.1662      |             |
| 150       | 29.94          | 165.60            |              | 65.66             | 0.1808      |             |
| 160       | 30.86          | 165.60            |              | 65.66             | 0.1864      |             |
| 170       | 32.63          | 165.60            |              | 65.66             | 0.1970      |             |
| 180       | 33.44          | 165.60            |              | 65.66             | 0.2019      |             |
| 190       | 36.60          | 165.60            |              | 65.66             | 0.2210      |             |
| 220       | 25.21          | 165.60            |              | 93.56             | 0.1522      |             |

Analysis of Load Case 15 : IP+VO+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 98.28             | -11.52       | 116.13            |             | 0.0992      |
| 20        |                | 98.28             | -18.42       | 99.30             |             | 0.1855      |
| 30        |                | 98.28             | -18.19       | 99.30             |             | 0.1832      |
| 40        |                | 98.28             | -19.76       | 95.65             |             | 0.2066      |
| 50        |                | 98.28             | -19.53       | 95.65             |             | 0.2042      |
| 60        |                | 98.28             | -21.45       | 90.90             |             | 0.2359      |
| 70        |                | 98.28             | -21.22       | 90.90             |             | 0.2334      |
| 80        |                | 98.28             | -23.61       | 80.83             |             | 0.2921      |
| 90        |                | 91.08             | -23.54       | 80.83             |             | 0.2912      |
| 100       | 17.57          | 165.60            |              | 97.46             | 0.1061      |             |
| 110       | 25.03          | 165.60            |              | 75.71             | 0.1511      |             |
| 120       | 25.43          | 165.60            |              | 75.71             | 0.1536      |             |
| 130       | 31.61          | 165.60            |              | 65.66             | 0.1909      |             |
| 140       | 33.12          | 165.60            |              | 65.66             | 0.2000      |             |
| 150       | 35.70          | 165.60            |              | 65.66             | 0.2156      |             |
| 160       | 36.68          | 165.60            |              | 65.66             | 0.2215      |             |
| 170       | 38.58          | 165.60            |              | 65.66             | 0.2330      |             |
| 180       | 39.42          | 165.60            |              | 65.66             | 0.2381      |             |
| 190       | 36.71          | 165.60            |              | 65.66             | 0.2217      |             |
| 220       | 25.19          | 165.60            |              | 93.56             | 0.1521      |             |

Analysis of Load Case 16 : IP+VE+EW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|





|     |       |        |        |        |        |
|-----|-------|--------|--------|--------|--------|
| 10  |       | 98.28  | -9.28  | 116.13 | 0.0799 |
| 20  |       | 98.28  | -14.76 | 99.30  | 0.1487 |
| 30  |       | 98.28  | -14.54 | 99.30  | 0.1464 |
| 40  |       | 98.28  | -15.74 | 95.65  | 0.1645 |
| 50  |       | 98.28  | -15.51 | 95.65  | 0.1621 |
| 60  |       | 98.28  | -16.98 | 90.90  | 0.1868 |
| 70  |       | 98.28  | -16.75 | 90.90  | 0.1842 |
| 80  |       | 98.28  | -18.58 | 80.83  | 0.2299 |
| 90  |       | 91.08  | -18.51 | 80.83  | 0.2290 |
| 100 | 18.28 | 165.60 |        | 97.46  | 0.1104 |
| 110 | 26.03 | 165.60 |        | 75.71  | 0.1572 |
| 120 | 26.43 | 165.60 |        | 75.71  | 0.1596 |
| 130 | 32.70 | 165.60 |        | 65.66  | 0.1975 |
| 140 | 33.88 | 165.60 |        | 65.66  | 0.2046 |
| 150 | 36.36 | 165.60 |        | 65.66  | 0.2195 |
| 160 | 37.17 | 165.60 |        | 65.66  | 0.2245 |
| 170 | 38.74 | 165.60 |        | 65.66  | 0.2340 |
| 180 | 39.59 | 165.60 |        | 65.66  | 0.2391 |
| 190 | 36.71 | 165.60 |        | 65.66  | 0.2217 |
| 220 | 25.19 | 165.60 |        | 93.56  | 0.1521 |

Analysis of Load Case 17 : NP+VO+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 98.28             | -11.52       | 116.13            |             | 0.0992      |
| 20        |                | 98.28             | -18.42       | 99.30             |             | 0.1855      |
| 30        |                | 98.28             | -18.19       | 99.30             |             | 0.1832      |
| 40        |                | 98.28             | -19.76       | 95.65             |             | 0.2066      |
| 50        |                | 98.28             | -19.53       | 95.65             |             | 0.2042      |
| 60        |                | 98.28             | -21.45       | 90.90             |             | 0.2359      |
| 70        |                | 98.28             | -21.22       | 90.90             |             | 0.2334      |
| 80        |                | 98.28             | -23.61       | 80.83             |             | 0.2921      |
| 90        |                | 91.08             | -23.54       | 80.83             |             | 0.2912      |
| 100       |                | 165.60            | -14.72       | 97.46             |             | 0.1511      |
| 110       |                | 165.60            | -20.22       | 75.71             |             | 0.2670      |
| 120       |                | 165.60            | -19.81       | 75.71             |             | 0.2617      |
| 130       |                | 165.60            | -20.27       | 65.66             |             | 0.3086      |
| 140       |                | 165.60            | -18.75       | 65.66             |             | 0.2855      |
| 150       |                | 165.60            | -16.14       | 65.66             |             | 0.2457      |
| 160       |                | 165.60            | -15.12       | 65.66             |             | 0.2303      |
| 170       |                | 165.60            | -13.22       | 65.66             |             | 0.2013      |
| 180       | 0.50           | 165.60            | -12.31       | 65.66             | 0.0030      | 0.1875      |
| 190       |                | 165.60            | -2.79        | 65.66             |             | 0.0425      |
| 220       |                | 165.60            | -1.54        | 93.56             |             | 0.0165      |

Analysis of Load Case 18 : FS+BS+IP+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 81.90             | -14.33       | 96.77             |             | 0.1481      |
| 20        |                | 81.90             | -22.83       | 82.75             |             | 0.2759      |
| 30        |                | 81.90             | -22.32       | 82.75             |             | 0.2697      |
| 40        |                | 81.90             | -23.99       | 79.71             |             | 0.3009      |
| 50        |                | 81.90             | -23.45       | 79.71             |             | 0.2942      |
| 60        |                | 81.90             | -25.45       | 75.75             |             | 0.3359      |
| 70        |                | 81.90             | -24.87       | 75.75             |             | 0.3283      |
| 80        |                | 81.90             | -27.33       | 67.36             |             | 0.4058      |
| 90        |                | 75.90             | -27.13       | 67.36             |             | 0.4027      |
| 100       | 20.74          | 138.00            |              | 81.21             | 0.1503      |             |
| 110       | 29.43          | 138.00            |              | 63.10             | 0.2133      |             |
| 120       | 29.31          | 138.00            |              | 63.10             | 0.2124      |             |
| 130       | 35.48          | 138.00            |              | 54.72             | 0.2571      |             |
| 140       | 36.46          | 138.00            |              | 54.72             | 0.2642      |             |
| 150       | 38.46          | 138.00            |              | 54.72             | 0.2787      |             |
| 160       | 38.92          | 138.00            |              | 54.72             | 0.2820      |             |
| 170       | 40.27          | 138.00            |              | 54.72             | 0.2918      |             |
| 180       | 40.60          | 138.00            |              | 54.72             | 0.2942      |             |
| 190       | 37.29          | 138.00            |              | 54.72             | 0.2702      |             |



220 25.41 138.00 77.97 0.1841

Analysis of Load Case 19 : FS+BS+EP+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 81.90             | -14.33       | 96.99             |             | 0.1477      |
| 20        |                | 81.90             | -22.83       | 82.90             |             | 0.2754      |
| 30        |                | 81.90             | -22.32       | 82.90             |             | 0.2692      |
| 40        |                | 81.90             | -23.99       | 79.84             |             | 0.3004      |
| 50        |                | 81.90             | -23.45       | 79.84             |             | 0.2937      |
| 60        |                | 81.90             | -25.45       | 75.80             |             | 0.3357      |
| 70        |                | 81.90             | -24.87       | 75.80             |             | 0.3281      |
| 80        |                | 81.90             | -27.33       | 67.40             |             | 0.4055      |
| 90        |                | 75.90             | -27.13       | 67.36             |             | 0.4027      |
| 100       |                | 138.00            | -24.70       | 81.21             |             | 0.3042      |
| 110       |                | 138.00            | -34.19       | 63.10             |             | 0.5418      |
| 120       |                | 138.00            | -33.46       | 63.10             |             | 0.5303      |
| 130       |                | 138.00            | -35.62       | 54.72             |             | 0.6510      |
| 140       |                | 138.00            | -33.66       | 54.72             |             | 0.6152      |
| 150       |                | 138.00            | -30.64       | 54.72             |             | 0.5599      |
| 160       |                | 138.00            | -29.17       | 54.72             |             | 0.5330      |
| 170       |                | 138.00            | -26.83       | 54.72             |             | 0.4904      |
| 180       |                | 138.00            | -25.47       | 54.72             |             | 0.4654      |
| 190       |                | 138.00            | -15.56       | 54.72             |             | 0.2844      |
| 220       |                | 138.00            | -10.14       | 77.97             |             | 0.1301      |

Analysis of Load Case 20 : BL+IP+OW

| From Node | Tensile Stress | All. Tens. Stress | Comp. Stress | All. Comp. Stress | Tens. Ratio | Comp. Ratio |
|-----------|----------------|-------------------|--------------|-------------------|-------------|-------------|
| 10        |                | 98.28             | -11.52       | 116.13            |             | 0.0992      |
| 20        |                | 98.28             | -18.42       | 99.30             |             | 0.1855      |
| 30        |                | 98.28             | -18.19       | 99.30             |             | 0.1832      |
| 40        |                | 98.28             | -19.76       | 95.65             |             | 0.2066      |
| 50        |                | 98.28             | -19.53       | 95.65             |             | 0.2042      |
| 60        |                | 98.28             | -21.45       | 90.90             |             | 0.2359      |
| 70        |                | 98.28             | -21.22       | 90.90             |             | 0.2334      |
| 80        |                | 98.28             | -23.61       | 80.83             |             | 0.2921      |
| 90        |                | 91.08             | -23.54       | 80.83             |             | 0.2912      |
| 100       | 17.57          | 165.60            |              | 97.46             | 0.1061      |             |
| 110       | 25.03          | 165.60            |              | 75.71             | 0.1511      |             |
| 120       | 25.43          | 165.60            |              | 75.71             | 0.1536      |             |
| 130       | 31.61          | 165.60            |              | 65.66             | 0.1909      |             |
| 140       | 33.12          | 165.60            |              | 65.66             | 0.2000      |             |
| 150       | 35.70          | 165.60            |              | 65.66             | 0.2156      |             |
| 160       | 36.68          | 165.60            |              | 65.66             | 0.2215      |             |
| 170       | 38.58          | 165.60            |              | 65.66             | 0.2330      |             |
| 180       | 39.42          | 165.60            |              | 65.66             | 0.2381      |             |
| 190       | 36.71          | 165.60            |              | 65.66             | 0.2217      |             |
| 220       | 25.19          | 165.60            |              | 93.56             | 0.1521      |             |

Absolute Maximum of the all of the Stress Ratio's 0.7879

Governing Element: Shell-3

Governing Load Case 10 : EP+OW+EQ+FS+BS

| Element Description | Long Pressure Stress | Weight + Axial Load Stress | Eccentric Moment Stress | Wind or Earthquake Stress | Combined Stress | Which Side |
|---------------------|----------------------|----------------------------|-------------------------|---------------------------|-----------------|------------|
| Skirt-9             | ...                  | -8.9                       | 2.4                     | 25.7                      | 19.2            | + Tens     |
| Skirt-9             | ...                  | -8.9                       | -2.4                    | -25.7                     | -37.0           | - Comp     |
| Skirt-8             | ...                  | -14.2                      | 3.9                     | 39.7                      | 29.5            | + Tens     |
| Skirt-8             | ...                  | -14.2                      | -3.9                    | -39.7                     | -57.8           | - Comp     |
| Skirt-7             | ...                  | -14.0                      | 3.9                     | 36.4                      | 26.4            | + Tens     |
| Skirt-7             | ...                  | -14.0                      | -3.9                    | -36.4                     | -54.3           | - Comp     |



|              |       |       |      |       |       |        |
|--------------|-------|-------|------|-------|-------|--------|
| Skirt-6      | ...   | -15.1 | 4.3  | 36.3  | 25.5  | + Tens |
| Skirt-6      | ...   | -15.1 | -4.3 | -36.3 | -55.8 | - Comp |
| Skirt-5      | ...   | -14.9 | 4.3  | 32.6  | 22.1  | + Tens |
| Skirt-5      | ...   | -14.9 | -4.3 | -32.6 | -51.9 | - Comp |
| Skirt-4      | ...   | -16.3 | 4.8  | 32.2  | 20.7  | + Tens |
| Skirt-4      | ...   | -16.3 | -4.8 | -32.2 | -53.3 | - Comp |
| Skirt-3      | ...   | -16.1 | 4.8  | 28.1  | 16.9  | + Tens |
| Skirt-3      | ...   | -16.1 | -4.8 | -28.1 | -49.0 | - Comp |
| Skirt-2      | ...   | -17.8 | 5.4  | 27.1  | 14.7  | + Tens |
| Skirt-2      | ...   | -17.8 | -5.4 | -27.1 | -50.3 | - Comp |
| Skirt-1      | ...   | -17.8 | 5.4  | 25.6  | 13.2  | + Tens |
| Skirt-1      | ...   | -17.8 | -5.4 | -25.6 | -48.8 | - Comp |
| Btm Dish End | -7.4  | -10.3 | 4.1  | 18.3  | 4.7   | + Tens |
| Btm Dish End | -7.4  | -10.3 | -4.1 | -18.3 | -40.1 | - Comp |
| Shell-1      | -10.3 | -14.1 | 5.8  | 25.6  | 6.9   | + Tens |
| Shell-1      | -10.3 | -14.1 | -5.8 | -25.6 | -55.7 | - Comp |
| Shell-2      | -10.3 | -13.8 | 5.8  | 21.1  | 2.8   | + Tens |
| Shell-2      | -10.3 | -13.8 | -5.8 | -21.1 | -51.0 | - Comp |
| Shell-3      | -11.8 | -13.6 | 6.5  | 19.8  | 0.8   | + Tens |
| Shell-3      | -11.8 | -13.6 | -6.5 | -19.8 | -51.7 | - Comp |
| Shell-4      | -11.8 | -12.1 | 6.5  | 15.7  | ...   | + Tens |
| Shell-4      | -11.8 | -12.1 | -6.5 | -15.7 | -46.1 | - Comp |
| Shell-5      | -11.8 | -9.6  | 6.5  | 12.0  | ...   | + Tens |
| Shell-5      | -11.8 | -9.6  | -6.5 | -12.0 | -39.9 | - Comp |
| Shell-6      | -11.8 | -8.7  | 6.4  | 8.7   | ...   | + Tens |
| Shell-6      | -11.8 | -8.7  | -6.4 | -8.7  | -35.6 | - Comp |
| Shell-7      | -11.8 | -6.8  | 6.4  | 5.7   | ...   | + Tens |
| Shell-7      | -11.8 | -6.8  | -6.4 | -5.7  | -30.8 | - Comp |
| Shell-8      | -11.8 | -6.0  | 6.4  | 3.2   | ...   | + Tens |
| Shell-8      | -11.8 | -6.0  | -6.4 | -3.2  | -27.4 | - Comp |
| Shell-9      | -11.8 | -2.7  | 0.3  | 1.3   | ...   | + Tens |
| Shell-9      | -11.8 | -2.7  | -0.3 | -1.3  | -16.1 | - Comp |
| Top Dish End | -8.1  | -1.6  | 0.1  | 0.4   | ...   | + Tens |
| Top Dish End | -8.1  | -1.6  | -0.1 | -0.4  | -10.2 | - Comp |



### Skirt Support Analysis Per EN 13445-3 Section 16.12:

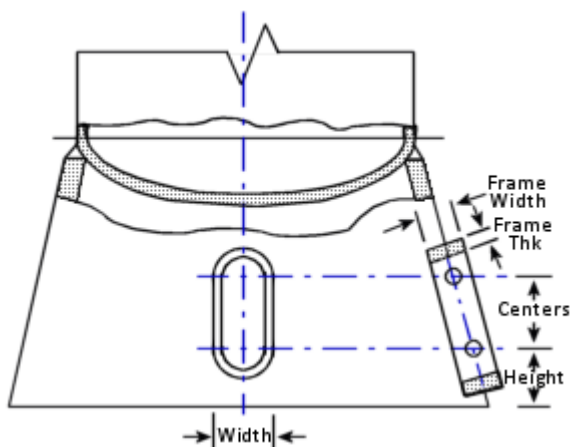
*The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.*

Item: Skirt-1

|                                       |         |                      |
|---------------------------------------|---------|----------------------|
| Total vessel (empty) weight           | Fg      | 4150788.500 N        |
| Head Weight below section 2-2         | DeltaFg | 115078.992 N         |
| Weight of vessel contents (liquid)    | Ff      | 1217193.500 N        |
| Vessel Gross Weight                   | Fg + Ff | 5367982.000 N        |
| Skirt Mean Diameter                   | Dz      | 5917.000 mm          |
| Skirt Analysis Thickness (Corroded)   | ez      | 16.000 mm            |
| Skirt Height                          | H       | 1000.000 mm          |
| Additional Axial Force at section 1-1 | F1      | -61874.996 N         |
| Moment at section 1-1                 | M1      | 10571771904.000 N-mm |
| Applied Moment - Bottom of Skirt      | Mb      | 11231203328.000 N-mm |
| Weight at Bottom of Skirt             | W       | 5390586.000 N        |

### Structural Type B Skirt per EN 13445 Figure 16.12-2:

|                                 |    |               |
|---------------------------------|----|---------------|
| Type of Head:                   |    | Torispherical |
| Head Mean Diameter at Skirt Top | Db | 5929.000 mm   |
| Head Corroded Thickness         | eb | 21.000 mm     |



There are no Access Openings in the Skirt - Plain Skirt Considered

|                    |       |                                   |
|--------------------|-------|-----------------------------------|
| Moment of Inertia  | Ixx = | 1301620064256.000 mm <sup>4</sup> |
| Moment of Inertia  | Iyy = | 1301620064256.000 mm <sup>4</sup> |
| Section Modulus    | W =   | 439959840.000 mm <sup>3</sup>     |
| Section Metal Area | A =   | 297420.906 mm <sup>2</sup>        |

### Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:  
 $= \sqrt{\min(I_{xx}, I_{yy}) / \text{Area}}$   
 $= \sqrt{\min(0.13016\text{E}+13, 0.13016\text{E}+13) / 297420.906}$   
 $= 2091.976 \text{ mm}$

Skirt Slenderness Ratio [rg/L]:  
 $= rg / L$   
 $= 2091.976 / 1000.0$   
 $= 2.092$

Correct the Moment [M4]:  
 top of skirt is the critical end  
 Controlling Moment [M4] = 0.10572E+11 N-mm



Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -0.10572E+11 / 1.17598E+10 - 5390586 / 297420.90625$$

$$= -24.131 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +.10572E+11 / 1.17598E+10 - 5390586 / 297420.90625$$

$$= -12.121 \text{ MPa}$$

#### 16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= -61874.996 - 4150789 - 1217194 + 4 \cdot 0.10572E+11 / 5917.0$$

$$= 1713956 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= -61874.996 - 4150789 - 1217194 - 4 \cdot 0.10572E+11 / 5917.0$$

$$= -12573671.0 \text{ N}$$

##### 16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [ $\sigma_{1pm}$ ]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (1713956 + 115078.992 + 1217194) / (\pi \cdot 5929.0 \cdot 21.0) + 0.343 \cdot 5929.0 / (4 \cdot 21.0)$$

$$= 32.02 \text{ MPa}$$

Stress [ $\sigma_{1qm}$ ]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-0.1E+08 + 115078.992 + 1217194) / (\pi \cdot 5929.0 \cdot 21.0) + 0.343 \cdot 5929.0 / (4 \cdot 21.0)$$

$$= -4.51 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [ $\sigma_{2pm}$  &  $\sigma_{2qm}$ ]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (1217194 + 115078.992) / (\pi \cdot 5929.0 \cdot 21.0) + 0.343 \cdot 5929.0 / (4 \cdot 21.0)$$

$$= 27.63 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [ $\sigma_{3pm}$ ]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 1713956 / (\pi \cdot 5917.0 \cdot 16.0)$$

$$= 5.76 \text{ MPa}$$

Stress [ $\sigma_{3qm}$ ]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -0.1E+08 / (\pi \cdot 5917.0 \cdot 16.0)$$

$$= -42.28 \text{ MPa}$$

##### 16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos( $\gamma$ )]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(5929.0 + 21.0 - 5917.0 + 16.0) / (357.94 + 21.0)$$

$$= 0.9353$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.935) = 20.716 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(21.0^2 + 16.0^2 + 2 \cdot 21.0 \cdot 16.0 \cdot 0.935)}$$

$$= 18.204 \text{ mm}$$



Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 18.20 \cdot 1713956.1 = 31.2136 \times 10^6 \text{ N-mm} \\ M_q &= a \cdot F_{zq} = 18.20 \cdot -12573671.0 = -2.3084 \times 10^8 \text{ N-mm} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left( \frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left( \frac{21.0}{16.0} \right)^2 \\ &= 0.532 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [ $\sigma_{1pb(o)}$  &  $\sigma_{2pb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.532 \cdot 6 \cdot 31213634 / (\pi \cdot 5929.0 \cdot 21.0^2) \\ &= 12.12 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{1qb(o)}$  &  $\sigma_{2qb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.532 \cdot 6 \cdot -0.2E+09 / (\pi \cdot 5929.0 \cdot 21.0^2) \\ &= -88.92 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{3pb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.532 \cdot 6 \cdot 31213634 / (\pi \cdot 5917.0 \cdot 16.0^2) \\ &= 20.92 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{3qb(o)}$ ]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.532 \cdot 6 \cdot -0.2E+09 / (\pi \cdot 5917.0 \cdot 16.0^2) \\ &= -153.49 \text{ MPa} \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/D_b &= 0.827/233.425 = 0.0035 \\ y &= 125 \cdot eb/D_b = 125 \cdot 21.0/5929.0 = 0.4427 \end{aligned}$$

Kloepper (torispherical) type head:

Geometric value [ $\alpha$ ]:

$$\begin{aligned} &= 6.37181 \cdot 2.71828 - 16.1 \cdot y + 3.6366 \cdot 2.71828 - 1.61536 \cdot y + 6.6736 \\ &= 6.37181 \cdot 2.71828 - 16.1 \cdot 0.443 + 3.6366 \cdot 2.71828 - 1.61536 \cdot 0.443 + 6.6736 \\ &= 8.4574 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [ $\sigma_{1b(p)}$  &  $\sigma_{2b(p)}$ ]:

$$\begin{aligned} &= (P + Ph) \cdot D_b / ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb) \\ &= (0.343 + 0.0) \cdot 5929.0 / ((20.72/40.0) \cdot 8.457 - 1) / (4 \cdot 21.0) \\ &= 81.89 \text{ MPa} \end{aligned}$$

#### 16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [ $\sigma_{1pitot}$ ]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 32.016 - 12.121 + 81.889 \\ &= 101.78 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{1potot}$ ]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 32.016 + 12.121 - 81.889 \\ &= -37.75 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{1qitot}$ ]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= -4.514 - 88.921 + 81.889 \\ &= 166.30 \text{ MPa} \end{aligned}$$

Stress [ $\sigma_{1qotot}$ ]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= -4.514 + 88.921 - 81.889 \end{aligned}$$



= -175.32 MPa

Stress [ $\sigma_{2pitot}$ ]:  
=  $\sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$   
= 27.634 + 12.121 + 81.889  
= 121.64 MPa

Stress [ $\sigma_{2potot}$ ]:  
=  $\sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$   
= 27.634 - 12.121 - 81.889  
= -66.38 MPa

Stress [ $\sigma_{2qitot}$ ]:  
=  $\sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$   
= 27.634 + -88.921 + 81.889  
= 20.60 MPa

Stress [ $\sigma_{2qotot}$ ]:  
=  $\sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$   
= 27.634 - -88.921 - 81.889  
= 34.67 MPa

Stress [ $\sigma_{3pitot}$ ]:  
=  $\sigma_{3pm} - \sigma_{3pb}$   
= 5.763 - 20.923  
= -15.16 MPa

Stress [ $\sigma_{3potot}$ ]:  
=  $\sigma_{3pm} + \sigma_{3pb}$   
= 5.763 + 20.923  
= 26.69 MPa

Stress [ $\sigma_{3qitot}$ ]:  
=  $\sigma_{3qm} - \sigma_{3qb}$   
= -42.279 - -153.491  
= 111.21 MPa

Stress [ $\sigma_{3qotot}$ ]:  
=  $\sigma_{3qm} + \sigma_{3qb}$   
= -42.279 + -153.491  
= -195.77 MPa

Allowable stresses:

For  $\sigma_{1pitot}$  and  $\sigma_{1potot}$ , the stress must not exceed:  
=  $fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$   
= 137.997(3 - (1/1.5)) • (32.016/137.997)<sup>2</sup>  
= 409.04 MPa

For  $\sigma_{1qitot}$  and  $\sigma_{1qotot}$ , the stress must not exceed:  
=  $fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$   
= 137.997(3 - (1/1.5)) • (-4.514/137.997)<sup>2</sup>  
= 413.89 MPa

For  $\sigma_{2pitot}$  and  $\sigma_{2potot}$ , the stress must not exceed:  
=  $fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$   
= 137.997(3 - (1/1.5)) • (27.634/137.997)<sup>2</sup>  
= 410.30 MPa

For  $\sigma_{2qitot}$  and  $\sigma_{2qotot}$ , the stress must not exceed:  
=  $fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$   
= 137.997(3 - (1/1.5)) • (27.634/137.997)<sup>2</sup>  
= 410.30 MPa

For  $\sigma_{3pitot}$  &  $\sigma_{3potot}$ , the stress must not exceed:  
=  $fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2$   
= 137.997(3 - (1/1.5)) • (5.763/137.997)<sup>2</sup>  
= 413.83 MPa

For  $\sigma_{3qitot}$  &  $\sigma_{3qotot}$ , the stress must not exceed:



$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2) \\ &= 137.997(3 - (1/1.5) \cdot (-42.279/137.997)^2) \\ &= 405.36 \text{ MPa} \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

**Summary of Stresses: (MPa)**

| Stress  | Actual  | Allowed | Result | Equation |
|---------|---------|---------|--------|----------|
| Stot1pi | 101.78  | 409.04  | Pass   | 46       |
| Stot1po | -37.75  | 409.04  | Pass   | 47       |
| Stot1qi | 166.30  | 413.89  | Pass   | 48       |
| Stot1qo | -175.32 | 413.89  | Pass   | 49       |
| Stot2pi | 121.64  | 410.30  | Pass   | 50       |
| Stot2po | -66.38  | 410.30  | Pass   | 51       |
| Stot2qi | 20.60   | 410.30  | Pass   | 52       |
| Stot2qo | 34.67   | 410.30  | Pass   | 53       |
| Stot3pi | -15.16  | 413.83  | Pass   | 54       |
| Stot3po | 26.69   | 413.83  | Pass   | 55       |
| Stot3qi | 111.21  | 405.36  | Pass   | 56       |
| Stot3qo | -195.77 | 405.36  | Pass   | 57       |

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Platform(s) installed in the Field before being lifted.

Packing is installed in the Field after being lifted.

Insulation is installed in the Shop.

Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

|                                                    |              |
|----------------------------------------------------|--------------|
| Center of Gravity of the Platforms                 | 44694.711 mm |
| Center of Gravity of the Packing                   | 34914.977 mm |
| Center of Gravity of the Liquid                    | 23730.268 mm |
| Center of Gravity of the Insulation                | 34794.336 mm |
| Center of Gravity of the Lining                    | 34800.000 mm |
| Center of Gravity of the Stiffening Rings          | 34998.496 mm |
| Center of Gravity of the Nozzles                   | 31571.014 mm |
| Center of Gravity of the Added Weights (Operating) | 37625.211 mm |
| Center of Gravity of the Added Weights (Empty)     | 38375.652 mm |
|                                                    |              |
| Center of Gravity of Bare Shell New and Cold       | 21075.260 mm |
| Center of Gravity of Bare Shell Corroded           | 20654.178 mm |
|                                                    |              |
| Vessel CG in the Operating Condition               | 30715.654 mm |
| Vessel CG in the Fabricated (Shop/Empty) Condition | 27582.920 mm |
| Vessel CG in the Test Condition                    | 32986.738 mm |

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#### Skirt Data :

|                                    |         |           |    |
|------------------------------------|---------|-----------|----|
| Skirt Outside Diameter at Base     | SOD, D6 | 5953.0000 | mm |
| Skirt Thickness                    | STHK    | 36.0000   | mm |
| Skirt Internal Corrosion Allowance | SCA     | 0.0000    | mm |
| Skirt External Corrosion Allowance |         | 0.0000    | mm |
| Skirt Material                     |         | E250BR    |    |

#### Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

|                                              |            |               |               |
|----------------------------------------------|------------|---------------|---------------|
| Thickness of Basing                          | TBA, ea4   | 66.0000       | mm            |
| Design Temperature of the Basing             |            | 38.00         | °C            |
| Basing Material                              |            | E250BR        |               |
| Basing Ope. All. Stress                      | BASOPE, f4 | 117.00        | MPa           |
| Basing Yield Stress                          |            | 230.00        | MPa           |
| Inside Diameter of Basing                    | DI, D4     | 5649.0000     | mm            |
| Outside Diameter of Basing                   | DOU, D5    | 6409.0000     | mm            |
| Nominal Diameter of Bolts                    | BND, d(b0) | 80.0000       | mm            |
| Bolt Corrosion Allowance                     | BCA        | 1.5000        | mm            |
| Bolt Material                                |            | SA-36         |               |
| Bolt Operating Allowable Stress              | SA, fB     | 138.00        | MPa           |
| Number of Bolts                              | RN, nb     | 36            |               |
| Diameter of Bolt Circle                      | DC, DBC    | 6181.0000     | mm            |
| Bolt Allowable Shear Stress                  |            | 82.800        | MPa           |
| Ultimate Comp. Strength of Concrete          | FPC        | 41.4          | MPa           |
| Allowable Comp. Strength of Concrete         | FC         | 22.5          | MPa           |
| Modular ratio Steel/Concrete                 |            | 9.833         |               |
| Thickness of Gusset Plates                   | TGA, e7    | 25.0000       | mm            |
| Width of Gussets at Top Plate                | TWDT       | 218.0000      | mm            |
| Width of Gussets at Base Plate               | BWDT       | 218.0000      | mm            |
| Gusset Plate Elastic Modulus                 | E          | 201746.0      | MPa           |
| Gusset Plate Yield Stress                    | SY         | 230.0         | MPa           |
| Height of Chair                              | h1         | 450.0000      | mm            |
| Distance between Gussets                     | RG, b6     | 140.0000      | mm            |
| Dist. from Bolt Center to Gusset (Rg/2)      | CG         | 70.0000       | mm            |
| Number of Gussets per bolt                   | NG         | 2             |               |
| Thickness of Top Plate or Ring               | TTA, e5    | 64.0000       | mm            |
| Radial Width of the Top Plate                | TOPWTH, b5 | 228.0000      | mm            |
| Anchor Bolt Hole Dia. in Top Plate           | dia,H      | 102.0000      | mm            |
| External Corrosion Allowance                 | CA         | 0.0000        | mm            |
| Dead Weight of Vessel                        | DW         | 3356197.5     | N             |
| Operating Weight of Vessel                   | ROW        | 6152565.5     | N             |
| Earthquake Moment on Basing                  | EQMOM      | 24622825472.0 | N-mm          |
| Wind Moment on Basing                        | WIMOM      | 7778143232.0  | N-mm          |
| User Moment at the Base [Earthquake Case]    |            | 2958450176.0  | N-mm (UEQMOM) |
| User Moment at the Base [WindLoad Case]      |            | 2958450176.0  | N-mm (UWIMOM) |
| Percent Bolt Preload                         | ppl        | 100.0         |               |
| Use AISC A5.2 Increase in Fc and Bolt Stress |            | No            |               |
| Use Allowable Weld Stress per AISC J2.5      |            | No            |               |
| Factor for Increase of Allowables            | Fact       | 1.0000        |               |

#### Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

#### Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:  

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.27581E+11/6181.0) - 5961562) / 36$$



= 330007.6 N

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.10737E+11/6181.0) - 3356198) / 36$$

$$= 99697.6 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/6181.0) - 6152566) / 36$$

$$= -170904.6 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 3744.3860 mm<sup>2</sup>  
 Area Available in all the Bolts Abss \* RN : 134797.8906 mm<sup>2</sup>

| Trial# | k     | knew  | Cc    | Ct    | z     | j     | Ft        | Fc        | Fs      |
|--------|-------|-------|-------|-------|-------|-------|-----------|-----------|---------|
| 2      | 0.300 | 0.451 | 1.511 | 2.441 | 0.438 | 0.781 | 2370708.0 | 8332269.5 | 29205.1 |
| 4      | 0.376 | 0.413 | 1.705 | 2.277 | 0.421 | 0.784 | 2488264.8 | 8449827.0 | 32856.2 |
| 6      | 0.394 | 0.404 | 1.751 | 2.236 | 0.417 | 0.784 | 2518687.5 | 8480249.0 | 33870.3 |
| 8      | 0.399 | 0.397 | 1.762 | 2.226 | 0.416 | 0.784 | 2526359.8 | 8487921.0 | 34131.0 |
| 10     | 0.398 | 0.397 | 1.759 | 2.229 | 0.416 | 0.784 | 2524439.0 | 8486001.0 | 34065.6 |
| 12     | 0.398 | 0.397 | 1.759 | 2.229 | 0.416 | 0.784 | 2523959.2 | 8485521.0 | 34049.3 |
| 13     | 0.397 | 0.397 | 1.758 | 2.229 | 0.416 | 0.784 | 2523719.0 | 8485280.0 | 34041.1 |

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 2523719 / (6.942 \cdot 6181.0 \cdot 2.229)$$

$$= 52.768 \text{ MPa, Should be } \leq 138.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 36 \cdot 3744.386 / (\pi \cdot 6181.0)$$

$$= 6.942 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc / [(2kd + t3) / (2kd)]$$

$$= 3.538 / [(2 \cdot 0.3974 \cdot 6181.0 + 380.0) / (2 \cdot 0.3974 \cdot 6181.0)]$$

$$= 3.812 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 228.0, b = 269.697, l/b = 0.845393

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(21625.541, 33505.938) = 33505.938 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 33505.938 / 151.8)} + 0.0$$

$$= 36.393 \text{ mm}$$

Nomenclature:

|                       |                               |
|-----------------------|-------------------------------|
| a = (Dc - Ds) / 2     | Skirt Distance to Bolt Circle |
| P = Sa * Abss         | Maximum Load on one Bolt      |
| l                     | Radial Top Plate Width        |
| lavg                  | Average Gusset Plate Width    |
| g1 = Gamma 1          | Constant Term f( b/l )        |
| g2 = Gamma 2          | Constant Term f( b/l )        |
| g = Flat distance / 2 | Nut 1/2 Dimension (from Tema) |
| Fb                    | Allowable Bending Stress      |

Values for table 10.6, l = 228.0, b = 140.0, b/l = 0.614035

As b/l ( 0.614) is less than 1, inverting b/l = 1.629.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [ 1.3 (\ln((2 \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1 ] - [(0.7 - g2)P / (4 \cdot \pi)]$$

$$= 516681 / (4 \cdot \pi) [ 1.3 (\ln((2 \cdot 228.0 \cdot \sin(\pi \cdot 114.0 / 228.0)) / (\pi \cdot 60.0))) + 1 ] - [(0.7 - 0.057) \cdot 516681 / (4 \cdot \pi)]$$

$$= 61886.5781 \text{ N}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(61886.58) / 153.33) + 0.0}$$

$$= 49.2128 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 516681 / (124.11 \cdot 218.0) + 0.0$$

$$= 19.098 \text{ (not less than } 9.525 + 0.0 \text{ ) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 6181.0 / 36 = 539.394 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76( P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}) )^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76(516681 \cdot 114.0 / (539.39 \cdot 386.0 \cdot 175))^{2/3} \cdot 2976.51^{1/3} + \text{Ca}$$

$$= 34.808 + 0.0 = 34.808 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 674885 / (3744.386 \cdot 36)$$

$$= 5.0 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

#### **Summary of Basing Thickness Calculations**

|                                              |          |    |
|----------------------------------------------|----------|----|
| Required Basing Thickness (tension)          | 36.3934  | mm |
| Actual Basing Thickness as entered by user   | 66.0000  | mm |
| Required Thickness of Chair Cap              | 49.2128  | mm |
| Actual Top Ring Thickness as entered by user | 64.0000  | mm |
| Required Gusset thickness, + CA              | 19.0984  | mm |
| Actual Gusset Thickness as entered by user   | 25.0000  | mm |
| Required Thickness of Skirt for Local Stress | 34.8083  | mm |
| Given Thickness of Skirt                     | 36.0000  | mm |
| Required Gusset Height to meet local stress  | 300.9927 | mm |

#### **Weld Size Calculations per Steel Plate Engineering Data - Vol. 2**

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 37.031 \cdot (36.0 - 0.0)$$

$$= 1333.01 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 1333 / [(0.4 \cdot 203) \cdot 2 \cdot 0.707]$$

$$= 11.611 \text{ mm}$$

#### **Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size**

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 516681 / (144.0 + 2 \cdot (320.0 + 64.0))$$

$$= 566.537 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 516681 \cdot 114.0 / (144.0 \cdot (384.0) + 0.6667 \cdot (384.0)^2)$$

$$= 383.473 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{566.54^2 + 383.47^2}$$

$$= 684.117 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$



$$= 684.12 / [(0.4 \cdot 203) \cdot 2 \cdot 0.707]$$
$$= 5.959 \text{ mm}$$

#### Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 516681 / (2 \cdot 218.0)$$
$$= 1185.049 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 516681 \cdot 114.0 / (2 \cdot 384.0 \cdot 218.0)$$
$$= 351.811 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{1185.05^2 + 351.81^2}$$
$$= 1236.168 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 1236.17 / [(0.4 \cdot 203) \cdot 2 \cdot 0.707]$$
$$= 10.767 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

#### **Summary of Required Weld Sizes:**

|                                                      |         |    |
|------------------------------------------------------|---------|----|
| Required Basing to Skirt Double Fillet Weld Size     | 11.6110 | mm |
| Required Gusset to Skirt Double Fillet Weld Size     | 6.3500  | mm |
| Required Top Plate to Skirt Weld Size                | 10.7675 | mm |
| Required Gusset to Top Plate Double Fillet Weld Size | 10.7675 | mm |

Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: N1 36"**

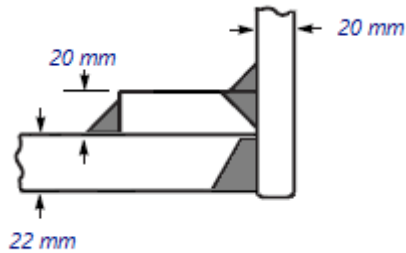
**From: 100**

|                                               |        |           |     |
|-----------------------------------------------|--------|-----------|-----|
| Pressure for Reinforcement Calculations       | P      | 0.5471    | MPa |
| Temperature for Internal Pressure             | Temp   | 150       | °C  |
| Design External Pressure                      | Pext   | 0.10      | MPa |
| Temperature for External Pressure             | Tempex | 150       | °C  |
| Parent Material                               |        | SA-516 70 |     |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | MPa |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | MPa |
| Inside Crown Radius of Torispherical Head     | L      | 5949.0000 | mm  |
| Inside Knuckle Radius of Torispherical Head   | r      | 356.9400  | mm  |
| Head Finished (Minimum) Thickness             | t      | 22.0000   | mm  |
| Head Internal Corrosion Allowance             | cas    | 1.0000    | mm  |
| Head External Corrosion Allowance             | co     | 0.0000    | mm  |
| Distance from Head Centerline                 | L1     | 0.0000    | mm  |
| User Entered Minimum Design Metal Temperature |        | -1.00     | °C  |

**Type of Element Connected to the Parent : Nozzle**

|                                               |      |                  |     |
|-----------------------------------------------|------|------------------|-----|
| Material                                      |      | SA-516 70        |     |
| Material UNS Number                           |      | K02700           |     |
| Material Specification/Type                   |      | Plate            |     |
| Allowable Stress at Temperature               | Sn   | 138.00           | MPa |
| Allowable Stress At Ambient                   | Sna  | 138.00           | MPa |
| Diameter Basis (for tr calc only)             |      | Outside          |     |
| Layout Angle                                  |      | 0.00             | deg |
| Diameter                                      |      | 36.0000          | in. |
| Size and Thickness Basis                      |      | Actual           |     |
| Actual Thickness                              | tn   | 20.0000          | mm  |
| Flange Material                               |      | SA-105           |     |
| Flange Type                                   |      | Weld Neck Flange |     |
| Corrosion Allowance                           | can  | 0.0000           | mm  |
| Joint Efficiency of Shell Seam at Nozzle      | E1   | 1.00             |     |
| Joint Efficiency of Nozzle Neck               | En   | 1.00             |     |
| Outside Projection                            | ho   | 400.0000         | mm  |
| Weld leg size between Nozzle and Pad/Shell    | Wo   | 10.0000          | mm  |
| Groove weld depth between Nozzle and Vessel   | Wgnv | 18.0000          | mm  |
| Inside Projection                             | h    | 0.0000           | mm  |
| Weld leg size, Inside Element to Shell        | Wi   | 0.0000           | mm  |
| Pad Material                                  |      | SA-516 70        |     |
| Pad Allowable Stress at Temperature           | Sp   | 138.00           | MPa |
| Pad Allowable Stress At Ambient               | Spa  | 138.00           | MPa |
| Diameter of Pad along vessel surface          | Dp   | 1050.0000        | mm  |
| Thickness of Pad                              | te   | 20.0000          | mm  |
| Weld leg size between Pad and Shell           | Wp   | 18.0000          | mm  |
| Groove weld depth between Pad and Nozzle      | Wgpn | 20.0000          | mm  |
| Reinforcing Pad Width                         |      | 67.8000          | mm  |
| Flange Class                                  |      | 150              |     |
| Flange Grade                                  |      | GR 1.1           |     |
| Flange Series                                 |      | A                |     |
| Piping Exit Angle [North Clockwise reference] |      | 180.0            | deg |
| Bend Radius Multiplier                        |      | 1.50             |     |
| Horizontal Run Length                         |      | 4250.0           | mm  |
| Centerline Distance to Tangent                |      | 2200.0           | mm  |

The Pressure Design option was MAWP + static head (bottom of element).



### Reinforcement CALCULATION, Description: N1 36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 36.000 in. |
| Actual Thickness Used in Calculation        | 0.787 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Torispherical Head, tr [Int. Press]  
 $= P \cdot L \cdot M / (2 \cdot S_v \cdot E - 0.2 \cdot P)$  Appendix 1-4 (d)  
 $= 0.55 \cdot 5950.0 \cdot 1.0 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 0.55)$   
 $= 11.7988 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.55 \cdot 457.2 / (138 \cdot 1.0 + 0.4 \cdot 0.55)$   
 $= 1.8096 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.0791 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

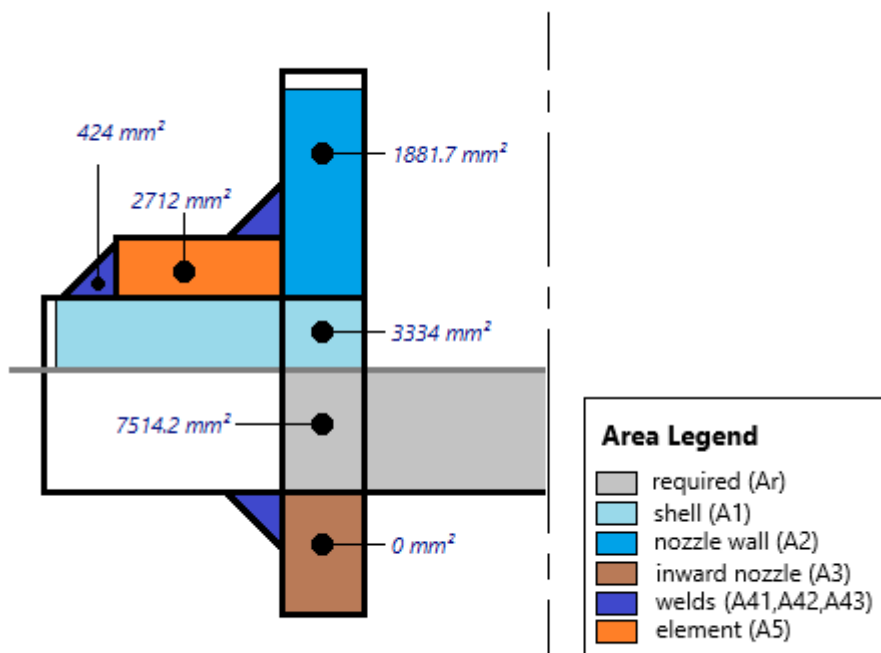
|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1748.8000 mm |
| Parallel to Vessel Wall, opening length                | d  | 874.4000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 52.5000 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External | Mapnc |
|--------------------------|-------------|-----------|----------|-------|
| Area Required            | Ar          | 10316.878 | 7514.198 | NA    |
| Area in Shell            | A1          | 8045.521  | 3334.004 | NA    |
| Area in Nozzle Wall      | A2          | 1909.989  | 1881.696 | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 424.000   | 424.000  | NA    |
| Area in Element          | A5          | 2712.001  | 2712.001 | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 13091.511 | 8351.700 | NA    |

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 1008.1249 20.0000 mm  
 Based on given Pad Diameter: 1050.0000 13.8237 mm  
 Based on Shell or Nozzle Thickness: 1008.1249 20.0000 mm

Area Required [A]:  
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$  per UG-37(d)  
 $= 0.5(874.4 \cdot 17.1871 \cdot 1 + 2 \cdot 20.0 \cdot 17.1871 \cdot 1(1 - 1.0))$   
 $= 7514.198 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 874.4(1.0 \cdot 21.0 - 1.0 \cdot 17.1871) - 2 \cdot 20.0$   
 $(1.0 \cdot 21.0 - 1.0 \cdot 17.1871) \cdot (1 - 1.0)$   
 $= 3334.004 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 52.5) \cdot (20.0 - 2.08) \cdot 1.0$





= 1881.696 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$   
= 424.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1050.0 - 914.4) \cdot 20.0 \cdot 1.0$   
= 2712.001 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 2.0791 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 2.5000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 21.8754 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 21.8754 mm |
| Wall Thickness, shell/head, external pressure  | trb2 = 4.9467 mm                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 4.9467 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.3312 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 8.331, max( 21.8754, 4.9467 ) ]  
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 2.0791, 8.3312 )  
= 8.3312 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 17.7, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 327.3 MPa | Passed |
| Occasional | : 5.8,  | Allowable | : 183.5 MPa | Passed |
| Shear      | : 9.1,  | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 20.0, tr = 1.81, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.09$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | 12 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 20.0, tr = 1.81, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.09$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -7 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 20.0, tr = 1.81, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.09$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -7 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |



Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20.0, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr * E^* / (tg - c) = 0.562$ , Temp. Reduction = 26 °C  
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 1.81, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr * E^* / (tg - c) = 0.09$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C  
 Gov. MDMT of the Reinforcement Pad : -33 °C  
 Gov. MDMT of the nozzle to shell joint welded assembly : -33 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $0.55/1.96 = 0.279$

#### Weld Size Calculations, Description: N1 36"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm  
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

#### Results Per UW-16.1:

|             | Required Thickness          | Actual Thickness         |
|-------------|-----------------------------|--------------------------|
| Nozzle Weld | 6.0000 = Min per Code       | 7.0700 = $0.7 * W_o$ mm  |
| Pad Weld    | 9.5000 = $0.5 * T_{minPad}$ | 12.7260 = $0.7 * W_p$ mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$   
 $= \max(0, (7514.1982 - 3334.0039 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 17.1871)) \cdot 138)$   
 $= 597852.44 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (1881.6959 + 2712.001 + 424.0 - 0.0 \cdot 1.0) \cdot 138$   
 $= 692370.75 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (1881.6959 + 0.0 + 100.0 + (840.0)) \cdot 138$   
 $= 389353.88 \text{ N}$

Weld Load [W3]:  
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (1881.6959 + 0.0 + 424.0 + 2712.001 + (840.0)) \cdot 138$   
 $= 808278.81 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 914.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 971168. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1050.0 \cdot 18.0 \cdot 0.49 \cdot 138$$

$$= 2007299. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 447.2) \cdot (20.0 - 0.0) \cdot 0.7 \cdot 138$$

$$= 2714075. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 914.4 \cdot 20.0 \cdot 0.74 \cdot 138$$

$$= 2933323. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 914.4 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 2639991. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2007299 + 2714075) = 4721375 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (971168 + 2933323 + 2639991 + 0) = 6544483 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (2007299 + 2639991 + 0) = 4647291 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                            |                  |
|----------------------|----------------------------|------------------|
| Path 1-1 = 4721374 N | , must exceed W = 597852 N | or W1 = 692370 N |
| Path 2-2 = 6544482 N | , must exceed W = 597852 N | or W2 = 389353 N |
| Path 3-3 = 4647290 N | , must exceed W = 597852 N | or W3 = 808278 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 17.5296 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 439.5294 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 2.236 %

#### Input Echo, WRC107/537 Item 1, Description: N1 36" :

|                                 |        |           |     |
|---------------------------------|--------|-----------|-----|
| Diameter Basis for Vessel       | Vbasis | ID        |     |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |     |
| Internal Corrosion Allowance    | Cas    | 1.0000    | mm  |
| Vessel Diameter                 | Dv     | 11898.000 | mm  |
| Vessel Thickness                | Tv     | 22.000    | mm  |
| Design Temperature              | T1     | 150.0     | C   |
| Vessel Material                 |        | SA-516 70 |     |
| Vessel UNS Number               |        | K02700    |     |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | MPa |



Vessel Hot S.I. Allowable Smh 138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |             |
|----------------------------------|--------|-------------|
| Attachment Type                  | Type   | Round       |
| WRC107 Attachment Classification | Holsol | Hollow      |
| Diameter Basis for Nozzle        | Nbasis | OD          |
| Corrosion Allowance for Nozzle   | Can    | 0.0000 mm   |
| Nozzle Diameter                  | Dn     | 914.400 mm  |
| Nozzle Thickness                 | Tn     | 20.000 mm   |
| Nozzle Material                  |        | SA-516 70   |
| Nozzle UNS Number                |        | K02700      |
| Nozzle Cold S.I. Allowable       | SNmc   | 138.00 MPa  |
| Nozzle Hot S.I. Allowable        | SNmh   | 138.00 MPa  |
| Thickness of Reinforcing Pad     | Tpad   | 20.000 mm   |
| Diameter of Reinforcing Pad      | Dpad   | 1050.000 mm |
| Design Internal Pressure         | Dp     | 0.547 MPa   |
| Include Pressure Thrust          |        | No          |

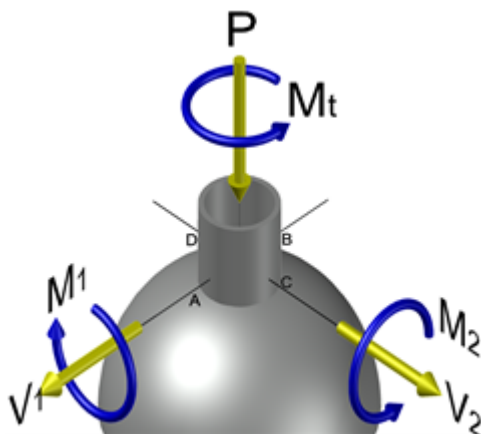
External Forces and Moments in WRC 107/537 Convention:

|                              |         |             |      |
|------------------------------|---------|-------------|------|
| Radial Load (SUS)            | P       | -67500.2    | N    |
| Longitudinal Shear (SUS)     | (V1) V1 | 45000.0     | N    |
| Circumferential Shear (SUS)  | (Vc) V2 | 45000.0     | N    |
| Circumferential Moment (SUS) | (Mc) M1 | 113399968.0 | N-mm |
| Longitudinal Moment (SUS)    | (Ml) M2 | 113399968.0 | N-mm |
| Torsional Moment (SUS)       | Mt      | 162000544.0 | N-mm |

Use Interactive Control No  
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No  
 Compute Pressure Stress per WRC-368 No  
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(5960.5 (22.0 - 1.0))}$$

$$= 2081.922 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:



Radial Load P -67500.2 N  
 Circumferential Shear (VC) V2 45000.0 N  
 Longitudinal Shear (VL) V1 45000.0 N  
 Circumferential Moment (MC) M1 113399968.0 N-mm  
 Longitudinal Moment (ML) M2 113399968.0 N-mm  
 Torsional Moment MT 162000544.0 N-mm

Dimensionless Param: U = 0.92 TAU = 22.36 RHO = 4.00 ( 2.05)

#### Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SP 7   | 0.02451 | (A,B,C,D) |
| M(x) / P                     | SP 7   | 0.01017 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SM 7   | 0.04362 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SM 7   | 0.01939 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SM 7   | 0.04362 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SM 7   | 0.01939 | (A,B,C,D) |
| N(y) * T / P                 | SP 7   | 0.21924 | (A,B,C,D) |
| M(y) / P                     | SP 7   | 0.03012 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SM 7   | 0.28746 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SM 7   | 0.07213 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SM 7   | 0.28746 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SM 7   | 0.07213 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |       |       |       |       |       |       |       |
|-----------------|------|----------------------------|-------|-------|-------|-------|-------|-------|-------|
|                 |      | Au                         | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad. Memb.      | P    | 1.0                        | 1.0   | 1.0   | 1.0   | 1.0   | 1.0   | 1.0   | 1.0   |
| Rad. Bend.      | P    | 2.5                        | -2.5  | 2.5   | -2.5  | 2.5   | -2.5  | 2.5   | -2.5  |
| Rad. Memb.      | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -5.9  | -5.9  | 5.9   | 5.9   |
| Rad. Bend.      | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -15.9 | 15.9  | 15.9  | -15.9 |
| Rad. Memb.      | ML   | -5.9                       | -5.9  | 5.9   | 5.9   | 0.0   | 0.0   | 0.0   | 0.0   |
| Rad. Bend.      | ML   | -15.9                      | 15.9  | 15.9  | -15.9 | 0.0   | 0.0   | 0.0   | 0.0   |
| Tot. Rad. Str.  |      | -18.4                      | 8.4   | 25.2  | -11.4 | -18.4 | 8.4   | 25.2  | -11.4 |
| Tang. Memb.     | P    | 8.8                        | 8.8   | 8.8   | 8.8   | 8.8   | 8.8   | 8.8   | 8.8   |
| Tang. Bend.     | P    | 7.3                        | -7.3  | 7.3   | -7.3  | 7.3   | -7.3  | 7.3   | -7.3  |
| Tang. Memb.     | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -39.2 | -39.2 | 39.2  | 39.2  |
| Tang. Bend.     | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -59.0 | 59.0  | 59.0  | -59.0 |
| Tang. Memb.     | ML   | -39.2                      | -39.2 | 39.2  | 39.2  | 0.0   | 0.0   | 0.0   | 0.0   |
| Tang. Bend.     | ML   | -59.0                      | 59.0  | 59.0  | -59.0 | 0.0   | 0.0   | 0.0   | 0.0   |
| Tot. Tang. Str. |      | -82.1                      | 21.4  | 114.2 | -18.3 | -82.1 | 21.4  | 114.2 | -18.3 |
| Shear VC        |      | 0.8                        | 0.8   | -0.8  | -0.8  | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear VL        |      | 0.0                        | 0.0   | 0.0   | 0.0   | -0.8  | -0.8  | 0.8   | 0.8   |
| Shear MT        |      | 3.0                        | 3.0   | 3.0   | 3.0   | 3.0   | 3.0   | 3.0   | 3.0   |
| Tot. Shear      |      | 3.8                        | 3.8   | 2.2   | 2.2   | 2.2   | 2.2   | 3.8   | 3.8   |
| Str. Int.       |      | 82.3                       | 22.4  | 114.3 | 18.9  | 82.2  | 21.7  | 114.4 | 19.9  |

Unitless Prm: U = 1.48 TAU = 0.00 ( 25.75) RHO = 0.00 ( 1.05)

#### Dimensionless Loads for Spherical Shells at Pad edge:

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SR 2   | 0.03874 | (A,B,C,D) |
| M(x) / P                     | SR 2   | 0.01895 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SR 3   | 0.03926 | (A,B,C,D) |



$M(x) * \sqrt{R_m * T} / MC$  SR 3 0.02922 (A,B,C,D)  
 $N(x) * T * \sqrt{R_m * T} / ML$  SR 3 0.03926 (A,B,C,D)  
 $M(x) * \sqrt{R_m * T} / ML$  SR 3 0.02922 (A,B,C,D)  
  
 $N(y) * T / P$  SR 2 0.01161 (A,B,C,D)  
 $M(y) / P$  SR 2 0.00571 (A,B,C,D)  
 $N(y) * T * \sqrt{R_m * T} / MC$  SR 3 0.01171 (A,B,C,D)  
 $M(y) * \sqrt{R_m * T} / MC$  SR 3 0.00874 (A,B,C,D)  
 $N(y) * T * \sqrt{R_m * T} / ML$  SR 3 0.01171 (A,B,C,D)  
 $M(y) * \sqrt{R_m * T} / ML$  SR 3 0.00874 (A,B,C,D)

Stress Concentration Factors:  $K_n = 1.00$ ,  $K_b = 1.00$

#### Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

| Type of Stress  | Load | Stress Intensity Values at |       |       |        |        |       |       |        |
|-----------------|------|----------------------------|-------|-------|--------|--------|-------|-------|--------|
|                 |      | Au                         | Al    | Bu    | Bl     | Cu     | Cl    | Du    | Dl     |
| Rad. Memb. P    |      | 5.9                        | 5.9   | 5.9   | 5.9    | 5.9    | 5.9   | 5.9   | 5.9    |
| Rad. Bend. P    |      | 17.4                       | -17.4 | 17.4  | -17.4  | 17.4   | -17.4 | 17.4  | -17.4  |
| Rad. Memb. MC   |      | 0.0                        | 0.0   | 0.0   | 0.0    | -28.5  | -28.5 | 28.5  | 28.5   |
| Rad. Bend. MC   |      | 0.0                        | 0.0   | 0.0   | 0.0    | -127.4 | 127.4 | 127.4 | -127.4 |
| Rad. Memb. ML   |      | -28.5                      | -28.5 | 28.5  | 28.5   | 0.0    | 0.0   | 0.0   | 0.0    |
| Rad. Bend. ML   |      | -127.4                     | 127.4 | 127.4 | -127.4 | 0.0    | 0.0   | 0.0   | 0.0    |
| Tot. Rad. Str.  |      | -132.6                     | 87.4  | 179.3 | -110.3 | -132.6 | 87.4  | 179.3 | -110.3 |
| Tang. Memb. P   |      | 1.8                        | 1.8   | 1.8   | 1.8    | 1.8    | 1.8   | 1.8   | 1.8    |
| Tang. Bend. P   |      | 5.2                        | -5.2  | 5.2   | -5.2   | 5.2    | -5.2  | 5.2   | -5.2   |
| Tang. Memb. MC  |      | 0.0                        | 0.0   | 0.0   | 0.0    | -8.5   | -8.5  | 8.5   | 8.5    |
| Tang. Bend. MC  |      | 0.0                        | 0.0   | 0.0   | 0.0    | -38.1  | 38.1  | 38.1  | -38.1  |
| Tang. Memb. ML  |      | -8.5                       | -8.5  | 8.5   | 8.5    | 0.0    | 0.0   | 0.0   | 0.0    |
| Tang. Bend. ML  |      | -38.1                      | 38.1  | 38.1  | -38.1  | 0.0    | 0.0   | 0.0   | 0.0    |
| Tot. Tang. Str. |      | -39.6                      | 26.1  | 53.6  | -33.1  | -39.6  | 26.1  | 53.6  | -33.1  |
| Shear VC        |      | 1.3                        | 1.3   | -1.3  | -1.3   | 0.0    | 0.0   | 0.0   | 0.0    |
| Shear VL        |      | 0.0                        | 0.0   | 0.0   | 0.0    | -1.3   | -1.3  | 1.3   | 1.3    |
| Shear MT        |      | 4.5                        | 4.5   | 4.5   | 4.5    | 4.5    | 4.5   | 4.5   | 4.5    |
| Tot. Shear      |      | 5.8                        | 5.8   | 3.2   | 3.2    | 3.2    | 3.2   | 5.8   | 5.8    |
| Str. Int.       |      | 132.9                      | 87.9  | 179.3 | 110.5  | 132.7  | 87.6  | 179.5 | 110.8  |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (MPa)

| Type of Stress | Load | Stress Intensity Values at |       |      |       |       |       |      |       |
|----------------|------|----------------------------|-------|------|-------|-------|-------|------|-------|
|                |      | Au                         | Al    | Bu   | Bl    | Cu    | Cl    | Du   | Dl    |
| Rad. Pm (SUS)  |      | 39.7                       | 39.7  | 39.7 | 39.7  | 39.7  | 39.7  | 39.7 | 39.7  |
| Rad. Pl (SUS)  |      | -5.0                       | -5.0  | 6.9  | 6.9   | -5.0  | -5.0  | 6.9  | 6.9   |
| Rad. Q (SUS)   |      | -13.4                      | 13.4  | 18.3 | -18.3 | -13.4 | 13.4  | 18.3 | -18.3 |
| Long. Pm (SUS) |      | 39.7                       | 39.7  | 39.7 | 39.7  | 39.7  | 39.7  | 39.7 | 39.7  |
| Long. Pl (SUS) |      | -30.4                      | -30.4 | 48.0 | 48.0  | -30.4 | -30.4 | 48.0 | 48.0  |
| Long. Q (SUS)  |      | -51.7                      | 51.7  | 66.2 | -66.2 | -51.7 | 51.7  | 66.2 | -66.2 |
| Shear Pm (SUS) |      | 0.0                        | 0.0   | 0.0  | 0.0   | 0.0   | 0.0   | 0.0  | 0.0   |
| Shear Pl (SUS) |      | 0.8                        | 0.8   | -0.8 | -0.8  | -0.8  | -0.8  | 0.8  | 0.8   |
| Shear Q (SUS)  |      | 3.0                        | 3.0   | 3.0  | 3.0   | 3.0   | 3.0   | 3.0  | 3.0   |
| Pm (SUS)       |      | 39.7                       | 39.7  | 39.7 | 39.7  | 39.7  | 39.7  | 39.7 | 39.7  |
| Pm+Pl (SUS)    |      | 34.8                       | 34.8  | 87.7 | 87.7  | 34.8  | 34.8  | 87.7 | 87.7  |



|                 |      |      |       |      |      |      |       |      |
|-----------------|------|------|-------|------|------|------|-------|------|
| Pm+Pl+Q (Total) | 64.2 | 62.1 | 154.0 | 29.0 | 63.9 | 61.4 | 154.1 | 30.0 |
|-----------------|------|------|-------|------|------|------|-------|------|

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 39.70     | 138.00         | Passed |
| Pm+Pl (SUS)         | 87.70     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 154.09    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Reinforcing Pad Edge (MPa)

| Type of Stress | Load            | Stress Intensity Values at |       |       |        |        |       |       |        |
|----------------|-----------------|----------------------------|-------|-------|--------|--------|-------|-------|--------|
|                |                 | Au                         | Al    | Bu    | Bl     | Cu     | Cl    | Du    | Dl     |
| Rad.           | Pm (SUS)        | 77.5                       | 77.5  | 77.5  | 77.5   | 77.5   | 77.5  | 77.5  | 77.5   |
| Rad.           | Pl (SUS)        | -22.6                      | -22.6 | 34.5  | 34.5   | -22.6  | -22.6 | 34.5  | 34.5   |
| Rad.           | Q (SUS)         | -110.0                     | 110.0 | 144.8 | -144.8 | -110.0 | 110.0 | 144.8 | -144.8 |
| Long.          | Pm (SUS)        | 77.5                       | 77.5  | 77.5  | 77.5   | 77.5   | 77.5  | 77.5  | 77.5   |
| Long.          | Pl (SUS)        | -6.7                       | -6.7  | 10.3  | 10.3   | -6.7   | -6.7  | 10.3  | 10.3   |
| Long.          | Q (SUS)         | -32.9                      | 32.9  | 43.4  | -43.4  | -32.9  | 32.9  | 43.4  | -43.4  |
| Shear          | Pm (SUS)        | 0.0                        | 0.0   | 0.0   | 0.0    | 0.0    | 0.0   | 0.0   | 0.0    |
| Shear          | Pl (SUS)        | 1.3                        | 1.3   | -1.3  | -1.3   | -1.3   | -1.3  | 1.3   | 1.3    |
| Shear          | Q (SUS)         | 4.5                        | 4.5   | 4.5   | 4.5    | 4.5    | 4.5   | 4.5   | 4.5    |
|                | Pm (SUS)        | 77.5                       | 77.5  | 77.5  | 77.5   | 77.5   | 77.5  | 77.5  | 77.5   |
|                | Pm+Pl (SUS)     | 70.9                       | 70.9  | 112.0 | 112.0  | 70.9   | 70.9  | 112.0 | 112.0  |
|                | Pm+Pl+Q (Total) | 93.7                       | 165.4 | 256.8 | 77.5   | 93.2   | 165.1 | 257.0 | 78.1   |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 77.50     | 138.00         | Passed |
| Pm+Pl (SUS)         | 112.03    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 257.02    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: N11 2"**

**From: 100**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5471 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                             |     |           |    |
|---------------------------------------------|-----|-----------|----|
| Inside Crown Radius of Torispherical Head   | L   | 5949.0000 | mm |
| Inside Knuckle Radius of Torispherical Head | r   | 356.9400  | mm |
| Head Finished (Minimum) Thickness           | t   | 22.0000   | mm |
| Head Internal Corrosion Allowance           | cas | 1.0000    | mm |
| Head External Corrosion Allowance           | co  | 0.0000    | mm |

|                               |    |          |    |
|-------------------------------|----|----------|----|
| Distance from Head Centerline | L1 | 650.0000 | mm |
|-------------------------------|----|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 180.00 | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Minimum |  |
| Nominal Thickness        |  | 80      |  |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

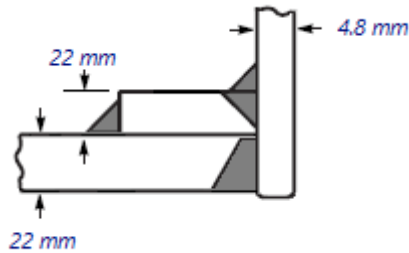
|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 22.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                          |      |               |     |
|------------------------------------------|------|---------------|-----|
| Pad Material                             |      | SA-516 70 +IT |     |
| Pad Allowable Stress at Temperature      | Sp   | 138.00        | MPa |
| Pad Allowable Stress At Ambient          | Spa  | 138.00        | MPa |
| Diameter of Pad along vessel surface     | Dp   | 160.3250      | mm  |
| Thickness of Pad                         | te   | 22.0000       | mm  |
| Weld leg size between Pad and Shell      | Wp   | 18.0000       | mm  |
| Groove weld depth between Pad and Nozzle | Wgpn | 22.0000       | mm  |
| Reinforcing Pad Width                    |      | 50.0000       | mm  |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 600    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).





### Reinforcement CALCULATION, Description: N11 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 1.993 in. |
| Actual Thickness Used in Calculation       | 0.191 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Torispherical Head, tr [Int. Press]  
 $= P \cdot L \cdot M / (2 \cdot S_v \cdot E - 0.2 \cdot P)$  Appendix 1-4 (d)  
 $= 0.55 \cdot 5950.0 \cdot 1.0 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 0.55)$   
 $= 11.7988 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.55 \cdot 25.32 / (118 \cdot 1.0 - 0.6 \cdot 0.55)$   
 $= 0.1177 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3246 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |         |             |
|--------------------------------------------------------|---------|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1      | 102.3250 mm |
| Parallel to Vessel Wall                                | Rn+tn+t | 51.1625 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |         | 34.1126 mm  |

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N11 2".  
 This calculation is valid for nozzles that meet all the requirements of  
 paragraph UG-36. Please check the Code carefully, especially for nozzles  
 that are not isolated or do not meet Code spacing requirements. To force  
 the computation of areas for small nozzles go to Tools->Configuration  
 and check the box to force the UG-37 small nozzle area calculation.*

### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 0.3246 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 2.5000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 21.8754 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 21.8754 mm |
| Wall Thickness, shell/head, external pressure  | trb2 = 4.9467 mm                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 4.9467 mm  |
| Wall Thickness per table UG-45                 | tb3 = 3.4200 mm                     |

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 3.42, \max( 21.8754, 4.9467 ) ]$   
 $= 3.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 0.3246, 3.42 )$   
 $= 3.4200 \text{ mm}$

Available Nozzle Neck Thickness = 4.8450 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,  
 (see 319.4.4 and 302.3.5):**



|              |       |             |           |        |
|--------------|-------|-------------|-----------|--------|
| Sustained :  | 42.5, | Allowable : | 118.0 MPa | Passed |
| Expansion :  | 0.0,  | Allowable : | 252.5 MPa | Passed |
| Occasional : | 1.3,  | Allowable : | 156.9 MPa | Passed |
| Shear :      | 34.8, | Allowable : | 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 4.845, tr = 0.118, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr * E* / (tg - c) = 0.024$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 4.845, tr = 0.118, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr * E* / (tg - c) = 0.024$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

*Note: This Element/Detail was specified as being Impact Tested.*

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22.0, tr = 11.799, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr * E* / (tg - c) = 0.562$ , Temp. Reduction = 26 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -5 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -30 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 4.845, tr = 0.118, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr * E* / (tg - c) = 0.024$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

|                                                        |   |         |
|--------------------------------------------------------|---|---------|
| Gov. MDMT of the Nozzle                                | : | -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : | -30 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : | -30 °C  |

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $0.55 / 10.20 = 0.054$

##### Weld Size Calculations, Description: N11 2"

|                                           |         |            |
|-------------------------------------------|---------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 4.8450 mm  |
| Intermediate Calc. for pad/shell Welds    | TminPad | 19.0000 mm |

##### Results Per UW-16.1:

|             |                          |                         |
|-------------|--------------------------|-------------------------|
|             | Required Thickness       | Actual Thickness        |
| Nozzle Weld | $3.3915 = 0.7 * t_{min}$ | $7.0700 = 0.7 * W_o$ mm |



Pad Weld  $9.5000 = 0.5 * T_{minPad} 12.7260 = 0.7 * W_p \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.0760 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 231.6258 mm

**Input Echo, WRC107/537 Item 1, Description: N11 2" :**

|                                 |        |           |     |
|---------------------------------|--------|-----------|-----|
| Diameter Basis for Vessel       | Vbasis | ID        |     |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |     |
| Internal Corrosion Allowance    | Cas    | 1.0000    | mm  |
| Vessel Diameter                 | Dv     | 11898.000 | mm  |
| Vessel Thickness                | Tv     | 22.000    | mm  |
| Design Temperature              | T1     | 150.0     | C   |
| Vessel Material                 |        | SA-516 70 |     |
| Vessel UNS Number               |        | K02700    |     |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | MPa |
| Vessel Hot S.I. Allowable       | Smh    | 138.00    | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |          |     |
|----------------------------------|--------|----------|-----|
| Attachment Type                  | Type   | Round    |     |
| WRC107 Attachment Classification | Holsol | Hollow   |     |
| Diameter Basis for Nozzle        | Nbasis | ID       |     |
| Corrosion Allowance for Nozzle   | Can    | 0.0000   | mm  |
| Nozzle Diameter                  | Dn     | 50.635   | mm  |
| Nozzle Thickness                 | Tn     | 4.845    | mm  |
| Nozzle Material                  |        | SA-106 B |     |
| Nozzle UNS Number                |        | K03006   |     |
| Nozzle Cold S.I. Allowable       | SNmc   | 118.00   | MPa |
| Nozzle Hot S.I. Allowable        | SNmh   | 118.00   | MPa |
| Thickness of Reinforcing Pad     | Tpad   | 22.000   | mm  |
| Diameter of Reinforcing Pad      | Dpad   | 160.325  | mm  |
| Design Internal Pressure         | Dp     | 0.547    | MPa |
| Include Pressure Thrust          |        | No       |     |

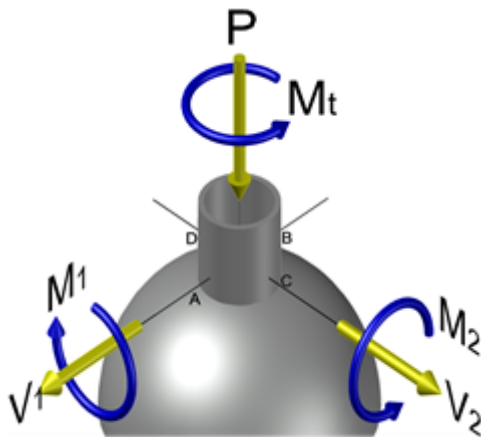
**External Forces and Moments in WRC 107/537 Convention:**

|                              |         |          |      |
|------------------------------|---------|----------|------|
| Radial Load (SUS)            | P       | -3750.0  | N    |
| Longitudinal Shear (SUS)     | (V1) V1 | 2500.0   | N    |
| Circumferential Shear (SUS)  | (Vc) V2 | 2500.0   | N    |
| Circumferential Moment (SUS) | (Mc) M1 | 350000.0 | N-mm |
| Longitudinal Moment (SUS)    | (Ml) M2 | 350000.0 | N-mm |
| Torsional Moment (SUS)       | Mt      | 500000.0 | N-mm |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



**Error/Warning Messages for WRC 107/537 Analysis: 1**

Warning - The ratio Ro/Rm, 0.0051, must be > 0.0571 and < 0.571.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(5960.5 (22.0 - 1.0))}$$

$$= 1227.847 \text{ mm}$$

WRC 107 Stress Calculation for SUSained loads:

|                        |      |    |          |      |
|------------------------|------|----|----------|------|
| Radial Load            |      | P  | -3750.0  | N    |
| Circumferential Shear  | (VC) | V2 | 2500.0   | N    |
| Longitudinal Shear     | (VL) | V1 | 2500.0   | N    |
| Circumferential Moment | (MC) | M1 | 350000.0 | N-mm |
| Longitudinal Moment    | (ML) | M2 | 350000.0 | N-mm |
| Torsional Moment       |      | MT | 500000.0 | N-mm |

Dimensionless Param: U = 0.06 TAU = 5.73 RHO = 4.00 ( 8.88)

**Dimensionless Loads for Spherical Shells at Attachment Junction:**

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SP 4   | 0.13459 | (A,B,C,D) |
| M(x) / P                     | SP 4   | 0.05253 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SM 4   | 0.91703 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SM 4   | 0.35351 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SM 4   | 0.91703 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SM 4   | 0.35351 | (A,B,C,D) |
| N(y) * T / P                 | SP 4   | 0.30479 | (A,B,C,D) |
| M(y) / P                     | SP 4   | 0.42441 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SM 4   | 0.10321 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SM 4   | 3.08833 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SM 4   | 0.10321 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SM 4   | 3.08833 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

**Stresses in the Vessel at the Attachment Junction (MPa)**

| Type of Stress | Load     | Stress Intensity Values at |      |     |      |      |      |     |      |
|----------------|----------|----------------------------|------|-----|------|------|------|-----|------|
|                |          | Au                         | Al   | Bu  | Bl   | Cu   | Cl   | Du  | Dl   |
| Rad.           | Memb. P  | 0.3                        | 0.3  | 0.3 | 0.3  | 0.3  | 0.3  | 0.3 | 0.3  |
| Rad.           | Bend. P  | 0.6                        | -0.6 | 0.6 | -0.6 | 0.6  | -0.6 | 0.6 | -0.6 |
| Rad.           | Memb. MC | 0.0                        | 0.0  | 0.0 | 0.0  | -0.3 | -0.3 | 0.3 | 0.3  |
| Rad.           | Bend. MC | 0.0                        | 0.0  | 0.0 | 0.0  | -0.8 | 0.8  | 0.8 | -0.8 |



|                 |      |      |      |       |      |      |      |       |
|-----------------|------|------|------|-------|------|------|------|-------|
| Rad. Memb. ML   | -0.3 | -0.3 | 0.3  | 0.3   | 0.0  | 0.0  | 0.0  | 0.0   |
| Rad. Bend. ML   | -0.8 | 0.8  | 0.8  | -0.8  | 0.0  | 0.0  | 0.0  | 0.0   |
| Tot. Rad. Str.  | -0.2 | 0.1  | 2.0  | -0.8  | -0.2 | 0.1  | 2.0  | -0.8  |
| Tang. Memb. P   | 0.6  | 0.6  | 0.6  | 0.6   | 0.6  | 0.6  | 0.6  | 0.6   |
| Tang. Bend. P   | 5.2  | -5.2 | 5.2  | -5.2  | 5.2  | -5.2 | 5.2  | -5.2  |
| Tang. Memb. MC  | 0.0  | 0.0  | 0.0  | 0.0   | -0.0 | -0.0 | 0.0  | 0.0   |
| Tang. Bend. MC  | 0.0  | 0.0  | 0.0  | 0.0   | -6.9 | 6.9  | 6.9  | -6.9  |
| Tang. Memb. ML  | -0.0 | -0.0 | 0.0  | 0.0   | 0.0  | 0.0  | 0.0  | 0.0   |
| Tang. Bend. ML  | -6.9 | 6.9  | 6.9  | -6.9  | 0.0  | 0.0  | 0.0  | 0.0   |
| Tot. Tang. Str. | -1.2 | 2.3  | 12.7 | -11.4 | -1.2 | 2.3  | 12.7 | -11.4 |
| Shear VC        | 0.6  | 0.6  | -0.6 | -0.6  | 0.0  | 0.0  | 0.0  | 0.0   |
| Shear VL        | 0.0  | 0.0  | 0.0  | 0.0   | -0.6 | -0.6 | 0.6  | 0.6   |
| Shear MT        | 2.0  | 2.0  | 2.0  | 2.0   | 2.0  | 2.0  | 2.0  | 2.0   |
| Tot. Shear      | 2.6  | 2.6  | 1.4  | 1.4   | 1.4  | 1.4  | 2.6  | 2.6   |
| Str. Int.       | 5.4  | 5.8  | 12.9 | 11.6  | 3.0  | 3.6  | 13.4 | 12.1  |

Unitless Prm: U = 0.23 TAU = 0.00 ( 16.05) RHO = 0.00 ( 4.33)

#### Dimensionless Loads for Spherical Shells at Pad edge:

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SR 2   | 0.21890 | (A,B,C,D) |
| M(x) / P                     | SR 2   | 0.17949 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SR 3   | 0.15974 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SR 3   | 0.66340 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SR 3   | 0.15974 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SR 3   | 0.66340 | (A,B,C,D) |
| N(y) * T / P                 | SR 2   | 0.06570 | (A,B,C,D) |
| M(y) / P                     | SR 2   | 0.05327 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SR 3   | 0.04870 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SR 3   | 0.19743 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SR 3   | 0.04870 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SR 3   | 0.19743 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

| Type of Stress  | Load | Stress Intensity Values at |      |      |       |      |      |      |       |
|-----------------|------|----------------------------|------|------|-------|------|------|------|-------|
|                 |      | Au                         | Al   | Bu   | Bl    | Cu   | Cl   | Du   | Dl    |
| Rad. Memb. P    |      | 1.9                        | 1.9  | 1.9  | 1.9   | 1.9  | 1.9  | 1.9  | 1.9   |
| Rad. Bend. P    |      | 9.2                        | -9.2 | 9.2  | -9.2  | 9.2  | -9.2 | 9.2  | -9.2  |
| Rad. Memb. MC   |      | 0.0                        | 0.0  | 0.0  | 0.0   | -0.4 | -0.4 | 0.4  | 0.4   |
| Rad. Bend. MC   |      | 0.0                        | 0.0  | 0.0  | 0.0   | -8.9 | 8.9  | 8.9  | -8.9  |
| Rad. Memb. ML   |      | -0.4                       | -0.4 | 0.4  | 0.4   | 0.0  | 0.0  | 0.0  | 0.0   |
| Rad. Bend. ML   |      | -8.9                       | 8.9  | 8.9  | -8.9  | 0.0  | 0.0  | 0.0  | 0.0   |
| Tot. Rad. Str.  |      | 1.7                        | 1.3  | 20.3 | -15.9 | 1.7  | 1.3  | 20.3 | -15.9 |
| Tang. Memb. P   |      | 0.6                        | 0.6  | 0.6  | 0.6   | 0.6  | 0.6  | 0.6  | 0.6   |
| Tang. Bend. P   |      | 2.7                        | -2.7 | 2.7  | -2.7  | 2.7  | -2.7 | 2.7  | -2.7  |
| Tang. Memb. MC  |      | 0.0                        | 0.0  | 0.0  | 0.0   | -0.1 | -0.1 | 0.1  | 0.1   |
| Tang. Bend. MC  |      | 0.0                        | 0.0  | 0.0  | 0.0   | -2.7 | 2.7  | 2.7  | -2.7  |
| Tang. Memb. ML  |      | -0.1                       | -0.1 | 0.1  | 0.1   | 0.0  | 0.0  | 0.0  | 0.0   |
| Tang. Bend. ML  |      | -2.7                       | 2.7  | 2.7  | -2.7  | 0.0  | 0.0  | 0.0  | 0.0   |
| Tot. Tang. Str. |      | 0.5                        | 0.4  | 6.0  | -4.7  | 0.5  | 0.4  | 6.0  | -4.7  |
| Shear VC        |      | 0.5                        | 0.5  | -0.5 | -0.5  | 0.0  | 0.0  | 0.0  | 0.0   |
| Shear VL        |      | 0.0                        | 0.0  | 0.0  | 0.0   | -0.5 | -0.5 | 0.5  | 0.5   |



|            |     |     |      |      |     |     |      |      |
|------------|-----|-----|------|------|-----|-----|------|------|
| Shear MT   | 0.6 | 0.6 | 0.6  | 0.6  | 0.6 | 0.6 | 0.6  | 0.6  |
| Tot. Shear | 1.1 | 1.1 | 0.1  | 0.1  | 0.1 | 0.1 | 1.1  | 1.1  |
| Str. Int.  | 2.5 | 2.3 | 20.3 | 15.9 | 1.7 | 1.3 | 20.4 | 16.0 |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |      |      |       |      |      |      |       |
|-----------------|------|----------------------------|------|------|-------|------|------|------|-------|
|                 |      | Au                         | Al   | Bu   | Bl    | Cu   | Cl   | Du   | Dl    |
| Rad. Pm (SUS)   |      | 37.9                       | 37.9 | 37.9 | 37.9  | 37.9 | 37.9 | 37.9 | 37.9  |
| Rad. Pl (SUS)   |      | -0.1                       | -0.1 | 0.6  | 0.6   | -0.1 | -0.1 | 0.6  | 0.6   |
| Rad. Q (SUS)    |      | -0.2                       | 0.2  | 1.4  | -1.4  | -0.2 | 0.2  | 1.4  | -1.4  |
| Long. Pm (SUS)  |      | 37.9                       | 37.9 | 37.9 | 37.9  | 37.9 | 37.9 | 37.9 | 37.9  |
| Long. Pl (SUS)  |      | 0.6                        | 0.6  | 0.7  | 0.7   | 0.6  | 0.6  | 0.7  | 0.7   |
| Long. Q (SUS)   |      | -1.8                       | 1.8  | 12.1 | -12.1 | -1.8 | 1.8  | 12.1 | -12.1 |
| Shear Pm (SUS)  |      | 0.0                        | 0.0  | 0.0  | 0.0   | 0.0  | 0.0  | 0.0  | 0.0   |
| Shear Pl (SUS)  |      | 0.6                        | 0.6  | -0.6 | -0.6  | -0.6 | -0.6 | 0.6  | 0.6   |
| Shear Q (SUS)   |      | 2.0                        | 2.0  | 2.0  | 2.0   | 2.0  | 2.0  | 2.0  | 2.0   |
| Pm (SUS)        |      | 37.9                       | 37.9 | 37.9 | 37.9  | 37.9 | 37.9 | 37.9 | 37.9  |
| Pm+Pl (SUS)     |      | 38.8                       | 38.8 | 39.1 | 39.1  | 38.8 | 38.8 | 39.1 | 39.1  |
| Pm+Pl+Q (Total) |      | 39.8                       | 41.9 | 50.8 | 37.2  | 38.7 | 40.9 | 51.2 | 37.7  |

##### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 37.85     | 138.00         | Passed |
| Pm+Pl (SUS)         | 39.10     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 51.21     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Reinforcing Pad Edge (MPa)

| Type of Stress | Load | Stress Intensity Values at |      |      |       |      |      |      |       |
|----------------|------|----------------------------|------|------|-------|------|------|------|-------|
|                |      | Au                         | Al   | Bu   | Bl    | Cu   | Cl   | Du   | Dl    |
| Rad. Pm (SUS)  |      | 77.5                       | 77.5 | 77.5 | 77.5  | 77.5 | 77.5 | 77.5 | 77.5  |
| Rad. Pl (SUS)  |      | 1.5                        | 1.5  | 2.2  | 2.2   | 1.5  | 1.5  | 2.2  | 2.2   |
| Rad. Q (SUS)   |      | 0.2                        | -0.2 | 18.1 | -18.1 | 0.2  | -0.2 | 18.1 | -18.1 |
| Long. Pm (SUS) |      | 77.5                       | 77.5 | 77.5 | 77.5  | 77.5 | 77.5 | 77.5 | 77.5  |
| Long. Pl (SUS) |      | 0.4                        | 0.4  | 0.7  | 0.7   | 0.4  | 0.4  | 0.7  | 0.7   |
| Long. Q (SUS)  |      | 0.1                        | -0.1 | 5.4  | -5.4  | 0.1  | -0.1 | 5.4  | -5.4  |
| Shear Pm (SUS) |      | 0.0                        | 0.0  | 0.0  | 0.0   | 0.0  | 0.0  | 0.0  | 0.0   |
| Shear Pl (SUS) |      | 0.5                        | 0.5  | -0.5 | -0.5  | -0.5 | -0.5 | 0.5  | 0.5   |
| Shear Q (SUS)  |      | 0.6                        | 0.6  | 0.6  | 0.6   | 0.6  | 0.6  | 0.6  | 0.6   |
| Pm (SUS)       |      | 77.5                       | 77.5 | 77.5 | 77.5  | 77.5 | 77.5 | 77.5 | 77.5  |
| Pm+Pl (SUS)    |      | 79.2                       | 79.2 | 79.9 | 79.9  | 79.2 | 79.2 | 79.9 | 79.9  |



|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Pm+Pl+Q (Total) | 79.9 | 79.5 | 97.8 | 72.8 | 79.3 | 78.8 | 97.9 | 72.9 |
|-----------------|------|------|------|------|------|------|------|------|

**Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 77.50     | 138.00         | Passed |
| Pm+Pl (SUS)         | 79.86     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 97.89     | 413.99         | Passed |

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.*

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**Input, Nozzle Desc: K1a**

**From: 110**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5371 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1332.0203 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |        |    |
|-----------------------------------|--|--------|----|
| Distance from Bottom/Left Tangent |  | 200.00 | mm |
|-----------------------------------|--|--------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

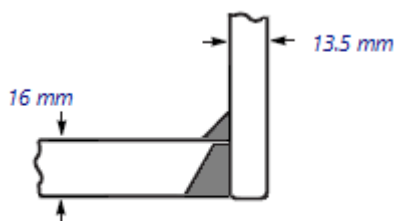
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K1a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45





Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.54 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.54)$   
= 11.5244 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.54 / (138.0 \cdot 1.0)] - 1))$   
= 0.0991 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 107.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1a.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 12.5244 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 12.5244 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 12.5244, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.099, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.54/1.96 = 0.274

Weld Size Calculations, Description: K1a

Intermediate Calc. for nozzle/shell Welds Tmin 13.4620 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: K2a**

**From: 110**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5371 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1332.0203 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |        |    |
|-----------------------------------|--|--------|----|
| Distance from Bottom/Left Tangent |  | 200.00 | mm |
|-----------------------------------|--|--------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 90.00  | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

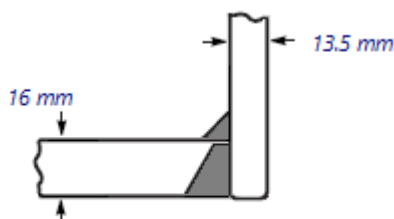
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K2a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.54 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.54)$   
= 11.5244 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.54 / (138.0 \cdot 1.0)] - 1))$   
= 0.0991 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2a.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 12.5244 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 12.5244 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 12.5244, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.099, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.54/1.96 = 0.274

Weld Size Calculations, Description: K2a

Intermediate Calc. for nozzle/shell Welds Tmin 13.4620 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: K3a**

**From: 110**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5371 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1332.0203 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |        |    |
|-----------------------------------|--|--------|----|
| Distance from Bottom/Left Tangent |  | 200.00 | mm |
|-----------------------------------|--|--------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

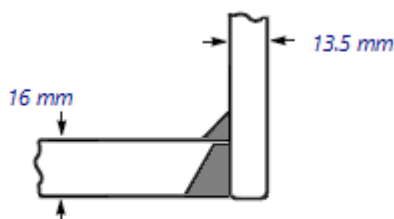
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K3a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.54 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.54)$   
= 11.5244 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.54 / (138.0 \cdot 1.0)] - 1))$   
= 0.0991 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3a.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 12.5244 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 12.5244 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 12.5244, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.099, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.54/1.96 = 0.274

Weld Size Calculations, Description: K3a

Intermediate Calc. for nozzle/shell Welds    Tmin    13.4620    mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: K4a**

**From: 110**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5371 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1332.0203 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |        |    |
|-----------------------------------|--|--------|----|
| Distance from Bottom/Left Tangent |  | 200.00 | mm |
|-----------------------------------|--|--------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

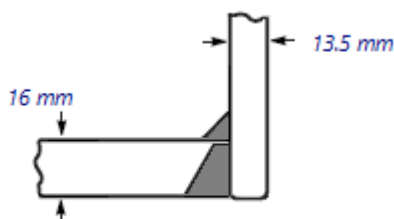
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K4a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.54 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.54)$   
= 11.5244 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.54 / (138.0 \cdot 1.0)] - 1))$   
= 0.0991 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4a.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 12.5244 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 12.5244 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 12.5244, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.099, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.54/1.96 = 0.274

Weld Size Calculations, Description: K4a

Intermediate Calc. for nozzle/shell Welds    Tmin    13.4620    mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: N9**

**From: 110**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5371 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1332.0203 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |        |    |
|-----------------------------------|--|--------|----|
| Distance from Bottom/Left Tangent |  | 200.00 | mm |
|-----------------------------------|--|--------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 315.00 | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

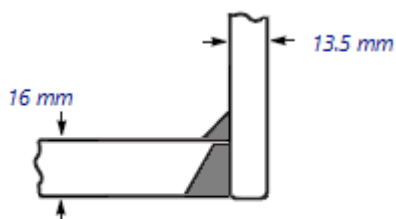
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: N9**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.54 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.54)$   
= 11.5244 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.54 / (138.0 \cdot 1.0)] - 1))$   
= 0.0991 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: N9.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 12.5244 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 12.5244 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 12.5244, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |           |        |
|------------|---|-------|-----------|---|-----------|--------|
| Sustained  | : | 12.2, | Allowable | : | 138.0 MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 332.8 MPa | Passed |
| Occasional | : | 0.4,  | Allowable | : | 183.5 MPa | Passed |
| Shear      | : | 12.5, | Allowable | : | 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:



### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.099, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\*/(tg - c) = 0.007, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -19 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 0.54/1.96 = 0.274

### Weld Size Calculations, Description: N9

Intermediate Calc. for nozzle/shell Welds Tmin 13.4620 mm

### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

### Input Echo, WRC297 Item 1, Description: N9 :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 16.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |         |            |
|------------------------------------|---------|------------|
| Diameter Basis for Nozzle          | ID      |            |
| Nozzle Corrosion Allowance         | 0.0000  | mm         |
| Nozzle Diameter                    | 50.800  | mm         |
| Nozzle Thickness                   | 13.4620 | mm         |
| Nozzle Stress Concentration Factor | 1.000   |            |
| Nozzle Material                    | SA-105  |            |
| Nozzle Cold S.I. Allowable         | SNmc    | 138.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh    | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

### Note:

External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUStained loads.

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.54      | MPa  |
| Radial Load              | P  | -3750.00  | N    |
| Circumferential Shear    | Vc | 2500.00   | N    |
| Longitudinal Shear       | Vl | 2500.00   | N    |
| Circumferential Moment   | Mc | 350000.00 | N-mm |
| Longitudinal Moment      | Ml | 350000.00 | N-mm |



Torsional Moment Mt 500000.00 N-mm

Include Axial Pressure Thrust No  
 Include Pressure Stress Indices per Div. 2 No  
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tv, (4.858) is < 5  
 check the Limitations of WRC bulletin 297.  
 Warning - The ratio, Dn/Tn (5.774) is < 10  
 check the Limitations of WRC bulletin 297.*

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(2961.5(16.0 - 1.0))}$   
 $= 773.253 \text{ mm}$

### WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5923.000 mm  
 Nozzle Outside Diameter (d) = 77.724 mm  
 Vessel Thickness used (T) = 15.000 mm  
 Nozzle Thickness used (t) = 13.462 mm  
 T / t = 1.114  
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.261$$

Nr/P = 0.227  
 Mr/P = 0.360  
 M0/P = 0.137  
 N0/P = 0.202  
 MrD/Mc = 0.288  
 NrDL/Mc = 0.148  
 M0d/Mc = 0.128  
 N0DL/Mc = -0.126  
 MrD/Ml = 0.280  
 NrDL/Ml = 0.151  
 M0D/Ml = 0.123  
 N0DL/Ml = -0.071

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 3                    | 3                   | 3                       | 3                      |
| Outplane Bending (P )                                            | 13                   | -13                 | 13                      | -13                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | 1                    | 1                   | -1                      | -1                     |
| Outplane Bending (ML)                                            | -14                  | 14                  | 14                      | -14                    |
| Normal Pressure Stress                                           | 105                  | 106                 | 105                     | 106                    |
| Outplane Stress Summary                                          | 109                  | 111                 | 135                     | 79                     |

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |
| Inplane Bending (P )                                               | 35                   | -35                 | 35                      | -35                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -3                   | -3                  | 3                       | 3                      |
| Inplane Bending (ML)                                               | -33                  | 33                  | 33                      | -33                    |
| Inplane Pressure Stress                                            | 52                   | 52                  | 52                      | 52                     |



Inplane Stress Summary 55 51 129 -10

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE                          |  | Au          | Al         | Bu             | B1            |
|---------------------------------------------|--|-------------|------------|----------------|---------------|
| (Shear stress normal to longitudinal plane) |  | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Shear (Vc)                         |  | 1           | 1          | -1             | -1            |
| Outplane Shear (Vl)                         |  | 0           | 0          | 0              | 0             |
| Outplane Shear (Mt)                         |  | 3           | 3          | 3              | 3             |

Shear Stress Summary 4 4 2 2

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE     |  | Au          | Al         | Bu             | B1            |
|------------------------|--|-------------|------------|----------------|---------------|
| (Stress Intensities)   |  | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Two * Max Shear Stress |  | 109         | 112        | 136            | 89            |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                      |  | Cu           | Cl          | Du            | D1           |
|--------------------------------------------|--|--------------|-------------|---------------|--------------|
| (Stresses Normal to circumferential plane) |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Membrane (P )                     |  | 3            | 3           | 3             | 3            |
| Outplane Bending (P )                      |  | 13           | -13         | 13            | -13          |
| Outplane Membrane (Mc)                     |  | 2            | 2           | -2            | -2           |
| Outplane Bending (Mc)                      |  | -15          | 15          | 15            | -15          |
| Outplane Membrane (ML)                     |  | 0            | 0           | 0             | 0            |
| Outplane Bending (ML)                      |  | 0            | 0           | 0             | 0            |
| Normal Pressure Stress                     |  | 52           | 52          | 52            | 52           |

Outplane Stress Summary 56 60 82 24

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                        |  | Cu           | Cl          | Du            | D1           |
|----------------------------------------------|--|--------------|-------------|---------------|--------------|
| (Stresses parallel to circumferential plane) |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Inplane Membrane (P )                        |  | 3            | 3           | 3             | 3            |
| Inplane Bending (P )                         |  | 35           | -35         | 35            | -35          |
| Inplane Membrane (Mc)                        |  | -2           | -2          | 2             | 2            |
| Inplane Bending (Mc)                         |  | -34          | 34          | 34            | -34          |
| Inplane Membrane (ML)                        |  | 0            | 0           | 0             | 0            |
| Inplane Bending (ML)                         |  | 0            | 0           | 0             | 0            |
| Inplane Pressure Stress                      |  | 105          | 106         | 105           | 106          |

Inplane Stress Summary 107 105 182 42

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                          |  | Cu           | Cl          | Du            | D1           |
|------------------------------------------------|--|--------------|-------------|---------------|--------------|
| (Shear stress normal to circumferential plane) |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Shear (Vc)                            |  | 0            | 0           | 0             | 0            |
| Outplane Shear (Vl)                            |  | -1           | -1          | 1             | 1            |
| Torsional Shear (Mt)                           |  | 3            | 3           | 3             | 3            |

Shear Stress Summary 2 2 4 4

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE  |  | Cu           | Cl          | Du            | D1           |
|------------------------|--|--------------|-------------|---------------|--------------|
| (Stress Intensities)   |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Two * Max Shear Stress |  | 107          | 105         | 183           | 43           |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE |  | Au | Al | Bu | B1 |
|--------------------|--|----|----|----|----|
|--------------------|--|----|----|----|----|





| (Stresses in the<br>hoop direction) | Top<br>Outside | Top<br>Inside | Bottom<br>Outside | Bottom<br>Inside |
|-------------------------------------|----------------|---------------|-------------------|------------------|
| Hoop Membrane (P )                  | 3              | 3             | 3                 | 3                |
| Hoop Bending (P )                   | 0              | 0             | 0                 | 0                |
| Hoop Membrane (Mc)                  | 0              | 0             | 0                 | 0                |
| Hoop Bending (Mc)                   | 0              | 0             | 0                 | 0                |
| Hoop Membrane (ML)                  | 1              | 1             | -1                | -1               |
| Hoop Bending (ML)                   | 0              | 0             | 0                 | 0                |
| Hoop Pressure Stress                | 0              | 1             | 0                 | 1                |
| Hoop Stress Summary                 | 5              | 6             | 2                 | 3                |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 1                    | 1                   | 1                       | 1                      |
| Axial Bending (P )                                               | 30                   | -30                 | 30                      | -30                    |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -9                   | -9                  | 9                       | 9                      |
| Axial Bending (ML)                                               | -30                  | 30                  | 30                      | -30                    |
| Axial Pressure Stress                                            | 0                    | 0                   | 0                       | 0                      |
| Axial Stress Summary                                             | -7                   | -7                  | 72                      | -49                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 1             | 1            | -1            | -1           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 3             | 3            | 3             | 3            |
| Shear Stress Summary                 | 5             | 5            | 2             | 2            |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 16            | 17           | 72            | 53           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 3                     | 3                    | 3                      | 3                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | 2                     | 2                    | -2                     | -2                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 0                     | 1                    | 0                      | 1                     |
| Hoop Stress Summary                                          | 6                     | 7                    | 1                      | 2                     |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 1                     | 1                    | 1                      | 1                     |
| Axial Bending (P )                                                  | 30                    | -30                  | 30                     | -30                   |
| Axial Membrane (Mc)                                                 | -9                    | -9                   | 9                      | 9                     |
| Axial Bending (Mc)                                                  | -31                   | 31                   | 31                     | -31                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |



Axial Pressure Stress 0 0 0 0

Axial Stress Summary -8 -6 73 -51

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -1            | -1           | 1             | 1            |
| Shear due to Torsion                    | 3             | 3            | 3             | 3            |
| Shear Stress Summary                    | 2             | 2            | 5             | 5            |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 16            | 14           | 74            | 54           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                                    | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)                     | 105              | 106   | 105   | 106   | 105   | 106   | 105   | 106   |
| Circ. Pl (SUS)                     | 4                | 4     | 2     | 2     | 1     | 1     | 5     | 5     |
| Circ. Q (SUS)                      | -1               | 1     | 27    | -27   | 1     | -1    | 69    | -69   |
| Long. Pm (SUS)                     | 52               | 52    | 52    | 52    | 52    | 52    | 52    | 52    |
| Long. Pl (SUS)                     | 0                | 0     | 6     | 6     | 5     | 5     | 1     | 1     |
| Long. Q (SUS)                      | 2                | -2    | 68    | -68   | -2    | 2     | 28    | -28   |
| Shear Pm (SUS)                     | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)                     | 1                | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                      | 3                | 3     | 3     | 3     | 3     | 3     | 3     | 3     |
| Pm (SUS)                           | 105.0            | 106.0 | 105.0 | 106.0 | 105.0 | 106.0 | 105.0 | 106.0 |
| Pm+Pl (SUS)                        | 109.0            | 110.0 | 107.0 | 108.0 | 106.0 | 107.0 | 110.0 | 111.0 |
| Pm+Pl+Q (Total)                    | 108.3            | 111.3 | 134.5 | 91.1  | 107.1 | 106.1 | 179.2 | 42.9  |

**Vessel Stress Summation Comparison (MPa):**

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 106.00    | 138.00         | Passed |
| Pm+Pl (SUS)            | 111.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 179.16    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |    |    |    |    |    |    |    |
|------------------------------------|------------------|----|----|----|----|----|----|----|
|                                    | Au               | Al | Bu | Bl | Cu | Cl | Du | Dl |
| Circ. Pm (SUS)                     | 0                | 1  | 0  | 1  | 0  | 1  | 0  | 1  |
| Circ. Pl (SUS)                     | 4                | 4  | 2  | 2  | 5  | 5  | 1  | 1  |
| Circ. Q (SUS)                      | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  |



|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Long. Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pl (SUS)  | -8   | -8   | 10   | 10   | -8   | -8   | 10   | 10   |
| Long. Q (SUS)   | 0    | 0    | 60   | -60  | -1   | 1    | 61   | -61  |
| Shear Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)  | 1    | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)   | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Pm (SUS)        | 0.0  | 1.0  | 0.0  | 1.0  | 0.0  | 1.0  | 0.0  | 1.0  |
| Pm+Pl (SUS)     | 12.2 | 13.2 | 10.1 | 10.1 | 13.2 | 14.1 | 10.1 | 10.1 |
| Pm+Pl+Q (Total) | 14.4 | 15.3 | 70.1 | 53.2 | 14.6 | 13.6 | 71.2 | 53.6 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 1.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 14.14     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 71.23     | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: N8 2"**

**From: 110**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5371 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1332.0203 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |        |    |
|-----------------------------------|--|--------|----|
| Distance from Bottom/Left Tangent |  | 450.00 | mm |
|-----------------------------------|--|--------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 215.00 | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

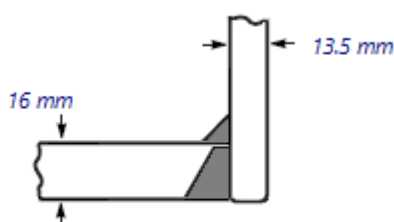
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105         |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 600    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: N8 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.54 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.54)$   
= 11.5244 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.54 / (118.0 \cdot 1.0)] - 1))$   
= 0.1159 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: N8 2".  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 12.5244 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 12.5244 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 12.5244, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |           |        |
|------------|---|-------|-----------|---|-----------|--------|
| Sustained  | : | 12.2, | Allowable | : | 118.0 MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 282.8 MPa | Passed |
| Occasional | : | 0.4,  | Allowable | : | 156.9 MPa | Passed |
| Shear      | : | 12.5, | Allowable | : | 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:



### Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.116, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio = tr \* E\*/(tg - c) = 0.009, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -19 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 0.54/10.20 = 0.053

### Weld Size Calculations, Description: N8 2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.4620 mm

### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.2558 mm  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

### Input Echo, WRC297 Item 1, Description: N8 2" :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 16.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

[See ASME VIII Div. 1, UG-23\(e\), for determination of the SPS allowable.](#)

|                                    |          |            |
|------------------------------------|----------|------------|
| Diameter Basis for Nozzle          | ID       |            |
| Nozzle Corrosion Allowance         | 0.0000   | mm         |
| Nozzle Diameter                    | 50.800   | mm         |
| Nozzle Thickness                   | 13.4620  | mm         |
| Nozzle Stress Concentration Factor | 1.000    |            |
| Nozzle Material                    | SA-106 B |            |
| Nozzle Cold S.I. Allowable         | SNmc     | 118.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh     | 118.00 MPa |

[See ASME VIII Div. 1, UG-23\(e\), for determination of the SPS allowable.](#)

### Note:

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUStained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.54      | MPa  |
| Radial Load              | P  | -3750.00  | N    |
| Circumferential Shear    | Vc | 2500.00   | N    |
| Longitudinal Shear       | Vl | 2500.00   | N    |
| Circumferential Moment   | Mc | 350000.00 | N-mm |
| Longitudinal Moment      | Ml | 350000.00 | N-mm |



Torsional Moment Mt 500000.00 N-mm

Include Axial Pressure Thrust No  
 Include Pressure Stress Indices per Div. 2 No  
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tv, (4.858) is < 5  
 check the Limitations of WRC bulletin 297.  
 Warning - The ratio, Dn/Tn (5.774) is < 10  
 check the Limitations of WRC bulletin 297.*

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(2961.5(16.0 - 1.0))}$   
 $= 773.253 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5923.000 mm  
 Nozzle Outside Diameter (d) = 77.724 mm  
 Vessel Thickness used (T) = 15.000 mm  
 Nozzle Thickness used (t) = 13.462 mm  
 T / t = 1.114  
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.261$$

Nr/P = 0.227  
 Mr/P = 0.360  
 M0/P = 0.137  
 N0/P = 0.202  
 MrD/Mc = 0.288  
 NrDL/Mc = 0.148  
 M0d/Mc = 0.128  
 N0DL/Mc = -0.126  
 MrD/Ml = 0.280  
 NrDL/Ml = 0.151  
 M0D/Ml = 0.123  
 N0DL/Ml = -0.071

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 3                    | 3                   | 3                       | 3                      |
| Outplane Bending (P )                                            | 13                   | -13                 | 13                      | -13                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | 1                    | 1                   | -1                      | -1                     |
| Outplane Bending (ML)                                            | -14                  | 14                  | 14                      | -14                    |
| Normal Pressure Stress                                           | 105                  | 106                 | 105                     | 106                    |
| Outplane Stress Summary                                          | 109                  | 111                 | 135                     | 79                     |

### Vessel Stresses (MPa):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |
| Inplane Bending (P )                                               | 35                   | -35                 | 35                      | -35                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -3                   | -3                  | 3                       | 3                      |
| Inplane Bending (ML)                                               | -33                  | 33                  | 33                      | -33                    |
| Inplane Pressure Stress                                            | 52                   | 52                  | 52                      | 52                     |



Inplane Stress Summary 55 51 129 -10

**Vessel Stresses (MPa):**

|                                             |  |             |            |                |               |
|---------------------------------------------|--|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          |  | Au          | Al         | Bu             | B1            |
| (Shear stress normal to longitudinal plane) |  | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Shear (Vc)                         |  | 1           | 1          | -1             | -1            |
| Outplane Shear (Vl)                         |  | 0           | 0          | 0              | 0             |
| Outplane Shear (Mt)                         |  | 3           | 3          | 3              | 3             |

Shear Stress Summary 4 4 2 2

**Vessel Stresses (MPa):**

|                        |  |             |            |                |               |
|------------------------|--|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE     |  | Au          | Al         | Bu             | B1            |
| (Stress Intensities)   |  | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Two * Max Shear Stress |  | 109         | 112        | 136            | 89            |

**Vessel Stresses (MPa):**

|                                            |  |              |             |               |              |
|--------------------------------------------|--|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      |  | Cu           | Cl          | Du            | D1           |
| (Stresses Normal to circumferential plane) |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Membrane (P )                     |  | 3            | 3           | 3             | 3            |
| Outplane Bending (P )                      |  | 13           | -13         | 13            | -13          |
| Outplane Membrane (Mc)                     |  | 2            | 2           | -2            | -2           |
| Outplane Bending (Mc)                      |  | -15          | 15          | 15            | -15          |
| Outplane Membrane (ML)                     |  | 0            | 0           | 0             | 0            |
| Outplane Bending (ML)                      |  | 0            | 0           | 0             | 0            |
| Normal Pressure Stress                     |  | 52           | 52          | 52            | 52           |

Outplane Stress Summary 56 60 82 24

**Vessel Stresses (MPa):**

|                                              |  |              |             |               |              |
|----------------------------------------------|--|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        |  | Cu           | Cl          | Du            | D1           |
| (Stresses parallel to circumferential plane) |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Inplane Membrane (P )                        |  | 3            | 3           | 3             | 3            |
| Inplane Bending (P )                         |  | 35           | -35         | 35            | -35          |
| Inplane Membrane (Mc)                        |  | -2           | -2          | 2             | 2            |
| Inplane Bending (Mc)                         |  | -34          | 34          | 34            | -34          |
| Inplane Membrane (ML)                        |  | 0            | 0           | 0             | 0            |
| Inplane Bending (ML)                         |  | 0            | 0           | 0             | 0            |
| Inplane Pressure Stress                      |  | 105          | 106         | 105           | 106          |

Inplane Stress Summary 107 105 182 42

**Vessel Stresses (MPa):**

|                                                |  |              |             |               |              |
|------------------------------------------------|--|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          |  | Cu           | Cl          | Du            | D1           |
| (Shear stress normal to circumferential plane) |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Shear (Vc)                            |  | 0            | 0           | 0             | 0            |
| Outplane Shear (Vl)                            |  | -1           | -1          | 1             | 1            |
| Torsional Shear (Mt)                           |  | 3            | 3           | 3             | 3            |

Shear Stress Summary 2 2 4 4

**Vessel Stresses (MPa):**

|                        |  |              |             |               |              |
|------------------------|--|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE  |  | Cu           | Cl          | Du            | D1           |
| (Stress Intensities)   |  | Left Outside | Left Inside | Right Outside | Right Inside |
| Two * Max Shear Stress |  | 107          | 105         | 183           | 43           |

**Nozzle Stresses (MPa):**

|                    |  |    |    |    |    |
|--------------------|--|----|----|----|----|
| LONGITUDINAL PLANE |  | Au | Al | Bu | B1 |
|--------------------|--|----|----|----|----|





| (Stresses in the<br>hoop direction) | Top<br>Outside | Top<br>Inside | Bottom<br>Outside | Bottom<br>Inside |
|-------------------------------------|----------------|---------------|-------------------|------------------|
| Hoop Membrane (P )                  | 3              | 3             | 3                 | 3                |
| Hoop Bending (P )                   | 0              | 0             | 0                 | 0                |
| Hoop Membrane (Mc)                  | 0              | 0             | 0                 | 0                |
| Hoop Bending (Mc)                   | 0              | 0             | 0                 | 0                |
| Hoop Membrane (ML)                  | 1              | 1             | -1                | -1               |
| Hoop Bending (ML)                   | 0              | 0             | 0                 | 0                |
| Hoop Pressure Stress                | 0              | 1             | 0                 | 1                |
| Hoop Stress Summary                 | 5              | 6             | 2                 | 3                |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 1                    | 1                   | 1                       | 1                      |
| Axial Bending (P )                                               | 30                   | -30                 | 30                      | -30                    |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -9                   | -9                  | 9                       | 9                      |
| Axial Bending (ML)                                               | -30                  | 30                  | 30                      | -30                    |
| Axial Pressure Stress                                            | 0                    | 0                   | 0                       | 0                      |
| Axial Stress Summary                                             | -7                   | -7                  | 72                      | -49                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 1             | 1            | -1            | -1           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 3             | 3            | 3             | 3            |
| Shear Stress Summary                 | 5             | 5            | 2             | 2            |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 16            | 17           | 72            | 53           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 3                     | 3                    | 3                      | 3                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | 2                     | 2                    | -2                     | -2                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 0                     | 1                    | 0                      | 1                     |
| Hoop Stress Summary                                          | 6                     | 7                    | 1                      | 2                     |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 1                     | 1                    | 1                      | 1                     |
| Axial Bending (P )                                                  | 30                    | -30                  | 30                     | -30                   |
| Axial Membrane (Mc)                                                 | -9                    | -9                   | 9                      | 9                     |
| Axial Bending (Mc)                                                  | -31                   | 31                   | 31                     | -31                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |



Axial Pressure Stress 0 0 0 0

Axial Stress Summary -8 -6 73 -51

**Nozzle Stresses (MPa):**

CIRCUMFERENTIAL PLANE  
 (Shear stress) Cu Cl Du Dl  
 Outside Inside Outside Inside

Shear due to (Vc) 0 0 0 0  
 Shear due to (Vl) -1 -1 1 1  
 Shear due to Torsion 3 3 3 3

Shear Stress Summary 2 2 5 5

**Nozzle Stresses (MPa):**

CIRCUMFERENTIAL PLANE  
 (Stress Intensities) Cu Cl Du Dl  
 Outside Inside Outside Inside

Two \* Max Shear Stress 16 14 74 54

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 105              | 106   | 105   | 106   | 105   | 106   | 105   | 106   |
| Circ. Pl (SUS)               | 4                | 4     | 2     | 2     | 1     | 1     | 5     | 5     |
| Circ. Q (SUS)                | -1               | 1     | 27    | -27   | 1     | -1    | 69    | -69   |
| Long. Pm (SUS)               | 52               | 52    | 52    | 52    | 52    | 52    | 52    | 52    |
| Long. Pl (SUS)               | 0                | 0     | 6     | 6     | 5     | 5     | 1     | 1     |
| Long. Q (SUS)                | 2                | -2    | 68    | -68   | -2    | 2     | 28    | -28   |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 1                | 1     | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                | 3                | 3     | 3     | 3     | 3     | 3     | 3     | 3     |
| Pm (SUS)                     | 105.0            | 106.0 | 105.0 | 106.0 | 105.0 | 106.0 | 105.0 | 106.0 |
| Pm+Pl (SUS)                  | 109.0            | 110.0 | 107.0 | 108.0 | 106.0 | 107.0 | 110.0 | 111.0 |
| Pm+Pl+Q (Total)              | 108.3            | 111.3 | 134.5 | 91.1  | 107.1 | 106.1 | 179.2 | 42.9  |

**Vessel Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 106.00    | 138.00         | Passed |
| Pm+Pl (SUS)         | 111.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 179.16    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |    |    |    |    |    |    |    |
|------------------------------|------------------|----|----|----|----|----|----|----|
|                              | Au               | Al | Bu | Bl | Cu | Cl | Du | Dl |
| Circ. Pm (SUS)               | 0                | 1  | 0  | 1  | 0  | 1  | 0  | 1  |
| Circ. Pl (SUS)               | 4                | 4  | 2  | 2  | 5  | 5  | 1  | 1  |
| Circ. Q (SUS)                | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  |



|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Long. Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pl (SUS)  | -8   | -8   | 10   | 10   | -8   | -8   | 10   | 10   |
| Long. Q (SUS)   | 0    | 0    | 60   | -60  | -1   | 1    | 61   | -61  |
| Shear Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)  | 1    | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)   | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Pm (SUS)        | 0.0  | 1.0  | 0.0  | 1.0  | 0.0  | 1.0  | 0.0  | 1.0  |
| Pm+Pl (SUS)     | 12.2 | 13.2 | 10.1 | 10.1 | 13.2 | 14.1 | 10.1 | 10.1 |
| Pm+Pl+Q (Total) | 14.4 | 15.3 | 70.1 | 53.2 | 14.6 | 13.6 | 71.2 | 53.6 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 1.00      | 118.00         | Passed |
| Pm+Pl (SUS)         | 14.14     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 71.23     | 354.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: K1b**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 6000.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

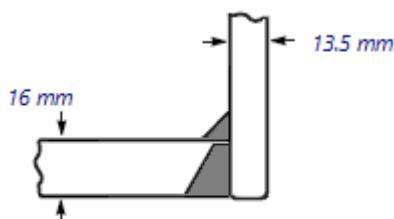
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K1b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
= 10.9001 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.51 / (138 \cdot 1.0)] - 1))$   
= 0.0937 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |             |
|-------------------------------------------------|---------|-------------|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620 mm  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550 mm  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1b.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |              |
|------------------------------------------------|------------------------|--------------|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776 mm  |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000 mm  |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.9001 mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.9001 mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158 mm  |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158 mm  |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000 mm  |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.9001, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.094, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

Weld Size Calculations, Description: K1b

Intermediate Calc. for nozzle/shell Welds Tmin 13.4620 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: K2b**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 4350.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

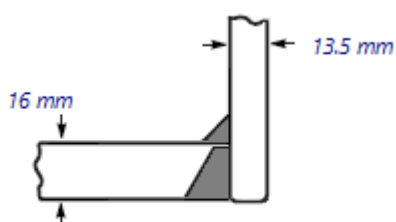
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K2b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
= 10.9001 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.51 / (138 \cdot 1.0)] - 1))$   
= 0.0937 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |             |
|-------------------------------------------------|---------|-------------|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620 mm  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550 mm  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2b.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |              |
|------------------------------------------------|------------------------|--------------|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776 mm  |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000 mm  |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.9001 mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.9001 mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158 mm  |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158 mm  |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000 mm  |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.9001, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.094, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C





**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

Weld Size Calculations, Description: K2b

Intermediate Calc. for nozzle/shell Welds    Tmin    13.4620    mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: K3b**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 4350.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

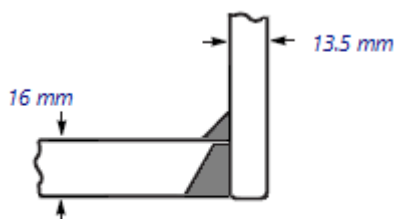
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K3b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
= 10.9001 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.51 / (138 \cdot 1.0)] - 1))$   
= 0.0937 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |             |
|-------------------------------------------------|---------|-------------|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620 mm  |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550 mm  |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3b.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |              |
|------------------------------------------------|------------------------|--------------|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776 mm  |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000 mm  |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.9001 mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.9001 mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158 mm  |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158 mm  |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000 mm  |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.9001, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.094, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

Weld Size Calculations, Description: K3b

Intermediate Calc. for nozzle/shell Welds    Tmin    13.4620    mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: K4b**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 4650.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

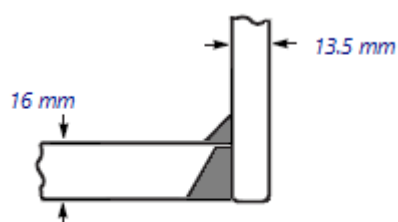
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K4b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
= 10.9001 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \cdot (\exp([P / (S_n \cdot E)] - 1))$  per Appendix 1-2 (a)(1)  
=  $25.4 \cdot (\exp([0.51 / (138 \cdot 1.0)] - 1))$   
= 0.0937 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 107.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 53.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 33.6550  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4b.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.9001 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.9001 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.9001, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.094, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

Weld Size Calculations, Description: K4b

Intermediate Calc. for nozzle/shell Welds    Tmin    13.4620    mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8058 mm

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**Input, Nozzle Desc: N2a**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 34.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                        |    |           |    |
|----------------------------------------|----|-----------|----|
| Distance from Cylinder/Cone Centerline | L1 | 1900.0001 | mm |
|----------------------------------------|----|-----------|----|

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 4000.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | -32.00  | deg |
| Diameter                          |  | 62.0000 | in. |

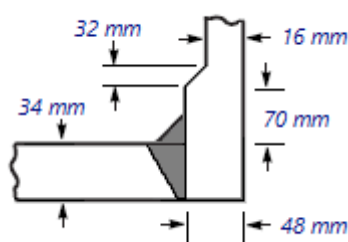
|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                                                 |    |         |    |
|-------------------------------------------------|----|---------|----|
| Hub Height of Integral Nozzle                   | h  | 70.0000 | mm |
| Height of Beveled Transition                    | L' | 32.0000 | mm |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |    | 48.0000 | mm |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 363.2720 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 22.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

The Pressure Design option was MAWP + static head (bottom of element).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.





## Reinforcement CALCULATION, Description: N2a

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 62.000 in.  
 Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
 $= 10.9001 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.51 \cdot 787.4 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
 $= 2.8947 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.7352 mm

### Intermediate Hub Nozzle Calculations:

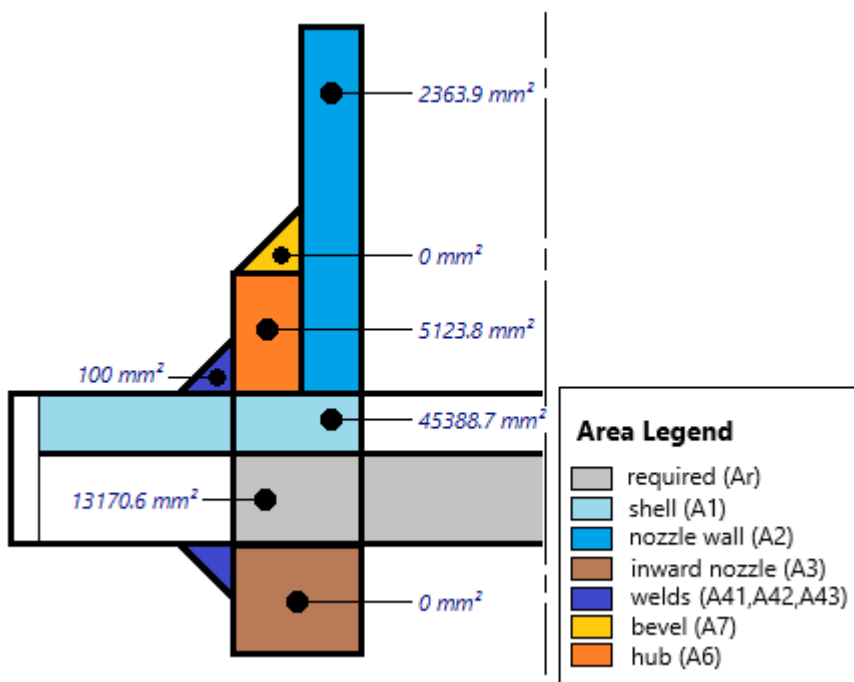
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$   
 $= 32.0 / 0.5773$   
 $= 55.4256 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 4347.2681 mm  
 Parallel to Vessel Wall, opening length d 2173.6340 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 82.5000 mm



### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |    | MAWP      | External  | Mapnc |
|--------------------------|----|-----------|-----------|-------|
| Area Required            | Ar | 11846.401 | 13170.588 | NA    |
| Area in Shell            | A1 | 59883.527 | 45388.746 | NA    |



|                       |             |           |           |    |
|-----------------------|-------------|-----------|-----------|----|
| Area in Nozzle Wall   | A2          | 2335.496  | 2363.925  | NA |
| Area in Inward Nozzle | A3          | 0.000     | 0.000     | NA |
| Area in Welds         | A41+A42+A43 | 100.000   | 100.000   | NA |
| Area in Element       | A5          | 0.000     | 0.000     | NA |
| Area in Hub + Bevel   | A6          | 5123.751  | 5123.751  | NA |
| TOTAL AREA AVAILABLE  | Atot        | 67442.773 | 52976.422 | NA |

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 45.22 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= 0.5( d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1) )$  per UG-37(d)  
 $= 0.5(2173.634 \cdot 12.1185 \cdot 1 + 2 \cdot 48.0 \cdot 12.1185 \cdot 1(1-1.0))$   
 $= 13170.588 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d( E1 \cdot t - F \cdot tr ) - 2 \cdot tn( E1 \cdot t - F \cdot tr ) \cdot ( 1 - fr1 )$   
 $= 2173.634( 1.0 \cdot 33.0 - 1.0 \cdot 12.118 ) - 2 \cdot 48.0$   
 $( 1.0 \cdot 33.0 - 1.0 \cdot 12.1185 ) \cdot ( 1 - 1.0 )$   
 $= 45388.746 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= ( 2 \cdot tlnp )( tn - trn ) fr2 / \sin( \alpha 3 )$   
 $= ( 2 \cdot 82.5 )( 16.0 - 2.74 ) 1.0 / \sin( 67.8 )$   
 $= 2363.925 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + ( Wi - can / 0.707 )^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + ( 0.0 )^2 \cdot 1.0$   
 $= 100.000 \text{ mm}^2$

Area Available in the Hub Section [A6]:  
 $= ( 2 \cdot \min(Tlnp, ho, Hubht) ) \cdot ( Hubtk - tn ) \cdot fr2$   
 $= ( 2 \cdot \min(82.5, 363.3, 70.0) ) \cdot ( 48.0 - 16.0 ) \cdot 1.0$   
 $= 5123.751 \text{ mm}^2$  Includes Bevel Area of 643.750 mm<sup>2</sup>

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 34.0, tr = 10.9, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.33$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 7 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

*Warning: Appendices 1-7 and 1-10 only apply to radial nozzles and this is a "hillside" nozzle. (see U-2(g)).*

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 32.0 / 0.5773$   
 $= 55.4256 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).



**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**

Parallel to Vessel Wall (Diameter Limit) D1 3260.4509 mm  
 Parallel to Vessel Wall, opening length d 1630.2255 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 82.5000 mm

**Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm<sup>2</sup> )**

| AREA AVAILABLE, A1 to A6          |      | No Pad    | With Pad  |
|-----------------------------------|------|-----------|-----------|
| Area Required [External Pressure] | AR   | 8780.831  | 8780.831  |
| Area Available in Shell           | A1   | 22694.373 | 22694.373 |
| Area Available in Nozzle Wall     | A2   | 2363.925  | 2363.925  |
| Area Available in Inward Nozzle   | A3   | 0.000     | 0.000     |
| Area Available in Welds           | A4   | 100.000   | 100.000   |
| Area Available in Pad             | A5   | 0.000     | 0.000     |
| Area Available in Hub             | A6   | 5123.751  | 5123.751  |
| TOTAL AREA AVAILABLE              | Atot | 30282.047 | 30282.047 |

The area available without a pad is Sufficient.

This is a non-Radial Large Nozzle, which is not addressed by App. 1-7.  
 This calculation must comply with U-2(g).

The following Calculation is performed using the internal design pressure (P)  
 in the corroded condition.

*Warning: The flange could not be considered in the moment of inertia calculations.*

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

|        | Area    | Distance | Inertia    | d       | I+Ad <sup>2</sup> |
|--------|---------|----------|------------|---------|-------------------|
| Nozzle | 4624.0  | 144.5    | 0.3218E+08 | 101.138 | 0.7948E+08        |
| Hub    | 2752.0  | 43.5     | 0.1813E+07 | 0.135   | 0.1813E+07        |
| Shell  | 17424.0 | 16.5     | 0.1581E+07 | 26.862  | 0.1415E+08        |
| Sum    | 24800.0 |          |            |         | 0.9545E+08        |

Membrane stress per Appendix 1-7 ( 1 or 2 ) Sm 110.94 MPa  
 Allowable Membrane Stress S 138.00 MPa  
 Bending stress at the inside of the Nozzle Sb 31.79 MPa  
 Combined Membrane plus Bending Stress Sm + Sb 142.73 MPa  
 Allowable Membrane + Bending Stress 1.5 \* S 207.00 MPa  
 M.A.W.P for the Appendix 1-7 Stress calc 0.63 MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 2314.735 mm<sup>2</sup>  
 Area from Shell = 10332.032 mm<sup>2</sup>  
 Total Area, As = 15398.769 mm<sup>2</sup>

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [ R ( R_n + t_n + \sqrt{(R \cdot m \cdot t)} ) + R_n ( t + \sqrt{(R \cdot m \cdot t_n)} ) ] / A_s$$

$$= 0.508 [ 2954.0 ( 771.4 + 16.0 + 313.092 ) + 771.4 ( 144.671 ) ] / 15398.769$$

$$= 110.943 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= ( R_n^3/6 + R \cdot R_n \cdot e ) P$$

$$= ( 771.4^3/6 + 2954.0 \cdot 771.4 \cdot 26.862 ) 0.508$$

$$= 69992032 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= ( M ( e + t/2 ) ) / I$$

$$= ( 69992032 ( 26.862 + 33.0/2 ) ) / 95448488$$

$$= 31.787 \text{ MPa}$$

Weld Size Calculations, Description: N2a

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm



### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
= max( 0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)  
= max( 0, (13170.5879 - 45388.7461 + 2 • 48.0 • 1.0 •  
(1.0 • 33.0 - 12.1185 ) )138)  
= max( 0, -4169037.75) N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A4-(Wi-Can/.707)<sup>2</sup>•fr2)•Sv  
= ( 7487.6758 + 100.0 - 0.0 • 1.0 ) • 138  
= 1046991.31 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv  
= ( 7487.6758 + 0.0 + 100.0 + ( 3168.0002 ) ) • 138  
= 1484130.25 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2•tn•t•fr1))•S  
= ( 7487.6758 + 0.0 + 100.0 + 0.0 + ( 3168.0002 ) ) • 138  
= 1484130.25 N

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) • Dlo • Wo • 0.49 • Snw  
= ( 3.1416/2.0 ) • 2218.7183 • 10.0 • 0.49 • 138  
= 2356418. N

Shear, Nozzle Wall [Snw]:  
= (π • (Dlr + Dlo )/4 ) • ( Thk - Can ) • 0.7 • Sn  
= (3.1416 • 1098.088) • ( 48.0 - 0.0 ) • 0.7 • 138  
= 15994417. N

Tension, Shell Groove Weld [Tngw]:  
= (π/2) • Dlo • Wgnvi • 0.74 • Sng  
= ( 3.1416/2.0 ) • 2218.7183 • ( 22.0 - 0.0 ) • 0.74 • 138  
= 7829222. N

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
= ( SONW + SNW ) = ( 2356419 + 15994417 ) = 18350836 N

Failure Path [Path 2-2]:  
= ( Sonw + Tpgw + Tngw + Sinw )  
= ( 2356419 + 0 + 7829222 + 0 ) = 10185641 N

Failure Path [Path 3-3]:  
= ( Sonw + Tngw + Sinw )  
= ( 2356419 + 7829222 + 0 ) = 10185641 N

### Summary of Failure Path Calculations:

|                      |                       |                   |
|----------------------|-----------------------|-------------------|
| Path 1-1 = 18350836N | , must exceed W = 0 N | or W1 = 1046991 N |
| Path 2-2 = 10185641N | , must exceed W = 0 N | or W2 = 1484130 N |
| Path 3-3 = 10185641N | , must exceed W = 0 N | or W3 = 1484130 N |

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

#### Reinforcement CALCULATION, Description: N2a

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 62.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
= 10.9001 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.51 \cdot 787.4 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
= 2.8947 mm

Required Nozzle thickness under External Pressure per UG-28 : 2.7352 mm

#### Intermediate Hub Nozzle Calculations:

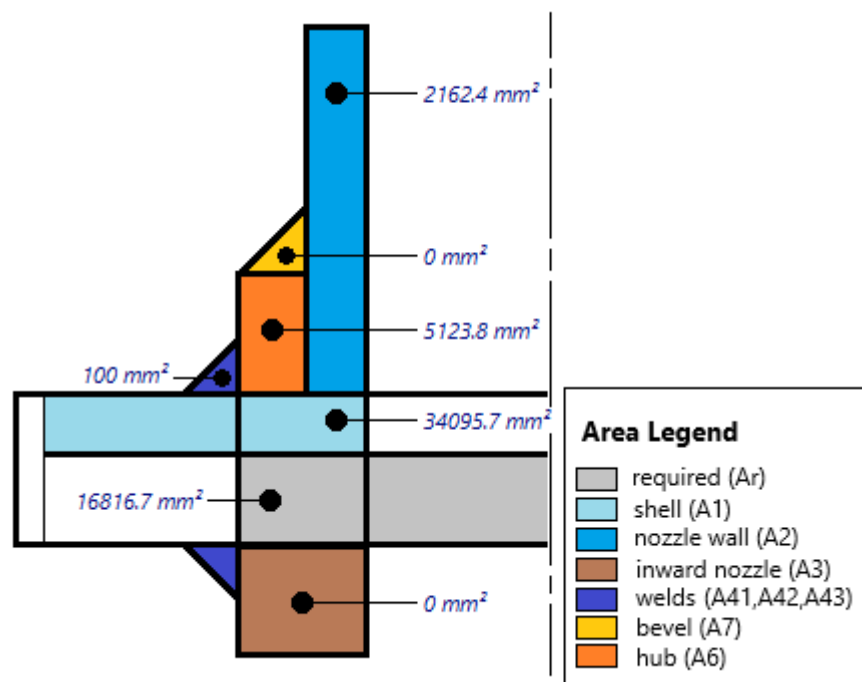
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
= ( Hub Thickness - Neck Thickness ) /  $\tan(30)$   
=  $32.0 / 0.5773$   
= 55.4256 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

#### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |              |
|-------------------------------------------------|------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 3085.6001 mm |
| Parallel to Vessel Wall, opening length         | d    | 1542.8000 mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 82.5000 mm   |



#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 16816.656 | 9348.209  | NA    |
| Area in Shell            | A1          | 34095.746 | 32215.986 | NA    |
| Area in Nozzle Wall      | A2          | 2162.370  | 2188.692  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 100.000   | 100.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| Area in Hub + Bevel      | A6          | 5123.751  | 5123.751  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 41481.867 | 39628.426 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (1542.8 \cdot 10.9001 \cdot 1.0 + 2 \cdot 48.0 \cdot 10.9001 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 16816.656 \text{ mm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 1542.8(1.0 \cdot 33.0 - 1.0 \cdot 10.9) - 2 \cdot 48.0$$

$$(1.0 \cdot 33.0 - 1.0 \cdot 10.9001) \cdot (1 - 1.0)$$

$$= 34095.746 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 82.5)(16.0 - 2.89)1.0$$

$$= 2162.370 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(82.5, 363.3, 70.0)) \cdot (48.0 - 16.0) \cdot 1.0$$

$$= 5123.751 \text{ mm}^2 \text{ Includes Bevel Area of } 643.750 \text{ mm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 2.8947 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 2.5000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 11.9001 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 11.9001 mm |
| Wall Thickness, shell/head, external pressure  | trb2 = 3.2158 mm                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.2158 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.3312 mm                     |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 8.331, \max( 11.9001, 3.2158 ) ]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 2.8947, 8.3312 )$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 26.4, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 318.6 MPa | Passed |
| Occasional | : 12.1, | Allowable | : 183.5 MPa | Passed |
| Shear      | : 10.9, | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$$

$$= 32.0 / 0.5773$$

$$= 55.4256 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

#### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |              |
|-------------------------------------------------|------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 2314.2000 mm |
| Parallel to Vessel Wall, opening length         | d    | 1157.1000 mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 82.5000 mm   |

#### Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm<sup>2</sup> )

| AREA AVAILABLE, A1 to A6          |      | No Pad    | With Pad  |
|-----------------------------------|------|-----------|-----------|
| Area Required [Internal Pressure] | AR   | 11211.665 | 11211.665 |
| Area Available in Shell           | A1   | 17047.873 | 17047.873 |
| Area Available in Nozzle Wall     | A2   | 2162.370  | 2162.370  |
| Area Available in Inward Nozzle   | A3   | 0.000     | 0.000     |
| Area Available in Welds           | A4   | 100.000   | 100.000   |
| Area Available in Pad             | A5   | 0.000     | 0.000     |
| Area Available in Hub             | A6   | 5123.751  | 5123.751  |
| TOTAL AREA AVAILABLE              | Atot | 24433.994 | 24433.994 |

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.



*Warning: The flange could not be considered in the moment of inertia calculations.*

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

|        | Area    | Distance | Inertia    | d       | I+Ad <sup>2</sup> |
|--------|---------|----------|------------|---------|-------------------|
| Nozzle | 4624.0  | 144.5    | 0.3218E+08 | 101.138 | 0.7948E+08        |
| Hub    | 2752.0  | 43.5     | 0.1813E+07 | 0.135   | 0.1813E+07        |
| Shell  | 17424.0 | 16.5     | 0.1581E+07 | 26.862  | 0.1415E+08        |
| Sum    | 24800.0 |          |            |         | 0.9545E+08        |

|                                                |        |     |
|------------------------------------------------|--------|-----|
| Membrane stress per Appendix 1-7 ( 1 or 2 ) Sm | 110.94 | MPa |
| Allowable Membrane Stress S                    | 138.00 | MPa |
| Bending stress at the inside of the Nozzle Sb  | 31.79  | MPa |
| Combined Membrane plus Bending Stress Sm + Sb  | 142.73 | MPa |
| Allowable Membrane + Bending Stress 1.5 * S    | 207.00 | MPa |
| M.A.W.P for the Appendix 1-7 Stress calc       | 0.63   | MPa |

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 2314.735 mm<sup>2</sup>  
 Area from Shell = 10332.032 mm<sup>2</sup>  
 Total Area, As = 15398.769 mm<sup>2</sup>

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [ R ( R_n + t_n + \sqrt{(r_m \cdot t)} ) + R_n ( t + \sqrt{(R_n m \cdot t_n)} ) ] / A_s$$

$$= 0.508 [ 2954.0 ( 771.4 + 16.0 + 313.092 ) + 771.4 ( 144.671 ) ] / 15398.769$$

$$= 110.943 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= ( R_n^3 / 6 + R \cdot R_n \cdot e ) P$$

$$= ( 771.4^3 / 6 + 2954.0 \cdot 771.4 \cdot 26.862 ) 0.508$$

$$= 69992032 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= ( M ( e + t/2 ) ) / I$$

$$= ( 69992032 ( 26.862 + 33.0/2 ) ) / 95448488$$

$$= 31.787 \text{ MPa}$$

Weld Size Calculations, Description: N2a

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

**Results Per UW-16.1:**

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:

$$= \max( 0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) S_v )$$

$$= \max( 0, (16816.6562 - 34095.7461 + 2 \cdot 48.0 \cdot 1.0 \cdot (1.0 \cdot 33.0 - 10.9001) ) 138 )$$

$$= \max( 0, -2091519.25 ) \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= ( 7286.1211 + 100.0 - 0.0 \cdot 1.0 ) \cdot 138$$

$$= 1019179.62 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= ( 7286.1211 + 0.0 + 100.0 + ( 3168.0002 ) ) \cdot 138$$





= 1456318.50 N

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\ &= (7286.1211 + 0.0 + 100.0 + 0.0 + (3168.0002)) \cdot 138 \\ &= 1456318.50 \text{ N} \end{aligned}$$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 1574.7999 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 1672537. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 779.4) \cdot (48.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 11352505. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 1574.7999 \cdot (22.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 5557018. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (1672537 + 11352505) = 13025042 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (1672537 + 0 + 5557018 + 0) = 7229555 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (1672537 + 5557018 + 0) = 7229555 \text{ N} \end{aligned}$$

#### Summary of Failure Path Calculations:

|                      |                       |                   |
|----------------------|-----------------------|-------------------|
| Path 1-1 = 13025042N | , must exceed W = 0 N | or W1 = 1019179 N |
| Path 2-2 = 7229554 N | , must exceed W = 0 N | or W2 = 1456318 N |
| Path 3-3 = 7229554 N | , must exceed W = 0 N | or W3 = 1456318 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1036.6124 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1444.1216 mm

*This nozzle or its diameter limit intersects weld seam # [7] on this element. The distance between the nozzle and seam is, 360.89 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1542.80 mm*

#### Warning:

*Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).*

#### Warning:

*The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.*

Input Echo, WRC297 Item 1, Description: N2a :

Diameter Basis for Cylindrical Shell ID



|                                    |     |           |     |
|------------------------------------|-----|-----------|-----|
| Shell Internal Corrosion Allowance |     | 1.0000    | mm  |
| Shell External Corrosion Allowance |     | 0.0000    | mm  |
| Shell Diameter                     |     | 5906.000  | mm  |
| Shell Thickness                    |     | 34.0000   | mm  |
| Shell Stress Concentration Factor  |     | 1.000     |     |
| Vessel Material                    |     | SA-516 70 |     |
| Vessel Cold S.I. Allowable         | Smc | 138.00    | MPa |
| Vessel Hot S.I. Allowable          | Smh | 138.00    | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |      |           |     |
|------------------------------------|------|-----------|-----|
| Diameter Basis for Nozzle          |      | OD        |     |
| Nozzle Corrosion Allowance         |      | 0.0000    | mm  |
| Nozzle Diameter                    |      | 1638.800  | mm  |
| Nozzle Thickness                   |      | 48.0000   | mm  |
| Nozzle Stress Concentration Factor |      | 1.000     |     |
| Nozzle Material                    |      | SA-516 70 |     |
| Nozzle Cold S.I. Allowable         | SNmc | 138.00    | MPa |
| Nozzle Hot S.I. Allowable          | SNmh | 138.00    | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUstained loads.*

|                                                 |    |            |      |
|-------------------------------------------------|----|------------|------|
| Design Internal Pressure                        | Dp | 0.51       | MPa  |
| Radial Load                                     | P  | -116249.82 | N    |
| Circumferential Shear                           | Vc | 77500.17   | N    |
| Longitudinal Shear                              | Vl | 77500.17   | N    |
| Circumferential Moment                          | Mc | 0.337E+09  | N-mm |
| Longitudinal Moment                             | Ml | 0.337E+09  | N-mm |
| Torsional Moment                                | Mt | 0.481E+09  | N-mm |
| Include Axial Pressure Thrust                   |    | No         |      |
| Include Pressure Stress Indices per Div. 2      |    | No         |      |
| Local Loads applied at end of Nozzle/Attachment |    | No         |      |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 1638.8 + 2 \cdot 1.65 \cdot \sqrt{(2970.5(34.0 - 1.0))}$   
 $= 2672.003 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |          |    |
|-------------------------|-------|----------|----|
| Vessel Mean Diameter    | (D) = | 5941.000 | mm |
| Nozzle Outside Diameter | (d) = | 1638.800 | mm |
| Vessel Thickness used   | (T) = | 33.000   | mm |
| Nozzle Thickness used   | (t) = | 48.000   | mm |
| T / t                   | =     | 0.688    |    |
| d / t                   | =     | 34.142   |    |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.701$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.036 |
| Mr/P    | = | 0.091 |
| M0/P    | = | 0.029 |
| N0/P    | = | 0.118 |
| MrD/Mc  | = | 0.234 |
| NrDL/Mc | = | 0.062 |
| M0d/Mc  | = | 0.078 |
| N0DL/Mc | = | 0.118 |
| MrD/Ml  | = | 0.079 |
| NrDL/Ml | = | 0.030 |
| M0D/Ml  | = | 0.024 |



NØDL/M1 = 0.123

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 12                   | 12                  | 12                      | 12                     |
| Outplane Bending (P )                                            | 18                   | -18                 | 18                      | -18                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -23                  | -23                 | 23                      | 23                     |
| Outplane Bending (ML)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 45                   | 45                  | 45                      | 45                     |
| Outplane Stress Summary                                          | 25                   | 44                  | 126                     | 35                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |
| Inplane Bending (P )                                               | 58                   | -58                 | 58                      | -58                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -5                   | -5                  | 5                       | 5                      |
| Inplane Bending (ML)                                               | -88                  | 88                  | 88                      | -88                    |
| Inplane Pressure Stress                                            | 22                   | 22                  | 22                      | 22                     |
| Inplane Stress Summary                                             | -9                   | 51                  | 179                     | -115                   |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 3                    | 3                   | 3                       | 3                      |
| Shear Stress Summary                                                 | 4                    | 4                   | 2                       | 2                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 36                   | 53                  | 179                     | 150                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 12                    | 12                   | 12                     | 12                    |
| Outplane Bending (P )                                                  | 18                    | -18                  | 18                     | -18                   |
| Outplane Membrane (Mc)                                                 | -22                   | -22                  | 22                     | 22                    |
| Outplane Bending (Mc)                                                  | -88                   | 88                   | 88                     | -88                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 22                    | 22                   | 22                     | 22                    |
| Outplane Stress Summary                                                | -57                   | 83                   | 164                    | -49                   |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|



|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 3    | 3   | 3   | 3    |
| Inplane Bending (P )    | 58   | -58 | 58  | -58  |
| Inplane Membrane (Mc)   | -11  | -11 | 11  | 11   |
| Inplane Bending (Mc)    | -265 | 265 | 265 | -265 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 45   | 45  | 45  | 45   |

|                        |      |     |     |      |
|------------------------|------|-----|-----|------|
| Inplane Stress Summary | -169 | 244 | 384 | -262 |
|------------------------|------|-----|-----|------|

**Vessel Stresses (MPa):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Shear (Vc)                            | 0            | 0           | 0             | 0            |
| Outplane Shear (Vl)                            | 0            | 0           | 0             | 0            |
| Torsional Shear (Mt)                           | 3            | 3           | 3             | 3            |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 4 | 4 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 169 | 244 | 384 | 262 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (MPa):**

|                                  |             |            |                |               |
|----------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE               | Au          | Al         | Bu             | Bl            |
| (Stresses in the hoop direction) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Hoop Membrane (P )               | 12          | 12         | 12             | 12            |
| Hoop Bending (P )                | 0           | 0          | 0              | 0             |
| Hoop Membrane (Mc)               | 0           | 0          | 0              | 0             |
| Hoop Bending (Mc)                | 0           | 0          | 0              | 0             |
| Hoop Membrane (ML)               | -23         | -23        | 23             | 23            |
| Hoop Bending (ML)                | 0           | 0          | 0              | 0             |
| Hoop Pressure Stress             | 9           | 10         | 9              | 10            |

|                     |    |   |    |    |
|---------------------|----|---|----|----|
| Hoop Stress Summary | -1 | 0 | 45 | 45 |
|---------------------|----|---|----|----|

**Nozzle Stresses (MPa):**

|                                         |             |            |                |               |
|-----------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | Bl            |
| (Stresses Normal to pipe cross-section) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Axial Membrane (P )                     | 0           | 0          | 0              | 0             |
| Axial Bending (P )                      | 22          | -22        | 22             | -22           |
| Axial Membrane (Mc)                     | 0           | 0          | 0              | 0             |
| Axial Bending (Mc)                      | 0           | 0          | 0              | 0             |
| Axial Membrane (ML)                     | -3          | -3         | 3              | 3             |
| Axial Bending (ML)                      | -33         | 33         | 33             | -33           |
| Axial Pressure Stress                   | 4           | 4          | 4              | 4             |

|                      |     |    |    |     |
|----------------------|-----|----|----|-----|
| Axial Stress Summary | -10 | 13 | 64 | -47 |
|----------------------|-----|----|----|-----|

**Nozzle Stresses (MPa):**

|                      |         |        |         |        |
|----------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE   | Au      | Al     | Bu      | Bl     |
| (Shear stress)       | Outside | Inside | Outside | Inside |
| Shear due to (Vc)    | 0       | 0      | 0       | 0      |
| Shear due to (Vl)    | 0       | 0      | 0       | 0      |
| Shear due to Torsion | 2       | 2      | 2       | 2      |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 1 | 1 |
|----------------------|---|---|---|---|

**Nozzle Stresses (MPa):**

|                    |    |    |    |    |
|--------------------|----|----|----|----|
| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|



| (Stress Intensities    | Outside | Inside | Outside | Inside |
|------------------------|---------|--------|---------|--------|
| Two * Max Shear Stress | 11      | 15     | 65      | 92     |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) |  | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|--|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           |  | 12                    | 12                   | 12                     | 12                    |
| Hoop Bending (P )                                            |  | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           |  | -22                   | -22                  | 22                     | 22                    |
| Hoop Bending (Mc)                                            |  | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           |  | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            |  | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         |  | 9                     | 10                   | 9                      | 10                    |
| Hoop Stress Summary                                          |  | 0                     | 0                    | 44                     | 44                    |

**Nozzle Stresses (MPa):**

| NOZZLE STRESSES (in MPa):                                           |                       |                      |                        |                       |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 22                    | -22                  | 22                     | -22                   |
| Axial Membrane (Mc)                                                 | -3                    | -3                   | 3                      | 3                     |
| Axial Bending (Mc)                                                  | -108                  | 108                  | 108                    | -108                  |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 4                     | 4                    | 4                      | 4                     |
| Axial Stress Summary                                                | -85                   | 88                   | 139                    | -122                  |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 2             | 2            | 2             | 2            |
| Shear Stress Summary                    | 1             | 1            | 2             | 2            |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 85            | 88           | 139           | 167          |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

[illegible]

|                 |      |      |       |       |       |       |       |       |
|-----------------|------|------|-------|-------|-------|-------|-------|-------|
| Pm+Pl (SUS)     | 34.0 | 34.0 | 80.0  | 80.0  | 37.0  | 37.0  | 59.0  | 59.0  |
| Pm+Pl+Q (Total) | 35.5 | 51.1 | 176.2 | 151.1 | 170.1 | 244.1 | 382.0 | 264.0 |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 45.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 80.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 382.04    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |       |       |
|------------------------------|------------------|------|------|------|------|------|-------|-------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du    | Dl    |
| Circ. Pm (SUS)               | 9                | 10   | 9    | 10   | 9    | 10   | 9     | 10    |
| Circ. Pl (SUS)               | -11              | -11  | 35   | 35   | -10  | -10  | 34    | 34    |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Long. Pm (SUS)               | 4                | 4    | 4    | 4    | 4    | 4    | 4     | 4     |
| Long. Pl (SUS)               | -3               | -3   | 3    | 3    | -3   | -3   | 3     | 3     |
| Long. Q (SUS)                | -11              | 11   | 55   | -55  | -86  | 86   | 130   | -130  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Shear Pl (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2     | 2     |
| Pm (SUS)                     | 9.0              | 10.0 | 9.0  | 10.0 | 9.0  | 10.0 | 9.0   | 10.0  |
| Pm+Pl (SUS)                  | 3.0              | 2.0  | 44.0 | 45.0 | 2.0  | 1.0  | 43.0  | 44.0  |
| Pm+Pl+Q (Total)              | 10.5             | 13.6 | 62.2 | 93.1 | 85.0 | 87.1 | 137.0 | 167.0 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 10.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 45.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 167.05    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: N2b**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 34.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                        |    |           |    |
|----------------------------------------|----|-----------|----|
| Distance from Cylinder/Cone Centerline | L1 | 1900.0001 | mm |
|----------------------------------------|----|-----------|----|

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 4000.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 32.00   | deg |
| Diameter                          |  | 62.0000 | in. |

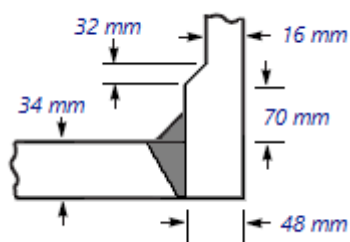
|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                                                 |    |         |    |
|-------------------------------------------------|----|---------|----|
| Hub Height of Integral Nozzle                   | h  | 70.0000 | mm |
| Height of Beveled Transition                    | L' | 32.0000 | mm |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |    | 48.0000 | mm |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 363.2720 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

The Pressure Design option was MAWP + static head (bottom of element).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



## Reinforcement CALCULATION, Description: N2b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 62.000 in.  
 Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
 $= 10.9001 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.51 \cdot 787.4 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
 $= 2.8947 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.7352 mm

### Intermediate Hub Nozzle Calculations:

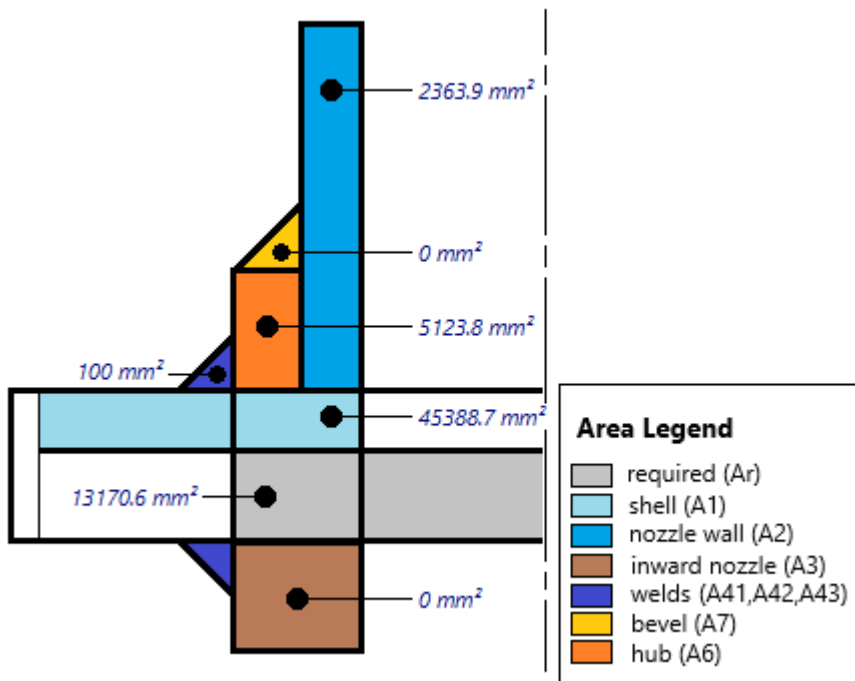
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$   
 $= 32.0 / 0.5773$   
 $= 55.4256 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 4347.2681 mm  
 Parallel to Vessel Wall, opening length d 2173.6340 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 82.5000 mm



### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |    | MAWP      | External  | Mapnc |
|--------------------------|----|-----------|-----------|-------|
| Area Required            | Ar | 11846.401 | 13170.588 | NA    |
| Area in Shell            | A1 | 59883.527 | 45388.746 | NA    |





|                       |             |           |           |    |
|-----------------------|-------------|-----------|-----------|----|
| Area in Nozzle Wall   | A2          | 2335.496  | 2363.925  | NA |
| Area in Inward Nozzle | A3          | 0.000     | 0.000     | NA |
| Area in Welds         | A41+A42+A43 | 100.000   | 100.000   | NA |
| Area in Element       | A5          | 0.000     | 0.000     | NA |
| Area in Hub + Bevel   | A6          | 5123.751  | 5123.751  | NA |
| TOTAL AREA AVAILABLE  | Atot        | 67442.773 | 52976.422 | NA |

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 45.22 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= 0.5( d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1) )$  per UG-37(d)  
 $= 0.5(2173.634 \cdot 12.1185 \cdot 1 + 2 \cdot 48.0 \cdot 12.1185 \cdot 1(1-1.0))$   
 $= 13170.588 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d( E1 \cdot t - F \cdot tr ) - 2 \cdot tn( E1 \cdot t - F \cdot tr ) \cdot ( 1 - fr1 )$   
 $= 2173.634( 1.0 \cdot 33.0 - 1.0 \cdot 12.118 ) - 2 \cdot 48.0$   
 $( 1.0 \cdot 33.0 - 1.0 \cdot 12.1185 ) \cdot ( 1 - 1.0 )$   
 $= 45388.746 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= ( 2 \cdot tlnp )( tn - trn ) fr2 / \sin( \alpha 3 )$   
 $= ( 2 \cdot 82.5 )( 16.0 - 2.74 ) 1.0 / \sin( 67.8 )$   
 $= 2363.925 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + ( Wi - can / 0.707 )^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + ( 0.0 )^2 \cdot 1.0$   
 $= 100.000 \text{ mm}^2$

Area Available in the Hub Section [A6]:  
 $= ( 2 \cdot \min(Tlnp, ho, Hubht) ) \cdot ( Hubtk - tn ) \cdot fr2$   
 $= ( 2 \cdot \min(82.5, 363.3, 70.0) ) \cdot ( 48.0 - 16.0 ) \cdot 1.0$   
 $= 5123.751 \text{ mm}^2$  Includes Bevel Area of 643.750 mm<sup>2</sup>

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 34.0, tr = 10.9, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.33$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 7 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

*Warning: Appendices 1-7 and 1-10 only apply to radial nozzles and this is a "hillside" nozzle. (see U-2(g)).*

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 32.0 / 0.5773$   
 $= 55.4256 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).



**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**

Parallel to Vessel Wall (Diameter Limit) D1 3260.4509 mm  
 Parallel to Vessel Wall, opening length d 1630.2255 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 82.5000 mm

**Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm<sup>2</sup> )**

| AREA AVAILABLE, A1 to A6          |      | No Pad    | With Pad  |
|-----------------------------------|------|-----------|-----------|
| Area Required [External Pressure] | AR   | 8780.831  | 8780.831  |
| Area Available in Shell           | A1   | 22694.373 | 22694.373 |
| Area Available in Nozzle Wall     | A2   | 2363.925  | 2363.925  |
| Area Available in Inward Nozzle   | A3   | 0.000     | 0.000     |
| Area Available in Welds           | A4   | 100.000   | 100.000   |
| Area Available in Pad             | A5   | 0.000     | 0.000     |
| Area Available in Hub             | A6   | 5123.751  | 5123.751  |
| TOTAL AREA AVAILABLE              | Atot | 30282.047 | 30282.047 |

The area available without a pad is Sufficient.

This is a non-Radial Large Nozzle, which is not addressed by App. 1-7.  
 This calculation must comply with U-2(g).

The following Calculation is performed using the internal design pressure (P)  
 in the corroded condition.

*Warning: The flange could not be considered in the moment of inertia calculations.*

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

|        | Area    | Distance | Inertia    | d       | I+Ad <sup>2</sup> |
|--------|---------|----------|------------|---------|-------------------|
| Nozzle | 4624.0  | 144.5    | 0.3218E+08 | 101.138 | 0.7948E+08        |
| Hub    | 2752.0  | 43.5     | 0.1813E+07 | 0.135   | 0.1813E+07        |
| Shell  | 17424.0 | 16.5     | 0.1581E+07 | 26.862  | 0.1415E+08        |
| Sum    | 24800.0 |          |            |         | 0.9545E+08        |

Membrane stress per Appendix 1-7 ( 1 or 2 ) Sm 110.94 MPa  
 Allowable Membrane Stress S 138.00 MPa  
 Bending stress at the inside of the Nozzle Sb 31.79 MPa  
 Combined Membrane plus Bending Stress Sm + Sb 142.73 MPa  
 Allowable Membrane + Bending Stress 1.5 \* S 207.00 MPa  
 M.A.W.P for the Appendix 1-7 Stress calc 0.63 MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 2314.735 mm<sup>2</sup>  
 Area from Shell = 10332.032 mm<sup>2</sup>  
 Total Area, As = 15398.769 mm<sup>2</sup>

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [ R ( R_n + t_n + \sqrt{(R \cdot t)} ) + R_n ( t + \sqrt{(R \cdot t_n)} ) ] / A_s$$

$$= 0.508 [ 2954.0 ( 771.4 + 16.0 + 313.092 ) + 771.4 ( 144.671 ) ] / 15398.769$$

$$= 110.943 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= ( R_n^3/6 + R \cdot R_n \cdot e ) P$$

$$= ( 771.4^3/6 + 2954.0 \cdot 771.4 \cdot 26.862 ) 0.508$$

$$= 69992032 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= ( M ( e + t/2 ) ) / I$$

$$= ( 69992032 ( 26.862 + 33.0/2 ) ) / 95448488$$

$$= 31.787 \text{ MPa}$$

Weld Size Calculations, Description: N2b

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm



### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (13170.5879 - 45388.7461 + 2 \cdot 48.0 \cdot 1.0 \cdot (1.0 \cdot 33.0 - 12.1185)) \cdot 138) \\
 &= \max(0, -4169037.75) \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (7487.6758 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\
 &= 1046991.31 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (7487.6758 + 0.0 + 100.0 + (3168.0002)) \cdot 138 \\
 &= 1484130.25 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (7487.6758 + 0.0 + 100.0 + 0.0 + (3168.0002)) \cdot 138 \\
 &= 1484130.25 \text{ N}
 \end{aligned}$$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 2218.7183 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 2356418. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 1098.088) \cdot (48.0 - 0.0) \cdot 0.7 \cdot 138 \\
 &= 15994417. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 2218.7183 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 5693980. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (2356419 + 15994417) = 18350836 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (2356419 + 0 + 5693980 + 0) = 8050398 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (2356419 + 5693980 + 0) = 8050398 \text{ N}
 \end{aligned}$$

### Summary of Failure Path Calculations:

|                      |                       |                   |
|----------------------|-----------------------|-------------------|
| Path 1-1 = 18350836N | , must exceed W = 0 N | or W1 = 1046991 N |
| Path 2-2 = 8050398 N | , must exceed W = 0 N | or W2 = 1484130 N |
| Path 3-3 = 8050398 N | , must exceed W = 0 N | or W3 = 1484130 N |

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

#### Reinforcement CALCULATION, Description: N2b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 62.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
= 10.9001 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.51 \cdot 787.4 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
= 2.8947 mm

Required Nozzle thickness under External Pressure per UG-28 : 2.7352 mm

#### Intermediate Hub Nozzle Calculations:

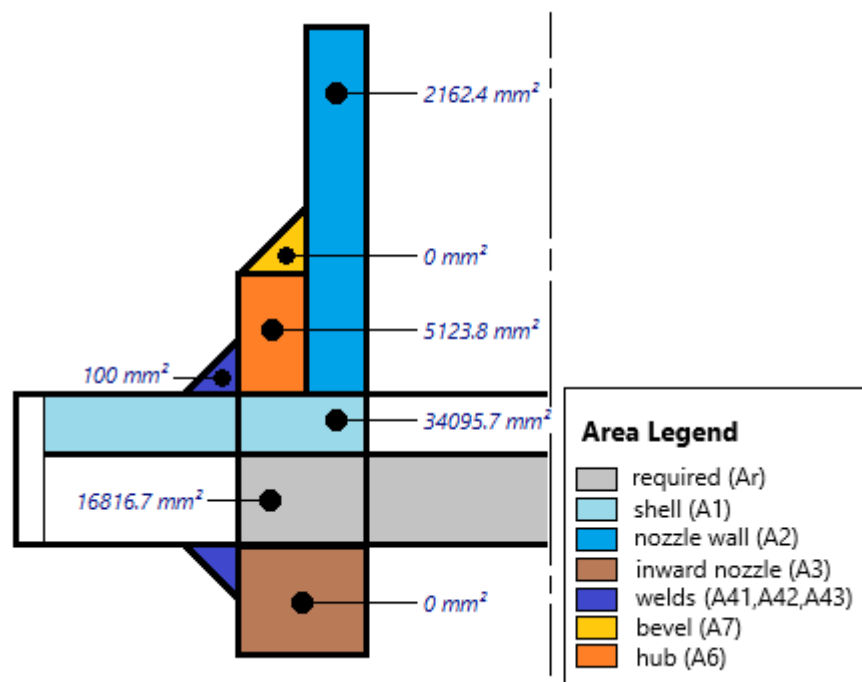
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
= ( Hub Thickness - Neck Thickness ) /  $\tan(30)$   
=  $32.0 / 0.5773$   
= 55.4256 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

#### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |              |
|-------------------------------------------------|------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 3085.6001 mm |
| Parallel to Vessel Wall, opening length         | d    | 1542.8000 mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 82.5000 mm   |



#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 16816.656 | 9348.209  | NA    |
| Area in Shell            | A1          | 34095.746 | 32215.986 | NA    |
| Area in Nozzle Wall      | A2          | 2162.370  | 2188.692  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 100.000   | 100.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| Area in Hub + Bevel      | A6          | 5123.751  | 5123.751  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 41481.867 | 39628.426 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (1542.8 \cdot 10.9001 \cdot 1.0 + 2 \cdot 48.0 \cdot 10.9001 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 16816.656 \text{ mm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 1542.8(1.0 \cdot 33.0 - 1.0 \cdot 10.9) - 2 \cdot 48.0$$

$$(1.0 \cdot 33.0 - 1.0 \cdot 10.9001) \cdot (1 - 1.0)$$

$$= 34095.746 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 82.5)(16.0 - 2.89)1.0$$

$$= 2162.370 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(82.5, 363.3, 70.0)) \cdot (48.0 - 16.0) \cdot 1.0$$

$$= 5123.751 \text{ mm}^2 \text{ Includes Bevel Area of } 643.750 \text{ mm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 2.8947 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 2.5000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 11.9001 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 11.9001 mm |
| Wall Thickness, shell/head, external pressure  | trb2 = 3.2158 mm                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.2158 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.3312 mm                     |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 8.331, \max( 11.9001, 3.2158 ) ]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 2.8947, 8.3312 )$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 26.4, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 318.6 MPa | Passed |
| Occasional | : 12.1, | Allowable | : 183.5 MPa | Passed |
| Shear      | : 10.9, | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$$

$$= 32.0 / 0.5773$$

$$= 55.4256 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

#### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |              |
|-------------------------------------------------|------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 2314.2000 mm |
| Parallel to Vessel Wall, opening length         | d    | 1157.1000 mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 82.5000 mm   |

#### Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm<sup>2</sup>)

| AREA AVAILABLE, A1 to A6          |      | No Pad    | With Pad  |
|-----------------------------------|------|-----------|-----------|
| Area Required [Internal Pressure] | AR   | 11211.665 | 11211.665 |
| Area Available in Shell           | A1   | 17047.873 | 17047.873 |
| Area Available in Nozzle Wall     | A2   | 2162.370  | 2162.370  |
| Area Available in Inward Nozzle   | A3   | 0.000     | 0.000     |
| Area Available in Welds           | A4   | 100.000   | 100.000   |
| Area Available in Pad             | A5   | 0.000     | 0.000     |
| Area Available in Hub             | A6   | 5123.751  | 5123.751  |
| TOTAL AREA AVAILABLE              | Atot | 24433.994 | 24433.994 |

The area available without a pad is Sufficient.

*The following Calculation is performed using the internal design pressure (P) in the corroded condition.*



*Warning: The flange could not be considered in the moment of inertia calculations.*

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

|        | Area    | Distance | Inertia    | d       | I+Ad <sup>2</sup> |
|--------|---------|----------|------------|---------|-------------------|
| Nozzle | 4624.0  | 144.5    | 0.3218E+08 | 101.138 | 0.7948E+08        |
| Hub    | 2752.0  | 43.5     | 0.1813E+07 | 0.135   | 0.1813E+07        |
| Shell  | 17424.0 | 16.5     | 0.1581E+07 | 26.862  | 0.1415E+08        |
| Sum    | 24800.0 |          |            |         | 0.9545E+08        |

|                                                |        |     |
|------------------------------------------------|--------|-----|
| Membrane stress per Appendix 1-7 ( 1 or 2 ) Sm | 110.94 | MPa |
| Allowable Membrane Stress S                    | 138.00 | MPa |
| Bending stress at the inside of the Nozzle Sb  | 31.79  | MPa |
| Combined Membrane plus Bending Stress Sm + Sb  | 142.73 | MPa |
| Allowable Membrane + Bending Stress 1.5 * S    | 207.00 | MPa |
| M.A.W.P for the Appendix 1-7 Stress calc       | 0.63   | MPa |

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 2314.735 mm<sup>2</sup>  
 Area from Shell = 10332.032 mm<sup>2</sup>  
 Total Area, As = 15398.769 mm<sup>2</sup>

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [ R ( R_n + t_n + \sqrt{(r_m \cdot t)} ) + R_n ( t + \sqrt{(R_n m \cdot t_n)} ) ] / A_s$$

$$= 0.508 [ 2954.0 ( 771.4 + 16.0 + 313.092 ) + 771.4 ( 144.671 ) ] / 15398.769$$

$$= 110.943 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= ( R_n^3 / 6 + R \cdot R_n \cdot e ) P$$

$$= ( 771.4^3 / 6 + 2954.0 \cdot 771.4 \cdot 26.862 ) 0.508$$

$$= 69992032 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= ( M ( e + t/2 ) ) / I$$

$$= ( 69992032 ( 26.862 + 33.0/2 ) ) / 95448488$$

$$= 31.787 \text{ MPa}$$

Weld Size Calculations, Description: N2b

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

**Results Per UW-16.1:**

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:

$$= \max( 0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) S_v )$$

$$= \max( 0, (16816.6562 - 34095.7461 + 2 \cdot 48.0 \cdot 1.0 \cdot (1.0 \cdot 33.0 - 10.9001) ) 138 )$$

$$= \max( 0, -2091519.25 ) \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= ( 7286.1211 + 100.0 - 0.0 \cdot 1.0 ) \cdot 138$$

$$= 1019179.62 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= ( 7286.1211 + 0.0 + 100.0 + ( 3168.0002 ) ) \cdot 138$$



= 1456318.50 N

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (7286.1211 + 0.0 + 100.0 + 0.0 + (3168.0002)) \cdot 138 \\ &= 1456318.50 \text{ N} \end{aligned}$$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 1574.7999 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 1672537. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 779.4) \cdot (48.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 11352505. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 1574.7999 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 4041468. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (1672537 + 11352505) = 13025042 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (1672537 + 0 + 4041468 + 0) = 5714004 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (1672537 + 4041468 + 0) = 5714004 \text{ N} \end{aligned}$$

#### Summary of Failure Path Calculations:

|                      |                       |                   |
|----------------------|-----------------------|-------------------|
| Path 1-1 = 13025042N | , must exceed W = 0 N | or W1 = 1019179 N |
| Path 2-2 = 5714004 N | , must exceed W = 0 N | or W2 = 1456318 N |
| Path 3-3 = 5714004 N | , must exceed W = 0 N | or W3 = 1456318 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1036.6124 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1444.1216 mm

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 670.23 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1542.80 mm*

*This nozzle or its diameter limit intersects weld seam # [8] on this element. The distance between the nozzle and seam is, 1288.90 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1542.80 mm*

#### Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

#### Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.





**Input Echo, WRC297 Item 1, Description: N2b :**

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 34.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |            |
|------------------------------------|-----------|------------|
| Diameter Basis for Nozzle          | OD        |            |
| Nozzle Corrosion Allowance         | 0.0000    | mm         |
| Nozzle Diameter                    | 1638.800  | mm         |
| Nozzle Thickness                   | 48.0000   | mm         |
| Nozzle Stress Concentration Factor | 1.000     |            |
| Nozzle Material                    | SA-516 70 |            |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh      | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUstained loads.*

|                          |    |            |      |
|--------------------------|----|------------|------|
| Design Internal Pressure | Dp | 0.51       | MPa  |
| Radial Load              | P  | -116249.82 | N    |
| Circumferential Shear    | Vc | 77500.17   | N    |
| Longitudinal Shear       | Vl | 77500.17   | N    |
| Circumferential Moment   | Mc | 0.337E+09  | N-mm |
| Longitudinal Moment      | Ml | 0.337E+09  | N-mm |
| Torsional Moment         | Mt | 0.481E+09  | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 1638.8 + 2 \cdot 1.65 \cdot \sqrt{(2970.5(34.0 - 1.0))}$   
 $= 2672.003 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |          |    |
|-------------------------|-------|----------|----|
| Vessel Mean Diameter    | (D) = | 5941.000 | mm |
| Nozzle Outside Diameter | (d) = | 1638.800 | mm |
| Vessel Thickness used   | (T) = | 33.000   | mm |
| Nozzle Thickness used   | (t) = | 48.000   | mm |
| T / t                   | =     | 0.688    |    |
| d / t                   | =     | 34.142   |    |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.701$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.036 |
| Mr/P    | = | 0.091 |
| M0/P    | = | 0.029 |
| N0/P    | = | 0.118 |
| MrD/Mc  | = | 0.234 |
| NrDL/Mc | = | 0.062 |
| M0d/Mc  | = | 0.078 |



NØDL/Mc = 0.118  
 MrD/Ml = 0.079  
 NrDL/Ml = 0.030  
 MØD/Ml = 0.024  
 NØDL/Ml = 0.123

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 12                   | 12                  | 12                      | 12                     |
| Outplane Bending (P )                                            | 18                   | -18                 | 18                      | -18                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -23                  | -23                 | 23                      | 23                     |
| Outplane Bending (ML)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 45                   | 45                  | 45                      | 45                     |
| Outplane Stress Summary                                          | 25                   | 44                  | 126                     | 35                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |
| Inplane Bending (P )                                               | 58                   | -58                 | 58                      | -58                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -5                   | -5                  | 5                       | 5                      |
| Inplane Bending (ML)                                               | -88                  | 88                  | 88                      | -88                    |
| Inplane Pressure Stress                                            | 22                   | 22                  | 22                      | 22                     |
| Inplane Stress Summary                                             | -9                   | 51                  | 179                     | -115                   |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 3                    | 3                   | 3                       | 3                      |
| Shear Stress Summary                                                 | 4                    | 4                   | 2                       | 2                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 36                   | 53                  | 179                     | 150                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 12                    | 12                   | 12                     | 12                    |
| Outplane Bending (P )                                                  | 18                    | -18                  | 18                     | -18                   |
| Outplane Membrane (Mc)                                                 | -22                   | -22                  | 22                     | 22                    |
| Outplane Bending (Mc)                                                  | -88                   | 88                   | 88                     | -88                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 22                    | 22                   | 22                     | 22                    |
| Outplane Stress Summary                                                | -57                   | 83                   | 164                    | -49                   |

**Vessel Stresses (MPa):**



| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 3                     | 3                    | 3                      | 3                     |
| Inplane Bending (P )                                                     | 58                    | -58                  | 58                     | -58                   |
| Inplane Membrane (Mc)                                                    | -11                   | -11                  | 11                     | 11                    |
| Inplane Bending (Mc)                                                     | -265                  | 265                  | 265                    | -265                  |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 45                    | 45                   | 45                     | 45                    |
| Inplane Stress Summary                                                   | -169                  | 244                  | 384                    | -262                  |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | 0                     | 0                    | 0                      | 0                     |
| Torsional Shear (Mt)                                                       | 3                     | 3                    | 3                      | 3                     |
| Shear Stress Summary                                                       | 2                     | 2                    | 4                      | 4                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 169                   | 244                  | 384                    | 262                   |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 12                   | 12                  | 12                      | 12                     |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -23                  | -23                 | 23                      | 23                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 9                    | 10                  | 9                       | 10                     |
| Hoop Stress Summary                                       | -1                   | 0                   | 45                      | 45                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (P )                                               | 22                   | -22                 | 22                      | -22                    |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -3                   | -3                  | 3                       | 3                      |
| Axial Bending (ML)                                               | -33                  | 33                  | 33                      | -33                    |
| Axial Pressure Stress                                            | 4                    | 4                   | 4                       | 4                      |
| Axial Stress Summary                                             | -10                  | 13                  | 64                      | -47                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 2             | 2            | 2             | 2            |



Shear Stress Summary 2 2 1 1

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 11            | 15           | 65            | 92           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 12                    | 12                   | 12                     | 12                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -22                   | -22                  | 22                     | 22                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 9                     | 10                   | 9                      | 10                    |
| Hoop Stress Summary                                          | 0                     | 0                    | 44                     | 44                    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 22                    | -22                  | 22                     | -22                   |
| Axial Membrane (Mc)                                                 | -3                    | -3                   | 3                      | 3                     |
| Axial Bending (Mc)                                                  | -108                  | 108                  | 108                    | -108                  |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 4                     | 4                    | 4                      | 4                     |
| Axial Stress Summary                                                | -85                   | 88                   | 139                    | -122                  |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 2             | 2            | 2             | 2            |
| Shear Stress Summary                    | 1             | 1            | 2             | 2            |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 85            | 88           | 139           | 167          |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |     |      |      |     |     |      |
|------------------------------------|------------------|-----|-----|------|------|-----|-----|------|
|                                    | Au               | Al  | Bu  | Bl   | Cu   | Cl  | Du  | Dl   |
| Circ. Pm (SUS)                     | 45               | 45  | 45  | 45   | 45   | 45  | 45  | 45   |
| Circ. Pl (SUS)                     | -11              | -11 | 35  | 35   | -8   | -8  | 14  | 14   |
| Circ. Q (SUS)                      | -9               | 9   | 45  | -45  | -207 | 207 | 323 | -323 |
| Long. Pm (SUS)                     | 22               | 22  | 22  | 22   | 22   | 22  | 22  | 22   |
| Long. Pl (SUS)                     | -2               | -2  | 8   | 8    | -10  | -10 | 34  | 34   |
| Long. Q (SUS)                      | -30              | 30  | 146 | -146 | -70  | 70  | 106 | -106 |

|                 |      |      |       |       |       |       |       |       |
|-----------------|------|------|-------|-------|-------|-------|-------|-------|
| Shear Pm (SUS)  | 0    | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)  | 0    | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Q (SUS)   | 3    | 3    | 3     | 3     | 3     | 3     | 3     | 3     |
| Pm (SUS)        | 45.0 | 45.0 | 45.0  | 45.0  | 45.0  | 45.0  | 45.0  | 45.0  |
| Pm+P1 (SUS)     | 34.0 | 34.0 | 80.0  | 80.0  | 37.0  | 37.0  | 59.0  | 59.0  |
| Pm+P1+Q (Total) | 35.5 | 51.1 | 176.2 | 151.1 | 170.1 | 244.1 | 382.0 | 264.0 |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 45.00     | 138.00         | Passed |
| Pm+P1 (SUS)         | 80.00     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 382.04    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |       |       |
|------------------------------|------------------|------|------|------|------|------|-------|-------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du    | Dl    |
| Circ. Pm (SUS)               | 9                | 10   | 9    | 10   | 9    | 10   | 9     | 10    |
| Circ. P1 (SUS)               | -11              | -11  | 35   | 35   | -10  | -10  | 34    | 34    |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Long. Pm (SUS)               | 4                | 4    | 4    | 4    | 4    | 4    | 4     | 4     |
| Long. P1 (SUS)               | -3               | -3   | 3    | 3    | -3   | -3   | 3     | 3     |
| Long. Q (SUS)                | -11              | 11   | 55   | -55  | -86  | 86   | 130   | -130  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Shear P1 (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2     | 2     |
| Pm (SUS)                     | 9.0              | 10.0 | 9.0  | 10.0 | 9.0  | 10.0 | 9.0   | 10.0  |
| Pm+P1 (SUS)                  | 3.0              | 2.0  | 44.0 | 45.0 | 2.0  | 1.0  | 43.0  | 44.0  |
| Pm+P1+Q (Total)              | 10.5             | 13.6 | 62.2 | 93.1 | 85.0 | 87.1 | 137.0 | 167.0 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 10.00     | 138.00         | Passed |
| Pm+P1 (SUS)         | 45.00     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 167.05    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: N3A**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 28.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 5620.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 180.00  | deg |
| Diameter                          |  | 32.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 34.0000 | mm |

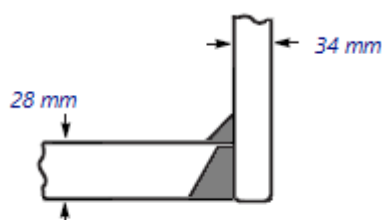
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 16.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 16.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 150    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 32.000 in.  
 Actual Thickness Used in Calculation 1.339 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
 $= 10.9001 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.51 \cdot 406.4 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
 $= 1.4941 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.9444 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1489.6000 mm  
 Parallel to Vessel Wall, opening length d 744.8000 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$

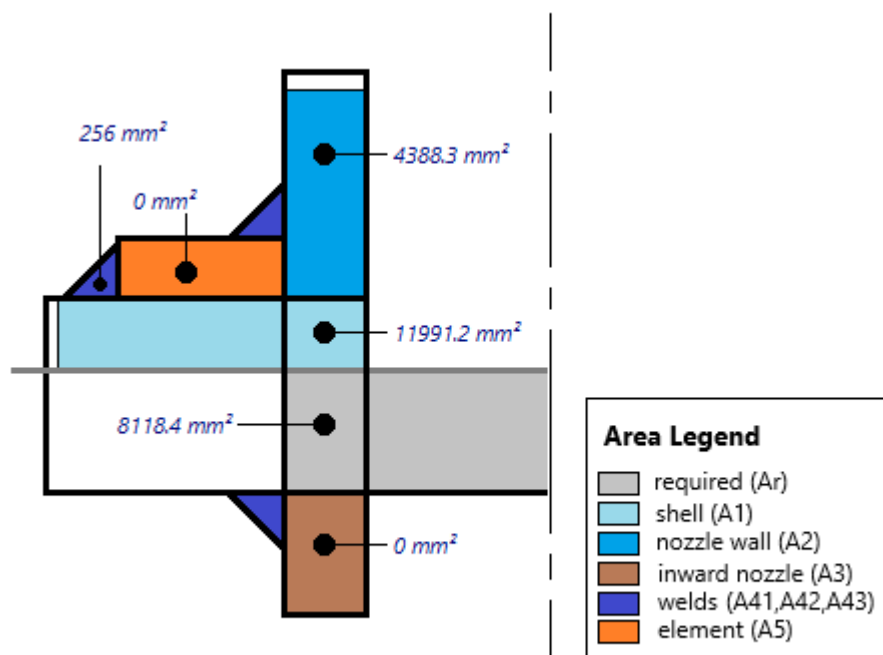


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)



| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 8118.385  | 4512.928  | NA    |
| Area in Shell            | A1          | 11991.216 | 11083.744 | NA    |
| Area in Nozzle Wall      | A2          | 4388.303  | 4327.509  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 256.000   | 256.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 16635.520 | 15667.253 | NA    |

#### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (744.8 \cdot 10.9001 \cdot 1.0 + 2 \cdot 34.0 \cdot 10.9001 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 8118.385 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 744.8(1.0 \cdot 27.0 - 1.0 \cdot 10.9) - 2 \cdot 34.0$   
 $(1.0 \cdot 27.0 - 1.0 \cdot 10.9001) \cdot (1 - 1.0)$   
 $= 11991.216 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 67.5)(34.0 - 1.49)1.0$   
 $= 4388.303 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 16.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$   
 $= 256.000 \text{ mm}^2$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 1.9444 \text{ mm}$   
 Wall Thickness per UG16(b),  $tr16b = 2.5000 \text{ mm}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 11.9001 \text{ mm}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 11.9001 \text{ mm}$   
 Wall Thickness, shell/head, external pressure  $trb2 = 3.2158 \text{ mm}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 3.2158 \text{ mm}$   
 Wall Thickness per table UG-45  $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 8.331, \max( 11.9001, 3.2158 ) ]$   
 $= 8.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 1.9444, 8.3312 )$   
 $= 8.3312 \text{ mm}$

Available Nozzle Neck Thickness = 34.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |           |        |
|------------|---|-------|-----------|---|-----------|--------|
| Sustained  | : | 10.1, | Allowable | : | 138.0 MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 334.9 MPa | Passed |
| Occasional | : | 2.7,  | Allowable | : | 183.5 MPa | Passed |
| Shear      | : | 5.7,  | Allowable | : | 96.6 MPa  | Passed |

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of





*the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 34.0, tr = 1.494, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr * E*/(tg - c) = 0.044$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | 27 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 28.0, tr = 10.9, c = 1.0 mm, E\* = 1.0  
Thickness Ratio =  $tr * E*/(tg - c) = 0.404$ , Temp. Reduction = 50 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | 2 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating =  $0.51/1.96 = 0.259$

##### Weld Size Calculations, Description: N3A

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

##### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 11.3120 = 0.7 * Wo mm |

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
=  $\max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$   
=  $\max(0, (8118.3853 - 11991.2158 + 2 \cdot 34.0 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 10.9001)) \cdot 138)$   
=  $\max(0, -383329.47) \text{ N}$

*Note: F is always set to 1.0 throughout the calculation.*

Weld Load [W1]:  
=  $(A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
=  $(4388.3032 + 0.0 + 256.0 - 0.0 \cdot 1.0) \cdot 138$   
= 640847.81 N

Weld Load [W2]:  
=  $(A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
=  $(4388.3032 + 0.0 + 256.0 + (1836.0002)) \cdot 138$   
= 894189.69 N

Weld Load [W3]:  
=  $(A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
=  $(4388.3032 + 0.0 + 256.0 + 0.0 + (1836.0002)) \cdot 138$   
= 894189.69 N

##### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
=  $(\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
=  $(3.1416/2.0) \cdot 812.8 \cdot 16.0 \cdot 0.49 \cdot 138$   
= 1381192. N



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 389.4) \cdot (34.0 - 0.0) \cdot 0.7 \cdot 138$$

$$= 4017584. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 812.8 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 2085919. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (1381192 + 4017584) = 5398776 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (1381192 + 0 + 2085919 + 0) = 3467111 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (1381192 + 2085919 + 0) = 3467111 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                       |                  |
|----------------------|-----------------------|------------------|
| Path 1-1 = 5398775 N | , must exceed W = 0 N | or W1 = 640847 N |
| Path 2-2 = 3467111 N | , must exceed W = 0 N | or W2 = 894189 N |
| Path 3-3 = 3467111 N | , must exceed W = 0 N | or W3 = 894189 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 28.0987 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 456.0987 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 4.366 %

*This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 154.67 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 744.80 mm*

#### Input Echo, WRC297 Item 1, Description: N3A :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 28.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |            |
|------------------------------------|-----------|------------|
| Diameter Basis for Nozzle          | OD        |            |
| Nozzle Corrosion Allowance         | 0.0000    | mm         |
| Nozzle Diameter                    | 812.800   | mm         |
| Nozzle Thickness                   | 34.0000   | mm         |
| Nozzle Stress Concentration Factor | 1.000     |            |
| Nozzle Material                    | SA-516 70 |            |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh      | 138.00 MPa |



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUstained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.51      | MPa  |
| Radial Load              | P  | -59999.96 | N    |
| Circumferential Shear    | Vc | 40000.02  | N    |
| Longitudinal Shear       | Vl | 40000.02  | N    |
| Circumferential Moment   | Mc | 0.896E+08 | N-mm |
| Longitudinal Moment      | Ml | 0.896E+08 | N-mm |
| Torsional Moment         | Mt | 0.128E+09 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 812.8 + 2 \cdot 1.65 \cdot \sqrt{(2967.5(28.0 - 1.0))}$$

$$= 1746.895 \text{ mm}$$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 5935.000 mm |
| Nozzle Outside Diameter | (d) = | 812.800 mm  |
| Vessel Thickness used   | (T) = | 27.000 mm   |
| Nozzle Thickness used   | (t) = | 34.000 mm   |
| T / t                   | =     | 0.794       |
| d / t                   | =     | 23.906      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.030$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.052 |
| Mr/P    | = | 0.130 |
| M0/P    | = | 0.042 |
| N0/P    | = | 0.114 |
| MrD/Mc  | = | 0.261 |
| NrDL/Mc | = | 0.082 |
| M0d/Mc  | = | 0.090 |
| N0DL/Mc | = | 0.104 |
| MrD/Ml  | = | 0.121 |
| NrDL/Ml | = | 0.055 |
| M0D/Ml  | = | 0.039 |
| N0DL/Ml | = | 0.166 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P)                                            | 9                    | 9                   | 9                       | 9                      |
| Outplane Bending (P)                                             | 20                   | -20                 | 20                      | -20                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -25                  | -25                 | 25                      | 25                     |
| Outplane Bending (Ml)                                            | -35                  | 35                  | 35                      | -35                    |
| Normal Pressure Stress                                           | 55                   | 55                  | 55                      | 55                     |
| Outplane Stress Summary                                          | 24                   | 54                  | 146                     | 33                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to | Au<br>Top | Al<br>Top | Bu<br>Bottom | Bl<br>Bottom |
|---------------------------------------------|-----------|-----------|--------------|--------------|
|---------------------------------------------|-----------|-----------|--------------|--------------|



| longitudinal plane)     | Outside | Inside | Outside | Inside |
|-------------------------|---------|--------|---------|--------|
| Inplane Membrane (P )   | 4       | 4      | 4       | 4      |
| Inplane Bending (P )    | 64      | -64    | 64      | -64    |
| Inplane Membrane (Mc)   | 0       | 0      | 0       | 0      |
| Inplane Bending (Mc)    | 0       | 0      | 0       | 0      |
| Inplane Membrane (ML)   | -8      | -8     | 8       | 8      |
| Inplane Bending (ML)    | -109    | 109    | 109     | -109   |
| Inplane Pressure Stress | 27      | 27     | 27      | 27     |
| Inplane Stress Summary  | -22     | 69     | 214     | -133   |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE                          | Au      | Al     | Bu      | Bl     |
|---------------------------------------------|---------|--------|---------|--------|
| (Shear stress normal to longitudinal plane) | Top     | Top    | Bottom  | Bottom |
|                                             | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)                         | 1       | 1      | -1      | -1     |
| Outplane Shear (Vl)                         | 0       | 0      | 0       | 0      |
| Outplane Shear (Mt)                         | 4       | 4      | 4       | 4      |
| Shear Stress Summary                        | 5       | 5      | 3       | 3      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE     | Au      | Al     | Bu      | Bl     |
|------------------------|---------|--------|---------|--------|
| (Stress Intensities    | Top     | Top    | Bottom  | Bottom |
|                        | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress | 48      | 71     | 214     | 167    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                      | Cu      | Cl     | Du      | Dl     |
|--------------------------------------------|---------|--------|---------|--------|
| (Stresses Normal to circumferential plane) | Left    | Left   | Right   | Right  |
|                                            | Outside | Inside | Outside | Inside |
| Outplane Membrane (P )                     | 9       | 9      | 9       | 9      |
| Outplane Bending (P )                      | 20      | -20    | 20      | -20    |
| Outplane Membrane (Mc)                     | -15     | -15    | 15      | 15     |
| Outplane Bending (Mc)                      | -82     | 82     | 82      | -82    |
| Outplane Membrane (ML)                     | 0       | 0      | 0       | 0      |
| Outplane Bending (ML)                      | 0       | 0      | 0       | 0      |
| Normal Pressure Stress                     | 27      | 27     | 27      | 27     |
| Outplane Stress Summary                    | -39     | 82     | 155     | -50    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                        | Cu      | Cl     | Du      | Dl     |
|----------------------------------------------|---------|--------|---------|--------|
| (Stresses parallel to circumferential plane) | Left    | Left   | Right   | Right  |
|                                              | Outside | Inside | Outside | Inside |
| Inplane Membrane (P )                        | 4       | 4      | 4       | 4      |
| Inplane Bending (P )                         | 64      | -64    | 64      | -64    |
| Inplane Membrane (Mc)                        | -12     | -12    | 12      | 12     |
| Inplane Bending (Mc)                         | -236    | 236    | 236     | -236   |
| Inplane Membrane (ML)                        | 0       | 0      | 0       | 0      |
| Inplane Bending (ML)                         | 0       | 0      | 0       | 0      |
| Inplane Pressure Stress                      | 55      | 55     | 55      | 55     |
| Inplane Stress Summary                       | -125    | 220    | 372     | -227   |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE                          | Cu      | Cl     | Du      | Dl     |
|------------------------------------------------|---------|--------|---------|--------|
| (Shear stress normal to circumferential plane) | Left    | Left   | Right   | Right  |
|                                                | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)                            | 0       | 0      | 0       | 0      |
| Outplane Shear (Vl)                            | -1      | -1     | 1       | 1      |
| Torsional Shear (Mt)                           | 4       | 4      | 4       | 4      |
| Shear Stress Summary                           | 3       | 3      | 5       | 5      |



**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 125                   | 220                  | 372                    | 228                   |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 9                    | 9                   | 9                       | 9                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -25                  | -25                 | 25                      | 25                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 6                    | 6                   | 6                       | 6                      |
| Hoop Stress Summary                                       | -9                   | -8                  | 40                      | 41                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (P )                                               | 32                   | -32                 | 32                      | -32                    |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -5                   | -5                  | 5                       | 5                      |
| Axial Bending (ML)                                               | -53                  | 53                  | 53                      | -53                    |
| Axial Pressure Stress                                            | 3                    | 3                   | 3                       | 3                      |
| Axial Stress Summary                                             | -23                  | 19                  | 95                      | -75                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 3             | 3            | 3             | 3            |
| Shear Stress Summary                 | 4             | 4            | 2             | 2            |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 24            | 29           | 95            | 117          |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 9                     | 9                    | 9                      | 9                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -15                   | -15                  | 15                     | 15                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 6                     | 6                    | 6                      | 6                     |
| Hoop Stress Summary                                          | 0                     | 0                    | 31                     | 32                    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 32                    | -32                  | 32                     | -32                   |
| Axial Membrane (Mc)                                                 | -5                    | -5                   | 5                      | 5                     |
| Axial Bending (Mc)                                                  | -125                  | 125                  | 125                    | -125                  |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 3                     | 3                    | 3                      | 3                     |
| Axial Stress Summary                                                | -95                   | 91                   | 167                    | -148                  |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 3             | 3            | 3             | 3            |
| Shear Stress Summary                    | 2             | 2            | 4             | 4            |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 95            | 91           | 167           | 180          |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |      |       |       |       |       |       |       |
|------------------------------------|------------------|------|-------|-------|-------|-------|-------|-------|
|                                    | Au               | Al   | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)                     | 55               | 55   | 55    | 55    | 55    | 55    | 55    | 55    |
| Circ. Pl (SUS)                     | -16              | -16  | 34    | 34    | -8    | -8    | 16    | 16    |
| Circ. Q (SUS)                      | -15              | 15   | 55    | -55   | -172  | 172   | 300   | -300  |
| Long. Pm (SUS)                     | 27               | 27   | 27    | 27    | 27    | 27    | 27    | 27    |
| Long. Pl (SUS)                     | -4               | -4   | 12    | 12    | -6    | -6    | 24    | 24    |
| Long. Q (SUS)                      | -45              | 45   | 173   | -173  | -62   | 62    | 102   | -102  |
| Shear Pm (SUS)                     | 0                | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)                     | 1                | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                      | 4                | 4    | 4     | 4     | 4     | 4     | 4     | 4     |
| Pm (SUS)                           | 55.0             | 55.0 | 55.0  | 55.0  | 55.0  | 55.0  | 55.0  | 55.0  |
| Pm+Pl (SUS)                        | 39.1             | 39.1 | 89.0  | 89.0  | 47.0  | 47.0  | 71.0  | 71.0  |
| Pm+Pl+Q (Total)                    | 47.1             | 69.6 | 212.1 | 168.1 | 125.1 | 219.1 | 371.1 | 229.1 |

**Vessel Stress Summation Comparison (MPa):**

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 55.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 89.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 371.11    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**



**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of Stress Int. Location | Stress Values at |      |      |       |      |      |       |       |
|------------------------------|------------------|------|------|-------|------|------|-------|-------|
|                              | Au               | Al   | Bu   | Bl    | Cu   | Cl   | Du    | Dl    |
| Circ. Pm (SUS)               | 6                | 6    | 6    | 6     | 6    | 6    | 6     | 6     |
| Circ. Pl (SUS)               | -16              | -16  | 34   | 34    | -6   | -6   | 24    | 24    |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0     | 0    | 0    | 0     | 0     |
| Long. Pm (SUS)               | 3                | 3    | 3    | 3     | 3    | 3    | 3     | 3     |
| Long. Pl (SUS)               | -5               | -5   | 5    | 5     | -5   | -5   | 5     | 5     |
| Long. Q (SUS)                | -21              | 21   | 85   | -85   | -93  | 93   | 157   | -157  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0     | 0    | 0    | 0     | 0     |
| Shear Pl (SUS)               | 0                | 0    | 0    | 0     | 0    | 0    | 0     | 0     |
| Shear Q (SUS)                | 3                | 3    | 3    | 3     | 3    | 3    | 3     | 3     |
| Pm (SUS)                     | 6.0              | 6.0  | 6.0  | 6.0   | 6.0  | 6.0  | 6.0   | 6.0   |
| Pm+Pl (SUS)                  | 10.0             | 10.0 | 40.0 | 40.0  | 2.0  | 2.0  | 30.0  | 30.0  |
| Pm+Pl+Q (Total)              | 23.7             | 29.6 | 93.2 | 117.2 | 95.2 | 91.2 | 165.1 | 179.1 |

**Nozzle Stress Summation Comparison (MPa):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 6.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 40.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 179.10    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: N4a**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 24.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                        |    |           |    |
|----------------------------------------|----|-----------|----|
| Distance from Cylinder/Cone Centerline | L1 | 1400.0000 | mm |
|----------------------------------------|----|-----------|----|

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 4000.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 242.00  | deg |
| Diameter                          |  | 24.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 20.0000 | mm |

|                 |  |        |  |
|-----------------|--|--------|--|
| Flange Material |  | SA-105 |  |
|-----------------|--|--------|--|

|                                                 |    |         |    |
|-------------------------------------------------|----|---------|----|
| Hub Height of Integral Nozzle                   | h  | 70.0000 | mm |
| Height of Beveled Transition                    | L' | 30.0000 | mm |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |    | 50.0000 | mm |

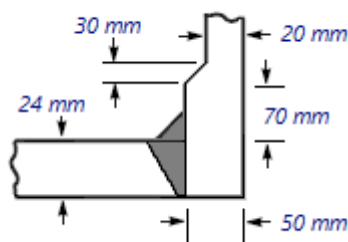
|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 347.6000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 150    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).





Note : Checking Nozzle 90 degrees to the Longitudinal axis.

### Reinforcement CALCULATION, Description: N4a

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 24.000 in. |
| Actual Thickness Used in Calculation        | 0.787 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
 $= 10.9001 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.51 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
 $= 1.1205 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.5514 mm

### Intermediate Hub Nozzle Calculations:

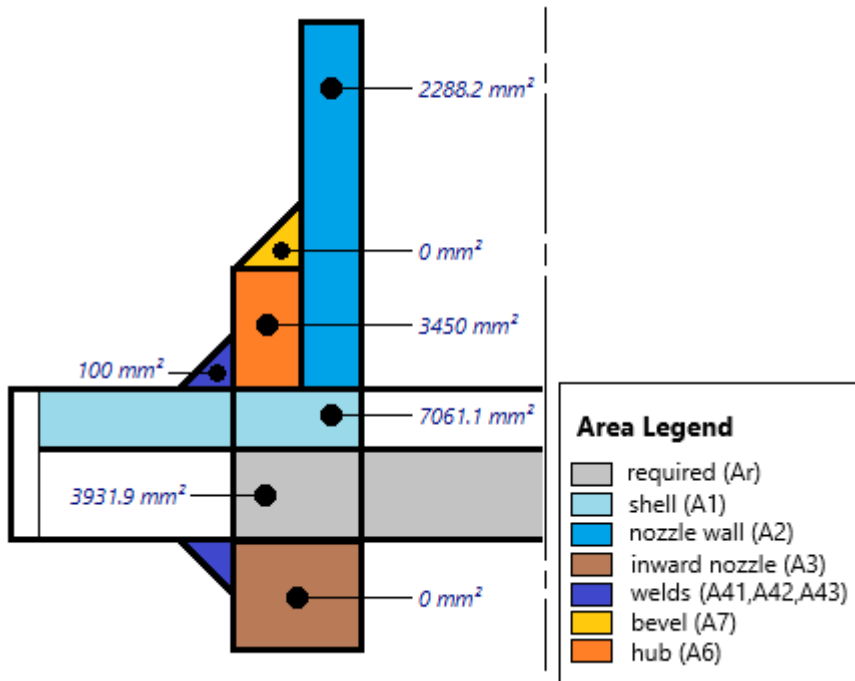
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$   
 $= 30.0 / 0.5773$   
 $= 51.9616 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |           |    |
|-------------------------------------------------|------|-----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl   | 1297.8151 | mm |
| Parallel to Vessel Wall, opening length         | d    | 648.9075  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 57.5000   | mm |



#### Results of Nozzle Reinforcement Area Calculations: (mm<sup>2</sup>)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 3536.575  | 3931.892  | NA    |
| Area in Shell            | A1          | 11388.299 | 7061.090  | NA    |
| Area in Nozzle Wall      | A2          | 2341.646  | 2288.201  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 100.000   | 100.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| Area in Hub              | A6          | 3450.000  | 3450.000  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 17279.945 | 12899.291 | NA    |

#### The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 61.38 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$  per UG-37(d)  
 $= 0.5(648.9075 \cdot 12.1185 \cdot 1 + 2 \cdot 50.0 \cdot 12.1185 \cdot 1(1 - 1.0))$   
 $= 3931.892 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 648.908(1.0 \cdot 23.0 - 1.0 \cdot 12.118) - 2 \cdot 50.0$   
 $(1.0 \cdot 23.0 - 1.0 \cdot 12.1185) \cdot (1 - 1.0)$   
 $= 7061.090 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$   
 $= (2 \cdot 57.5)(20.0 - 1.55)1.0/\sin(68.0)$   
 $= 2288.201 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$



= 100.000 mm<sup>2</sup>

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(57.5, 347.6, 70.0)) \cdot (50.0 - 20.0) \cdot 1.0 \\ &= 3450.000 \text{ mm}^2 \end{aligned}$$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 20.0, tr = 1.121, c = 0.0 mm, E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | 12 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24.0, tr = 10.9, c = 1.0 mm, E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.474, Temp. Reduction = 36 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -2 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -38 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -38 °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

##### Weld Size Calculations, Description: N4a

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

##### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (3931.8918 - 7061.0898 + 2 \cdot 50.0 \cdot 1.0 \cdot (1.0 \cdot 23.0 - 12.1185)) \cdot 138) \\ &= \max(0, -281635.53) \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i \cdot Can/.707)^2 \cdot fr_2) \cdot Sv \\ &= (5738.2017 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 805588.75 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv \\ &= (5738.2017 + 0.0 + 100.0 + (2300.0)) \cdot 138 \\ &= 1122956.00 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (5738.2017 + 0.0 + 100.0 + 0.0 + (2300.0)) \cdot 138 \\ &= 1122956.00 \text{ N} \end{aligned}$$



### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 694.4769 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 737578. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 335.8461) \cdot (50.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 5095660. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 694.4769 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1782262. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (737578 + 5095660) = 5833237 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (737578 + 0 + 1782262 + 0) = 2519840 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (737578 + 1782262 + 0) = 2519840 \text{ N} \end{aligned}$$

### Summary of Failure Path Calculations:

|                      |                       |                   |
|----------------------|-----------------------|-------------------|
| Path 1-1 = 5833237 N | , must exceed W = 0 N | or W1 = 805588 N  |
| Path 2-2 = 2519840 N | , must exceed W = 0 N | or W2 = 1122956 N |
| Path 3-3 = 2519840 N | , must exceed W = 0 N | or W3 = 1122956 N |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

### Reinforcement CALCULATION, Description: N4a

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 24.000 in. |
| Actual Thickness Used in Calculation        | 0.787 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51) \\ &= 10.9001 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\ &= 0.51 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.51) \\ &= 1.1205 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.5514 mm

### Intermediate Hub Nozzle Calculations:



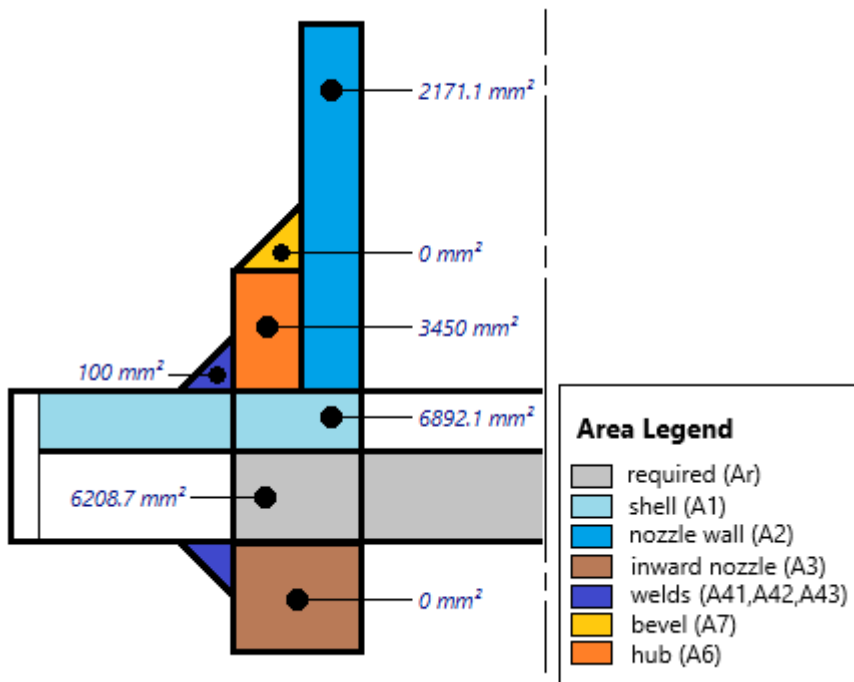
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 30.0/0.5773$   
 $= 51.9616 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

#### UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1139.2001 mm  
 Parallel to Vessel Wall, opening length d 569.6000 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 57.5000 mm



#### Results of Nozzle Reinforcement Area Calculations: (mm<sup>2</sup>)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 6208.690  | 3451.348  | NA    |
| Area in Shell            | A1          | 6892.110  | 6198.104  | NA    |
| Area in Nozzle Wall      | A2          | 2171.138  | 2121.585  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 100.000   | 100.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| Area in Hub              | A6          | 3450.000  | 3450.000  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 12613.249 | 11869.689 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= ( d \cdot \text{tr} \cdot F + 2 \cdot \text{tn} \cdot \text{tr} \cdot F \cdot (1 - \text{fr1}) ) \text{ UG-37(c)}$   
 $= ( 569.6 \cdot 10.9001 \cdot 1.0 + 2 \cdot 50.0 \cdot 10.9001 \cdot 1.0 \cdot (1 - 1.0) )$   
 $= 6208.690 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d( E1 \cdot t - F \cdot \text{tr} ) - 2 \cdot \text{tn}( E1 \cdot t - F \cdot \text{tr} ) \cdot ( 1 - \text{fr1} )$



$$= 569.6(1.0 \cdot 23.0 - 1.0 \cdot 10.9) - 2 \cdot 50.0$$

$$(1.0 \cdot 23.0 - 1.0 \cdot 10.9001) \cdot (1 - 1.0)$$

$$= 6892.110 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2$$

$$= (2 \cdot 57.5)(20.0 - 1.12)1.0$$

$$= 2171.138 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(57.5, 347.6, 70.0)) \cdot (50.0 - 20.0) \cdot 1.0$$

$$= 3450.000 \text{ mm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 1.5514 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 2.5000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 11.9001 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 11.9001 mm |
| Wall Thickness, shell/head, external pressure  | trb2 = 3.2158 mm                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.2158 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.3312 mm                     |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 8.331, \max( 11.9001, 3.2158 ) ]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 1.5514, 8.3312 )$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 15.8, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 329.2 MPa | Passed |
| Occasional | : 3.5,  | Allowable | : 183.5 MPa | Passed |
| Shear      | : 9.4,  | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Weld Size Calculations, Description: N4a

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max( 0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr))S_v )$$

$$= \max( 0, (6208.6899 - 6892.1104 + 2 \cdot 50.0 \cdot 1.0 \cdot (1.0 \cdot 23.0 - 10.9001) )138 )$$

$$= 72659.32 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.



Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (5621.1382 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 789435.69 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (5621.1382 + 0.0 + 100.0 + (2300.0)) \cdot 138 \\ &= 1106803.00 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (5621.1382 + 0.0 + 100.0 + 0.0 + (2300.0)) \cdot 138 \\ &= 1106803.00 \text{ N} \end{aligned}$$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 647434. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 294.8) \cdot (50.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 4472883. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 609.6 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1564439. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (647434 + 4472883) = 5120317 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (647434 + 0 + 1564439 + 0) = 2211873 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (647434 + 1564439 + 0) = 2211873 \text{ N} \end{aligned}$$

### Summary of Failure Path Calculations:

|                      |                           |                   |
|----------------------|---------------------------|-------------------|
| Path 1-1 = 5120316 N | , must exceed W = 72659 N | or W1 = 789435 N  |
| Path 2-2 = 2211872 N | , must exceed W = 72659 N | or W2 = 1106803 N |
| Path 3-3 = 2211872 N | , must exceed W = 72659 N | or W3 = 1106803 N |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 188.8443 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 563.6707 mm

### Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: N4a :



|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 24.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |            |
|------------------------------------|-----------|------------|
| Diameter Basis for Nozzle          | OD        |            |
| Nozzle Corrosion Allowance         | 0.0000    | mm         |
| Nozzle Diameter                    | 669.600   | mm         |
| Nozzle Thickness                   | 50.0000   | mm         |
| Nozzle Stress Concentration Factor | 1.000     |            |
| Nozzle Material                    | SA-516 70 |            |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh      | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSstained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.51      | MPa  |
| Radial Load              | P  | -44999.97 | N    |
| Circumferential Shear    | Vc | 29999.98  | N    |
| Longitudinal Shear       | Vl | 29999.98  | N    |
| Circumferential Moment   | Mc | 0.504E+08 | N-mm |
| Longitudinal Moment      | Ml | 0.504E+08 | N-mm |
| Torsional Moment         | Mt | 0.720E+08 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 669.6 + 2 \cdot 1.65 \cdot \sqrt{(2965.5(24.0 - 1.0))}$$

$$= 1531.440 \text{ mm}$$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 5931.000 mm |
| Nozzle Outside Diameter | (d) = | 669.600 mm  |
| Vessel Thickness used   | (T) = | 23.000 mm   |
| Nozzle Thickness used   | (t) = | 50.000 mm   |
| T / t                   | =     | 0.460       |
| d / t                   | =     | 13.392      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.813$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.067 |
| Mr/P    | = | 0.153 |
| M0/P    | = | 0.045 |
| N0/P    | = | 0.103 |
| MrD/Mc  | = | 0.287 |
| NrDL/Mc | = | 0.088 |
| M0d/Mc  | = | 0.090 |
| N0DL/Mc | = | 0.074 |
| MrD/Ml  | = | 0.125 |
| NrDL/Ml | = | 0.069 |





M0D/M1 = 0.040  
 N0DL/M1 = 0.134

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 8                    | 8                   | 8                       | 8                      |
| Outplane Bending (P )                                            | 22                   | -22                 | 22                      | -22                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -19                  | -19                 | 19                      | 19                     |
| Outplane Bending (ML)                                            | -33                  | 33                  | 33                      | -33                    |
| Normal Pressure Stress                                           | 64                   | 65                  | 64                      | 65                     |
| Outplane Stress Summary                                          | 43                   | 66                  | 149                     | 36                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 5                    | 5                   | 5                       | 5                      |
| Inplane Bending (P )                                               | 77                   | -77                 | 77                      | -77                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -9                   | -9                  | 9                       | 9                      |
| Inplane Bending (ML)                                               | -106                 | 106                 | 106                     | -106                   |
| Inplane Pressure Stress                                            | 32                   | 32                  | 32                      | 32                     |
| Inplane Stress Summary                                             | 0                    | 56                  | 232                     | -136                   |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 4                    | 4                   | 4                       | 4                      |
| Shear Stress Summary                                                 | 5                    | 5                   | 3                       | 3                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 45                   | 69                  | 232                     | 172                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 8                     | 8                    | 8                      | 8                     |
| Outplane Bending (P )                                                  | 22                    | -22                  | 22                     | -22                   |
| Outplane Membrane (Mc)                                                 | -10                   | -10                  | 10                     | 10                    |
| Outplane Bending (Mc)                                                  | -77                   | 77                   | 77                     | -77                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 32                    | 32                   | 32                     | 32                    |
| Outplane Stress Summary                                                | -23                   | 84                   | 151                    | -48                   |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|



|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 5    | 5   | 5   | 5    |
| Inplane Bending (P )    | 77   | -77 | 77  | -77  |
| Inplane Membrane (Mc)   | -12  | -12 | 12  | 12   |
| Inplane Bending (Mc)    | -244 | 244 | 244 | -244 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 64   | 65  | 64  | 65   |
| Inplane Stress Summary  | -108 | 225 | 405 | -238 |

**Vessel Stresses (MPa):**

|                         |         |        |         |        |
|-------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE   | Cu      | Cl     | Du      | Dl     |
| (Shear stress normal to | Left    | Left   | Right   | Right  |
| circumferential plane)  | Outside | Inside | Outside | Inside |
| Outplane Shear (Vc)     | 0       | 0      | 0       | 0      |
| Outplane Shear (Vl)     | -1      | -1     | 1       | 1      |
| Torsional Shear (Mt)    | 4       | 4      | 4       | 4      |
| Shear Stress Summary    | 3       | 3      | 5       | 5      |

**Vessel Stresses (MPa):**

|                        |         |        |         |        |
|------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE  | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities    | Left    | Left   | Right   | Right  |
|                        | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress | 108     | 225    | 405     | 239    |

**Nozzle Stresses (MPa):**

|                      |         |        |         |        |
|----------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE   | Au      | Al     | Bu      | Bl     |
| (Stresses in the     | Top     | Top    | Bottom  | Bottom |
| hoop direction)      | Outside | Inside | Outside | Inside |
| Hoop Membrane (P )   | 8       | 8      | 8       | 8      |
| Hoop Bending (P )    | 0       | 0      | 0       | 0      |
| Hoop Membrane (Mc)   | 0       | 0      | 0       | 0      |
| Hoop Bending (Mc)    | 0       | 0      | 0       | 0      |
| Hoop Membrane (ML)   | -19     | -19    | 19      | 19     |
| Hoop Bending (ML)    | 0       | 0      | 0       | 0      |
| Hoop Pressure Stress | 3       | 3      | 3       | 3      |
| Hoop Stress Summary  | -7      | -6     | 30      | 31     |

**Nozzle Stresses (MPa):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE    | Au      | Al     | Bu      | Bl     |
| (Stresses Normal to   | Top     | Top    | Bottom  | Bottom |
| pipe cross-section)   | Outside | Inside | Outside | Inside |
| Axial Membrane (P )   | 0       | 0      | 0       | 0      |
| Axial Bending (P )    | 12      | -12    | 12      | -12    |
| Axial Membrane (Mc)   | 0       | 0      | 0       | 0      |
| Axial Bending (Mc)    | 0       | 0      | 0       | 0      |
| Axial Membrane (ML)   | -3      | -3     | 3       | 3      |
| Axial Bending (ML)    | -16     | 16     | 16      | -16    |
| Axial Pressure Stress | 1       | 1      | 1       | 1      |
| Axial Stress Summary  | -4      | 1      | 34      | -23    |

**Nozzle Stresses (MPa):**

|                      |         |        |         |        |
|----------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE   | Au      | Al     | Bu      | Bl     |
| (Shear stress)       | Outside | Inside | Outside | Inside |
| Shear due to (Vc)    | 0       | 0      | 0       | 0      |
| Shear due to (Vl)    | 0       | 0      | 0       | 0      |
| Shear due to Torsion | 2       | 2      | 2       | 2      |
| Shear Stress Summary | 2       | 2      | 1       | 1      |

**Nozzle Stresses (MPa):**



| LONGITUDINAL PLANE<br>(Stress Intensities | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|-------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                    | 8             | 9            | 35            | 55           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 8                     | 8                    | 8                      | 8                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -10                   | -10                  | 10                     | 10                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 3                     | 3                    | 3                      | 3                     |
| Hoop Stress Summary                                          | 1                     | 1                    | 22                     | 23                    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 12                    | -12                  | 12                     | -12                   |
| Axial Membrane (Mc)                                                 | -3                    | -3                   | 3                      | 3                     |
| Axial Bending (Mc)                                                  | -43                   | 43                   | 43                     | -43                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 1                     | 1                    | 1                      | 1                     |
| Axial Stress Summary                                                | -32                   | 29                   | 62                     | -51                   |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 2             | 2            | 2             | 2            |
| Shear Stress Summary                    | 1             | 1            | 2             | 2            |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                       | 33            | 29           | 62            | 74           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |     |      |      |     |     |      |
|------------------------------------|------------------|-----|-----|------|------|-----|-----|------|
|                                    | Au               | Al  | Bu  | Bl   | Cu   | Cl  | Du  | Dl   |
| Circ. Pm (SUS)                     | 64               | 65  | 64  | 65   | 64   | 65  | 64  | 65   |
| Circ. Pl (SUS)                     | -11              | -11 | 27  | 27   | -7   | -7  | 17  | 17   |
| Circ. Q (SUS)                      | -11              | 11  | 55  | -55  | -167 | 167 | 321 | -321 |
| Long. Pm (SUS)                     | 32               | 32  | 32  | 32   | 32   | 32  | 32  | 32   |
| Long. Pl (SUS)                     | -4               | -4  | 14  | 14   | -2   | -2  | 18  | 18   |
| Long. Q (SUS)                      | -29              | 29  | 183 | -183 | -55  | 55  | 99  | -99  |
| Shear Pm (SUS)                     | 0                | 0   | 0   | 0    | 0    | 0   | 0   | 0    |
| Shear Pl (SUS)                     | 1                | 1   | -1  | -1   | -1   | -1  | 1   | 1    |
| Shear Q (SUS)                      | 4                | 4   | 4   | 4    | 4    | 4   | 4   | 4    |



|                 |      |      |       |       |       |       |       |       |
|-----------------|------|------|-------|-------|-------|-------|-------|-------|
| Pm (SUS)        | 64.0 | 65.0 | 64.0  | 65.0  | 64.0  | 65.0  | 64.0  | 65.0  |
| Pm+Pl (SUS)     | 53.0 | 54.0 | 91.0  | 92.0  | 57.0  | 58.0  | 81.0  | 82.0  |
| Pm+Pl+Q (Total) | 44.1 | 67.4 | 229.1 | 174.1 | 110.1 | 225.1 | 402.1 | 239.1 |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 65.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 92.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 402.10    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 3                | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Circ. Pl (SUS)               | -11              | -11  | 27   | 27   | -2   | -2   | 18   | 18   |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 1                | 1    | 1    | 1    | 1    | 1    | 1    | 1    |
| Long. Pl (SUS)               | -3               | -3   | 3    | 3    | -3   | -3   | 3    | 3    |
| Long. Q (SUS)                | -4               | 4    | 28   | -28  | -31  | 31   | 55   | -55  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)                     | 3.0              | 3.0  | 3.0  | 3.0  | 3.0  | 3.0  | 3.0  | 3.0  |
| Pm+Pl (SUS)                  | 8.0              | 8.0  | 30.0 | 30.0 | 3.0  | 3.0  | 21.0 | 21.0 |
| Pm+Pl+Q (Total)              | 9.2              | 10.8 | 33.2 | 54.1 | 34.2 | 29.1 | 59.1 | 72.1 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 3.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 30.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 72.11     | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: N4b**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 24.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                        |    |           |    |
|----------------------------------------|----|-----------|----|
| Distance from Cylinder/Cone Centerline | L1 | 1400.0000 | mm |
|----------------------------------------|----|-----------|----|

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 4000.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 280.00  | deg |
| Diameter                          |  | 24.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 20.0000 | mm |

|                 |  |        |  |
|-----------------|--|--------|--|
| Flange Material |  | SA-105 |  |
|-----------------|--|--------|--|

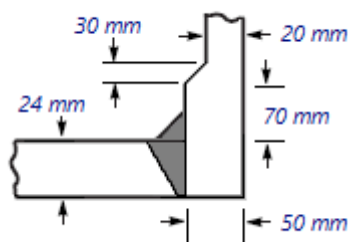
|                                                 |    |         |    |
|-------------------------------------------------|----|---------|----|
| Hub Height of Integral Nozzle                   | h  | 70.0000 | mm |
| Height of Beveled Transition                    | L' | 30.0000 | mm |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |    | 50.0000 | mm |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 347.6000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

### Reinforcement CALCULATION, Description: N4b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 24.000 in. |
| Actual Thickness Used in Calculation        | 0.787 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
 $= 10.9001 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.51 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
 $= 1.1205 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.5514 mm

### Intermediate Hub Nozzle Calculations:

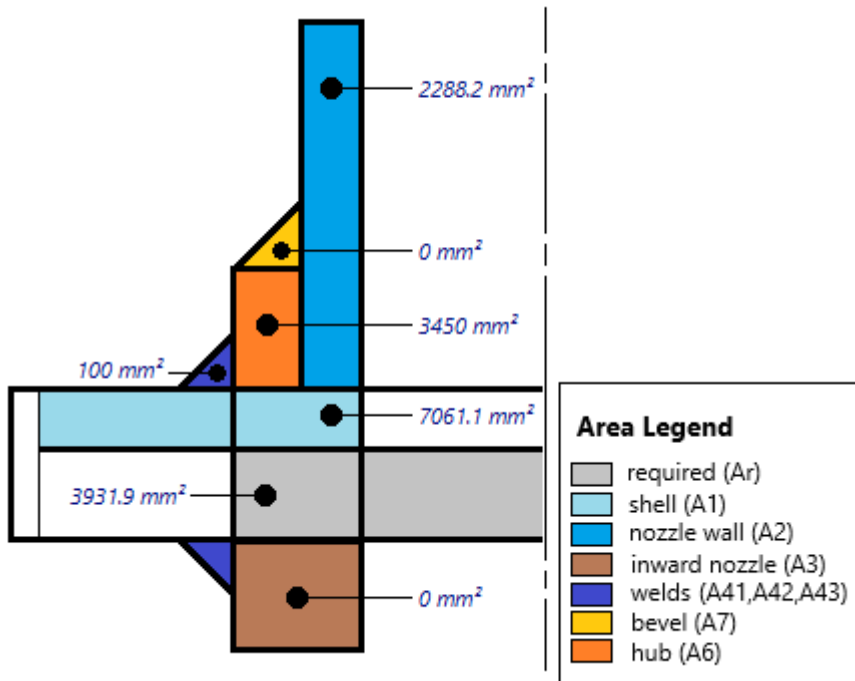
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 30.0 / 0.5773$   
 $= 51.9616 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |           |    |
|-------------------------------------------------|------|-----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl   | 1297.8151 | mm |
| Parallel to Vessel Wall, opening length         | d    | 648.9075  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 57.5000   | mm |



#### Results of Nozzle Reinforcement Area Calculations: (mm<sup>2</sup>)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 3536.575  | 3931.892  | NA    |
| Area in Shell            | A1          | 11388.299 | 7061.090  | NA    |
| Area in Nozzle Wall      | A2          | 2341.646  | 2288.201  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 100.000   | 100.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| Area in Hub              | A6          | 3450.000  | 3450.000  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 17279.945 | 12899.291 | NA    |

#### The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 61.38 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$  per UG-37(d)  
 $= 0.5(648.9075 \cdot 12.1185 \cdot 1 + 2 \cdot 50.0 \cdot 12.1185 \cdot 1(1 - 1.0))$   
 $= 3931.892 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 648.908(1.0 \cdot 23.0 - 1.0 \cdot 12.118) - 2 \cdot 50.0$   
 $(1.0 \cdot 23.0 - 1.0 \cdot 12.1185) \cdot (1 - 1.0)$   
 $= 7061.090 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$   
 $= (2 \cdot 57.5)(20.0 - 1.55)1.0/\sin(68.0)$   
 $= 2288.201 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$



= 100.000 mm<sup>2</sup>

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(57.5, 347.6, 70.0)) \cdot (50.0 - 20.0) \cdot 1.0 \\ &= 3450.000 \text{ mm}^2 \end{aligned}$$

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 20.0, tr = 1.121, c = 0.0 mm, E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | 12 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24.0, tr = 10.9, c = 1.0 mm, E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.474, Temp. Reduction = 36 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -2 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -38 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -38 °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

##### Weld Size Calculations, Description: N4b

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

##### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (3931.8918 - 7061.0898 + 2 \cdot 50.0 \cdot 1.0 \cdot (1.0 \cdot 23.0 - 12.1185)) \cdot 138) \\ &= \max(0, -281635.53) \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - C_{an}/.707)^2 \cdot fr_2) \cdot S_v \\ &= (5738.2017 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 805588.75 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (5738.2017 + 0.0 + 100.0 + (2300.0)) \cdot 138 \\ &= 1122956.00 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (5738.2017 + 0.0 + 100.0 + 0.0 + (2300.0)) \cdot 138 \\ &= 1122956.00 \text{ N} \end{aligned}$$





### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 694.4769 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 737578. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 335.8461) \cdot (50.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 5095660. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 694.4769 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1782262. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (737578 + 5095660) = 5833237 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (737578 + 0 + 1782262 + 0) = 2519840 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (737578 + 1782262 + 0) = 2519840 \text{ N} \end{aligned}$$

### Summary of Failure Path Calculations:

|                      |                       |                   |
|----------------------|-----------------------|-------------------|
| Path 1-1 = 5833237 N | , must exceed W = 0 N | or W1 = 805588 N  |
| Path 2-2 = 2519840 N | , must exceed W = 0 N | or W2 = 1122956 N |
| Path 3-3 = 2519840 N | , must exceed W = 0 N | or W3 = 1122956 N |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

### Reinforcement CALCULATION, Description: N4b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 24.000 in. |
| Actual Thickness Used in Calculation        | 0.787 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51) \\ &= 10.9001 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\ &= 0.51 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.51) \\ &= 1.1205 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.5514 mm

### Intermediate Hub Nozzle Calculations:

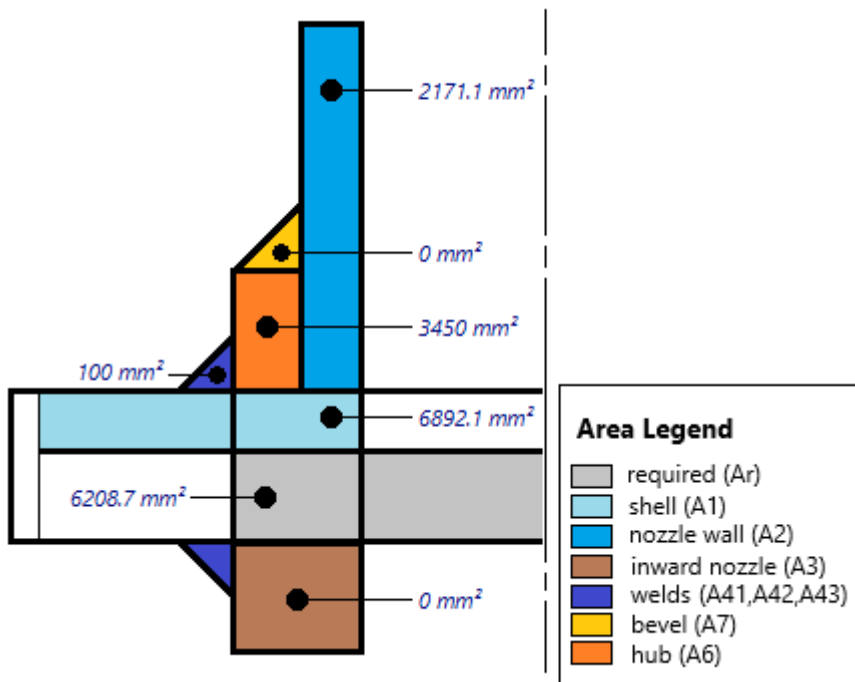
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 30.0/0.5773$   
 $= 51.9616 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

#### UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1139.2001 mm  
 Parallel to Vessel Wall, opening length d 569.6000 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 57.5000 mm



#### Results of Nozzle Reinforcement Area Calculations: (mm<sup>2</sup>)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 6208.690  | 3451.348  | NA    |
| Area in Shell            | A1          | 6892.110  | 6198.104  | NA    |
| Area in Nozzle Wall      | A2          | 2171.138  | 2121.585  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 100.000   | 100.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| Area in Hub              | A6          | 3450.000  | 3450.000  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 12613.249 | 11869.689 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= ( d \cdot \text{tr} \cdot F + 2 \cdot \text{tn} \cdot \text{tr} \cdot F \cdot (1 - \text{fr1}) ) \text{ UG-37(c)}$   
 $= ( 569.6 \cdot 10.9001 \cdot 1.0 + 2 \cdot 50.0 \cdot 10.9001 \cdot 1.0 \cdot (1 - 1.0) )$   
 $= 6208.690 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d( E1 \cdot t - F \cdot \text{tr} ) - 2 \cdot \text{tn}( E1 \cdot t - F \cdot \text{tr} ) \cdot ( 1 - \text{fr1} )$



$$= 569.6(1.0 \cdot 23.0 - 1.0 \cdot 10.9) - 2 \cdot 50.0$$

$$(1.0 \cdot 23.0 - 1.0 \cdot 10.9001) \cdot (1 - 1.0)$$

$$= 6892.110 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2$$

$$= (2 \cdot 57.5)(20.0 - 1.12)1.0$$

$$= 2171.138 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(57.5, 347.6, 70.0)) \cdot (50.0 - 20.0) \cdot 1.0$$

$$= 3450.000 \text{ mm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 1.5514 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 2.5000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 11.9001 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 11.9001 mm |
| Wall Thickness, shell/head, external pressure  | trb2 = 3.2158 mm                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 3.2158 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.3312 mm                     |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 8.331, \max( 11.9001, 3.2158 ) ]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 1.5514, 8.3312 )$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |           |        |
|------------|---|-------|-----------|---|-----------|--------|
| Sustained  | : | 15.8, | Allowable | : | 138.0 MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 329.2 MPa | Passed |
| Occasional | : | 3.5,  | Allowable | : | 183.5 MPa | Passed |
| Shear      | : | 9.4,  | Allowable | : | 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Weld Size Calculations, Description: N4b

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

#### Results Per UW-16.1:

|             |                       |                      |
|-------------|-----------------------|----------------------|
|             | Required Thickness    | Actual Thickness     |
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max( 0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr))S_v )$$

$$= \max( 0, (6208.6899 - 6892.1104 + 2 \cdot 50.0 \cdot 1.0 \cdot (1.0 \cdot 23.0 - 10.9001) )138 )$$

$$= 72659.32 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.



Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (5621.1382 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 789435.69 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (5621.1382 + 0.0 + 100.0 + (2300.0)) \cdot 138 \\ &= 1106803.00 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (5621.1382 + 0.0 + 100.0 + 0.0 + (2300.0)) \cdot 138 \\ &= 1106803.00 \text{ N} \end{aligned}$$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 647434. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 294.8) \cdot (50.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 4472883. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 609.6 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1564439. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (647434 + 4472883) = 5120317 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (647434 + 0 + 1564439 + 0) = 2211873 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (647434 + 1564439 + 0) = 2211873 \text{ N} \end{aligned}$$

### Summary of Failure Path Calculations:

|                      |                           |                   |
|----------------------|---------------------------|-------------------|
| Path 1-1 = 5120316 N | , must exceed W = 72659 N | or W1 = 789435 N  |
| Path 2-2 = 2211872 N | , must exceed W = 72659 N | or W2 = 1106803 N |
| Path 3-3 = 2211872 N | , must exceed W = 72659 N | or W3 = 1106803 N |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 188.8443 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 563.6707 mm

*This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 257.78 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 569.60 mm*

### Warning:

*The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific*



analysis code for all assumptions and limitations.

**Input Echo, WRC297 Item 1, Description: N4b :**

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 24.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |            |
|------------------------------------|-----------|------------|
| Diameter Basis for Nozzle          | OD        |            |
| Nozzle Corrosion Allowance         | 0.0000    | mm         |
| Nozzle Diameter                    | 669.600   | mm         |
| Nozzle Thickness                   | 50.0000   | mm         |
| Nozzle Stress Concentration Factor | 1.000     |            |
| Nozzle Material                    | SA-516 70 |            |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 MPa |
| Nozzle Hot S.I. Allowable          | SNmh      | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUStained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.51      | MPa  |
| Radial Load              | P  | -44999.97 | N    |
| Circumferential Shear    | Vc | 29999.98  | N    |
| Longitudinal Shear       | Vl | 29999.98  | N    |
| Circumferential Moment   | Mc | 0.504E+08 | N-mm |
| Longitudinal Moment      | Ml | 0.504E+08 | N-mm |
| Torsional Moment         | Mt | 0.720E+08 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$   
 $= 669.6 + 2 \cdot 1.65 \cdot \sqrt{(2965.5(24.0 - 1.0))}$   
 $= 1531.440 \text{ mm}$

WRC 297 Curve Access Parameters:

|                         |       |          |    |
|-------------------------|-------|----------|----|
| Vessel Mean Diameter    | (D) = | 5931.000 | mm |
| Nozzle Outside Diameter | (d) = | 669.600  | mm |
| Vessel Thickness used   | (T) = | 23.000   | mm |
| Nozzle Thickness used   | (t) = | 50.000   | mm |
| T / t                   | =     | 0.460    |    |
| d / t                   | =     | 13.392   |    |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.813$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.067 |
| Mr/P    | = | 0.153 |
| M0/P    | = | 0.045 |
| N0/P    | = | 0.103 |
| MrD/Mc  | = | 0.287 |
| NrDL/Mc | = | 0.088 |



M0d/Mc = 0.090  
 N0DL/Mc = 0.074  
 MrD/Ml = 0.125  
 NrDL/Ml = 0.069  
 M0D/Ml = 0.040  
 N0DL/Ml = 0.134

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 8                    | 8                   | 8                       | 8                      |
| Outplane Bending (P )                                            | 22                   | -22                 | 22                      | -22                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -19                  | -19                 | 19                      | 19                     |
| Outplane Bending (ML)                                            | -33                  | 33                  | 33                      | -33                    |
| Normal Pressure Stress                                           | 64                   | 65                  | 64                      | 65                     |
| Outplane Stress Summary                                          | 43                   | 66                  | 149                     | 36                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 5                    | 5                   | 5                       | 5                      |
| Inplane Bending (P )                                               | 77                   | -77                 | 77                      | -77                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -9                   | -9                  | 9                       | 9                      |
| Inplane Bending (ML)                                               | -106                 | 106                 | 106                     | -106                   |
| Inplane Pressure Stress                                            | 32                   | 32                  | 32                      | 32                     |
| Inplane Stress Summary                                             | 0                    | 56                  | 232                     | -136                   |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 4                    | 4                   | 4                       | 4                      |
| Shear Stress Summary                                                 | 5                    | 5                   | 3                       | 3                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 45                   | 69                  | 232                     | 172                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 8                     | 8                    | 8                      | 8                     |
| Outplane Bending (P )                                                  | 22                    | -22                  | 22                     | -22                   |
| Outplane Membrane (Mc)                                                 | -10                   | -10                  | 10                     | 10                    |
| Outplane Bending (Mc)                                                  | -77                   | 77                   | 77                     | -77                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 32                    | 32                   | 32                     | 32                    |
| Outplane Stress Summary                                                | -23                   | 84                   | 151                    | -48                   |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 5                     | 5                    | 5                      | 5                     |
| Inplane Bending (P )                                                     | 77                    | -77                  | 77                     | -77                   |
| Inplane Membrane (Mc)                                                    | -12                   | -12                  | 12                     | 12                    |
| Inplane Bending (Mc)                                                     | -244                  | 244                  | 244                    | -244                  |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 64                    | 65                   | 64                     | 65                    |
| Inplane Stress Summary                                                   | -108                  | 225                  | 405                    | -238                  |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                       | 4                     | 4                    | 4                      | 4                     |
| Shear Stress Summary                                                       | 3                     | 3                    | 5                      | 5                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 108                   | 225                  | 405                    | 239                   |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 8                    | 8                   | 8                       | 8                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -19                  | -19                 | 19                      | 19                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 3                    | 3                   | 3                       | 3                      |
| Hoop Stress Summary                                       | -7                   | -6                  | 30                      | 31                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (P )                                               | 12                   | -12                 | 12                      | -12                    |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -3                   | -3                  | 3                       | 3                      |
| Axial Bending (ML)                                               | -16                  | 16                  | 16                      | -16                    |
| Axial Pressure Stress                                            | 1                    | 1                   | 1                       | 1                      |
| Axial Stress Summary                                             | -4                   | 1                   | 34                      | -23                    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 2             | 2            | 2             | 2            |



Shear Stress Summary 2 2 1 1

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 8             | 9            | 35            | 55           |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 8                     | 8                    | 8                      | 8                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -10                   | -10                  | 10                     | 10                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 3                     | 3                    | 3                      | 3                     |

Hoop Stress Summary 1 1 22 23

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 12                    | -12                  | 12                     | -12                   |
| Axial Membrane (Mc)                                                 | -3                    | -3                   | 3                      | 3                     |
| Axial Bending (Mc)                                                  | -43                   | 43                   | 43                     | -43                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 1                     | 1                    | 1                      | 1                     |

Axial Stress Summary -32 29 62 -51

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 2             | 2            | 2             | 2            |

Shear Stress Summary 1 1 2 2

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 33            | 29           | 62            | 74           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |     |      |      |     |     |      |
|------------------------------------|------------------|-----|-----|------|------|-----|-----|------|
|                                    | Au               | Al  | Bu  | Bl   | Cu   | Cl  | Du  | Dl   |
| Circ. Pm (SUS)                     | 64               | 65  | 64  | 65   | 64   | 65  | 64  | 65   |
| Circ. Pl (SUS)                     | -11              | -11 | 27  | 27   | -7   | -7  | 17  | 17   |
| Circ. Q (SUS)                      | -11              | 11  | 55  | -55  | -167 | 167 | 321 | -321 |
| Long. Pm (SUS)                     | 32               | 32  | 32  | 32   | 32   | 32  | 32  | 32   |
| Long. Pl (SUS)                     | -4               | -4  | 14  | 14   | -2   | -2  | 18  | 18   |
| Long. Q (SUS)                      | -29              | 29  | 183 | -183 | -55  | 55  | 99  | -99  |



|                 |      |      |       |       |       |       |       |       |
|-----------------|------|------|-------|-------|-------|-------|-------|-------|
| Shear Pm (SUS)  | 0    | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear P1 (SUS)  | 1    | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 4    | 4    | 4     | 4     | 4     | 4     | 4     | 4     |
| Pm (SUS)        | 64.0 | 65.0 | 64.0  | 65.0  | 64.0  | 65.0  | 64.0  | 65.0  |
| Pm+P1 (SUS)     | 53.0 | 54.0 | 91.0  | 92.0  | 57.0  | 58.0  | 81.0  | 82.0  |
| Pm+P1+Q (Total) | 44.1 | 67.4 | 229.1 | 174.1 | 110.1 | 225.1 | 402.1 | 239.1 |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 65.00     | 138.00         | Passed |
| Pm+P1 (SUS)         | 92.02     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 402.10    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 3                | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Circ. P1 (SUS)               | -11              | -11  | 27   | 27   | -2   | -2   | 18   | 18   |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 1                | 1    | 1    | 1    | 1    | 1    | 1    | 1    |
| Long. P1 (SUS)               | -3               | -3   | 3    | 3    | -3   | -3   | 3    | 3    |
| Long. Q (SUS)                | -4               | 4    | 28   | -28  | -31  | 31   | 55   | -55  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear P1 (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)                     | 3.0              | 3.0  | 3.0  | 3.0  | 3.0  | 3.0  | 3.0  | 3.0  |
| Pm+P1 (SUS)                  | 8.0              | 8.0  | 30.0 | 30.0 | 3.0  | 3.0  | 21.0 | 21.0 |
| Pm+P1+Q (Total)              | 9.2              | 10.8 | 33.2 | 54.1 | 34.2 | 29.1 | 59.1 | 72.1 |

##### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 3.00      | 138.00         | Passed |
| Pm+P1 (SUS)         | 30.00     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 72.11     | 414.00         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: A130"**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 16.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 5650.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 120.00  | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

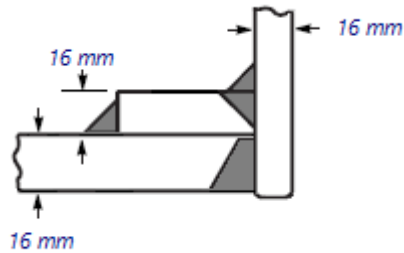
|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                          |      |           |     |
|------------------------------------------|------|-----------|-----|
| Pad Material                             |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature      | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient          | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface     | Dp   | 1382.0000 | mm  |
| Thickness of Pad                         | te   | 16.0000   | mm  |
| Weld leg size between Pad and Shell      | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle | Wgpn | 16.0000   | mm  |
| Reinforcing Pad Width                    |      | 310.0000  | mm  |

This is a Manway, Coupling or Access Opening.

|              |  |        |
|--------------|--|--------|
| Flange Class |  | 150    |
| Flange Grade |  | GR 1.1 |

The Pressure Design option was MAWP + static head (bottom of element).



### Reinforcement CALCULATION, Description: A130"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 30.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

### Messages regarding the Nozzle Input Data:

*This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
 $= 10.9001 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.51 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
 $= 1.4007 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

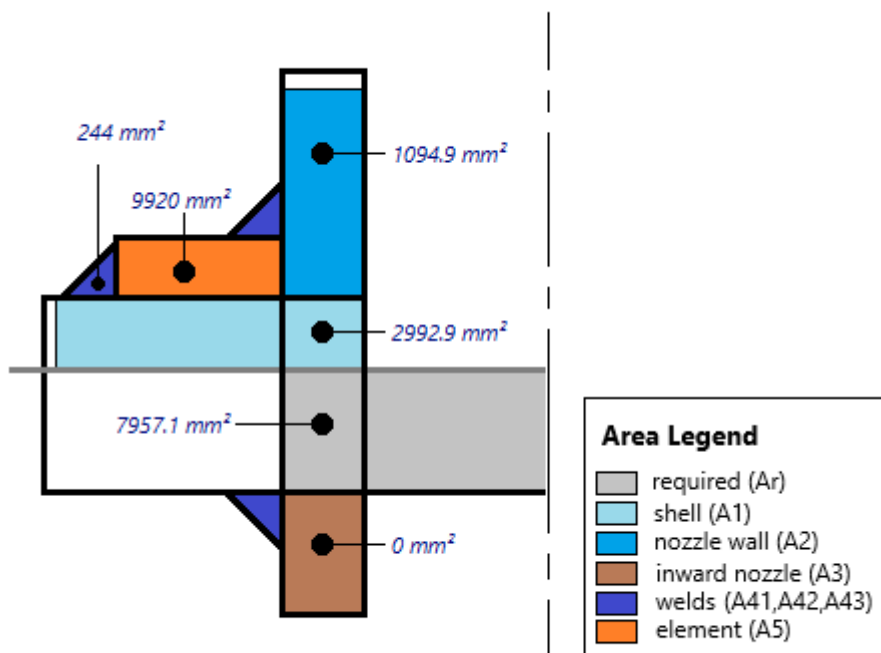
|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1460.0000 mm |
| Parallel to Vessel Wall, opening length                | d  | 730.0000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 37.5000 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5  |      | MAWP      | External  | Mapnc |
|---------------------------|------|-----------|-----------|-------|
| Area Required             | Ar   | 7957.064  | 4423.251  | NA    |
| Area in Shell             | A1   | 2992.936  | 2103.498  | NA    |
| Area in Nozzle Wall       | A2   | 1094.949  | 1059.332  | NA    |
| Area in Inward Nozzle     | A3   | 0.000     | 0.000     | NA    |
| Area in Welds A41+A42+A43 |      | 244.000   | 244.000   | NA    |
| Area in Element           | A5   | 9920.000  | 9920.000  | NA    |
| TOTAL AREA AVAILABLE      | Atot | 14251.886 | 13326.829 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 988.5737 16.0000 mm  
 Based on given Pad Diameter: 1382.0000 5.8471 mm  
 Based on Shell or Nozzle Thickness: 988.5737 16.0000 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (730.0 \cdot 10.9001 \cdot 1.0 + 2 \cdot 16.0 \cdot 10.9001 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 7957.064 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 730.0(1.0 \cdot 15.0 - 1.0 \cdot 10.9) - 2 \cdot 16.0$   
 $(1.0 \cdot 15.0 - 1.0 \cdot 10.9001) \cdot (1 - 1.0)$   
 $= 2992.936 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 37.5) \cdot (16.0 - 1.4) \cdot 1.0$



= 1094.949 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$   
= 244.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1382.0 - 762.0) \cdot 16.0 \cdot 1.0$   
= 9920.000 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 1.8756$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

=  $t_a$   
= 1.8756 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.401$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.088$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.401$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.088$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.401$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.088$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 10.9$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.727$ , Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -29 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 10.9$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.727$ , Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -29 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -29 °C  
Gov. MDMT of the Reinforcement Pad : -29 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C



### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

### Weld Size Calculations, Description: A130"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm  
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 7.5000 = 0.5*TminPad  | 8.4840 = 0.7 * Wp mm |

### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (7957.0645 - 2992.9355 + 2 \cdot 16.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 10.9001)) \cdot 138)$   
 $= 703082.50 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (1094.9493 + 9920.0 + 244.0 - 0.0 \cdot 1.0) \cdot 138$   
 $= 1553574.88 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (1094.9493 + 0.0 + 100.0 + (480.0)) \cdot 138$   
 $= 231119.19 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (1094.9493 + 0.0 + 244.0 + 9920.0 + (480.0)) \cdot 138$   
 $= 1619808.12 \text{ N}$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 809306. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$   
 $= (3.1416/2.0) \cdot 1382.0 \cdot 12.0 \cdot 0.49 \cdot 138$   
 $= 1761325. \text{ N}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 373.0) \cdot (16.0 - 0.0) \cdot 0.7 \cdot 138$   
 $= 1811002. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:  
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$   
 $= (3.1416/2) \cdot 762.0 \cdot 16.0 \cdot 0.74 \cdot 138$   
 $= 1955549. \text{ N}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 762.0 \cdot (16.0 - 1.0) \cdot 0.74 \cdot 138$   
 $= 1833327. \text{ N}$



See UG-41 and UW-15 for clarification of allowable stresses.

**Strength of Failure Paths:**

Failure Path [Path 1-1]:

$$= ( \text{SPEW} + \text{SNW} ) = ( 1761325 + 1811002 ) = 3572327 \text{ N}$$

Failure Path [Path 2-2]:

$$= ( \text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw} ) \\ = ( 809307 + 1955549 + 1833327 + 0 ) = 4598183 \text{ N}$$

Failure Path [Path 3-3]:

$$= ( \text{Spew} + \text{Tngw} + \text{Sinw} ) \\ = ( 1761325 + 1833327 + 0 ) = 3594652 \text{ N}$$

**Summary of Failure Path Calculations:**

|                      |                            |                   |
|----------------------|----------------------------|-------------------|
| Path 1-1 = 3572327 N | , must exceed W = 703082 N | or W1 = 1553574 N |
| Path 2-2 = 4598182 N | , must exceed W = 703082 N | or W2 = 231119 N  |
| Path 3-3 = 3594652 N | , must exceed W = 703082 N | or W3 = 1619808 N |

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.6818 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 440.6818 mm

**Percent Elongation Calculations:**

% Elongation per Table UG-79-1 (50\*tnom/Rf\*(1-Rf/Ro))    2.145 %

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**Input, Nozzle Desc: N3B**

**From: 120**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.5081 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 28.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 5620.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 240.00  | deg |
| Diameter                          |  | 32.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 24.0000 | mm |

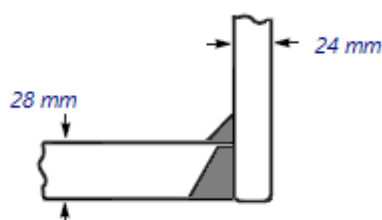
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 16.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 150    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).







ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 32.000 in.  
 Actual Thickness Used in Calculation 0.945 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.51 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.51)$   
 $= 10.9001 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.51 \cdot 406.4 / (138 \cdot 1.0 + 0.4 \cdot 0.51)$   
 $= 1.4941 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.9444 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |           |    |
|-------------------------------------------------|------|-----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 1529.6000 | mm |
| Parallel to Vessel Wall, opening length         | d    | 764.8000  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 60.0000   | mm |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$

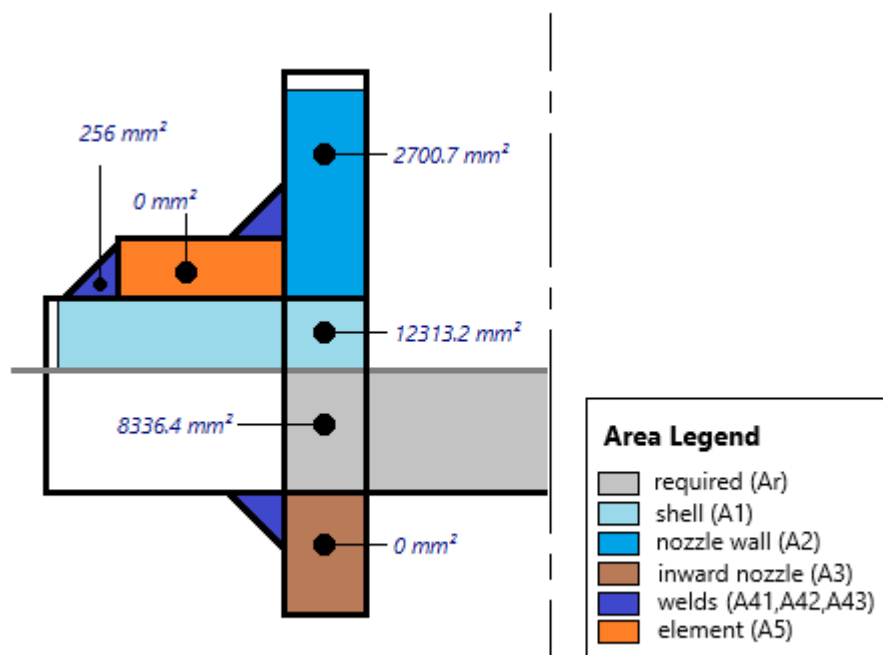


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)



| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 8336.388  | 4634.113  | NA    |
| Area in Shell            | A1          | 12313.214 | 11381.375 | NA    |
| Area in Nozzle Wall      | A2          | 2700.714  | 2646.677  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 256.000   | 256.000   | NA    |
| Area in Element          | A5          | 0.000     | 0.000     | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 15269.928 | 14284.052 | NA    |

#### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (764.8 \cdot 10.9001 \cdot 1.0 + 2 \cdot 24.0 \cdot 10.9001 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 8336.388 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 764.8(1.0 \cdot 27.0 - 1.0 \cdot 10.9) - 2 \cdot 24.0$   
 $(1.0 \cdot 27.0 - 1.0 \cdot 10.9001) \cdot (1 - 1.0)$   
 $= 12313.214 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 60.0)(24.0 - 1.49)1.0$   
 $= 2700.714 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 16.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$   
 $= 256.000 \text{ mm}^2$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 1.9444 \text{ mm}$   
 Wall Thickness per UG16(b),  $tr16b = 2.5000 \text{ mm}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 11.9001 \text{ mm}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 11.9001 \text{ mm}$   
 Wall Thickness, shell/head, external pressure  $trb2 = 3.2158 \text{ mm}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 3.2158 \text{ mm}$   
 Wall Thickness per table UG-45  $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 8.331, \max( 11.9001, 3.2158 ) ]$   
 $= 8.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 1.9444, 8.3312 )$   
 $= 8.3312 \text{ mm}$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |           |        |
|------------|---|-------|-----------|---|-----------|--------|
| Sustained  | : | 14.0, | Allowable | : | 138.0 MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 331.0 MPa | Passed |
| Occasional | : | 3.9,  | Allowable | : | 183.5 MPa | Passed |
| Shear      | : | 7.8,  | Allowable | : | 96.6 MPa  | Passed |

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of



*the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 24.0, tr = 1.494, c = 0.0 mm, E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.062, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | 18 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 24.0, tr = 1.494, c = 0.0 mm, E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.062, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -2 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.51/1.96 = 0.259

##### Weld Size Calculations, Description: N3B

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

##### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 11.3120 = 0.7 * Wo mm |

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
= max( 0, (A-A1+2\*tn\*fr1\*(E1\*t-tr))Sv)  
= max( 0, (8336.3877 - 12313.2139 + 2 \* 24.0 \* 1.0 \*  
(1.0 \* 27.0 - 10.9001 ) )138)  
= max( 0, -442110.53) N

*Note: F is always set to 1.0 throughout the calculation.*

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 2700.7136 + 0.0 + 256.0 - 0.0 \* 1.0 ) \* 138  
= 407984.41 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 \* tn \* t \* fr1)) \* Sv  
= ( 2700.7136 + 0.0 + 256.0 + ( 1296.0 ) ) \* 138  
= 586814.00 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2\*tn\*t\*fr1))\*S  
= ( 2700.7136 + 0.0 + 256.0 + 0.0 + ( 1296.0 ) ) \* 138  
= 586814.00 N

##### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 812.8 \* 16.0 \* 0.49 \* 138  
= 1381192. N



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 394.4) \cdot (24.0 - 0.0) \cdot 0.7 \cdot 138$$

$$= 2872356. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 812.8 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 2085919. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (1381192 + 2872356) = 4253548 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (1381192 + 0 + 2085919 + 0) = 3467111 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (1381192 + 2085919 + 0) = 3467111 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                       |                  |
|----------------------|-----------------------|------------------|
| Path 1-1 = 4253547 N | , must exceed W = 0 N | or W1 = 407984 N |
| Path 2-2 = 3467111 N | , must exceed W = 0 N | or W2 = 586814 N |
| Path 3-3 = 3467111 N | , must exceed W = 0 N | or W3 = 586814 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 28.0987 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 456.0987 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 3.043 %

*This nozzle or its diameter limit intersects weld seam # [5] on this element. The distance between the nozzle and seam is, 567.12 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 764.80 mm*

#### Input Echo, WRC297 Item 1, Description: N3B :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 28.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |            |
|------------------------------------|-----------|------------|
| Diameter Basis for Nozzle          | OD        |            |
| Nozzle Corrosion Allowance         | 0.0000    | mm         |
| Nozzle Diameter                    | 812.800   | mm         |
| Nozzle Thickness                   | 24.0000   | mm         |
| Nozzle Stress Concentration Factor | 1.000     |            |
| Nozzle Material                    | SA-516 70 |            |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 MPa |



Nozzle Hot S.I. Allowable SNmh 138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSstained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.51      | MPa  |
| Radial Load              | P  | -59999.96 | N    |
| Circumferential Shear    | Vc | 40000.02  | N    |
| Longitudinal Shear       | Vl | 40000.02  | N    |
| Circumferential Moment   | Mc | 0.896E+08 | N-mm |
| Longitudinal Moment      | Ml | 0.896E+08 | N-mm |
| Torsional Moment         | Mt | 0.128E+09 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$   
 $= 812.8 + 2 \cdot 1.65 \cdot \sqrt{(2967.5(28.0 - 1.0))}$   
 $= 1746.895 \text{ mm}$

**WRC 297 Curve Access Parameters:**

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 5935.000 mm |
| Nozzle Outside Diameter | (d) = | 812.800 mm  |
| Vessel Thickness used   | (T) = | 27.000 mm   |
| Nozzle Thickness used   | (t) = | 24.000 mm   |
| T / t                   | =     | 1.125       |
| d / t                   | =     | 33.867      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.030$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.052 |
| Mr/P    | = | 0.124 |
| M0/P    | = | 0.043 |
| N0/P    | = | 0.126 |
| MrD/Mc  | = | 0.245 |
| NrDL/Mc | = | 0.083 |
| M0d/Mc  | = | 0.095 |
| N0DL/Mc | = | 0.133 |
| MrD/Ml  | = | 0.120 |
| NrDL/Ml | = | 0.052 |
| M0D/Ml  | = | 0.042 |
| N0DL/Ml | = | 0.198 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 10                   | 10                  | 10                      | 10                     |
| Outplane Bending (P )                                            | 21                   | -21                 | 21                      | -21                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -29                  | -29                 | 29                      | 29                     |
| Outplane Bending (Ml)                                            | -38                  | 38                  | 38                      | -38                    |
| Normal Pressure Stress                                           | 55                   | 55                  | 55                      | 55                     |
| Outplane Stress Summary                                          | 18                   | 52                  | 155                     | 36                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|
|--------------------|----|----|----|----|



| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
|-------------------------------------------|-------------|------------|----------------|---------------|
| Inplane Membrane (P )                     | 4           | 4          | 4              | 4             |
| Inplane Bending (P )                      | 61          | -61        | 61             | -61           |
| Inplane Membrane (Mc)                     | 0           | 0          | 0              | 0             |
| Inplane Bending (Mc)                      | 0           | 0          | 0              | 0             |
| Inplane Membrane (ML)                     | -7          | -7         | 7              | 7             |
| Inplane Bending (ML)                      | -108        | 108        | 108            | -108          |
| Inplane Pressure Stress                   | 27          | 27         | 27             | 27            |
| Inplane Stress Summary                    | -23         | 71         | 209            | -130          |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to longitudinal plane) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|-------------------------------------------------------------------|----------------|---------------|-------------------|------------------|
| Outplane Shear (Vc)                                               | 1              | 1             | -1                | -1               |
| Outplane Shear (Vl)                                               | 0              | 0             | 0                 | 0                |
| Outplane Shear (Mt)                                               | 4              | 4             | 4                 | 4                |
| Shear Stress Summary                                              | 5              | 5             | 3                 | 3                |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au Top Outside | Al Top Inside | Bu Bottom Outside | Bl Bottom Inside |
|--------------------------------------------|----------------|---------------|-------------------|------------------|
| Two * Max Shear Stress                     | 44             | 73            | 210               | 166              |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|---------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Outplane Membrane (P )                                              | 10              | 10             | 10               | 10              |
| Outplane Bending (P )                                               | 21              | -21            | 21               | -21             |
| Outplane Membrane (Mc)                                              | -20             | -20            | 20               | 20              |
| Outplane Bending (Mc)                                               | -86             | 86             | 86               | -86             |
| Outplane Membrane (ML)                                              | 0               | 0              | 0                | 0               |
| Outplane Bending (ML)                                               | 0               | 0              | 0                | 0               |
| Normal Pressure Stress                                              | 27              | 27             | 27               | 27              |
| Outplane Stress Summary                                             | -47             | 82             | 166              | -49             |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|-----------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Inplane Membrane (P )                                                 | 4               | 4              | 4                | 4               |
| Inplane Bending (P )                                                  | 61              | -61            | 61               | -61             |
| Inplane Membrane (Mc)                                                 | -12             | -12            | 12               | 12              |
| Inplane Bending (Mc)                                                  | -222            | 222            | 222              | -222            |
| Inplane Membrane (ML)                                                 | 0               | 0              | 0                | 0               |
| Inplane Bending (ML)                                                  | 0               | 0              | 0                | 0               |
| Inplane Pressure Stress                                               | 55              | 55             | 55               | 55              |
| Inplane Stress Summary                                                | -113            | 208            | 355              | -210            |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to circumferential plane) | Cu Left Outside | Cl Left Inside | Du Right Outside | Dl Right Inside |
|-------------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
| Outplane Shear (Vc)                                                     | 0               | 0              | 0                | 0               |
| Outplane Shear (Vl)                                                     | -1              | -1             | 1                | 1               |
| Torsional Shear (Mt)                                                    | 4               | 4              | 4                | 4               |



Shear Stress Summary 3 3 5 5

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities | Cu      | Cl     | Du      | Dl     |
|----------------------------------------------|---------|--------|---------|--------|
|                                              | Left    | Left   | Right   | Right  |
|                                              | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                       | 114     | 208    | 355     | 210    |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au      | Al     | Bu      | Bl     |
|-----------------------------------------------------------|---------|--------|---------|--------|
|                                                           | Top     | Top    | Bottom  | Bottom |
|                                                           | Outside | Inside | Outside | Inside |
| Hoop Membrane (P )                                        | 10      | 10     | 10      | 10     |
| Hoop Bending (P )                                         | 0       | 0      | 0       | 0      |
| Hoop Membrane (Mc)                                        | 0       | 0      | 0       | 0      |
| Hoop Bending (Mc)                                         | 0       | 0      | 0       | 0      |
| Hoop Membrane (ML)                                        | -29     | -29    | 29      | 29     |
| Hoop Bending (ML)                                         | 0       | 0      | 0       | 0      |
| Hoop Pressure Stress                                      | 9       | 10     | 9       | 10     |

Hoop Stress Summary -10 -9 49 50

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au      | Al     | Bu      | Bl     |
|------------------------------------------------------------------|---------|--------|---------|--------|
|                                                                  | Top     | Top    | Bottom  | Bottom |
|                                                                  | Outside | Inside | Outside | Inside |
| Axial Membrane (P )                                              | 1       | 1      | 1       | 1      |
| Axial Bending (P )                                               | 61      | -61    | 61      | -61    |
| Axial Membrane (Mc)                                              | 0       | 0      | 0       | 0      |
| Axial Bending (Mc)                                               | 0       | 0      | 0       | 0      |
| Axial Membrane (ML)                                              | -7      | -7     | 7       | 7      |
| Axial Bending (ML)                                               | -107    | 107    | 107     | -107   |
| Axial Pressure Stress                                            | 4       | 4      | 4       | 4      |

Axial Stress Summary -48 44 182 -155

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au      | Al     | Bu      | Bl     |
|--------------------------------------|---------|--------|---------|--------|
|                                      | Outside | Inside | Outside | Inside |
|                                      |         |        |         |        |
| Shear due to (Vc)                    | 1       | 1      | -1      | -1     |
| Shear due to (Vl)                    | 0       | 0      | 0       | 0      |
| Shear due to Torsion                 | 5       | 5      | 5       | 5      |

Shear Stress Summary 6 6 3 3

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities | Au      | Al     | Bu      | Bl     |
|-------------------------------------------|---------|--------|---------|--------|
|                                           | Outside | Inside | Outside | Inside |
|                                           |         |        |         |        |
| Two * Max Shear Stress                    | 49      | 55     | 182     | 205    |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu      | Cl     | Du      | Dl     |
|--------------------------------------------------------------|---------|--------|---------|--------|
|                                                              | Left    | Left   | Right   | Right  |
|                                                              | Outside | Inside | Outside | Inside |
| Hoop Membrane (P )                                           | 10      | 10     | 10      | 10     |
| Hoop Bending (P )                                            | 0       | 0      | 0       | 0      |
| Hoop Membrane (Mc)                                           | -20     | -20    | 20      | 20     |
| Hoop Bending (Mc)                                            | 0       | 0      | 0       | 0      |
| Hoop Membrane (ML)                                           | 0       | 0      | 0       | 0      |
| Hoop Bending (ML)                                            | 0       | 0      | 0       | 0      |
| Hoop Pressure Stress                                         | 9       | 10     | 9       | 10     |

Hoop Stress Summary 0 0 39 40

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu              | Cl             | Du               | Dl              |
|---------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
|                                                                     | Left<br>Outside | Left<br>Inside | Right<br>Outside | Right<br>Inside |
| Axial Membrane (P )                                                 | 1               | 1              | 1                | 1               |
| Axial Bending (P )                                                  | 61              | -61            | 61               | -61             |
| Axial Membrane (Mc)                                                 | -7              | -7             | 7                | 7               |
| Axial Bending (Mc)                                                  | -233            | 233            | 233              | -233            |
| Axial Membrane (ML)                                                 | 0               | 0              | 0                | 0               |
| Axial Bending (ML)                                                  | 0               | 0              | 0                | 0               |
| Axial Pressure Stress                                               | 4               | 4              | 4                | 4               |
| Axial Stress Summary                                                | -174            | 169            | 308              | -280            |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------|---------|--------|---------|--------|
|                                         | Outside | Inside | Outside | Inside |
| Shear due to (Vc)                       | 0       | 0      | 0       | 0      |
| Shear due to (Vl)                       | -1      | -1     | 1       | 1      |
| Shear due to Torsion                    | 5       | 5      | 5       | 5      |
| Shear Stress Summary                    | 3       | 3      | 6       | 6      |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------------|---------|--------|---------|--------|
|                                               | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                        | 174     | 169    | 308     | 321    |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |      |       |       |       |       |       |       |
|------------------------------------|------------------|------|-------|-------|-------|-------|-------|-------|
|                                    | Au               | Al   | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)                     | 55               | 55   | 55    | 55    | 55    | 55    | 55    | 55    |
| Circ. Pl (SUS)                     | -19              | -19  | 39    | 39    | -8    | -8    | 16    | 16    |
| Circ. Q (SUS)                      | -17              | 17   | 59    | -59   | -161  | 161   | 283   | -283  |
| Long. Pm (SUS)                     | 27               | 27   | 27    | 27    | 27    | 27    | 27    | 27    |
| Long. Pl (SUS)                     | -3               | -3   | 11    | 11    | -10   | -10   | 30    | 30    |
| Long. Q (SUS)                      | -47              | 47   | 169   | -169  | -65   | 65    | 107   | -107  |
| Shear Pm (SUS)                     | 0                | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)                     | 1                | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                      | 4                | 4    | 4     | 4     | 4     | 4     | 4     | 4     |
| Pm (SUS)                           | 55.0             | 55.0 | 55.0  | 55.0  | 55.0  | 55.0  | 55.0  | 55.0  |
| Pm+Pl (SUS)                        | 36.1             | 36.1 | 94.0  | 94.0  | 47.0  | 47.0  | 71.1  | 71.1  |
| Pm+Pl+Q (Total)                    | 43.2             | 72.3 | 207.2 | 166.1 | 114.1 | 208.1 | 354.1 | 212.2 |

**Vessel Stress Summation Comparison (MPa):**

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 55.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 94.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 354.13    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**





### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |       |       |       |       |       |       |
|------------------------------|------------------|------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al   | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 9                | 10   | 9     | 10    | 9     | 10    | 9     | 10    |
| Circ. Pl (SUS)               | -19              | -19  | 39    | 39    | -10   | -10   | 30    | 30    |
| Circ. Q (SUS)                | 0                | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)               | 4                | 4    | 4     | 4     | 4     | 4     | 4     | 4     |
| Long. Pl (SUS)               | -6               | -6   | 8     | 8     | -6    | -6    | 8     | 8     |
| Long. Q (SUS)                | -46              | 46   | 168   | -168  | -172  | 172   | 294   | -294  |
| Shear Pm (SUS)               | 0                | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 1                | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                | 5                | 5    | 5     | 5     | 5     | 5     | 5     | 5     |
| Pm (SUS)                     | 9.0              | 10.0 | 9.0   | 10.0  | 9.0   | 10.0  | 9.0   | 10.0  |
| Pm+Pl (SUS)                  | 10.1             | 9.1  | 48.0  | 49.0  | 2.6   | 2.8   | 39.0  | 40.0  |
| Pm+Pl+Q (Total)              | 48.9             | 54.3 | 180.1 | 205.2 | 174.1 | 170.2 | 306.1 | 322.2 |

### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 10.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 49.03     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 322.22    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: K5a**

**From: 130**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 8770.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

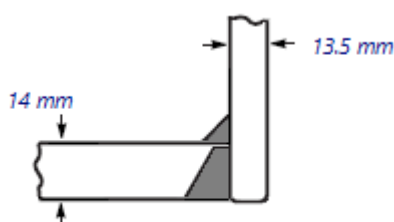
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 14.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K5a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \left( \exp \left( \frac{P}{S_n \cdot E} \right) - 1 \right)$  per Appendix 1-2 (a)(1)  
=  $25.4 \left( \exp \left( \frac{0.5}{138.0 \cdot 1.0} \right) - 1 \right)$   
= 0.0919 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 103.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 51.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 32.5000  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5a.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.6895, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.092, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: K5a

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.8057 mm

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**Input, Nozzle Desc: K6a**

**From: 130**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 8770.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

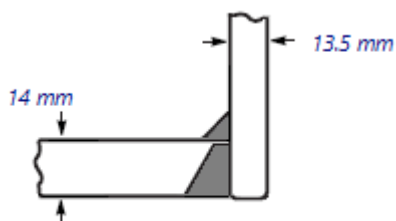
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 14.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K6a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \left( \exp \left( \frac{P}{S_n \cdot E} \right) - 1 \right)$  per Appendix 1-2 (a)(1)  
=  $25.4 \left( \exp \left( \frac{0.5}{138.0 \cdot 1.0} \right) - 1 \right)$   
= 0.0919 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 103.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 51.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 32.5000  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6a.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.6895, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govern. thk, tg = 13.462, tr = 0.092, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: K6a

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.8057 mm

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**Input, Nozzle Desc: K5b**

**From: 140**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 9820.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

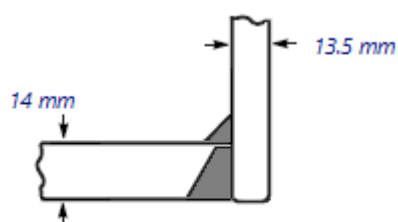
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 14.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K5b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45





Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \left( \exp \left( \frac{P}{S_n \cdot E} \right) - 1 \right)$  per Appendix 1-2 (a)(1)  
=  $25.4 \left( \exp \left( \frac{0.5}{138.0 \cdot 1.0} \right) - 1 \right)$   
= 0.0919 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 103.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 51.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 32.5000  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5b.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.6895, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.092, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: K5b

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.8057 mm

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**Input, Nozzle Desc: K6b**

**From: 140**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |         |    |
|-----------------------------------|--|---------|----|
| Distance from Bottom/Left Tangent |  | 9820.00 | mm |
|-----------------------------------|--|---------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

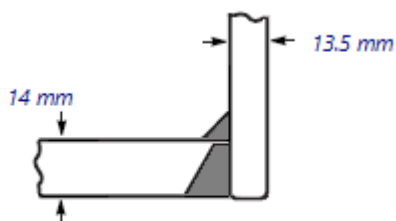
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 14.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K6b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \left( \exp \left( \frac{P}{S_n \cdot E} \right) - 1 \right)$  per Appendix 1-2 (a)(1)  
=  $25.4 \left( \exp \left( \frac{0.5}{138.0 \cdot 1.0} \right) - 1 \right)$   
= 0.0919 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1      | 103.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 51.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 32.5000  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6b.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 4.8, max( 11.6895, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.092, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: K6b

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.8057 mm

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**Input, Nozzle Desc: K7a**

**From: 140**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 10370.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-105 +IT |     |
| Material UNS Number             |     | K03504     |     |
| Material Specification/Type     |     | Forgings   |     |
| Allowable Stress at Temperature | Sn  | 138.00     | MPa |
| Allowable Stress At Ambient     | Sna | 138.00     | MPa |

|                                   |  |        |     |
|-----------------------------------|--|--------|-----|
| Diameter Basis (for tr calc only) |  | Inside |     |
| Layout Angle                      |  | 0.00   | deg |
| Diameter                          |  | 2.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 13.4620 | mm |

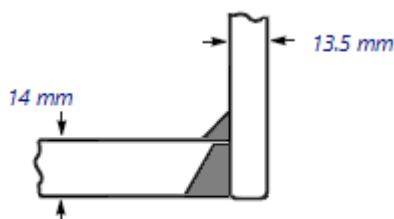
|                 |  |                |  |
|-----------------|--|----------------|--|
| Flange Material |  | SA-105 +IT     |  |
| Flange Type     |  | Long Weld Neck |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                                  |    |          |    |
|--------------------------------------------------|----|----------|----|
| Outside Projection                               | ho | 209.5500 | mm |
| Weld leg size between Nozzle and Pad/Shell       | Wo | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel Wgnv |    | 14.0000  | mm |
| Inside Projection                                | h  | 0.0000   | mm |
| Weld leg size, Inside Element to Shell           | Wi | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K7a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.  
Actual Thickness Used in Calculation 0.530 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]  
=  $R \left( \exp \left( \frac{P}{S_n \cdot E} \right) - 1 \right)$  per Appendix 1-2 (a)(1)  
=  $25.4 \left( \exp \left( \frac{0.5}{138.0 \cdot 1.0} \right) - 1 \right)$   
= 0.0919 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3776 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |         |          |    |
|-------------------------------------------------|---------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl      | 103.7240 | mm |
| Parallel to Vessel Wall                         | Rn+tn+t | 51.8620  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp    | 32.5000  | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7a.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3776  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 4.8000  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1,tb2 ) ]  
= min[ 4.8, max( 11.6895, 3.2158 ) ]  
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3776, 4.8 )  
= 4.8000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.092, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.007$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -19 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: K7a

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2558 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.8057 mm

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**Input, Nozzle Desc: A230"**

**From: 140**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 10100.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 30.00   | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

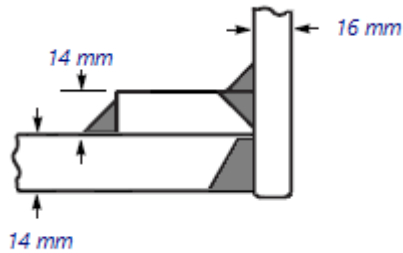
|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                          |      |           |     |
|------------------------------------------|------|-----------|-----|
| Pad Material                             |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature      | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient          | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface     | Dp   | 1382.0000 | mm  |
| Thickness of Pad                         | te   | 14.0000   | mm  |
| Weld leg size between Pad and Shell      | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle | Wgpn | 14.0000   | mm  |
| Reinforcing Pad Width                    |      | 310.0000  | mm  |

This is a Manway, Coupling or Access Opening.

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



### Reinforcement CALCULATION, Description: A230"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 30.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

*This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 1.3737 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

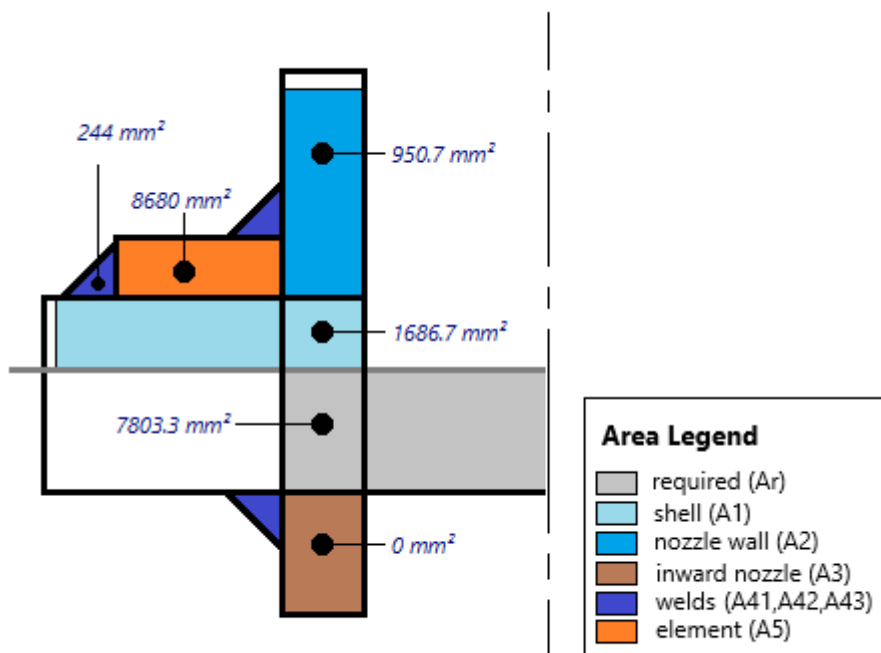
|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1460.0000 mm |
| Parallel to Vessel Wall, opening length                | d  | 730.0000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 32.5000 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 7803.346  | 4421.450  | NA    |
| Area in Shell            | A1          | 1686.655  | 647.100   | NA    |
| Area in Nozzle Wall      | A2          | 950.709   | 918.087   | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 244.000   | 244.000   | NA    |
| Area in Element          | A5          | 8680.001  | 8680.001  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 11561.364 | 10489.188 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 1113.5702 14.0000 mm  
 Based on given Pad Diameter: 1382.0000 7.9387 mm  
 Based on Shell or Nozzle Thickness: 1113.5702 14.0000 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (730.0 \cdot 10.6895 \cdot 1.0 + 2 \cdot 16.0 \cdot 10.6895 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 7803.346 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 730.0(1.0 \cdot 13.0 - 1.0 \cdot 10.69) - 2 \cdot 16.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.6895) \cdot (1 - 1.0)$   
 $= 1686.655 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 32.5) \cdot (16.0 - 1.37) \cdot 1.0$



= 950.709 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$   
= 244.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1382.0 - 762.0) \cdot 14.0 \cdot 1.0$   
= 8680.001 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 1.8756$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

=  $t_a$   
= 1.8756 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.374$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.086$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | 6 °C    |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -18 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -28 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -18 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -28 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -18 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -28 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -18 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -28 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

Gov. MDMT of the Nozzle : -29 °C



Gov. MDMT of the Reinforcement Pad : -29 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: A230"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm  
Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 6.5000 = 0.5*TminPad  | 8.4840 = 0.7 * Wp mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
= max( 0, (A-A1+2\*tn\*fr1\*(E1\*t-tr))Sv)  
= max( 0, (7803.3462 - 1686.6548 + 2 \* 16.0 \* 1.0 \*  
(1.0 \* 13.0 - 10.6895 ) )138)  
= 854218.38 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 950.7087 + 8680.001 + 244.0 - 0.0 \* 1.0 ) \* 138  
= 1362569.50 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 \* tn \* t \* fr1)) \* Sv  
= ( 950.7087 + 0.0 + 100.0 + ( 416.0 ) ) \* 138  
= 202384.94 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2\*tn\*t\*fr1))\*S  
= ( 950.7087 + 0.0 + 244.0 + 8680.001 + ( 416.0 ) ) \* 138  
= 1419971.50 N

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 762.0 \* 10.0 \* 0.49 \* 138  
= 809306. N

Shear, Pad Element Weld [Spew]:  
= (π/2) \* DP \* WP \* 0.49 \* SEW  
= ( 3.1416/2.0 ) \* 1382.0 \* 12.0 \* 0.49 \* 138  
= 1761325. N

Shear, Nozzle Wall [Snw]:  
= (π \* ( Dlr + Dlo )/4 ) \* ( Thk - Can ) \* 0.7 \* Sn  
= (3.1416 \* 373.0) \* ( 16.0 - 0.0 ) \* 0.7 \* 138  
= 1811002. N

Tension, Pad Groove Weld [Tpgw]:  
= ( π/2 ) \* Dlo \* Wgpn \* 0.74 \* Seg  
= (3.1416/2 ) \* 762.0 \* 14.0 \* 0.74 \* 138  
= 1711105. N

Tension, Shell Groove Weld [Tngw]:  
= (π/2) \* Dlo \* ( T - cas ) \* 0.74 \* Sng



$$= (3.1416/2.0) \cdot 762.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 138 \\ = 1588884. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1761325 + 1811002) = 3572327 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (809307 + 1711105 + 1588884 + 0) = 4109296 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1761325 + 1588884 + 0) = 3350209 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                            |                   |
|----------------------|----------------------------|-------------------|
| Path 1-1 = 3572327 N | , must exceed W = 854218 N | or W1 = 1362569 N |
| Path 2-2 = 4109295 N | , must exceed W = 854218 N | or W2 = 202384 N  |
| Path 3-3 = 3350209 N | , must exceed W = 854218 N | or W3 = 1419971 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.6818 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 438.6817 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1  $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$  2.145 %

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**Input, Nozzle Desc: A330"**

**From: 150**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 13700.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 30.00   | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

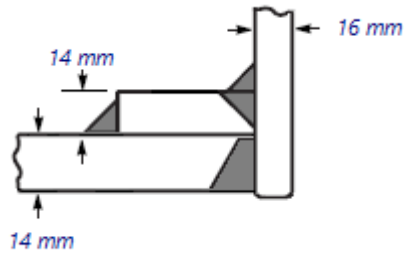
|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                          |      |           |     |
|------------------------------------------|------|-----------|-----|
| Pad Material                             |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature      | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient          | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface     | Dp   | 1382.0000 | mm  |
| Thickness of Pad                         | te   | 14.0000   | mm  |
| Weld leg size between Pad and Shell      | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle | Wgpn | 14.0000   | mm  |
| Reinforcing Pad Width                    |      | 310.0000  | mm  |

This is a Manway, Coupling or Access Opening.

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



### Reinforcement CALCULATION, Description: A330"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 30.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

*This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 1.3737 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1460.0000 mm |
| Parallel to Vessel Wall, opening length                | d  | 730.0000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 32.5000 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



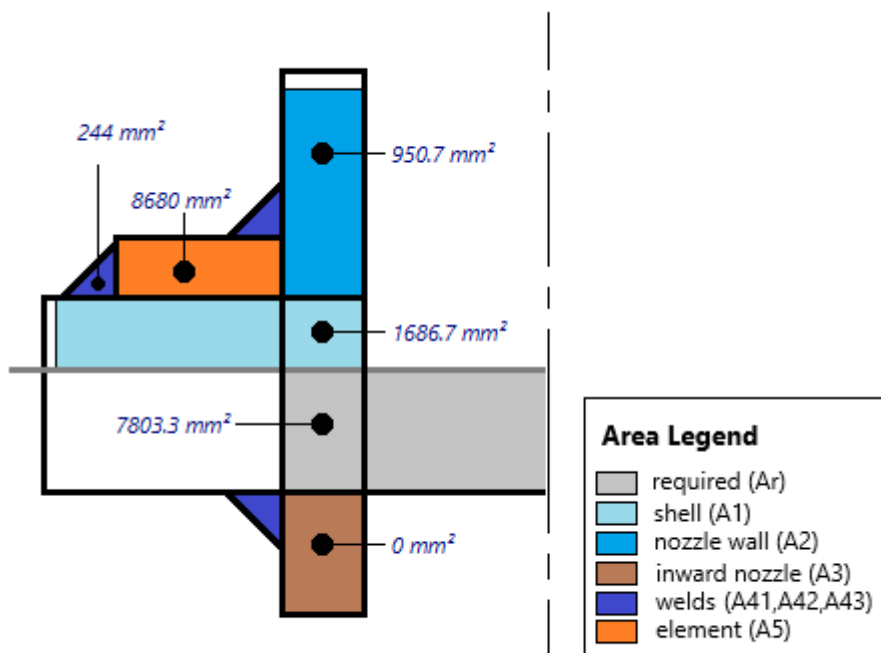


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 7803.346  | 4421.450  | NA    |
| Area in Shell            | A1          | 1686.655  | 647.100   | NA    |
| Area in Nozzle Wall      | A2          | 950.709   | 918.087   | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 244.000   | 244.000   | NA    |
| Area in Element          | A5          | 8680.001  | 8680.001  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 11561.364 | 10489.188 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter  | Thickness  |
|-----------------------------------------|-----------|------------|
| Based on given Pad Thickness:           | 1113.5702 | 14.0000 mm |
| Based on given Pad Diameter:            | 1382.0000 | 7.9387 mm  |
| Based on Shell or Nozzle Thickness:     | 1113.5702 | 14.0000 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (730.0 \cdot 10.6895 \cdot 1.0 + 2 \cdot 16.0 \cdot 10.6895 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 7803.346 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 730.0(1.0 \cdot 13.0 - 1.0 \cdot 10.69) - 2 \cdot 16.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.6895) \cdot (1 - 1.0)$   
 $= 1686.655 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 32.5) \cdot (16.0 - 1.37) \cdot 1.0$



= 950.709 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$   
= 244.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1382.0 - 762.0) \cdot 14.0 \cdot 1.0$   
= 8680.001 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 1.8756$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

=  $t_a$   
= 1.8756 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.374$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.086$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -29 °C



Gov. MDMT of the Reinforcement Pad : -29 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: A330"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm  
Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 6.5000 = 0.5*TminPad  | 8.4840 = 0.7 * Wp mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
= max( 0, (A-A1+2\*tn\*fr1\*(E1\*t-tr))Sv)  
= max( 0, (7803.3462 - 1686.6548 + 2 \* 16.0 \* 1.0 \*  
(1.0 \* 13.0 - 10.6895 ) )138)  
= 854218.38 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 950.7087 + 8680.001 + 244.0 - 0.0 \* 1.0 ) \* 138  
= 1362569.50 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 \* tn \* t \* fr1)) \* Sv  
= ( 950.7087 + 0.0 + 100.0 + ( 416.0 ) ) \* 138  
= 202384.94 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2\*tn\*t\*fr1))\*S  
= ( 950.7087 + 0.0 + 244.0 + 8680.001 + ( 416.0 ) ) \* 138  
= 1419971.50 N

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 762.0 \* 10.0 \* 0.49 \* 138  
= 809306. N

Shear, Pad Element Weld [Spew]:  
= (π/2) \* DP \* WP \* 0.49 \* SEW  
= ( 3.1416/2.0 ) \* 1382.0 \* 12.0 \* 0.49 \* 138  
= 1761325. N

Shear, Nozzle Wall [Snw]:  
= (π \* ( Dlr + Dlo )/4 ) \* ( Thk - Can ) \* 0.7 \* Sn  
= (3.1416 \* 373.0) \* ( 16.0 - 0.0 ) \* 0.7 \* 138  
= 1811002. N

Tension, Pad Groove Weld [Tpgw]:  
= ( π/2 ) \* Dlo \* Wgpn \* 0.74 \* Seg  
= (3.1416/2 ) \* 762.0 \* 14.0 \* 0.74 \* 138  
= 1711105. N

Tension, Shell Groove Weld [Tngw]:  
= (π/2) \* Dlo \* ( T - cas ) \* 0.74 \* Sng



$$= (3.1416/2.0) \cdot 762.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 138 \\ = 1588884. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1761325 + 1811002) = 3572327 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (809307 + 1711105 + 1588884 + 0) = 4109296 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1761325 + 1588884 + 0) = 3350209 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                            |                   |
|----------------------|----------------------------|-------------------|
| Path 1-1 = 3572327 N | , must exceed W = 854218 N | or W1 = 1362569 N |
| Path 2-2 = 4109295 N | , must exceed W = 854218 N | or W2 = 202384 N  |
| Path 3-3 = 3350209 N | , must exceed W = 854218 N | or W3 = 1419971 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.6818 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 438.6817 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1  $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$  2.145 %

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**Input, Nozzle Desc: K7b**

**From: 170**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 18535.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 150.00  | deg |
| Diameter                          |  | 2.0000  | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Nominal |  |
| Nominal Thickness        |  | STD     |  |

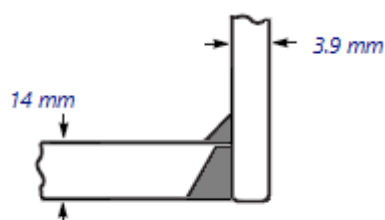
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 150.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 300    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.  
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.5 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
= 0.1272 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2828 mm

**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |    |
|-------------------------------------------------|------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 105.0036 | mm |
| Parallel to Vessel Wall, opening length         | d    | 52.5018  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 9.7790   | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7b.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.2828  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 3.4200  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1,tb2) ]  
= min[ 3.42, max( 11.6895, 3.2158 ) ]  
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.2828, 3.42 )  
= 3.4200 mm

Available Nozzle Neck Thickness =  $0.875 \cdot 3.912 = 3.423$  mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)**

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

**Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B**

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/5.10 = 0.098

Weld Size Calculations, Description: K7b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

**Results Per UW-16.1:**

|             | Required Thickness       | Actual Thickness        |
|-------------|--------------------------|-------------------------|
| Nozzle Weld | $2.7381 = 0.7 * t_{min}$ | $7.0700 = 0.7 * W_o$ mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1541 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.1540 mm

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**Input, Nozzle Desc: A430"**

**From: 170**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 19570.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 120.00  | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

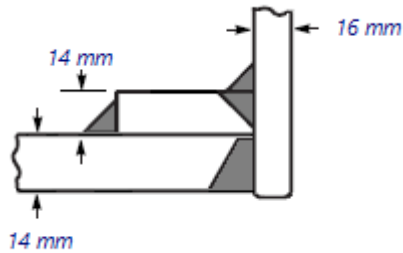
|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 14.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                               |      |           |     |
|-----------------------------------------------|------|-----------|-----|
| Pad Material                                  |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature           | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient               | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface          | Dp   | 1382.0000 | mm  |
| Thickness of Pad                              | te   | 14.0000   | mm  |
| Weld leg size between Pad and Shell           | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle      | Wgpn | 14.0000   | mm  |
| Reinforcing Pad Width                         |      | 310.0000  | mm  |
| This is a Manway, Coupling or Access Opening. |      |           |     |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 150    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | B      |  |

The Pressure Design option was MAWP + static head (bottom of element).





#### Reinforcement CALCULATION, Description: A430"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 30.000 in.  
 Actual Thickness Used in Calculation 0.630 in.

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 1.3737 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

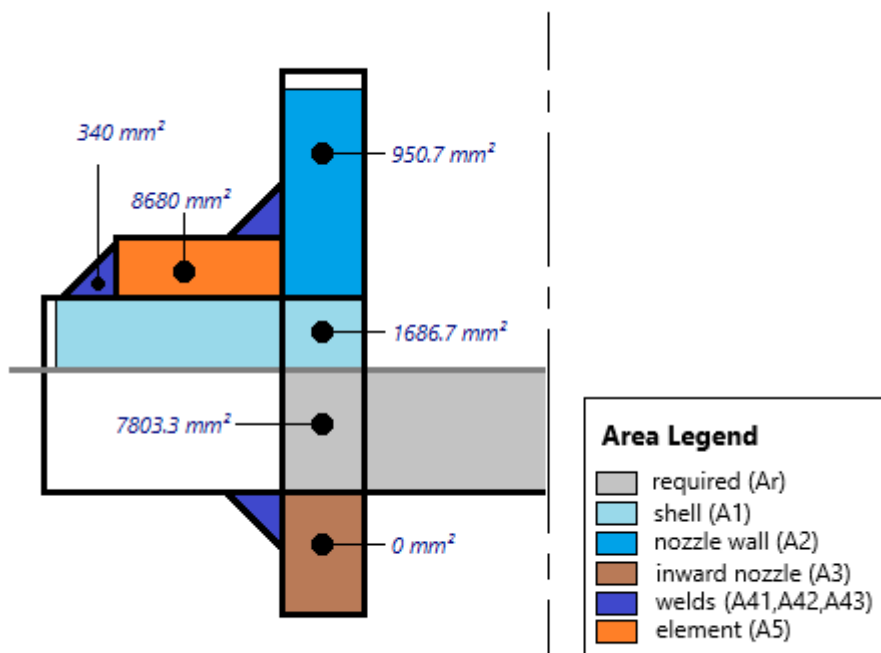
|                                                        |    |           |    |
|--------------------------------------------------------|----|-----------|----|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1460.0000 | mm |
| Parallel to Vessel Wall, opening length                | d  | 730.0000  | mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 32.5000   | mm |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 7803.346  | 4421.450  | NA    |
| Area in Shell            | A1          | 1686.655  | 647.100   | NA    |
| Area in Nozzle Wall      | A2          | 950.709   | 918.087   | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 340.000   | 340.000   | NA    |
| Area in Element          | A5          | 8680.001  | 8680.001  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 11657.365 | 10585.188 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter  | Thickness  |
|-----------------------------------------|-----------|------------|
| Based on given Pad Thickness:           | 1106.7130 | 14.0000 mm |
| Based on given Pad Diameter:            | 1382.0000 | 7.7838 mm  |
| Based on Shell or Nozzle Thickness:     | 1106.7130 | 14.0000 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (730.0 \cdot 10.6895 \cdot 1.0 + 2 \cdot 16.0 \cdot 10.6895 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 7803.346 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 730.0(1.0 \cdot 13.0 - 1.0 \cdot 10.69) - 2 \cdot 16.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.6895) \cdot (1 - 1.0)$   
 $= 1686.655 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 32.5) \cdot (16.0 - 1.37) \cdot 1.0$



= 950.709 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $14.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$   
= 340.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1382.0 - 762.0) \cdot 14.0 \cdot 1.0$   
= 8680.001 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 1.8756$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

=  $t_a$   
= 1.8756 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.374$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.086$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -29 °C



Gov. MDMT of the Reinforcement Pad : -29 °C  
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

#### Weld Size Calculations, Description: A430"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm  
 Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 9.8980 = 0.7 * Wo mm |
| Pad Weld    | 6.5000 = 0.5*TminPad  | 8.4840 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (7803.3462 - 1686.6548 + 2 \cdot 16.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 10.6895)) \cdot 138)$   
 $= 854218.38 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$   
 $= (950.7087 + 8680.001 + 340.0 - 0.0 \cdot 1.0) \cdot 138$   
 $= 1375816.12 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$   
 $= (950.7087 + 0.0 + 196.0 + (416.0)) \cdot 138$   
 $= 215631.58 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$   
 $= (950.7087 + 0.0 + 340.0 + 8680.001 + (416.0)) \cdot 138$   
 $= 1433218.25 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 762.0 \cdot 14.0 \cdot 0.49 \cdot 138$   
 $= 1133029. \text{ N}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$   
 $= (3.1416/2.0) \cdot 1382.0 \cdot 12.0 \cdot 0.49 \cdot 138$   
 $= 1761325. \text{ N}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 373.0) \cdot (16.0 - 0.0) \cdot 0.7 \cdot 138$   
 $= 1811002. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$   
 $= (3.1416/2) \cdot 762.0 \cdot 14.0 \cdot 0.74 \cdot 138$   
 $= 1711105. \text{ N}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$



$$= (3.1416/2.0) \cdot 762.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 138 \\ = 1588884. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1761325 + 1811002) = 3572327 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (1133029 + 1711105 + 1588884 + 0) = 4433019 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1761325 + 1588884 + 0) = 3350209 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                            |                   |
|----------------------|----------------------------|-------------------|
| Path 1-1 = 3572327 N | , must exceed W = 854218 N | or W1 = 1375816 N |
| Path 2-2 = 4433018 N | , must exceed W = 854218 N | or W2 = 215631 N  |
| Path 3-3 = 3350209 N | , must exceed W = 854218 N | or W3 = 1433218 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.6818 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 438.6817 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1  $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$  2.145 %

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**Input, Nozzle Desc: N5**

**From: 180**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 30.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 22050.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 180.00  | deg |
| Diameter                          |  | 46.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |        |  |
|-----------------|--|--------|--|
| Flange Material |  | SA-105 |  |
|-----------------|--|--------|--|

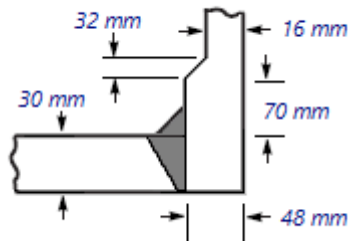
|                                                 |    |         |    |
|-------------------------------------------------|----|---------|----|
| Hub Height of Integral Nozzle                   | h  | 70.0000 | mm |
| Height of Beveled Transition                    | L' | 32.0000 | mm |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |    | 48.0000 | mm |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 150    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).



### Reinforcement CALCULATION, Description: N5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 46.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 584.2 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 2.1064 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.4000 mm

### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$   
 $= 32.0 / 0.5773$   
 $= 55.4256 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

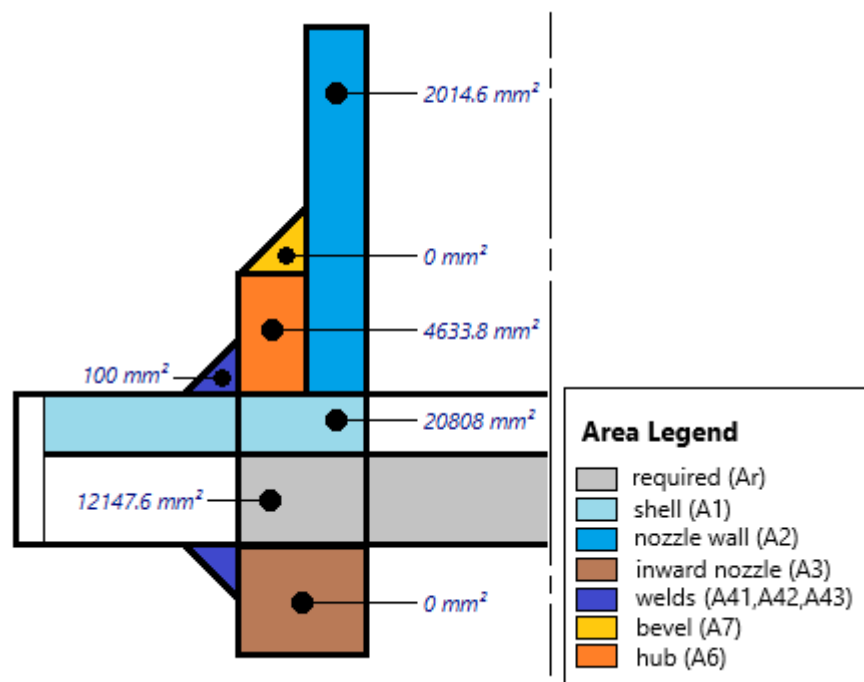
### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |              |
|-------------------------------------------------|------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 2272.8000 mm |
| Parallel to Vessel Wall, opening length         | d    | 1136.4000 mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 72.5000 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5  |      | MAWP      | External  | Mapnc |
|---------------------------|------|-----------|-----------|-------|
| Area Required             | Ar   | 12147.564 | 6882.926  | NA    |
| Area in Shell             | A1   | 20808.035 | 19189.748 | NA    |
| Area in Nozzle Wall       | A2   | 2014.578  | 1971.997  | NA    |
| Area in Inward Nozzle     | A3   | 0.000     | 0.000     | NA    |
| Area in Welds A41+A42+A43 |      | 100.000   | 100.000   | NA    |
| Area in Element           | A5   | 0.000     | 0.000     | NA    |
| Area in Hub + Bevel       | A6   | 4633.752  | 4633.752  | NA    |
| TOTAL AREA AVAILABLE      | Atot | 27556.365 | 25895.498 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (1136.4 \cdot 10.6895 \cdot 1.0 + 2 \cdot 48.0 \cdot 10.6895 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 12147.564 \text{ mm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 1136.4 (1.0 \cdot 29.0 - 1.0 \cdot 10.69) - 2 \cdot 48.0$$

$$(1.0 \cdot 29.0 - 1.0 \cdot 10.6895) \cdot (1 - 1.0)$$

$$= 20808.035 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 72.5) (16.0 - 2.11) 1.0$$

$$= 2014.578 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$





Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(72.5, 400.0, 70.0)) \cdot (48.0 - 16.0) \cdot 1.0 \\ &= 4633.752 \text{ mm}^2 \text{ Includes Bevel Area of } 153.751 \text{ mm}^2 \end{aligned}$$

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 2.4000 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 2.5000 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 11.6895 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 11.6895 \text{ mm} \\ \text{Wall Thickness, shell/head, external pressure} & \quad t_{rb2} = 3.2158 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.2158 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 8.3312 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[ t_{b3}, \max( t_{b1}, t_{b2} ) ] \\ &= \min[ 8.331, \max( 11.6895, 3.2158 ) ] \\ &= 8.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max( t_a, t_b ) \\ &= \max( 2.4, 8.3312 ) \\ &= 8.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |           |        |
|------------|---|-------|-----------|---|-----------|--------|
| Sustained  | : | 23.1, | Allowable | : | 138.0 MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 321.9 MPa | Passed |
| Occasional | : | 8.7,  | Allowable | : | 183.5 MPa | Passed |
| Shear      | : | 11.0, | Allowable | : | 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)**

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 2.106$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.132$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | 6 °C    |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

**Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B**

Govrn. thk,  $t_g = 30.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.369$ , Temp. Reduction = 64 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | 4 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating =  $0.50/1.96 = 0.254$

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:



$$= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$$

$$= 32.0/0.5773$$

$$= 55.4256 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**

Parallel to Vessel Wall (Diameter Limit) D1 1704.6000 mm  
 Parallel to Vessel Wall, opening length d 852.3000 mm  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 72.5000 mm

**Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm<sup>2</sup> )**

| AREA AVAILABLE, A1 to A6          |      | No Pad    | With Pad  |
|-----------------------------------|------|-----------|-----------|
| Area Required [Internal Pressure] | AR   | 8098.781  | 8098.781  |
| Area Available in Shell           | A1   | 10404.018 | 10404.018 |
| Area Available in Nozzle Wall     | A2   | 2014.578  | 2014.578  |
| Area Available in Inward Nozzle   | A3   | 0.000     | 0.000     |
| Area Available in Welds           | A4   | 100.000   | 100.000   |
| Area Available in Pad             | A5   | 0.000     | 0.000     |
| Area Available in Hub             | A6   | 4633.752  | 4633.752  |
| TOTAL AREA AVAILABLE              | Atot | 17152.348 | 17152.348 |

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

*Warning: The flange could not be considered in the moment of inertia calculations.*

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

|        | Area    | Distance | Inertia    | d      | I+Ad <sup>2</sup> |
|--------|---------|----------|------------|--------|-------------------|
| Nozzle | 4560.0  | 142.5    | 0.3087E+08 | 96.053 | 0.7294E+08        |
| Hub    | 2752.0  | 43.5     | 0.1813E+07 | 2.951  | 0.1837E+07        |
| Shell  | 13456.0 | 14.5     | 0.9430E+06 | 31.947 | 0.1468E+08        |
| Sum    | 20768.0 |          |            |        | 0.8945E+08        |

|                                                |        |     |
|------------------------------------------------|--------|-----|
| Membrane stress per Appendix 1-7 ( 1 or 2 ) Sm | 100.08 | MPa |
| Allowable Membrane Stress S                    | 138.00 | MPa |
| Bending stress at the inside of the Nozzle Sb  | 21.78  | MPa |
| Combined Membrane plus Bending Stress Sm + Sb  | 121.86 | MPa |
| Allowable Membrane + Bending Stress 1.5 * S    | 207.00 | MPa |
| M.A.W.P for the Appendix 1-7 Stress calc       | 0.69   | MPa |

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 2000.267 mm<sup>2</sup>  
 Area from Shell = 8508.745 mm<sup>2</sup>  
 Total Area, As = 13261.013 mm<sup>2</sup>

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [ R ( Rn + tn + \sqrt{rm \cdot t} ) + Rn ( t + \sqrt{Rnm \cdot tn} ) ] / As$$

$$= 0.498 [ 2954.0 ( 568.2 + 16.0 + 293.405 ) + 568.2 ( 125.017 ) ] / 13261.013$$

$$= 100.08 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= ( Rn^3/6 + R \cdot Rn \cdot e ) P$$

$$= ( 568.2^3/6 + 2954.0 \cdot 568.2 \cdot 31.947 ) 0.498$$

$$= 41966960 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= ( M ( e + t/2 ) ) / I$$

$$= ( 41966960 ( 31.947 + 29.0/2 ) ) / 89450016$$

$$= 21.784 \text{ MPa}$$



Weld Size Calculations, Description: N5

Intermediate Calc. for nozzle/shell welds T<sub>min</sub> 19.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness                 |
|-------------|-----------------------|----------------------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * W <sub>o</sub> mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
= max( 0, (A-A1+2•t<sub>n</sub>•f<sub>r1</sub>•(E1•t-tr))S<sub>v</sub>)  
= max( 0, (12147.5645 - 20808.0352 + 2 • 48.0 • 1.0 •  
(1.0 • 29.0 - 10.6895 ) )138)  
= max( 0, -952469.50) N

For hub type nozzles, A<sub>2</sub> includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A<sub>2</sub>+A<sub>4</sub>-(W<sub>i</sub>-Can/.707)<sup>2</sup>•f<sub>r2</sub>)•S<sub>v</sub>  
= ( 6648.3296 + 100.0 - 0.0 • 1.0 ) • 138  
= 931173.50 N

Weld Load [W2]:  
= (A<sub>2</sub> + A<sub>3</sub> + A<sub>4</sub> + (2 • t<sub>n</sub> • t • f<sub>r1</sub>)) • S<sub>v</sub>  
= ( 6648.3296 + 0.0 + 100.0 + ( 2784.0 ) ) • 138  
= 1315325.88 N

Weld Load [W3]:  
= (A<sub>2</sub>+A<sub>3</sub>+A<sub>4</sub>+A<sub>5</sub>+(2•t<sub>n</sub>•t•f<sub>r1</sub>))•S  
= ( 6648.3296 + 0.0 + 100.0 + 0.0 + ( 2784.0 ) ) • 138  
= 1315325.88 N

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) • D<sub>lo</sub> • W<sub>o</sub> • 0.49 • S<sub>nw</sub>  
= ( 3.1416/2.0 ) • 1168.4 • 10.0 • 0.49 • 138  
= 1240914. N

Shear, Nozzle Wall [Snw]:  
= (π • (D<sub>lr</sub> + D<sub>lo</sub>)/4 ) • ( Thk - Can ) • 0.7 • S<sub>n</sub>  
= (3.1416 • 576.2) • ( 48.0 - 0.0 ) • 0.7 • 138  
= 8392755. N

Tension, Shell Groove Weld [Tngw]:  
= (π/2) • D<sub>lo</sub> • W<sub>gnvi</sub> • 0.74 • S<sub>ng</sub>  
= ( 3.1416/2.0 ) • 1168.4 • ( 16.0 - 0.0 ) • 0.74 • 138  
= 2998508. N

See UG-41 and UW-15 for clarification of allowable stresses.

**Strength of Failure Paths:**

Failure Path [Path 1-1]:  
= ( SONW + SNW ) = ( 1240914 + 8392755 ) = 9633670 N

Failure Path [Path 2-2]:  
= ( Sonw + Tpgw + Tngw + Sinw )  
= ( 1240914 + 0 + 2998508 + 0 ) = 4239423 N

Failure Path [Path 3-3]:  
= ( Sonw + Tngw + Sinw )  
= ( 1240914 + 2998508 + 0 ) = 4239423 N

**Summary of Failure Path Calculations:**

|                                            |                   |
|--------------------------------------------|-------------------|
| Path 1-1 = 9633670 N , must exceed W = 0 N | or W1 = 931173 N  |
| Path 2-2 = 4239422 N , must exceed W = 0 N | or W2 = 1315325 N |



Path 3-3 = 4239422 N , must exceed W = 0 N or W3 = 1315325 N

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

**Note:** The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 58.3637 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 488.3636 mm

*This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 246.77 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1136.40 mm*

**Warning:**

*Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).*

**Input Echo, WRC297 Item 1, Description: N5 :**

|                                      |           |        |     |
|--------------------------------------|-----------|--------|-----|
| Diameter Basis for Cylindrical Shell | ID        |        |     |
| Shell Internal Corrosion Allowance   | 1.0000    | mm     |     |
| Shell External Corrosion Allowance   | 0.0000    | mm     |     |
| Shell Diameter                       | 5906.000  | mm     |     |
| Shell Thickness                      | 30.0000   | mm     |     |
| Shell Stress Concentration Factor    | 1.000     |        |     |
| Vessel Material                      | SA-516 70 |        |     |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 | MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |           |        |     |
|------------------------------------|-----------|--------|-----|
| Diameter Basis for Nozzle          | OD        |        |     |
| Nozzle Corrosion Allowance         | 0.0000    | mm     |     |
| Nozzle Diameter                    | 1232.400  | mm     |     |
| Nozzle Thickness                   | 48.0000   | mm     |     |
| Nozzle Stress Concentration Factor | 1.000     |        |     |
| Nozzle Material                    | SA-516 70 |        |     |
| Nozzle Cold S.I. Allowable         | SNmc      | 138.00 | MPa |
| Nozzle Hot S.I. Allowable          | SNmh      | 138.00 | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:*

*These loads are assumed to be SUstained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.50      | MPa  |
| Radial Load              | P  | -86249.83 | N    |
| Circumferential Shear    | Vc | 57500.19  | N    |
| Longitudinal Shear       | Vl | 57500.19  | N    |
| Circumferential Moment   | Mc | 0.185E+09 | N-mm |
| Longitudinal Moment      | Ml | 0.185E+09 | N-mm |
| Torsional Moment         | Mt | 0.265E+09 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 1232.4 + 2 \cdot 1.65 \cdot \sqrt{(2968.5(30.0 - 1.0))} \\
 &= 2200.636 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5937.000 mm  
 Nozzle Outside Diameter (d) = 1232.400 mm  
 Vessel Thickness used (T) = 29.000 mm  
 Nozzle Thickness used (t) = 48.000 mm  
 T / t = 0.604  
 d / t = 25.675

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.970$$

Nr/P = 0.042  
 Mr/P = 0.102  
 M0/P = 0.030  
 N0/P = 0.108  
 MrD/Mc = 0.247  
 NrDL/Mc = 0.068  
 M0d/Mc = 0.081  
 N0DL/Mc = 0.109  
 MrD/M1 = 0.093  
 NrDL/M1 = 0.038  
 M0D/M1 = 0.027  
 N0DL/M1 = 0.129

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 11                   | 11                  | 11                      | 11                     |
| Outplane Bending (P )                                            | 18                   | -18                 | 18                      | -18                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -23                  | -23                 | 23                      | 23                     |
| Outplane Bending (ML)                                            | -29                  | 29                  | 29                      | -29                    |
| Normal Pressure Stress                                           | 50                   | 51                  | 50                      | 51                     |
| Outplane Stress Summary                                          | 28                   | 49                  | 132                     | 37                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 4                    | 4                   | 4                       | 4                      |
| Inplane Bending (P )                                               | 62                   | -62                 | 62                      | -62                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -6                   | -6                  | 6                       | 6                      |
| Inplane Bending (ML)                                               | -100                 | 100                 | 100                     | -100                   |
| Inplane Pressure Stress                                            | 25                   | 25                  | 25                      | 25                     |
| Inplane Stress Summary                                             | -14                  | 60                  | 198                     | -126                   |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 3                    | 3                   | 3                       | 3                      |
| Shear Stress Summary                                                 | 4                    | 4                   | 2                       | 2                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|



Two \* Max Shear Stress 43 62 199 163

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 11                    | 11                   | 11                     | 11                    |
| Outplane Bending (P )                                                  | 18                    | -18                  | 18                     | -18                   |
| Outplane Membrane (Mc)                                                 | -19                   | -19                  | 19                     | 19                    |
| Outplane Bending (Mc)                                                  | -87                   | 87                   | 87                     | -87                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 25                    | 25                   | 25                     | 25                    |
| Outplane Stress Summary                                                | -51                   | 85                   | 161                    | -50                   |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 4                     | 4                    | 4                      | 4                     |
| Inplane Bending (P )                                                     | 62                    | -62                  | 62                     | -62                   |
| Inplane Membrane (Mc)                                                    | -12                   | -12                  | 12                     | 12                    |
| Inplane Bending (Mc)                                                     | -264                  | 264                  | 264                    | -264                  |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 50                    | 51                   | 50                     | 51                    |
| Inplane Stress Summary                                                   | -159                  | 245                  | 394                    | -260                  |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                       | 3                     | 3                    | 3                      | 3                     |
| Shear Stress Summary                                                       | 2                     | 2                    | 4                      | 4                     |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 159                   | 245                  | 394                    | 260                   |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 11                   | 11                  | 11                      | 11                     |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -23                  | -23                 | 23                      | 23                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 6                    | 7                   | 6                       | 7                      |
| Hoop Stress Summary                                       | -5                   | -4                  | 40                      | 41                     |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|



|                       |     |     |    |     |
|-----------------------|-----|-----|----|-----|
| Axial Membrane (P )   | 0   | 0   | 0  | 0   |
| Axial Bending (P )    | 18  | -18 | 18 | -18 |
| Axial Membrane (Mc)   | 0   | 0   | 0  | 0   |
| Axial Bending (Mc)    | 0   | 0   | 0  | 0   |
| Axial Membrane (ML)   | -3  | -3  | 3  | 3   |
| Axial Bending (ML)    | -29 | 29  | 29 | -29 |
| Axial Pressure Stress | 3   | 3   | 3  | 3   |

|                      |     |    |    |     |
|----------------------|-----|----|----|-----|
| Axial Stress Summary | -10 | 11 | 54 | -39 |
|----------------------|-----|----|----|-----|

**Nozzle Stresses (MPa):**

|                                      |               |              |               |              |
|--------------------------------------|---------------|--------------|---------------|--------------|
| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 2             | 2            | 2             | 2            |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 1 | 1 |
|----------------------|---|---|---|---|

**Nozzle Stresses (MPa):**

|                                            |               |              |               |              |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Two * Max Shear Stress                     | 12            | 16           | 54            | 81           |

**Nozzle Stresses (MPa):**

|                                                              |                       |                      |                        |                       |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Hoop Membrane (P )                                           | 11                    | 11                   | 11                     | 11                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -19                   | -19                  | 19                     | 19                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 6                     | 7                    | 6                      | 7                     |

|                     |    |   |    |    |
|---------------------|----|---|----|----|
| Hoop Stress Summary | -1 | 0 | 37 | 37 |
|---------------------|----|---|----|----|

**Nozzle Stresses (MPa):**

|                                                                     |                       |                      |                        |                       |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 18                    | -18                  | 18                     | -18                   |
| Axial Membrane (Mc)                                                 | -3                    | -3                   | 3                      | 3                     |
| Axial Bending (Mc)                                                  | -83                   | 83                   | 83                     | -83                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 3                     | 3                    | 3                      | 3                     |

|                      |     |    |     |     |
|----------------------|-----|----|-----|-----|
| Axial Stress Summary | -65 | 65 | 109 | -94 |
|----------------------|-----|----|-----|-----|

**Nozzle Stresses (MPa):**

|                                         |               |              |               |              |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 2             | 2            | 2             | 2            |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 1 | 1 | 2 | 2 |
|----------------------|---|---|---|---|

**Nozzle Stresses (MPa):**

|                                               |               |              |               |              |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|

Two \* Max Shear Stress 65 66 109 132

### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |       |       |       |       |       |       |
|------------------------------|------------------|------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al   | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 50               | 51   | 50    | 51    | 50    | 51    | 50    | 51    |
| Circ. Pl (SUS)               | -12              | -12  | 34    | 34    | -8    | -8    | 16    | 16    |
| Circ. Q (SUS)                | -11              | 11   | 47    | -47   | -202  | 202   | 326   | -326  |
| Long. Pm (SUS)               | 25               | 25   | 25    | 25    | 25    | 25    | 25    | 25    |
| Long. Pl (SUS)               | -2               | -2   | 10    | 10    | -8    | -8    | 30    | 30    |
| Long. Q (SUS)                | -38              | 38   | 162   | -162  | -69   | 69    | 105   | -105  |
| Shear Pm (SUS)               | 0                | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 1                | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                | 3                | 3    | 3     | 3     | 3     | 3     | 3     | 3     |
| Pm (SUS)                     | 50.0             | 51.0 | 50.0  | 51.0  | 50.0  | 51.0  | 50.0  | 51.0  |
| Pm+Pl (SUS)                  | 38.1             | 39.1 | 84.0  | 85.0  | 42.0  | 43.0  | 66.1  | 67.1  |
| Pm+Pl+Q (Total)              | 42.8             | 62.3 | 197.1 | 165.0 | 160.0 | 245.0 | 392.1 | 259.1 |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 51.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 85.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 392.07    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |       |       |
|------------------------------|------------------|------|------|------|------|------|-------|-------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du    | Dl    |
| Circ. Pm (SUS)               | 6                | 7    | 6    | 7    | 6    | 7    | 6     | 7     |
| Circ. Pl (SUS)               | -12              | -12  | 34   | 34   | -8   | -8   | 30    | 30    |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Long. Pm (SUS)               | 3                | 3    | 3    | 3    | 3    | 3    | 3     | 3     |
| Long. Pl (SUS)               | -3               | -3   | 3    | 3    | -3   | -3   | 3     | 3     |
| Long. Q (SUS)                | -11              | 11   | 47   | -47  | -65  | 65   | 101   | -101  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Shear Pl (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0     | 0     |
| Shear Q (SUS)                | 2                | 2    | 2    | 2    | 2    | 2    | 2     | 2     |
| Pm (SUS)                     | 6.0              | 7.0  | 6.0  | 7.0  | 6.0  | 7.0  | 6.0   | 7.0   |
| Pm+Pl (SUS)                  | 6.0              | 5.0  | 40.0 | 41.0 | 2.0  | 1.0  | 36.0  | 37.0  |
| Pm+Pl+Q (Total)              | 11.7             | 16.5 | 53.3 | 82.1 | 65.1 | 66.1 | 107.1 | 132.1 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of | Max. S.I. | S.I. Allowable | Result |
|---------|-----------|----------------|--------|
|---------|-----------|----------------|--------|





| Stress Int.     |        |        |        |
|-----------------|--------|--------|--------|
| Pm (SUS)        | 7.00   | 138.00 | Passed |
| Pm+Pl (SUS)     | 41.00  | 207.00 | Passed |
| Pm+Pl+Q (TOTAL) | 132.06 | 414.00 | Passed |

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.*

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**Input, Nozzle Desc: K8a**

**From: 180**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 22250.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 270.00  | deg |
| Diameter                          |  | 2.0000  | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Nominal |  |
| Nominal Thickness        |  | STD     |  |

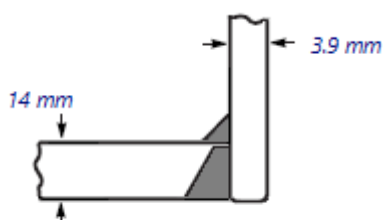
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 150.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 300    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K8a**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.  
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.5 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
= 0.1272 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2828 mm

**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |    |
|-------------------------------------------------|------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 105.0036 | mm |
| Parallel to Vessel Wall, opening length         | d    | 52.5018  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 9.7790   | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8a.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.2828  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 3.4200  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1,tb2) ]  
= min[ 3.42, max( 11.6895, 3.2158 ) ]  
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.2828, 3.42 )  
= 3.4200 mm

Available Nozzle Neck Thickness =  $0.875 \cdot 3.912 = 3.423$  mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)**

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

**Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B**

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/5.10 = 0.098

Weld Size Calculations, Description: K8a

Intermediate Calc. for nozzle/shell Welds    Tmin    3.9116    mm

**Results Per UW-16.1:**

|             | Required Thickness       | Actual Thickness        |
|-------------|--------------------------|-------------------------|
| Nozzle Weld | $2.7381 = 0.7 * t_{min}$ | $7.0700 = 0.7 * W_o$ mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1541 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.1540 mm

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**Input, Nozzle Desc: A530"**

**From: 180**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 1999.9998 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 22350.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 45.00   | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

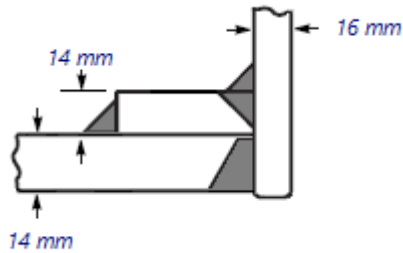
|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                          |      |           |     |
|------------------------------------------|------|-----------|-----|
| Pad Material                             |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature      | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient          | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface     | Dp   | 1382.0000 | mm  |
| Thickness of Pad                         | te   | 14.0000   | mm  |
| Weld leg size between Pad and Shell      | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle | Wgpn | 14.0000   | mm  |
| Reinforcing Pad Width                    |      | 310.0000  | mm  |

This is a Manway, Coupling or Access Opening.

|               |  |        |
|---------------|--|--------|
| Flange Class  |  | 150    |
| Flange Grade  |  | GR 1.1 |
| Flange Series |  | A      |

The Pressure Design option was MAWP + static head (bottom of element).



#### Reinforcement CALCULATION, Description: A530"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 30.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 1.3737 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1460.0000 mm |
| Parallel to Vessel Wall, opening length                | d  | 730.0000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 32.5000 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$

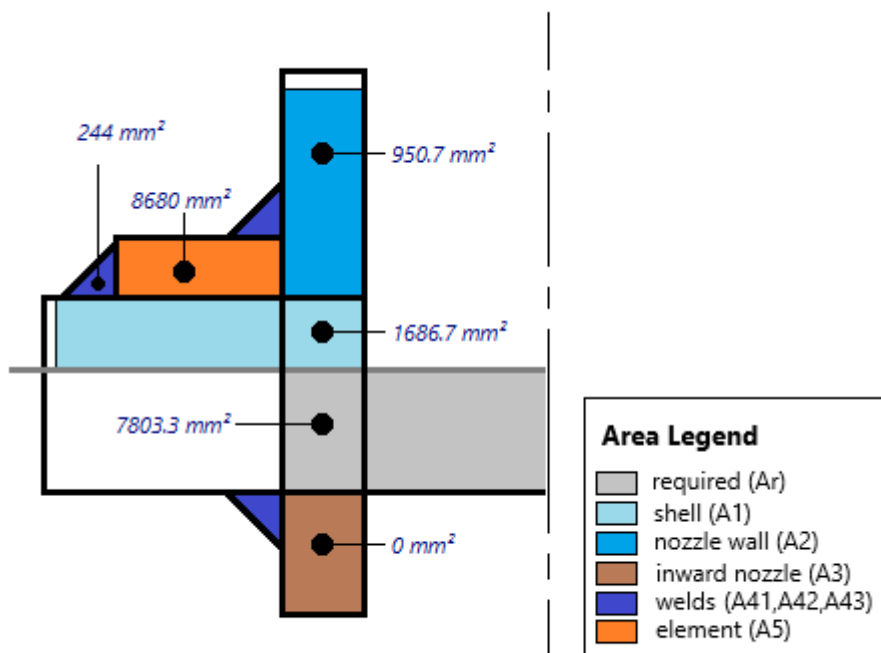


Figure UG 37.1 Nozzle Areas

#### Results of Nozzle Reinforcement Area Calculations: (mm²)

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 7803.346  | 4421.450  | NA    |
| Area in Shell            | A1          | 1686.655  | 647.100   | NA    |
| Area in Nozzle Wall      | A2          | 950.709   | 918.087   | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 244.000   | 244.000   | NA    |
| Area in Element          | A5          | 8680.001  | 8680.001  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 11561.364 | 10489.188 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter  | Thickness  |
|-----------------------------------------|-----------|------------|
| Based on given Pad Thickness:           | 1113.5702 | 14.0000 mm |
| Based on given Pad Diameter:            | 1382.0000 | 7.9387 mm  |
| Based on Shell or Nozzle Thickness:     | 1113.5702 | 14.0000 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (730.0 \cdot 10.6895 \cdot 1.0 + 2 \cdot 16.0 \cdot 10.6895 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 7803.346 \text{ mm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 730.0(1.0 \cdot 13.0 - 1.0 \cdot 10.69) - 2 \cdot 16.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.6895) \cdot (1 - 1.0)$   
 $= 1686.655 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 32.5) \cdot (16.0 - 1.37) \cdot 1.0$



= 950.709 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$   
= 244.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1382.0 - 762.0) \cdot 14.0 \cdot 1.0$   
= 8680.001 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 1.8756$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

=  $t_a$   
= 1.8756 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.374$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.086$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -29 °C





Gov. MDMT of the Reinforcement Pad : -29 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: A530"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm  
Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 6.5000 = 0.5*TminPad  | 8.4840 = 0.7 * Wp mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
= max( 0, (A-A1+2\*tn\*fr1\*(E1\*t-tr))Sv)  
= max( 0, (7803.3462 - 1686.6548 + 2 \* 16.0 \* 1.0 \*  
(1.0 \* 13.0 - 10.6895 ) )138)  
= 854218.38 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 950.7087 + 8680.001 + 244.0 - 0.0 \* 1.0 ) \* 138  
= 1362569.50 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 \* tn \* t \* fr1)) \* Sv  
= ( 950.7087 + 0.0 + 100.0 + ( 416.0 ) ) \* 138  
= 202384.94 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2\*tn\*t\*fr1))\*S  
= ( 950.7087 + 0.0 + 244.0 + 8680.001 + ( 416.0 ) ) \* 138  
= 1419971.50 N

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 762.0 \* 10.0 \* 0.49 \* 138  
= 809306. N

Shear, Pad Element Weld [Spew]:  
= (π/2) \* DP \* WP \* 0.49 \* SEW  
= ( 3.1416/2.0 ) \* 1382.0 \* 12.0 \* 0.49 \* 138  
= 1761325. N

Shear, Nozzle Wall [Snw]:  
= (π \* ( Dlr + Dlo )/4 ) \* ( Thk - Can ) \* 0.7 \* Sn  
= (3.1416 \* 373.0) \* ( 16.0 - 0.0 ) \* 0.7 \* 138  
= 1811002. N

Tension, Pad Groove Weld [Tpgw]:  
= ( π/2 ) \* Dlo \* Wgpn \* 0.74 \* Seg  
= (3.1416/2 ) \* 762.0 \* 14.0 \* 0.74 \* 138  
= 1711105. N

Tension, Shell Groove Weld [Tngw]:  
= (π/2) \* Dlo \* ( T - cas ) \* 0.74 \* Sng



$$= (3.1416/2.0) \cdot 762.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 138 \\ = 1588884. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1761325 + 1811002) = 3572327 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (809307 + 1711105 + 1588884 + 0) = 4109296 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1761325 + 1588884 + 0) = 3350209 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                            |                   |
|----------------------|----------------------------|-------------------|
| Path 1-1 = 3572327 N | , must exceed W = 854218 N | or W1 = 1362569 N |
| Path 2-2 = 4109295 N | , must exceed W = 854218 N | or W2 = 202384 N  |
| Path 3-3 = 3350209 N | , must exceed W = 854218 N | or W3 = 1419971 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.6818 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 438.6817 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1  $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$  2.145 %

*This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 98.71 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 730.00 mm*

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**Input, Nozzle Desc: K9a**

**From: 190**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |          |    |
|--------------------------------------|-----|----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00  | mm |
| Design Length of Section             | L   | 932.1440 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000  | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000   | mm |
| Shell External Corrosion Allowance   | co  | 0.0000   | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 25320.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 150.00  | deg |
| Diameter                          |  | 2.0000  | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Nominal |  |
| Nominal Thickness        |  | STD     |  |

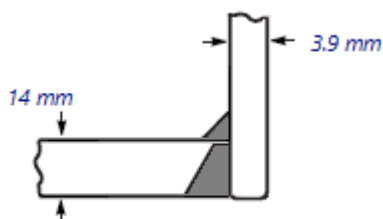
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 150.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 300    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K9a**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.  
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.5 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
= 0.1272 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2828 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |    |
|-------------------------------------------------|------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 105.0036 | mm |
| Parallel to Vessel Wall, opening length         | d    | 52.5018  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 9.7790   | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K9a.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.2828  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 3.4200  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 3.42, max( 11.6895, 3.2158 ) ]  
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.2828, 3.42 )  
= 3.4200 mm

Available Nozzle Neck Thickness =  $0.875 \cdot 3.912 = 3.423$  mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating =  $0.50/5.10 = 0.098$

Weld Size Calculations, Description: K9a

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

**Results Per UW-16.1:**

|             | Required Thickness       | Actual Thickness        |
|-------------|--------------------------|-------------------------|
| Nozzle Weld | $2.7381 = 0.7 * t_{min}$ | $7.0700 = 0.7 * W_o$ mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1541 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.1540 mm

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**Input, Nozzle Desc: N6**

**From: 190**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |          |    |
|--------------------------------------|-----|----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00  | mm |
| Design Length of Section             | L   | 932.1440 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000  | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000   | mm |
| Shell External Corrosion Allowance   | co  | 0.0000   | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 25310.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 300.00  | deg |
| Diameter                          |  | 4.0000  | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Nominal |  |
| Nominal Thickness        |  | STD     |  |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

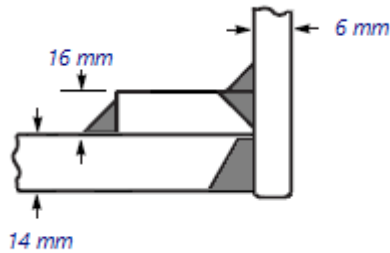
|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 150.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                          |      |           |     |
|------------------------------------------|------|-----------|-----|
| Pad Material                             |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature      | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient          | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface     | Dp   | 314.3000  | mm  |
| Thickness of Pad                         | te   | 16.0000   | mm  |
| Weld leg size between Pad and Shell      | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle | Wgpn | 16.0000   | mm  |
| Reinforcing Pad Width                    |      | 100.0000  | mm  |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 300    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



### Reinforcement CALCULATION, Description: N6

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |           |
|---------------------------------------------|-----------|
| Actual Outside Diameter Used in Calculation | 4.500 in. |
| Actual Thickness Used in Calculation        | 0.237 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 0.2409 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4105 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |             |
|--------------------------------------------------------|----|-------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 204.5208 mm |
| Parallel to Vessel Wall, opening length                | d  | 102.2604 mm |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 31.0495 mm  |

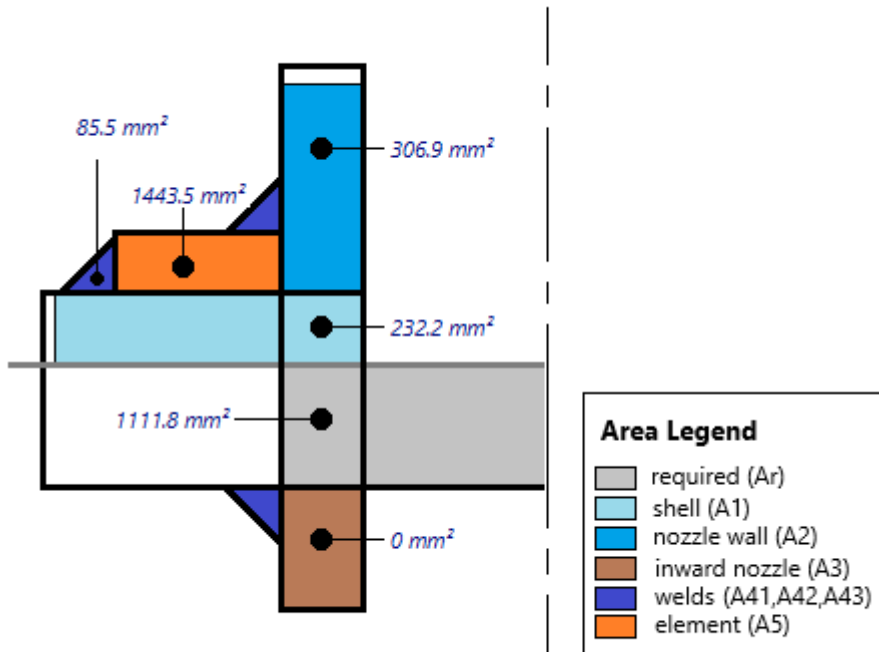
The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP     | External | Mapnc |
|--------------------------|-------------|----------|----------|-------|
| Area Required            | Ar          | 1111.764 | 451.663  | NA    |
| Area in Shell            | A1          | 232.240  | 433.685  | NA    |
| Area in Nozzle Wall      | A2          | 306.859  | 297.855  | NA    |
| Area in Inward Nozzle    | A3          | 0.000    | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 85.509   | 85.509   | NA    |
| Area in Element          | A5          | 1443.533 | 1443.533 | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 2068.141 | 2260.582 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 144.7472 16.0000 mm  
 Based on given Pad Diameter: 314.3000 5.3996 mm  
 Based on Shell or Nozzle Thickness: 195.2256 6.0198 mm

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (102.2604 \cdot 10.6895 \cdot 1.0 + 2 \cdot 6.0198 \cdot 10.6895 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 1111.764 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 102.26(1.0 \cdot 13.0 - 1.0 \cdot 10.69) - 2 \cdot 6.02$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.6895) \cdot (1 - 0.855)$   
 $= 232.240 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 31.05) \cdot (6.02 - 0.24) \cdot 0.8551$





= 306.859 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
 =  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
 =  $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$   
 = 85.509 mm<sup>2</sup>

Area Available in Element [A5]:  
 =  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
 =  $(204.5208 - 114.3) \cdot 16.0 \cdot 1.0$   
 = 1443.533 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 0.4105$  mm  
 Wall Thickness per UG16(b),  $tr16b = 2.5000$  mm  
 Wall Thickness, shell/head, internal pressure  $trb1 = 11.6895$  mm  
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 11.6895$  mm  
 Wall Thickness, shell/head, external pressure  $trb2 = 3.2158$  mm  
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 3.2158$  mm  
 Wall Thickness per table UG-45  $tb3 = 5.2578$  mm

Determine Nozzle Thickness candidate [tb]:  
 =  $\min[ tb3, \max( tb1, tb2 ) ]$   
 =  $\min[ 5.258, \max( 11.6895, 3.2158 ) ]$   
 = 5.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 =  $\max( ta, tb )$   
 =  $\max( 0.4105, 5.2578 )$   
 = 5.2578 mm

Available Nozzle Neck Thickness =  $0.875 \cdot 6.020 = 5.267$  mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |           |        |
|------------|---|-------|-----------|---|-----------|--------|
| Sustained  | : | 35.9, | Allowable | : | 118.0 MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 259.1 MPa | Passed |
| Occasional | : | 2.0,  | Allowable | : | 156.9 MPa | Passed |
| Shear      | : | 27.5, | Allowable | : | 82.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 5.267$ ,  $t_r = 0.241$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.046$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 5.267$ ,  $t_r = 0.241$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.046$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 5.267$ ,  $t_r = 0.241$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.046$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |



### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 14.0, tr = 10.69, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.822$ , Temp. Reduction = 10 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -18 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -28 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

### Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 5.267, tr = 0.241, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.046$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

|                                                        |           |
|--------------------------------------------------------|-----------|
| Gov. MDMT of the Nozzle                                | : -104 °C |
| Gov. MDMT of the Reinforcement Pad                     | : -29 °C  |
| Gov. MDMT of the nozzle to shell joint welded assembly | : -29 °C  |

### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $0.50 / 5.10 = 0.098$

### Weld Size Calculations, Description: N6

|                                           |         |            |
|-------------------------------------------|---------|------------|
| Intermediate Calc. for nozzle/shell Welds | Tmin    | 6.0198 mm  |
| Intermediate Calc. for pad/shell Welds    | TminPad | 13.0000 mm |

### Results Per UW-16.1:

|             | Required Thickness              | Actual Thickness            |
|-------------|---------------------------------|-----------------------------|
| Nozzle Weld | $4.2139 = 0.7 \cdot t_{min}$    | $7.0700 = 0.7 \cdot W_o$ mm |
| Pad Weld    | $6.5000 = 0.5 \cdot T_{minPad}$ | $8.4840 = 0.7 \cdot W_p$ mm |

### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$   
 $= \max(0, (1111.7639 - 232.2401 + 2 \cdot 6.0198 \cdot 0.8551 \cdot (1.0 \cdot 13.0 - 10.6895)) \cdot 138)$   
 $= 124643.93 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2 + A5 + A4 - (W_i - Can / 7.07)^2 \cdot f_{r2}) \cdot S_v$   
 $= (306.859 + 1443.5328 + 85.5088 - 0.0 \cdot 0.86) \cdot 138$   
 $= 253328.16 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (306.859 + 0.0 + 85.5088 + (133.8339)) \cdot 138$   
 $= 72608.35 \text{ N}$

Weld Load [W3]:  
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (306.859 + 0.0 + 85.5088 + 1443.5328 + (133.8339)) \cdot 138$   
 $= 271795.34 \text{ N}$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$



= 103802. N

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 314.3 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 400568. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 54.1401) \cdot (6.0198 - 0.0) \cdot 0.7 \cdot 118$$

$$= 84566. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 114.3 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 293332. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 138$$

$$= 238328. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (400568 + 84566) = 485133 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (103802 + 293332 + 238328 + 0) = 635463 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (400568 + 238328 + 0) = 638896 \text{ N}$$

#### Summary of Failure Path Calculations:

|                     |                            |                  |
|---------------------|----------------------------|------------------|
| Path 1-1 = 485133 N | , must exceed W = 124643 N | or W1 = 253328 N |
| Path 2-2 = 635462 N | , must exceed W = 124643 N | or W2 = 72608 N  |
| Path 3-3 = 638896 N | , must exceed W = 124643 N | or W3 = 271795 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.5531 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.5530 mm

#### Input Echo, WRC297 Item 1, Description: N6 :

|                                      |           |            |
|--------------------------------------|-----------|------------|
| Diameter Basis for Cylindrical Shell | ID        |            |
| Shell Internal Corrosion Allowance   | 1.0000    | mm         |
| Shell External Corrosion Allowance   | 0.0000    | mm         |
| Shell Diameter                       | 5906.000  | mm         |
| Shell Thickness                      | 14.0000   | mm         |
| Shell Stress Concentration Factor    | 1.000     |            |
| Vessel Material                      | SA-516 70 |            |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 MPa |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                            |         |    |
|----------------------------|---------|----|
| Diameter Basis for Nozzle  | OD      |    |
| Nozzle Corrosion Allowance | 0.0000  | mm |
| Nozzle Diameter            | 114.300 | mm |



|                                    |      |          |     |
|------------------------------------|------|----------|-----|
| Nozzle Thickness                   |      | 6.0198   | mm  |
| Nozzle Stress Concentration Factor |      | 1.000    |     |
| Nozzle Material                    |      | SA-106 B |     |
| Nozzle Cold S.I. Allowable         | SNmc | 118.00   | MPa |
| Nozzle Hot S.I. Allowable          | SNmh | 118.00   | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                              |         |    |
|------------------------------|---------|----|
| Diameter of Reinforcing Pad  | 314.300 | mm |
| Thickness of Reinforcing Pad | 16.0000 | mm |

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSained loads.*

|                          |    |           |      |
|--------------------------|----|-----------|------|
| Design Internal Pressure | Dp | 0.50      | MPa  |
| Radial Load              | P  | -7500.00  | N    |
| Circumferential Shear    | Vc | 5000.00   | N    |
| Longitudinal Shear       | Vl | 5000.00   | N    |
| Circumferential Moment   | Mc | 0.140E+07 | N-mm |
| Longitudinal Moment      | Ml | 0.140E+07 | N-mm |
| Torsional Moment         | Mt | 0.200E+07 | N-mm |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2968.5(14.0 - 1.0))}$   
 $= 762.567 \text{ mm}$

**WRC 297 Curve Access Parameters:**

|                         |       |          |    |
|-------------------------|-------|----------|----|
| Vessel Mean Diameter    | (D) = | 5937.000 | mm |
| Nozzle Outside Diameter | (d) = | 114.300  | mm |
| Vessel Thickness used   | (T) = | 29.000   | mm |
| Nozzle Thickness used   | (t) = | 6.020    | mm |
| T / t                   | =     | 4.817    |    |
| d / t                   | =     | 18.987   |    |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.275$$

|         |   |        |
|---------|---|--------|
| Nr/P    | = | 0.149  |
| Mr/P    | = | 0.113  |
| M0/P    | = | 0.268  |
| N0/P    | = | 0.315  |
| MrD/Mc  | = | 0.114  |
| NrDL/Mc | = | 0.131  |
| M0d/Mc  | = | 0.300  |
| N0DL/Mc | = | -0.148 |
| MrD/Ml  | = | 0.114  |
| NrDL/Ml | = | 0.132  |
| M0D/Ml  | = | 0.292  |
| N0DL/Ml | = | -0.101 |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 2                    | 2                   | 2                       | 2                      |
| Outplane Bending (P )                                            | 14                   | -14                 | 14                      | -14                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | 1                    | 1                   | -1                      | -1                     |



|                        |     |    |    |     |
|------------------------|-----|----|----|-----|
| Outplane Bending (ML)  | -25 | 25 | 25 | -25 |
| Normal Pressure Stress | 50  | 51 | 50 | 51  |

|                         |    |    |    |    |
|-------------------------|----|----|----|----|
| Outplane Stress Summary | 43 | 66 | 91 | 12 |
|-------------------------|----|----|----|----|

**Vessel Stresses (MPa):**

|                                           |             |            |                |               |
|-------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                        | Au          | Al         | Bu             | Bl            |
| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Inplane Membrane (P )                     | 1           | 1          | 1              | 1             |
| Inplane Bending (P )                      | 6           | -6         | 6              | -6            |
| Inplane Membrane (Mc)                     | 0           | 0          | 0              | 0             |
| Inplane Bending (Mc)                      | 0           | 0          | 0              | 0             |
| Inplane Membrane (ML)                     | -1          | -1         | 1              | 1             |
| Inplane Bending (ML)                      | -9          | 9          | 9              | -9            |
| Inplane Pressure Stress                   | 25          | 25         | 25             | 25            |
| Inplane Stress Summary                    | 20          | 28         | 44             | 12            |

**Vessel Stresses (MPa):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Shear (Vc)                         | 0           | 0          | 0              | 0             |
| Outplane Shear (Vl)                         | 0           | 0          | 0              | 0             |
| Outplane Shear (Mt)                         | 3           | 3          | 3              | 3             |
| Shear Stress Summary                        | 4           | 4          | 2              | 2             |

**Vessel Stresses (MPa):**

|                        |             |            |                |               |
|------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE     | Au          | Al         | Bu             | Bl            |
| (Stress Intensities)   | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Two * Max Shear Stress | 44          | 66         | 91             | 14            |

**Vessel Stresses (MPa):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Membrane (P )                     | 2            | 2           | 2             | 2            |
| Outplane Bending (P )                      | 14           | -14         | 14            | -14          |
| Outplane Membrane (Mc)                     | 2            | 2           | -2            | -2           |
| Outplane Bending (Mc)                      | -26          | 26          | 26            | -26          |
| Outplane Membrane (ML)                     | 0            | 0           | 0             | 0            |
| Outplane Bending (ML)                      | 0            | 0           | 0             | 0            |
| Normal Pressure Stress                     | 25           | 25          | 25            | 25           |
| Outplane Stress Summary                    | 18           | 42          | 66            | -14          |

**Vessel Stresses (MPa):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Inplane Membrane (P )                        | 1            | 1           | 1             | 1            |
| Inplane Bending (P )                         | 6            | -6          | 6             | -6           |
| Inplane Membrane (Mc)                        | -1           | -1          | 1             | 1            |
| Inplane Bending (Mc)                         | -9           | 9           | 9             | -9           |
| Inplane Membrane (ML)                        | 0            | 0           | 0             | 0            |
| Inplane Bending (ML)                         | 0            | 0           | 0             | 0            |
| Inplane Pressure Stress                      | 50           | 51          | 50            | 51           |
| Inplane Stress Summary                       | 46           | 54          | 69            | 38           |

**Vessel Stresses (MPa):**

|                       |    |    |    |    |
|-----------------------|----|----|----|----|
| CIRCUMFERENTIAL PLANE | Cu | Cl | Du | Dl |
|-----------------------|----|----|----|----|



| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| Outplane Shear (Vc)                            | 0            | 0           | 0             | 0            |
| Outplane Shear (Vl)                            | 0            | 0           | 0             | 0            |
| Torsional Shear (Mt)                           | 3            | 3           | 3             | 3            |
| Shear Stress Summary                           | 2            | 2           | 4             | 4            |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left Outside | Cl<br>Left Inside | Du<br>Right Outside | Dl<br>Right Inside |
|-----------------------------------------------|--------------------|-------------------|---------------------|--------------------|
| Two * Max Shear Stress                        | 46                 | 54                | 72                  | 53                 |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses in the hoop direction) | Au<br>Top Outside | Al<br>Top Inside | Bu<br>Bottom Outside | Bl<br>Bottom Inside |
|--------------------------------------------------------|-------------------|------------------|----------------------|---------------------|
| Hoop Membrane (P )                                     | 2                 | 2                | 2                    | 2                   |
| Hoop Bending (P )                                      | 0                 | 0                | 0                    | 0                   |
| Hoop Membrane (Mc)                                     | 0                 | 0                | 0                    | 0                   |
| Hoop Bending (Mc)                                      | 0                 | 0                | 0                    | 0                   |
| Hoop Membrane (ML)                                     | 1                 | 1                | -1                   | -1                  |
| Hoop Bending (ML)                                      | 0                 | 0                | 0                    | 0                   |
| Hoop Pressure Stress                                   | 4                 | 5                | 4                    | 5                   |
| Hoop Stress Summary                                    | 9                 | 9                | 6                    | 6                   |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to pipe cross-section) | Au<br>Top Outside | Al<br>Top Inside | Bu<br>Bottom Outside | Bl<br>Bottom Inside |
|---------------------------------------------------------------|-------------------|------------------|----------------------|---------------------|
| Axial Membrane (P )                                           | 3                 | 3                | 3                    | 3                   |
| Axial Bending (P )                                            | 47                | -47              | 47                   | -47                 |
| Axial Membrane (Mc)                                           | 0                 | 0                | 0                    | 0                   |
| Axial Bending (Mc)                                            | 0                 | 0                | 0                    | 0                   |
| Axial Membrane (ML)                                           | -26               | -26              | 26                   | 26                  |
| Axial Bending (ML)                                            | -97               | 97               | 97                   | -97                 |
| Axial Pressure Stress                                         | 2                 | 2                | 2                    | 2                   |
| Axial Stress Summary                                          | -70               | 29               | 177                  | -112                |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 4             | 4            | -4            | -4           |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 16            | 16           | 16            | 16           |
| Shear Stress Summary                 | 20            | 20           | 11            | 11           |

**Nozzle Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 90            | 46           | 178           | 121          |

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the hoop direction) | Cu<br>Left Outside | Cl<br>Left Inside | Du<br>Right Outside | Dl<br>Right Inside |
|-----------------------------------------------------------|--------------------|-------------------|---------------------|--------------------|
| Hoop Membrane (P )                                        | 2                  | 2                 | 2                   | 2                  |
| Hoop Bending (P )                                         | 0                  | 0                 | 0                   | 0                  |
| Hoop Membrane (Mc)                                        | 2                  | 2                 | -2                  | -2                 |



|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Hoop Bending (Mc)    | 0 | 0 | 0 | 0 |
| Hoop Membrane (ML)   | 0 | 0 | 0 | 0 |
| Hoop Bending (ML)    | 0 | 0 | 0 | 0 |
| Hoop Pressure Stress | 4 | 5 | 4 | 5 |

|                     |   |    |   |   |
|---------------------|---|----|---|---|
| Hoop Stress Summary | 9 | 10 | 5 | 6 |
|---------------------|---|----|---|---|

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                       |     |     |    |     |
|-----------------------|-----|-----|----|-----|
| Axial Membrane (P )   | 3   | 3   | 3  | 3   |
| Axial Bending (P )    | 47  | -47 | 47 | -47 |
| Axial Membrane (Mc)   | -26 | -26 | 26 | 26  |
| Axial Bending (Mc)    | -97 | 97  | 97 | -97 |
| Axial Membrane (ML)   | 0   | 0   | 0  | 0   |
| Axial Bending (ML)    | 0   | 0   | 0  | 0   |
| Axial Pressure Stress | 2   | 2   | 2  | 2   |

|                      |     |    |     |      |
|----------------------|-----|----|-----|------|
| Axial Stress Summary | -71 | 30 | 177 | -112 |
|----------------------|-----|----|-----|------|

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
|-----------------------------------------|---------------|--------------|---------------|--------------|

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear due to (Vc)    | 0  | 0  | 0  | 0  |
| Shear due to (Vl)    | -4 | -4 | 4  | 4  |
| Shear due to Torsion | 16 | 16 | 16 | 16 |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear Stress Summary | 11 | 11 | 20 | 20 |
|----------------------|----|----|----|----|

**Nozzle Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
|-----------------------------------------------|---------------|--------------|---------------|--------------|

|                        |    |    |     |     |
|------------------------|----|----|-----|-----|
| Two * Max Shear Stress | 84 | 35 | 180 | 125 |
|------------------------|----|----|-----|-----|

**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$   
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{2960.5(14.0 - 1.0)}$   
 $= 761.693 \text{ mm}$

WRC 297 Curve Access Parameters:

|                             |   |             |
|-----------------------------|---|-------------|
| Vessel Mean Diameter (D)    | = | 5921.000 mm |
| Nozzle Outside Diameter (d) | = | 314.300 mm  |
| Vessel Thickness used (T)   | = | 13.000 mm   |
| Nozzle Thickness used (t)   | = | 106.020 mm  |
| T / t                       | = | 0.123       |
| d / t                       | = | 10.000      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.133$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.121 |
| Mr/P    | = | 0.213 |
| M0/P    | = | 0.061 |
| N0/P    | = | 0.084 |
| MrD/Mc  | = | 0.310 |
| NrDL/Mc | = | 0.090 |
| M0d/Mc  | = | 0.094 |
| N0DL/Mc | = | 0.046 |
| MrD/M1  | = | 0.177 |
| NrDL/M1 | = | 0.118 |
| M0D/M1  | = | 0.054 |



NØDL/M1 = 0.085

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 3                    | 3                   | 3                       | 3                      |
| Outplane Bending (P )                                            | 16                   | -16                 | 16                      | -16                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -2                   | -2                  | 2                       | 2                      |
| Outplane Bending (ML)                                            | -8                   | 8                   | 8                       | -8                     |
| Normal Pressure Stress                                           | 112                  | 113                 | 112                     | 113                    |
| Outplane Stress Summary                                          | 122                  | 107                 | 143                     | 94                     |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 5                    | 5                   | 5                       | 5                      |
| Inplane Bending (P )                                               | 56                   | -56                 | 56                      | -56                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -3                   | -3                  | 3                       | 3                      |
| Inplane Bending (ML)                                               | -27                  | 27                  | 27                      | -27                    |
| Inplane Pressure Stress                                            | 56                   | 56                  | 56                      | 56                     |
| Inplane Stress Summary                                             | 87                   | 30                  | 149                     | -19                    |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 0                    | 0                   | 0                       | 0                      |
| Shear Stress Summary                                                 | 1                    | 1                   | 0                       | 0                      |

**Vessel Stresses (MPa):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 122                  | 107                 | 149                     | 114                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 3                     | 3                    | 3                      | 3                     |
| Outplane Bending (P )                                                  | 16                    | -16                  | 16                     | -16                   |
| Outplane Membrane (Mc)                                                 | -1                    | -1                   | 1                      | 1                     |
| Outplane Bending (Mc)                                                  | -14                   | 14                   | 14                     | -14                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 56                    | 56                   | 56                     | 56                    |
| Outplane Stress Summary                                                | 60                    | 57                   | 92                     | 30                    |

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|





|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Inplane Membrane (P )   | 5   | 5   | 5   | 5   |
| Inplane Bending (P )    | 56  | -56 | 56  | -56 |
| Inplane Membrane (Mc)   | -2  | -2  | 2   | 2   |
| Inplane Bending (Mc)    | -49 | 49  | 49  | -49 |
| Inplane Membrane (ML)   | 0   | 0   | 0   | 0   |
| Inplane Bending (ML)    | 0   | 0   | 0   | 0   |
| Inplane Pressure Stress | 112 | 113 | 112 | 113 |

|                        |     |     |     |    |
|------------------------|-----|-----|-----|----|
| Inplane Stress Summary | 123 | 108 | 226 | 15 |
|------------------------|-----|-----|-----|----|

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu              | Cl             | Du               | Dl              |
|----------------------------------------------------------------------------|-----------------|----------------|------------------|-----------------|
|                                                                            | Left<br>Outside | Left<br>Inside | Right<br>Outside | Right<br>Inside |
| Outplane Shear (Vc)                                                        | 0               | 0              | 0                | 0               |
| Outplane Shear (Vl)                                                        | 0               | 0              | 0                | 0               |
| Torsional Shear (Mt)                                                       | 0               | 0              | 0                | 0               |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 0 | 0 | 1 | 1 |
|----------------------|---|---|---|---|

**Vessel Stresses (MPa):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu              | Cl             | Du               | Dl              |
|-----------------------------------------------|-----------------|----------------|------------------|-----------------|
|                                               | Left<br>Outside | Left<br>Inside | Right<br>Outside | Right<br>Inside |
| Two * Max Shear Stress                        | 123             | 108            | 226              | 30              |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

| Type of<br>Stress Int.<br>Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------------|------------------|------|------|------|------|------|------|------|
|                                    | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)                     | 50               | 51   | 50   | 51   | 50   | 51   | 50   | 51   |
| Circ. Pl (SUS)                     | 3                | 3    | 1    | 1    | 0    | 0    | 2    | 2    |
| Circ. Q (SUS)                      | -11              | 11   | 39   | -39  | -3   | 3    | 15   | -15  |
| Long. Pm (SUS)                     | 25               | 25   | 25   | 25   | 25   | 25   | 25   | 25   |
| Long. Pl (SUS)                     | 0                | 0    | 2    | 2    | 4    | 4    | 0    | 0    |
| Long. Q (SUS)                      | -3               | 3    | 15   | -15  | -12  | 12   | 40   | -40  |
| Shear Pm (SUS)                     | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)                     | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                      | 3                | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Pm (SUS)                           | 50.0             | 51.0 | 50.0 | 51.0 | 50.0 | 51.0 | 50.0 | 51.0 |
| Pm+Pl (SUS)                        | 53.0             | 54.0 | 51.0 | 52.0 | 50.0 | 51.0 | 52.0 | 53.0 |
| Pm+Pl+Q (Total)                    | 42.4             | 65.2 | 90.2 | 15.5 | 47.3 | 54.7 | 69.2 | 53.3 |

**Vessel Stress Summation Comparison (MPa):**

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 51.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 54.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 90.19     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

| Type of Stress Int. Location | Stress Values at |       |       |       |       |       |       |       |
|------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                              | Au               | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 112              | 113   | 112   | 113   | 112   | 113   | 112   | 113   |
| Circ. Pl (SUS)               | 1                | 1     | 5     | 5     | 3     | 3     | 7     | 7     |
| Circ. Q (SUS)                | 8                | -8    | 24    | -24   | 7     | -7    | 105   | -105  |
| Long. Pm (SUS)               | 56               | 56    | 56    | 56    | 56    | 56    | 56    | 56    |
| Long. Pl (SUS)               | 2                | 2     | 8     | 8     | 2     | 2     | 4     | 4     |
| Long. Q (SUS)                | 29               | -29   | 83    | -83   | 2     | -2    | 30    | -30   |
| Shear Pm (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Q (SUS)                | 0                | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Pm (SUS)                     | 112.0            | 113.0 | 112.0 | 113.0 | 112.0 | 113.0 | 112.0 | 113.0 |
| Pm+Pl (SUS)                  | 113.0            | 114.0 | 117.0 | 118.0 | 115.0 | 116.0 | 119.0 | 120.0 |
| Pm+Pl+Q (Total)              | 121.0            | 106.0 | 147.0 | 113.0 | 122.0 | 109.0 | 224.0 | 30.0  |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 113.00    | 138.00         | Passed |
| Pm+Pl (SUS)         | 120.00    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 224.00    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

| Type of Stress Int. Location | Stress Values at |      |       |       |      |      |       |       |
|------------------------------|------------------|------|-------|-------|------|------|-------|-------|
|                              | Au               | Al   | Bu    | Bl    | Cu   | Cl   | Du    | Dl    |
| Circ. Pm (SUS)               | 4                | 5    | 4     | 5     | 4    | 5    | 4     | 5     |
| Circ. Pl (SUS)               | 3                | 3    | 1     | 1     | 4    | 4    | 0     | 0     |
| Circ. Q (SUS)                | 0                | 0    | 0     | 0     | 0    | 0    | 0     | 0     |
| Long. Pm (SUS)               | 2                | 2    | 2     | 2     | 2    | 2    | 2     | 2     |
| Long. Pl (SUS)               | -23              | -23  | 29    | 29    | -23  | -23  | 29    | 29    |
| Long. Q (SUS)                | -50              | 50   | 144   | -144  | -50  | 50   | 144   | -144  |
| Shear Pm (SUS)               | 0                | 0    | 0     | 0     | 0    | 0    | 0     | 0     |
| Shear Pl (SUS)               | 4                | 4    | -4    | -4    | -4   | -4   | 4     | 4     |
| Shear Q (SUS)                | 16               | 16   | 16    | 16    | 16   | 16   | 16    | 16    |
| Pm (SUS)                     | 4.0              | 5.0  | 4.0   | 5.0   | 4.0  | 5.0  | 4.0   | 5.0   |
| Pm+Pl (SUS)                  | 29.1             | 30.1 | 31.6  | 31.6  | 30.1 | 31.0 | 31.6  | 31.6  |
| Pm+Pl+Q (Total)              | 87.7             | 45.2 | 175.8 | 121.4 | 82.6 | 34.6 | 177.3 | 124.6 |

#### Nozzle Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 5.00      | 118.00         | Passed |
| Pm+Pl (SUS)         | 31.62     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 177.31    | 354.00         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.





**Input, Nozzle Desc: K8b**

**From: 190**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 24710.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 270.00  | deg |
| Diameter                          |  | 2.0000  | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Nominal |  |
| Nominal Thickness        |  | STD     |  |

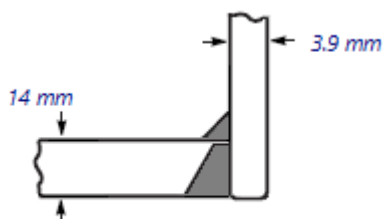
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 150.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 300    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K8b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.  
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.5 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
= 0.1272 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2828 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |    |
|-------------------------------------------------|------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 105.0036 | mm |
| Parallel to Vessel Wall, opening length         | d    | 52.5018  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 9.7790   | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8b.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.2828  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 3.4200  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 3.42, max( 11.6895, 3.2158 ) ]  
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.2828, 3.42 )  
= 3.4200 mm

Available Nozzle Neck Thickness =  $0.875 \cdot 3.912 = 3.423$  mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/5.10 = 0.098

Weld Size Calculations, Description: K8b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

**Results Per UW-16.1:**

|             | Required Thickness       | Actual Thickness        |
|-------------|--------------------------|-------------------------|
| Nozzle Weld | $2.7381 = 0.7 * t_{min}$ | $7.0700 = 0.7 * W_o$ mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1541 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.1540 mm

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**Input, Nozzle Desc: K9b**

**From: 190**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 24050.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 230.00  | deg |
| Diameter                          |  | 2.0000  | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Nominal |  |
| Nominal Thickness        |  | STD     |  |

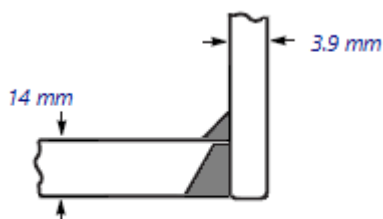
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 150.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 300    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



**Reinforcement CALCULATION, Description: K9b**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.  
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
= 10.6895 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.5 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
= 0.1272 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2828 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |    |
|-------------------------------------------------|------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 105.0036 | mm |
| Parallel to Vessel Wall, opening length         | d    | 52.5018  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 9.7790   | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: K9b.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.2828  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 11.6895 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 11.6895 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 3.2158  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 3.2158  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 3.4200  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 3.42, max( 11.6895, 3.2158 ) ]  
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.2828, 3.42 )  
= 3.4200 mm

Available Nozzle Neck Thickness =  $0.875 \cdot 3.912 = 3.423$  mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |





Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)    -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)    -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating =  $0.50/5.10 = 0.098$

Weld Size Calculations, Description: K9b

Intermediate Calc. for nozzle/shell Welds    Tmin    3.9116    mm

**Results Per UW-16.1:**

|             | Required Thickness       | Actual Thickness        |
|-------------|--------------------------|-------------------------|
| Nozzle Weld | $2.7381 = 0.7 * t_{min}$ | $7.0700 = 0.7 * W_o$ mm |

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1541 mm  
The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.1540 mm

*This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 11.01 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 52.50 mm*

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**Input, Nozzle Desc: A630"**

**From: 190**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 24200.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 45.00   | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

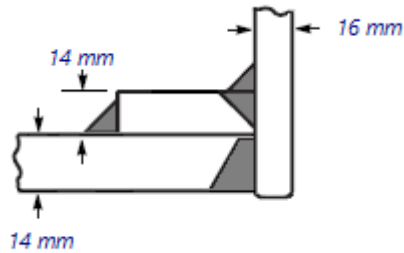
|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 14.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                               |      |           |     |
|-----------------------------------------------|------|-----------|-----|
| Pad Material                                  |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature           | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient               | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface          | Dp   | 1500.0000 | mm  |
| Thickness of Pad                              | te   | 14.0000   | mm  |
| Weld leg size between Pad and Shell           | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle      | Wgpn | 14.0000   | mm  |
| Reinforcing Pad Width                         |      | 369.0000  | mm  |
| This is a Manway, Coupling or Access Opening. |      |           |     |

|              |  |        |  |
|--------------|--|--------|--|
| Flange Class |  | 150    |  |
| Flange Grade |  | GR 1.1 |  |

The Pressure Design option was MAWP + static head (bottom of element).



### Reinforcement CALCULATION, Description: A630"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 30.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

*This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 1.3737 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1460.0000 mm |
| Parallel to Vessel Wall, opening length                | d  | 730.0000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 32.5000 mm   |

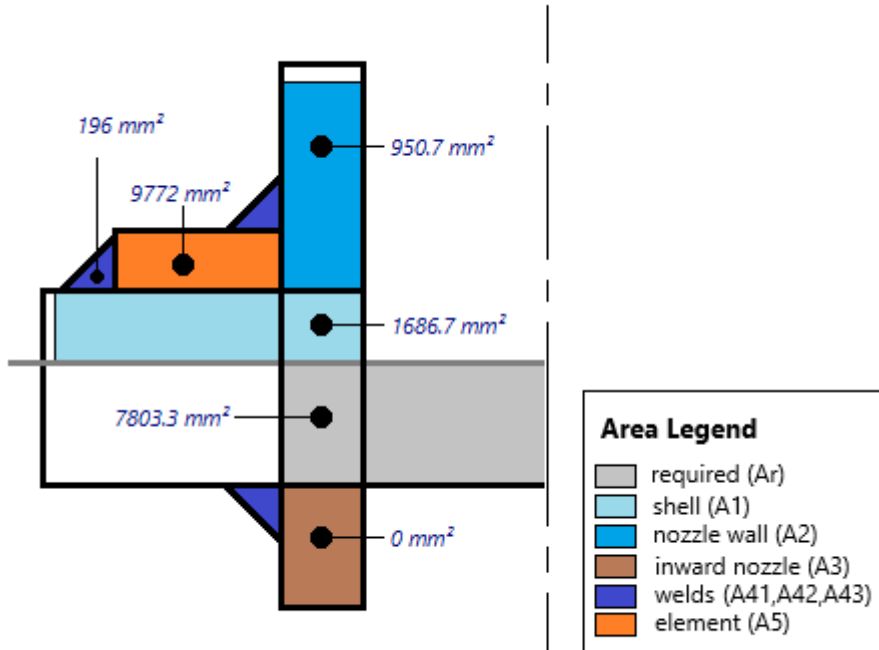
The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5    |             | MAWP             | External         | Mapnc     |
|-----------------------------|-------------|------------------|------------------|-----------|
| Area Required               | Ar          | 7803.346         | 4421.450         | NA        |
| Area in Shell               | A1          | 1686.655         | 647.100          | NA        |
| Area in Nozzle Wall         | A2          | 950.709          | 918.087          | NA        |
| Area in Inward Nozzle       | A3          | 0.000            | 0.000            | NA        |
| Area in Welds               | A41+A42+A43 | 196.000          | 196.000          | NA        |
| Area in Element             | A5          | 9772.001         | 9772.001         | NA        |
| <b>TOTAL AREA AVAILABLE</b> | <b>Atot</b> | <b>12605.364</b> | <b>11533.188</b> | <b>NA</b> |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter  | Thickness  |
|-----------------------------------------|-----------|------------|
| Based on given Pad Thickness:           | 1116.9987 | 14.0000 mm |
| Based on given Pad Diameter:            | 1500.0000 | 7.1203 mm  |
| Based on Shell or Nozzle Thickness:     | 1116.9987 | 14.0000 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (730.0 \cdot 10.6895 \cdot 1.0 + 2 \cdot 16.0 \cdot 10.6895 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 7803.346 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 730.0(1.0 \cdot 13.0 - 1.0 \cdot 10.69) - 2 \cdot 16.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.6895) \cdot (1 - 1.0)$   
 $= 1686.655 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 32.5) \cdot (16.0 - 1.37) \cdot 1.0$



= 950.709 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $14.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$   
= 196.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1460.0 - 762.0) \cdot 14.0 \cdot 1.0$   
= 9772.001 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 1.8756$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

=  $t_a$   
= 1.8756 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.374$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.086$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -29 °C



Gov. MDMT of the Reinforcement Pad : -29 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: A630"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm  
Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 9.8980 = 0.7 * Wo mm |
| Pad Weld    | 6.5000 = 0.5*TminPad  | 8.4840 = 0.7 * Wp mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
= max( 0, (A-A1+2\*tn\*fr1\*(E1\*t-tr))Sv)  
= max( 0, (7803.3462 - 1686.6548 + 2 \* 16.0 \* 1.0 \*  
(1.0 \* 13.0 - 10.6895 ) )138)  
= 854218.38 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 950.7087 + 9772.001 + 196.0 - 0.0 \* 1.0 ) \* 138  
= 1506626.50 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 \* tn \* t \* fr1)) \* Sv  
= ( 950.7087 + 0.0 + 196.0 + ( 416.0 ) ) \* 138  
= 215631.58 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2\*tn\*t\*fr1))\*S  
= ( 950.7087 + 0.0 + 196.0 + 9772.001 + ( 416.0 ) ) \* 138  
= 1564028.62 N

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 762.0 \* 14.0 \* 0.49 \* 138  
= 1133029. N

Shear, Pad Element Weld [Spew]:  
= (π/2) \* DP \* WP \* 0.49 \* SEW  
= ( 3.1416/2.0 ) \* 1500.0 \* 12.0 \* 0.49 \* 138  
= 1911713. N

Shear, Nozzle Wall [Snw]:  
= (π \* ( Dlr + Dlo )/4 ) \* ( Thk - Can ) \* 0.7 \* Sn  
= (3.1416 \* 373.0) \* ( 16.0 - 0.0 ) \* 0.7 \* 138  
= 1811002. N

Tension, Pad Groove Weld [Tpgw]:  
= ( π/2 ) \* Dlo \* Wgpn \* 0.74 \* Seg  
= (3.1416/2 ) \* 762.0 \* 14.0 \* 0.74 \* 138  
= 1711105. N

Tension, Shell Groove Weld [Tngw]:  
= (π/2) \* Dlo \* ( T - cas ) \* 0.74 \* Sng



$$= (3.1416/2.0) \cdot 762.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 138 \\ = 1588884. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1911713 + 1811002) = 3722715 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (1133029 + 1711105 + 1588884 + 0) = 4433019 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1911713 + 1588884 + 0) = 3500597 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                            |                   |
|----------------------|----------------------------|-------------------|
| Path 1-1 = 3722715 N | , must exceed W = 854218 N | or W1 = 1506626 N |
| Path 2-2 = 4433018 N | , must exceed W = 854218 N | or W2 = 215631 N  |
| Path 3-3 = 3500597 N | , must exceed W = 854218 N | or W3 = 1564028 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.6818 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 438.6817 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 2.145 %

*This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 49.35 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 730.00 mm*

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**Input, Nozzle Desc: A7 30"**

**From: 190**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                      |     |           |    |
|--------------------------------------|-----|-----------|----|
| Inside Diameter of Cylindrical Shell | D   | 5906.00   | mm |
| Design Length of Section             | L   | 2000.0001 | mm |
| Shell Finished (Minimum) Thickness   | t   | 14.0000   | mm |
| Shell Internal Corrosion Allowance   | cas | 1.0000    | mm |
| Shell External Corrosion Allowance   | co  | 0.0000    | mm |

|                                   |  |          |    |
|-----------------------------------|--|----------|----|
| Distance from Bottom/Left Tangent |  | 24550.00 | mm |
|-----------------------------------|--|----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 0.00    | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 16.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

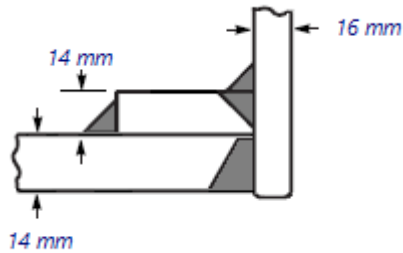
|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 14.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 14.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                               |      |           |     |
|-----------------------------------------------|------|-----------|-----|
| Pad Material                                  |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature           | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient               | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface          | Dp   | 1500.0000 | mm  |
| Thickness of Pad                              | te   | 14.0000   | mm  |
| Weld leg size between Pad and Shell           | Wp   | 12.0000   | mm  |
| Groove weld depth between Pad and Nozzle      | Wgpn | 14.0000   | mm  |
| Reinforcing Pad Width                         |      | 369.0000  | mm  |
| This is a Manway, Coupling or Access Opening. |      |           |     |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 150    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).





### Reinforcement CALCULATION, Description: A7 30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 30.000 in. |
| Actual Thickness Used in Calculation        | 0.630 in.  |

### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 0.5 \cdot 2954.0 / (138 \cdot 1.0 - 0.6 \cdot 0.5)$   
 $= 10.6895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 1.3737 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1460.0000 mm |
| Parallel to Vessel Wall, opening length                | d  | 730.0000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 32.5000 mm   |

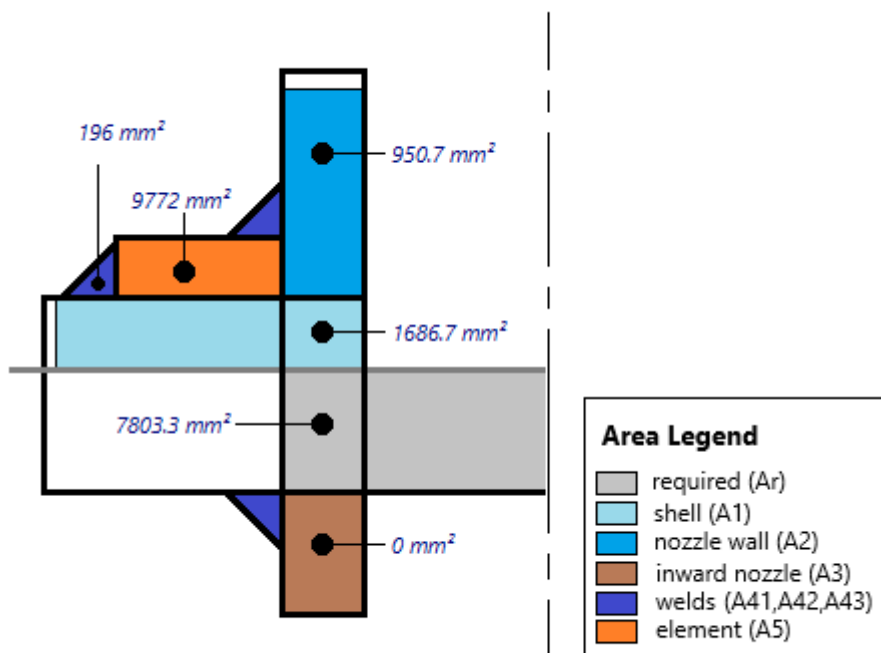
The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External  | Mapnc |
|--------------------------|-------------|-----------|-----------|-------|
| Area Required            | Ar          | 7803.346  | 4421.450  | NA    |
| Area in Shell            | A1          | 1686.655  | 647.100   | NA    |
| Area in Nozzle Wall      | A2          | 950.709   | 918.087   | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000     | NA    |
| Area in Welds            | A41+A42+A43 | 196.000   | 196.000   | NA    |
| Area in Element          | A5          | 9772.001  | 9772.001  | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 12605.364 | 11533.188 | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

| SELECTION OF POSSIBLE REINFORCING PADS: | Diameter  | Thickness  |
|-----------------------------------------|-----------|------------|
| Based on given Pad Thickness:           | 1116.9987 | 14.0000 mm |
| Based on given Pad Diameter:            | 1500.0000 | 7.1203 mm  |
| Based on Shell or Nozzle Thickness:     | 1116.9987 | 14.0000 mm |

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (730.0 \cdot 10.6895 \cdot 1.0 + 2 \cdot 16.0 \cdot 10.6895 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 7803.346 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 730.0(1.0 \cdot 13.0 - 1.0 \cdot 10.69) - 2 \cdot 16.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.6895) \cdot (1 - 1.0)$   
 $= 1686.655 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 32.5) \cdot (16.0 - 1.37) \cdot 1.0$



= 950.709 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$   
=  $14.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$   
= 196.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$   
=  $(1460.0 - 762.0) \cdot 14.0 \cdot 1.0$   
= 9772.001 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 1.8756$  mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

=  $t_a$   
= 1.8756 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 16.0$ ,  $t_r = 1.374$ ,  $c = 0.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.086$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C  
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 10.69$ ,  $c = 1.0$  mm,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.822$ , Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -29 °C



Gov. MDMT of the Reinforcement Pad : -29 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 0.50/1.96 = 0.254

Weld Size Calculations, Description: A7 30"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm  
Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

**Results Per UW-16.1:**

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 9.8980 = 0.7 * Wo mm |
| Pad Weld    | 6.5000 = 0.5*TminPad  | 8.4840 = 0.7 * Wp mm |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
= max( 0, (A-A1+2\*tn\*fr1\*(E1\*t-tr))Sv)  
= max( 0, (7803.3462 - 1686.6548 + 2 \* 16.0 \* 1.0 \*  
(1.0 \* 13.0 - 10.6895 ) )138)  
= 854218.38 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 950.7087 + 9772.001 + 196.0 - 0.0 \* 1.0 ) \* 138  
= 1506626.50 N

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 \* tn \* t \* fr1)) \* Sv  
= ( 950.7087 + 0.0 + 196.0 + ( 416.0 ) ) \* 138  
= 215631.58 N

Weld Load [W3]:  
= (A2+A3+A4+A5+(2\*tn\*t\*fr1))\*S  
= ( 950.7087 + 0.0 + 196.0 + 9772.001 + ( 416.0 ) ) \* 138  
= 1564028.62 N

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 762.0 \* 14.0 \* 0.49 \* 138  
= 1133029. N

Shear, Pad Element Weld [Spew]:  
= (π/2) \* DP \* WP \* 0.49 \* SEW  
= ( 3.1416/2.0 ) \* 1500.0 \* 12.0 \* 0.49 \* 138  
= 1911713. N

Shear, Nozzle Wall [Snw]:  
= (π \* ( Dlr + Dlo )/4 ) \* ( Thk - Can ) \* 0.7 \* Sn  
= (3.1416 \* 373.0) \* ( 16.0 - 0.0 ) \* 0.7 \* 138  
= 1811002. N

Tension, Pad Groove Weld [Tpgw]:  
= ( π/2 ) \* Dlo \* Wgpn \* 0.74 \* Seg  
= (3.1416/2 ) \* 762.0 \* 14.0 \* 0.74 \* 138  
= 1711105. N

Tension, Shell Groove Weld [Tngw]:  
= (π/2) \* Dlo \* ( T - cas ) \* 0.74 \* Sng



$$= (3.1416/2.0) \cdot 762.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 138 \\ = 1588884. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1911713 + 1811002) = 3722715 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (1133029 + 1711105 + 1588884 + 0) = 4433019 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1911713 + 1588884 + 0) = 3500597 \text{ N}$$

#### Summary of Failure Path Calculations:

|                      |                            |                   |
|----------------------|----------------------------|-------------------|
| Path 1-1 = 3722715 N | , must exceed W = 854218 N | or W1 = 1506626 N |
| Path 2-2 = 4433018 N | , must exceed W = 854218 N | or W2 = 215631 N  |
| Path 3-3 = 3500597 N | , must exceed W = 854218 N | or W3 = 1564028 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.6818 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 438.6817 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 2.145 %

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 730.00 mm*

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**Input, Nozzle Desc: N7**

**From: 220**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                             |     |           |    |
|---------------------------------------------|-----|-----------|----|
| Inside Crown Radius of Torispherical Head   | L   | 5945.0000 | mm |
| Inside Knuckle Radius of Torispherical Head | r   | 356.7000  | mm |
| Head Finished (Minimum) Thickness           | t   | 20.0000   | mm |
| Head Internal Corrosion Allowance           | cas | 1.0000    | mm |
| Head External Corrosion Allowance           | co  | 0.0000    | mm |

|                               |    |        |    |
|-------------------------------|----|--------|----|
| Distance from Head Centerline | L1 | 0.0000 | mm |
|-------------------------------|----|--------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |           |     |
|---------------------------------|-----|-----------|-----|
| Material                        |     | SA-516 70 |     |
| Material UNS Number             |     | K02700    |     |
| Material Specification/Type     |     | Plate     |     |
| Allowable Stress at Temperature | Sn  | 138.00    | MPa |
| Allowable Stress At Ambient     | Sna | 138.00    | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 0.00    | deg |
| Diameter                          |  | 30.0000 | in. |

|                          |    |         |    |
|--------------------------|----|---------|----|
| Size and Thickness Basis |    | Actual  |    |
| Actual Thickness         | tn | 12.0000 | mm |

|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

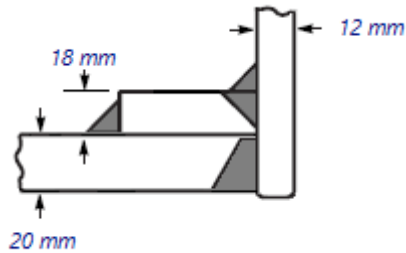
|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 400.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|                                          |      |           |     |
|------------------------------------------|------|-----------|-----|
| Pad Material                             |      | SA-516 70 |     |
| Pad Allowable Stress at Temperature      | Sp   | 138.00    | MPa |
| Pad Allowable Stress At Ambient          | Spa  | 138.00    | MPa |
| Diameter of Pad along vessel surface     | Dp   | 1062.0000 | mm  |
| Thickness of Pad                         | te   | 18.0000   | mm  |
| Weld leg size between Pad and Shell      | Wp   | 14.0000   | mm  |
| Groove weld depth between Pad and Nozzle | Wgpn | 18.0000   | mm  |
| Reinforcing Pad Width                    |      | 150.0000  | mm  |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 150    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).



#### Reinforcement CALCULATION, Description: N7

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                             |            |
|---------------------------------------------|------------|
| Actual Outside Diameter Used in Calculation | 30.000 in. |
| Actual Thickness Used in Calculation        | 0.472 in.  |

#### Messages regarding the Nozzle Input Data:

*For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.*

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Torispherical Head, tr [Int. Press]  
 $= P \cdot L \cdot M / (2 \cdot S_v \cdot E \cdot 0.2 \cdot P)$  Appendix 1-4 (d)  
 $= 0.5 \cdot 5946.0 \cdot 1.0 / (2 \cdot 138 \cdot 1.0 \cdot 0.2 \cdot 0.5)$   
 $= 10.7388 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 0.5 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.5)$   
 $= 1.3737 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8756 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

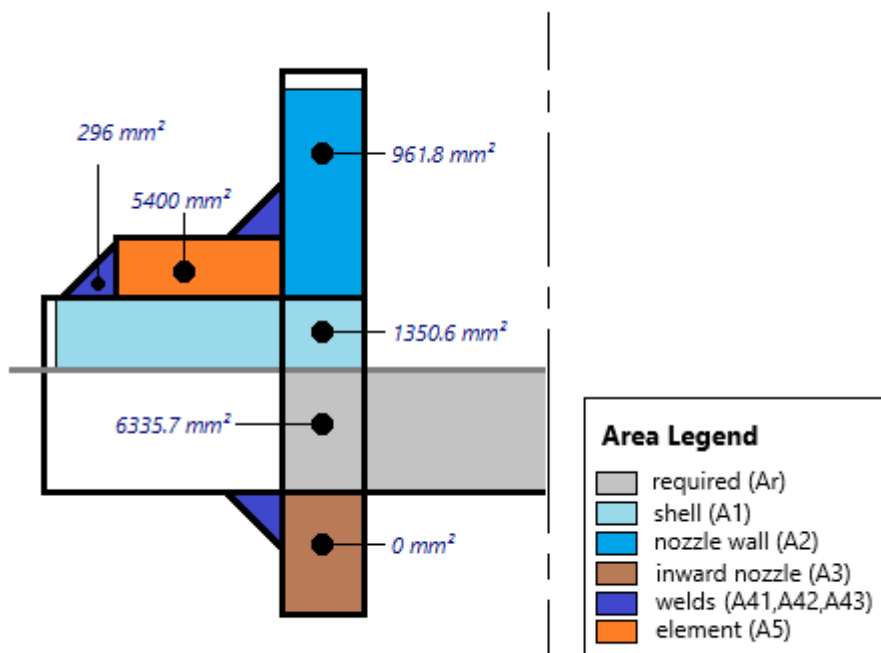
|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 1476.0000 mm |
| Parallel to Vessel Wall, opening length                | d  | 738.0000 mm  |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 47.5000 mm   |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (mm²)**

| AREA AVAILABLE, A1 to A5 |             | MAWP      | External | Mapnc |
|--------------------------|-------------|-----------|----------|-------|
| Area Required            | Ar          | 7925.263  | 6335.680 | NA    |
| Area in Shell            | A1          | 6096.736  | 1350.640 | NA    |
| Area in Nozzle Wall      | A2          | 1009.497  | 961.822  | NA    |
| Area in Inward Nozzle    | A3          | 0.000     | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 296.000   | 296.000  | NA    |
| Area in Element          | A5          | 5400.000  | 5400.000 | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 12802.233 | 8008.462 | NA    |

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.  
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness  
 Based on given Pad Thickness: 969.0677 18.0000 mm  
 Based on given Pad Diameter: 1062.0000 12.4241 mm  
 Based on Shell or Nozzle Thickness: 1072.6014 12.0000 mm

Area Required [A]:  
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$  per UG-37(d)  
 $= 0.5(738.0 \cdot 17.1699 \cdot 1 + 2 \cdot 12.0 \cdot 17.1699 \cdot 1(1 - 1.0))$   
 $= 6335.680 \text{ mm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 738.0(1.0 \cdot 19.0 - 1.0 \cdot 17.17) - 2 \cdot 12.0$   
 $(1.0 \cdot 19.0 - 1.0 \cdot 17.1699) \cdot (1 - 1.0)$   
 $= 1350.640 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 47.5) \cdot (12.0 - 1.88) \cdot 1.0$





= 961.822 mm<sup>2</sup>

Area Available in Welds [A41 + A42 + A43]:  
=  $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$   
=  $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$   
= 296.000 mm<sup>2</sup>

Area Available in Element [A5]:  
=  $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$   
=  $(1062.0 - 762.0) \cdot 18.0 \cdot 1.0$   
= 5400.000 mm<sup>2</sup>

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 1.8756 mm                      |
| Wall Thickness per UG16(b),                    | tr16b = 2.5000 mm                   |
| Wall Thickness, shell/head, internal pressure  | trb1 = 20.0000 mm                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 20.0000 mm |
| Wall Thickness, shell/head, external pressure  | trb2 = 4.9440 mm                    |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 4.9440 mm  |
| Wall Thickness per table UG-45                 | tb3 = 8.3312 mm                     |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 8.331, max( 20.0, 4.944 ) ]  
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 1.8756, 8.3312 )  
= 8.3312 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |         |           |             |        |
|------------|---------|-----------|-------------|--------|
| Sustained  | : 29.8, | Allowable | : 138.0 MPa | Passed |
| Expansion  | : 0.0,  | Allowable | : 315.2 MPa | Passed |
| Occasional | : 7.5,  | Allowable | : 183.5 MPa | Passed |
| Shear      | : 11.5, | Allowable | : 96.6 MPa  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 12.0, tr = 1.374, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.114$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -3 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, tr = 1.374, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.114$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -23 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, tr = 1.374, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.114$ , Temp. Reduction = 78 °C

|                                                |        |
|------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B | -23 °C |
|------------------------------------------------|--------|



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, c = 1.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.565$ , Temp. Reduction = 25 °C  
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 1.374, c = 0.0 mm, E\* = 1.0  
 Thickness Ratio =  $tr \cdot E^*/(tg - c) = 0.114$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C  
 Gov. MDMT of the Reinforcement Pad : -36 °C  
 Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $0.50/1.96 = 0.254$

#### Weld Size Calculations, Description: N7

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm  
 Intermediate Calc. for pad/shell Welds TminPad 18.0000 mm

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness     |
|-------------|-----------------------|----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm |
| Pad Weld    | 9.0000 = 0.5*TminPad  | 9.8980 = 0.7 * Wp mm |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (6335.6802 - 1350.6404 + 2 \cdot 12.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 17.1699)) \cdot 138)$   
 $= 693925.31 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (961.822 + 5400.0 + 296.0 - 0.0 \cdot 1.0) \cdot 138$   
 $= 918684.69 \text{ N}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (961.822 + 0.0 + 100.0 + (456.0)) \cdot 138$   
 $= 209437.83 \text{ N}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (961.822 + 0.0 + 296.0 + 5400.0 + (456.0)) \cdot 138$   
 $= 981606.19 \text{ N}$

#### Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 809306. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 1062.0 \cdot 14.0 \cdot 0.49 \cdot 138 \\ &= 1579075. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 375.0) \cdot (12.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 1365534. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 762.0 \cdot 18.0 \cdot 0.74 \cdot 138 \\ &= 2199992. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 762.0 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1955549. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (1579075 + 1365534) = 2944609 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (809307 + 2199993 + 1955549 + 0) = 4964848 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (S_{pew} + T_{ngw} + S_{inw}) \\ &= (1579075 + 1955549 + 0) = 3534624 \text{ N} \end{aligned}$$

#### Summary of Failure Path Calculations:

|                      |                            |                  |
|----------------------|----------------------------|------------------|
| Path 1-1 = 2944609 N | , must exceed W = 693925 N | or W1 = 918684 N |
| Path 2-2 = 4964848 N | , must exceed W = 693925 N | or W2 = 209437 N |
| Path 3-3 = 3534624 N | , must exceed W = 693925 N | or W3 = 981606 N |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 12.1803 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 432.1802 mm

#### Percent Elongation Calculations:

% Elongation per Table UG-79-1 ( $50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$ ) 1.600 %

#### Input Echo, WRC107/537 Item 1, Description: N7 :

|                                 |        |              |
|---------------------------------|--------|--------------|
| Diameter Basis for Vessel       | Vbasis | ID           |
| Cylindrical or Spherical Vessel | Cylsph | Spherical    |
| Internal Corrosion Allowance    | Cas    | 1.0000 mm    |
| Vessel Diameter                 | Dv     | 11890.000 mm |
| Vessel Thickness                | Tv     | 20.000 mm    |
| Design Temperature              | T1     | 150.0 C      |
| Vessel Material                 |        | SA-516 70    |
| Vessel UNS Number               |        | K02700       |



|                            |     |        |     |
|----------------------------|-----|--------|-----|
| Vessel Cold S.I. Allowable | Smc | 138.00 | MPa |
| Vessel Hot S.I. Allowable  | Smh | 138.00 | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |             |
|----------------------------------|--------|-------------|
| Attachment Type                  | Type   | Round       |
| WRC107 Attachment Classification | Holsol | Hollow      |
| Diameter Basis for Nozzle        | Nbasis | OD          |
| Corrosion Allowance for Nozzle   | Can    | 0.0000 mm   |
| Nozzle Diameter                  | Dn     | 762.000 mm  |
| Nozzle Thickness                 | Tn     | 12.000 mm   |
| Nozzle Material                  |        | SA-516 70   |
| Nozzle UNS Number                |        | K02700      |
| Nozzle Cold S.I. Allowable       | SNmc   | 138.00 MPa  |
| Nozzle Hot S.I. Allowable        | SNmh   | 138.00 MPa  |
| Thickness of Reinforcing Pad     | Tpad   | 18.000 mm   |
| Diameter of Reinforcing Pad      | Dpad   | 1062.000 mm |
| Design Internal Pressure         | Dp     | 0.498 MPa   |
| Include Pressure Thrust          |        | No          |

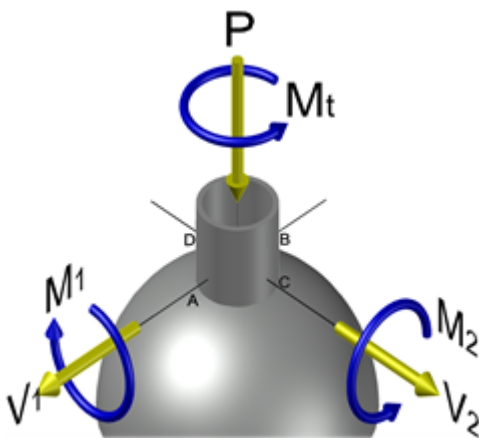
External Forces and Moments in WRC 107/537 Convention:

|                              |         |            |      |
|------------------------------|---------|------------|------|
| Radial Load (SUS)            | P       | -56249.9   | N    |
| Longitudinal Shear (SUS)     | (V1) V1 | 37500.0    | N    |
| Circumferential Shear (SUS)  | (Vc) V2 | 37500.0    | N    |
| Circumferential Moment (SUS) | (Mc) M1 | 99000272.0 | N-mm |
| Longitudinal Moment (SUS)    | (Ml) M2 | 78750032.0 | N-mm |
| Torsional Moment (SUS)       | Mt      | 78750032.0 | N-mm |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 762.0 + 2 \cdot 1.65 \cdot \sqrt{(5955.5 (20.0 - 1.0))}$$

$$= 1872.069 \text{ mm}$$



WRC 107 Stress Calculation for SUSTained loads:

|                        |         |            |      |
|------------------------|---------|------------|------|
| Radial Load            | P       | -56249.9   | N    |
| Circumferential Shear  | (VC) V2 | 37500.0    | N    |
| Longitudinal Shear     | (VL) V1 | 37500.0    | N    |
| Circumferential Moment | (MC) M1 | 99000272.0 | N-mm |
| Longitudinal Moment    | (ML) M2 | 78750032.0 | N-mm |
| Torsional Moment       | MT      | 78750032.0 | N-mm |

Dimensionless Param: U = 0.81 TAU = 31.25 RHO = 4.00 ( 3.08)

Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979         | Figure | Value   | Location  |
|------------------------------|--------|---------|-----------|
| N(x) * T / P                 | SP 7   | 0.02235 | (A,B,C,D) |
| M(x) / P                     | SP 7   | 0.01307 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / MC | SM 7   | 0.04354 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / MC     | SM 7   | 0.02758 | (A,B,C,D) |
| N(x) * T * SQRT(Rm * T) / ML | SM 7   | 0.04354 | (A,B,C,D) |
| M(x) * SQRT(Rm * T) / ML     | SM 7   | 0.02758 | (A,B,C,D) |
| N(y) * T / P                 | SP 7   | 0.25168 | (A,B,C,D) |
| M(y) / P                     | SP 7   | 0.03810 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / MC | SM 7   | 0.32500 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / MC     | SM 7   | 0.09504 | (A,B,C,D) |
| N(y) * T * SQRT(Rm * T) / ML | SM 7   | 0.32500 | (A,B,C,D) |
| M(y) * SQRT(Rm * T) / ML     | SM 7   | 0.09504 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |       |       |       |        |       |       |       |
|-----------------|------|----------------------------|-------|-------|-------|--------|-------|-------|-------|
|                 |      | Au                         | Al    | Bu    | Bl    | Cu     | Cl    | Du    | Dl    |
| Rad. Memb.      | P    | 0.9                        | 0.9   | 0.9   | 0.9   | 0.9    | 0.9   | 0.9   | 0.9   |
| Rad. Bend.      | P    | 3.2                        | -3.2  | 3.2   | -3.2  | 3.2    | -3.2  | 3.2   | -3.2  |
| Rad. Memb.      | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -6.7   | -6.7  | 6.7   | 6.7   |
| Rad. Bend.      | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -25.5  | 25.5  | 25.5  | -25.5 |
| Rad. Memb.      | ML   | -5.3                       | -5.3  | 5.3   | 5.3   | 0.0    | 0.0   | 0.0   | 0.0   |
| Rad. Bend.      | ML   | -20.3                      | 20.3  | 20.3  | -20.3 | 0.0    | 0.0   | 0.0   | 0.0   |
| Tot. Rad. Str.  |      | -21.4                      | 12.6  | 29.7  | -17.2 | -28.0  | 16.5  | 36.3  | -21.1 |
| Tang. Memb.     | P    | 10.3                       | 10.3  | 10.3  | 10.3  | 10.3   | 10.3  | 10.3  | 10.3  |
| Tang. Bend.     | P    | 9.4                        | -9.4  | 9.4   | -9.4  | 9.4    | -9.4  | 9.4   | -9.4  |
| Tang. Memb.     | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -50.0  | -50.0 | 50.0  | 50.0  |
| Tang. Bend.     | MC   | 0.0                        | 0.0   | 0.0   | 0.0   | -87.8  | 87.8  | 87.8  | -87.8 |
| Tang. Memb.     | ML   | -39.8                      | -39.8 | 39.8  | 39.8  | 0.0    | 0.0   | 0.0   | 0.0   |
| Tang. Bend.     | ML   | -69.8                      | 69.8  | 69.8  | -69.8 | 0.0    | 0.0   | 0.0   | 0.0   |
| Tot. Tang. Str. |      | -89.9                      | 31.0  | 129.3 | -29.1 | -118.0 | 38.7  | 157.5 | -36.8 |
| Shear VC        |      | 0.8                        | 0.8   | -0.8  | -0.8  | 0.0    | 0.0   | 0.0   | 0.0   |
| Shear VL        |      | 0.0                        | 0.0   | 0.0   | 0.0   | -0.8   | -0.8  | 0.8   | 0.8   |
| Shear MT        |      | 2.3                        | 2.3   | 2.3   | 2.3   | 2.3    | 2.3   | 2.3   | 2.3   |
| Tot. Shear      |      | 3.2                        | 3.2   | 1.5   | 1.5   | 1.5    | 1.5   | 3.2   | 3.2   |
| Str. Int.       |      | 90.0                       | 31.5  | 129.3 | 29.3  | 118.1  | 38.8  | 157.6 | 37.4  |

Unitless Prm: U = 1.58 TAU = 0.00 ( 43.75) RHO = 0.00 ( 1.58)

Dimensionless Loads for Spherical Shells at Pad edge:

| Curves read for 1979 | Figure | Value   | Location  |
|----------------------|--------|---------|-----------|
| N(x) * T / P         | SR 2   | 0.03544 | (A,B,C,D) |
| M(x) / P             | SR 2   | 0.01722 | (A,B,C,D) |



$N(x) * T * \sqrt{R_m * T} / MC$  SR 3 0.03469 (A,B,C,D)  
 $M(x) * \sqrt{R_m * T} / MC$  SR 3 0.02500 (A,B,C,D)  
 $N(x) * T * \sqrt{R_m * T} / ML$  SR 3 0.03469 (A,B,C,D)  
 $M(x) * \sqrt{R_m * T} / ML$  SR 3 0.02500 (A,B,C,D)  
  
 $N(y) * T / P$  SR 2 0.01058 (A,B,C,D)  
 $M(y) / P$  SR 2 0.00516 (A,B,C,D)  
 $N(y) * T * \sqrt{R_m * T} / MC$  SR 3 0.01037 (A,B,C,D)  
 $M(y) * \sqrt{R_m * T} / MC$  SR 3 0.00741 (A,B,C,D)  
 $N(y) * T * \sqrt{R_m * T} / ML$  SR 3 0.01037 (A,B,C,D)  
 $M(y) * \sqrt{R_m * T} / ML$  SR 3 0.00741 (A,B,C,D)

Stress Concentration Factors:  $K_n = 1.00$ ,  $K_b = 1.00$

#### Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

| Type of Stress  | Load     | Stress Intensity Values at |       |       |       |        |       |       |        |
|-----------------|----------|----------------------------|-------|-------|-------|--------|-------|-------|--------|
|                 |          | Au                         | Al    | Bu    | Bl    | Cu     | Cl    | Du    | Dl     |
| Rad.            | Memb. P  | 5.5                        | 5.5   | 5.5   | 5.5   | 5.5    | 5.5   | 5.5   | 5.5    |
| Rad.            | Bend. P  | 16.1                       | -16.1 | 16.1  | -16.1 | 16.1   | -16.1 | 16.1  | -16.1  |
| Rad.            | Memb. MC | 0.0                        | 0.0   | 0.0   | 0.0   | -28.3  | -28.3 | 28.3  | 28.3   |
| Rad.            | Bend. MC | 0.0                        | 0.0   | 0.0   | 0.0   | -122.3 | 122.3 | 122.3 | -122.3 |
| Rad.            | Memb. ML | -22.5                      | -22.5 | 22.5  | 22.5  | 0.0    | 0.0   | 0.0   | 0.0    |
| Rad.            | Bend. ML | -97.3                      | 97.3  | 97.3  | -97.3 | 0.0    | 0.0   | 0.0   | 0.0    |
| Tot. Rad. Str.  |          | -98.1                      | 64.2  | 141.4 | -85.4 | -128.9 | 83.4  | 172.2 | -104.6 |
| Tang.           | Memb. P  | 1.6                        | 1.6   | 1.6   | 1.6   | 1.6    | 1.6   | 1.6   | 1.6    |
| Tang.           | Bend. P  | 4.8                        | -4.8  | 4.8   | -4.8  | 4.8    | -4.8  | 4.8   | -4.8   |
| Tang.           | Memb. MC | 0.0                        | 0.0   | 0.0   | 0.0   | -8.5   | -8.5  | 8.5   | 8.5    |
| Tang.           | Bend. MC | 0.0                        | 0.0   | 0.0   | 0.0   | -36.2  | 36.2  | 36.2  | -36.2  |
| Tang.           | Memb. ML | -6.7                       | -6.7  | 6.7   | 6.7   | 0.0    | 0.0   | 0.0   | 0.0    |
| Tang.           | Bend. ML | -28.8                      | 28.8  | 28.8  | -28.8 | 0.0    | 0.0   | 0.0   | 0.0    |
| Tot. Tang. Str. |          | -29.1                      | 18.9  | 42.0  | -25.3 | -38.2  | 24.6  | 51.2  | -31.0  |
| Shear VC        |          | 1.2                        | 1.2   | -1.2  | -1.2  | 0.0    | 0.0   | 0.0   | 0.0    |
| Shear VL        |          | 0.0                        | 0.0   | 0.0   | 0.0   | -1.2   | -1.2  | 1.2   | 1.2    |
| Shear MT        |          | 2.3                        | 2.3   | 2.3   | 2.3   | 2.3    | 2.3   | 2.3   | 2.3    |
| Tot. Shear      |          | 3.5                        | 3.5   | 1.2   | 1.2   | 1.2    | 1.2   | 3.5   | 3.5    |
| Str. Int.       |          | 98.3                       | 64.5  | 141.4 | 85.4  | 128.9  | 83.4  | 172.3 | 104.7  |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (MPa)

| Type of Stress | Load     | Stress Intensity Values at |       |      |       |       |       |       |       |
|----------------|----------|----------------------------|-------|------|-------|-------|-------|-------|-------|
|                |          | Au                         | Al    | Bu   | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad.           | Pm (SUS) | 40.0                       | 40.0  | 40.0 | 40.0  | 40.0  | 40.0  | 40.0  | 40.0  |
| Rad.           | Pl (SUS) | -4.4                       | -4.4  | 6.2  | 6.2   | -5.8  | -5.8  | 7.6   | 7.6   |
| Rad.           | Q (SUS)  | -17.0                      | 17.0  | 23.5 | -23.5 | -22.2 | 22.2  | 28.7  | -28.7 |
| Long.          | Pm (SUS) | 40.0                       | 40.0  | 40.0 | 40.0  | 40.0  | 40.0  | 40.0  | 40.0  |
| Long.          | Pl (SUS) | -29.4                      | -29.4 | 50.1 | 50.1  | -39.7 | -39.7 | 60.4  | 60.4  |
| Long.          | Q (SUS)  | -60.4                      | 60.4  | 79.2 | -79.2 | -78.4 | 78.4  | 97.1  | -97.1 |
| Shear          | Pm (SUS) | 0.0                        | 0.0   | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear          | Pl (SUS) | 0.8                        | 0.8   | -0.8 | -0.8  | -0.8  | -0.8  | 0.8   | 0.8   |
| Shear          | Q (SUS)  | 2.3                        | 2.3   | 2.3  | 2.3   | 2.3   | 2.3   | 2.3   | 2.3   |
| Pm (SUS)       |          | 40.0                       | 40.0  | 40.0 | 40.0  | 40.0  | 40.0  | 40.0  | 40.0  |
| Pm+Pl (SUS)    |          | 35.7                       | 35.7  | 90.2 | 90.2  | 34.3  | 34.3  | 100.4 | 100.4 |



|                 |      |      |       |      |      |      |       |      |
|-----------------|------|------|-------|------|------|------|-------|------|
| Pm+Pl+Q (Total) | 68.7 | 71.5 | 169.4 | 23.0 | 90.1 | 78.8 | 197.6 | 19.6 |
|-----------------|------|------|-------|------|------|------|-------|------|

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 40.04     | 138.00         | Passed |
| Pm+Pl (SUS)         | 100.41    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 197.62    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Reinforcing Pad Edge (MPa)

| Type of Stress  | Load     | Stress Intensity Values at |       |       |        |        |       |       |        |
|-----------------|----------|----------------------------|-------|-------|--------|--------|-------|-------|--------|
|                 |          | Au                         | Al    | Bu    | Bl     | Cu     | Cl    | Du    | Dl     |
| Rad.            | Pm (SUS) | 78.0                       | 78.0  | 78.0  | 78.0   | 78.0   | 78.0  | 78.0  | 78.0   |
| Rad.            | Pl (SUS) | -17.0                      | -17.0 | 28.0  | 28.0   | -22.8  | -22.8 | 33.8  | 33.8   |
| Rad.            | Q (SUS)  | -81.2                      | 81.2  | 113.4 | -113.4 | -106.2 | 106.2 | 138.4 | -138.4 |
| Long.           | Pm (SUS) | 78.0                       | 78.0  | 78.0  | 78.0   | 78.0   | 78.0  | 78.0  | 78.0   |
| Long.           | Pl (SUS) | -5.1                       | -5.1  | 8.4   | 8.4    | -6.8   | -6.8  | 10.1  | 10.1   |
| Long.           | Q (SUS)  | -24.0                      | 24.0  | 33.6  | -33.6  | -31.4  | 31.4  | 41.1  | -41.1  |
| Shear           | Pm (SUS) | 0.0                        | 0.0   | 0.0   | 0.0    | 0.0    | 0.0   | 0.0   | 0.0    |
| Shear           | Pl (SUS) | 1.2                        | 1.2   | -1.2  | -1.2   | -1.2   | -1.2  | 1.2   | 1.2    |
| Shear           | Q (SUS)  | 2.3                        | 2.3   | 2.3   | 2.3    | 2.3    | 2.3   | 2.3   | 2.3    |
| Pm (SUS)        |          | 78.0                       | 78.0  | 78.0  | 78.0   | 78.0   | 78.0  | 78.0  | 78.0   |
| Pm+Pl (SUS)     |          | 73.0                       | 73.0  | 106.1 | 106.1  | 71.3   | 71.3  | 111.8 | 111.8  |
| Pm+Pl+Q (Total) |          | 69.4                       | 142.4 | 219.4 | 60.1   | 90.7   | 161.4 | 250.2 | 74.0   |

#### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 77.97     | 138.00         | Passed |
| Pm+Pl (SUS)         | 111.82    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 250.24    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: N10**

**From: 220**

|                                         |        |        |     |
|-----------------------------------------|--------|--------|-----|
| Pressure for Reinforcement Calculations | P      | 0.4983 | MPa |
| Temperature for Internal Pressure       | Temp   | 150    | °C  |
| Design External Pressure                | Pext   | 0.10   | MPa |
| Temperature for External Pressure       | Tempex | 150    | °C  |

|                                        |     |           |     |
|----------------------------------------|-----|-----------|-----|
| Parent Material                        |     | SA-516 70 |     |
| Parent Allowable Stress at Temperature | Sv  | 138.00    | MPa |
| Parent Allowable Stress At Ambient     | Sva | 138.00    | MPa |

|                                             |     |           |    |
|---------------------------------------------|-----|-----------|----|
| Inside Crown Radius of Torispherical Head   | L   | 5945.0000 | mm |
| Inside Knuckle Radius of Torispherical Head | r   | 356.7000  | mm |
| Head Finished (Minimum) Thickness           | t   | 20.0000   | mm |
| Head Internal Corrosion Allowance           | cas | 1.0000    | mm |
| Head External Corrosion Allowance           | co  | 0.0000    | mm |

|                               |    |           |    |
|-------------------------------|----|-----------|----|
| Distance from Head Centerline | L1 | 1500.0000 | mm |
|-------------------------------|----|-----------|----|

|                                               |  |       |    |
|-----------------------------------------------|--|-------|----|
| User Entered Minimum Design Metal Temperature |  | -1.00 | °C |
|-----------------------------------------------|--|-------|----|

**Type of Element Connected to the Parent : Nozzle**

|                                 |     |            |     |
|---------------------------------|-----|------------|-----|
| Material                        |     | SA-106 B   |     |
| Material UNS Number             |     | K03006     |     |
| Material Specification/Type     |     | Smls. pipe |     |
| Allowable Stress at Temperature | Sn  | 118.00     | MPa |
| Allowable Stress At Ambient     | Sna | 118.00     | MPa |

|                                   |  |         |     |
|-----------------------------------|--|---------|-----|
| Diameter Basis (for tr calc only) |  | Outside |     |
| Layout Angle                      |  | 180.00  | deg |
| Diameter                          |  | 2.0000  | in. |

|                          |  |         |  |
|--------------------------|--|---------|--|
| Size and Thickness Basis |  | Nominal |  |
| Nominal Thickness        |  | STD     |  |

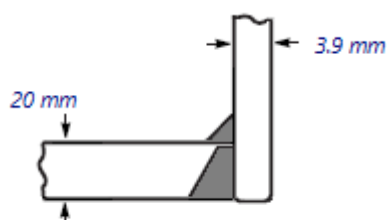
|                 |  |                  |  |
|-----------------|--|------------------|--|
| Flange Material |  | SA-105           |  |
| Flange Type     |  | Weld Neck Flange |  |

|                                          |     |        |    |
|------------------------------------------|-----|--------|----|
| Corrosion Allowance                      | can | 0.0000 | mm |
| Joint Efficiency of Shell Seam at Nozzle | E1  | 1.00   |    |
| Joint Efficiency of Nozzle Neck          | En  | 1.00   |    |

|                                             |      |          |    |
|---------------------------------------------|------|----------|----|
| Outside Projection                          | ho   | 200.0000 | mm |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000  | mm |
| Groove weld depth between Nozzle and Vessel | Wgnv | 16.0000  | mm |
| Inside Projection                           | h    | 0.0000   | mm |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000   | mm |

|               |  |        |  |
|---------------|--|--------|--|
| Flange Class  |  | 600    |  |
| Flange Grade  |  | GR 1.1 |  |
| Flange Series |  | A      |  |

The Pressure Design option was MAWP + static head (bottom of element).



Note : Checking Nozzle in the Meridional direction.





### Reinforcement CALCULATION, Description: N10

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.  
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Torispherical Head, tr [Int. Press]  
=  $P \cdot L \cdot M / (2 \cdot S_v \cdot E \cdot 0.2 \cdot P)$  Appendix 1-4 (d)  
=  $0.5 \cdot 5946.0 \cdot 1.0 / (2 \cdot 138 \cdot 1.0 \cdot 0.2 \cdot 0.5)$   
= 10.7388 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.5 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
= 0.1272 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3186 mm

### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |    |
|-------------------------------------------------|------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 108.5144 | mm |
| Parallel to Vessel Wall, opening length         | d    | 54.2572  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 9.7790   | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: N10.  
This calculation is valid for nozzles that meet all the requirements of  
paragraph UG-36. Please check the Code carefully, especially for nozzles  
that are not isolated or do not meet Code spacing requirements. To force  
the computation of areas for small nozzles go to Tools->Configuration  
and check the box to force the UG-37 small nozzle area calculation.*

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

#### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 3.423, tr = 0.127, c = 0.0 mm, E\* = 1.0  
Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.037$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating =  $0.50 / 10.20 = 0.049$

### Weld Size Calculations, Description: N10

|                                           |      |        |    |
|-------------------------------------------|------|--------|----|
| Intermediate Calc. for nozzle/shell Welds | Tmin | 3.9116 | mm |
|-------------------------------------------|------|--------|----|

### Results Per UW-16.1:

|             | Required Thickness           | Actual Thickness            |
|-------------|------------------------------|-----------------------------|
| Nozzle Weld | $2.7381 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o$ mm |



Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

#### Reinforcement CALCULATION, Description: N10

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.  
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Torispherical Head, tr [Int. Press]  
=  $P \cdot L \cdot M / (2 \cdot S_v \cdot E \cdot 0.2 \cdot P)$  Appendix 1-4 (d)  
=  $0.5 \cdot 5946.0 \cdot 1.0 / (2 \cdot 138 \cdot 1.0 \cdot 0.2 \cdot 0.5)$   
= 10.7388 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
=  $0.5 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$   
= 0.1272 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3186 mm

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |    |
|-------------------------------------------------|------|----------|----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 105.0036 | mm |
| Parallel to Vessel Wall, opening length         | d    | 52.5018  | mm |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 9.7790   | mm |

*Taking a UG-36(c)(3)(a) exemption for nozzle: N10.  
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                        |           |    |
|------------------------------------------------|------------------------|-----------|----|
| Wall Thickness for Internal/External pressures | ta                     | = 0.3186  | mm |
| Wall Thickness per UG16(b),                    | tr16b                  | = 2.5000  | mm |
| Wall Thickness, shell/head, internal pressure  | trb1                   | = 20.0000 | mm |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) | = 20.0000 | mm |
| Wall Thickness, shell/head, external pressure  | trb2                   | = 4.9440  | mm |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) | = 4.9440  | mm |
| Wall Thickness per table UG-45                 | tb3                    | = 3.4200  | mm |

Determine Nozzle Thickness candidate [tb]:  
= min[ tb3, max( tb1, tb2 ) ]  
= min[ 3.42, max( 20.0, 4.944 ) ]  
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
= max( ta, tb )  
= max( 0.3186, 3.42 )  
= 3.4200 mm

Available Nozzle Neck Thickness =  $0.875 \cdot 3.912 = 3.423$  mm -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |       |     |        |
|------------|---|-------|-----------|---|-------|-----|--------|
| Sustained  | : | 72.0, | Allowable | : | 118.0 | MPa | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 223.0 | MPa | Passed |
| Occasional | : | 1.6,  | Allowable | : | 156.9 | MPa | Passed |
| Shear      | : | 48.2, | Allowable | : | 82.6  | MPa | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*



Weld Size Calculations, Description: N10

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

**Results Per UW-16.1:**

Required Thickness Actual Thickness  
 Nozzle Weld 2.7381 = 0.7 \* tmin. 7.0700 = 0.7 \* Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 7.9208 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.5824 mm

**Warning:**

*The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.*

**Input Echo, WRC107/537 Item 1, Description: N10 :**

|                                 |        |           |     |
|---------------------------------|--------|-----------|-----|
| Diameter Basis for Vessel       | Vbasis | ID        |     |
| Cylindrical or Spherical Vessel | Cylsph | Spherical |     |
| Internal Corrosion Allowance    | Cas    | 1.0000    | mm  |
| Vessel Diameter                 | Dv     | 11890.000 | mm  |
| Vessel Thickness                | Tv     | 20.000    | mm  |
| Design Temperature              | T1     | 150.0     | C   |
| Vessel Material                 |        | SA-516 70 |     |
| Vessel UNS Number               |        | K02700    |     |
| Vessel Cold S.I. Allowable      | Smc    | 138.00    | MPa |
| Vessel Hot S.I. Allowable       | Smh    | 138.00    | MPa |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                  |        |          |     |
|----------------------------------|--------|----------|-----|
| Attachment Type                  | Type   | Round    |     |
| WRC107 Attachment Classification | Holsol | Hollow   |     |
| Diameter Basis for Nozzle        | Nbasis | OD       |     |
| Corrosion Allowance for Nozzle   | Can    | 0.0000   | mm  |
| Nozzle Diameter                  | Dn     | 60.325   | mm  |
| Nozzle Thickness                 | Tn     | 3.912    | mm  |
| Nozzle Material                  |        | SA-106 B |     |
| Nozzle UNS Number                |        | K03006   |     |
| Nozzle Cold S.I. Allowable       | SNmc   | 118.00   | MPa |
| Nozzle Hot S.I. Allowable        | SNmh   | 118.00   | MPa |
| Design Internal Pressure         | Dp     | 0.498    | MPa |
| Include Pressure Thrust          |        | No       |     |

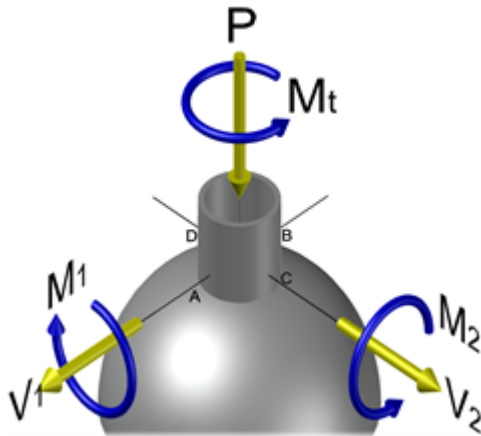
**External Forces and Moments in WRC 107/537 Convention:**

|                              |         |          |      |
|------------------------------|---------|----------|------|
| Radial Load (SUS)            | P       | -4500.0  | N    |
| Longitudinal Shear (SUS)     | (V1) V1 | 3250.0   | N    |
| Circumferential Shear (SUS)  | (Vc) V2 | 3250.0   | N    |
| Circumferential Moment (SUS) | (Mc) M1 | 500000.0 | N-mm |
| Longitudinal Moment (SUS)    | (Ml) M2 | 500000.0 | N-mm |
| Torsional Moment (SUS)       | Mt      | 574999.3 | N-mm |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".*



**Error/Warning Messages for WRC 107/537 Analysis: 1**  
 Warning - The ratio Ro/Rm, 0.0051, must be > 0.0571 and < 0.571.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(5955.5 (20.0 - 1.0))}$   
 $= 1170.394 \text{ mm}$

WRC 107 Stress Calculation for SUSained loads:

|                        |         |          |      |
|------------------------|---------|----------|------|
| Radial Load            | P       | -4500.0  | N    |
| Circumferential Shear  | (VC) V2 | 3250.0   | N    |
| Longitudinal Shear     | (VL) V1 | 3250.0   | N    |
| Circumferential Moment | (MC) M1 | 500000.0 | N-mm |
| Longitudinal Moment    | (ML) M2 | 500000.0 | N-mm |
| Torsional Moment       | MT      | 574999.3 | N-mm |

Dimensionless Param: U = 0.09 TAU = 7.21 RHO = 4.00 ( 4.86)

#### Dimensionless Loads for Spherical Shells at Attachment Junction:

| Curves read for 1979                  | Figure | Value   | Location  |
|---------------------------------------|--------|---------|-----------|
| $N(x) * T / P$                        | SP 4   | 0.11032 | (A,B,C,D) |
| $M(x) / P$                            | SP 4   | 0.04854 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(Rm * T) / MC$ | SM 4   | 0.57944 | (A,B,C,D) |
| $M(x) * \text{SQRT}(Rm * T) / MC$     | SM 4   | 0.27905 | (A,B,C,D) |
| $N(x) * T * \text{SQRT}(Rm * T) / ML$ | SM 4   | 0.57944 | (A,B,C,D) |
| $M(x) * \text{SQRT}(Rm * T) / ML$     | SM 4   | 0.27905 | (A,B,C,D) |
| $N(y) * T / P$                        | SP 4   | 0.32540 | (A,B,C,D) |
| $M(y) / P$                            | SP 4   | 0.34578 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(Rm * T) / MC$ | SM 4   | 0.10259 | (A,B,C,D) |
| $M(y) * \text{SQRT}(Rm * T) / MC$     | SM 4   | 2.19722 | (A,B,C,D) |
| $N(y) * T * \text{SQRT}(Rm * T) / ML$ | SM 4   | 0.10259 | (A,B,C,D) |
| $M(y) * \text{SQRT}(Rm * T) / ML$     | SM 4   | 2.19722 | (A,B,C,D) |

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (MPa)

| Type of Stress | Load     | Stress Intensity Values at |      |     |      |      |      |     |      |
|----------------|----------|----------------------------|------|-----|------|------|------|-----|------|
|                |          | Au                         | Al   | Bu  | Bl   | Cu   | Cl   | Du  | Dl   |
| Rad.           | Memb. P  | 1.4                        | 1.4  | 1.4 | 1.4  | 1.4  | 1.4  | 1.4 | 1.4  |
| Rad.           | Bend. P  | 3.6                        | -3.6 | 3.6 | -3.6 | 3.6  | -3.6 | 3.6 | -3.6 |
| Rad.           | Memb. MC | 0.0                        | 0.0  | 0.0 | 0.0  | -2.4 | -2.4 | 2.4 | 2.4  |



|                 |       |       |      |       |       |       |      |       |
|-----------------|-------|-------|------|-------|-------|-------|------|-------|
| Rad. Bend. MC   | 0.0   | 0.0   | 0.0  | 0.0   | -6.9  | 6.9   | 6.9  | -6.9  |
| Rad. Memb. ML   | -2.4  | -2.4  | 2.4  | 2.4   | 0.0   | 0.0   | 0.0  | 0.0   |
| Rad. Bend. ML   | -6.9  | 6.9   | 6.9  | -6.9  | 0.0   | 0.0   | 0.0  | 0.0   |
| Tot. Rad. Str.  | -4.3  | 2.3   | 14.3 | -6.8  | -4.3  | 2.3   | 14.3 | -6.8  |
| Tang. Memb. P   | 4.1   | 4.1   | 4.1  | 4.1   | 4.1   | 4.1   | 4.1  | 4.1   |
| Tang. Bend. P   | 25.9  | -25.9 | 25.9 | -25.9 | 25.9  | -25.9 | 25.9 | -25.9 |
| Tang. Memb. MC  | 0.0   | 0.0   | 0.0  | 0.0   | -0.4  | -0.4  | 0.4  | 0.4   |
| Tang. Bend. MC  | 0.0   | 0.0   | 0.0  | 0.0   | -54.3 | 54.3  | 54.3 | -54.3 |
| Tang. Memb. ML  | -0.4  | -0.4  | 0.4  | 0.4   | 0.0   | 0.0   | 0.0  | 0.0   |
| Tang. Bend. ML  | -54.3 | 54.3  | 54.3 | -54.3 | 0.0   | 0.0   | 0.0  | 0.0   |
| Tot. Tang. Str. | -24.8 | 32.0  | 84.6 | -75.6 | -24.8 | 32.0  | 84.6 | -75.6 |
| Shear VC        | 1.8   | 1.8   | -1.8 | -1.8  | 0.0   | 0.0   | 0.0  | 0.0   |
| Shear VL        | 0.0   | 0.0   | 0.0  | 0.0   | -1.8  | -1.8  | 1.8  | 1.8   |
| Shear MT        | 5.3   | 5.3   | 5.3  | 5.3   | 5.3   | 5.3   | 5.3  | 5.3   |
| Tot. Shear      | 7.1   | 7.1   | 3.5  | 3.5   | 3.5   | 3.5   | 7.1  | 7.1   |
| Str. Int.       | 27.0  | 33.6  | 84.8 | 75.8  | 25.3  | 32.4  | 85.3 | 76.4  |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (MPa)

| Type of Stress  | Load | Stress Intensity Values at |       |       |       |       |       |       |       |
|-----------------|------|----------------------------|-------|-------|-------|-------|-------|-------|-------|
|                 |      | Au                         | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Rad. Pm (SUS)   |      | 78.0                       | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  |
| Rad. Pl (SUS)   |      | -1.0                       | -1.0  | 3.8   | 3.8   | -1.0  | -1.0  | 3.8   | 3.8   |
| Rad. Q (SUS)    |      | -3.3                       | 3.3   | 10.5  | -10.5 | -3.3  | 3.3   | 10.5  | -10.5 |
| Long. Pm (SUS)  |      | 78.0                       | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  |
| Long. Pl (SUS)  |      | 3.6                        | 3.6   | 4.5   | 4.5   | 3.6   | 3.6   | 4.5   | 4.5   |
| Long. Q (SUS)   |      | -28.4                      | 28.4  | 80.1  | -80.1 | -28.4 | 28.4  | 80.1  | -80.1 |
| Shear Pm (SUS)  |      | 0.0                        | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| Shear Pl (SUS)  |      | 1.8                        | 1.8   | -1.8  | -1.8  | -1.8  | -1.8  | 1.8   | 1.8   |
| Shear Q (SUS)   |      | 5.3                        | 5.3   | 5.3   | 5.3   | 5.3   | 5.3   | 5.3   | 5.3   |
| Pm (SUS)        |      | 78.0                       | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  | 78.0  |
| Pm+Pl (SUS)     |      | 82.2                       | 82.2  | 83.9  | 83.9  | 82.2  | 82.2  | 83.9  | 83.9  |
| Pm+Pl+Q (Total) |      | 75.9                       | 111.6 | 162.7 | 71.4  | 74.3  | 110.4 | 163.3 | 71.9  |

##### Vessel Stress Summation Comparison (MPa):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 77.97     | 138.00         | Passed |
| Pm+Pl (SUS)         | 83.93     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 163.28    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Noted issues for the current model:**

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**\*\* Warning: One or more nozzles do not meet proper Spacing Requirements!**

**Please review all reports carefully!**

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